A History of Modern Philosophy

Introduction, Third Book, Ninth Book and Tenth Book

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translated from *Den nyere Filosofis Historie*, 2nd edition (1903)

1894–1895 / 1903

Translator's note. This volume contains four parts of Høffding's two-volume history, in the order of the work. From the first volume: the Introduction, in which he sets out his conception of philosophy and of what a history of philosophy is for, and the Third Book, on the great systems of the seventeenth century (Descartes and the Cartesians, Gassendi, Hobbes, Spinoza, Leibniz and Wolff). From the second volume: the Ninth Book, on positivism (French philosophy after the Revolution and Comte; English philosophy before 1840 and John Stuart Mill; Darwin and Spencer), and the Tenth Book, on the philosophy of Germany from 1850 to 1880 — Robert Mayer and the conservation of energy, and the idealist and critical reactions to materialism. The Third and Ninth Books carry the history of the mechanical conception of nature and of the principles of conservation, of motion, of force and of energy, from Descartes and Leibniz to Spencer, which the Tenth Book presupposes. Høffding's notes to the Third and Ninth Books stand, as in the original, at the end of each volume; those to the Tenth Book are given as footnotes. The numbers in the margin are the pages of the second Danish edition (1903–04); each volume is paged separately. The translation is made from the diplomatic transcription of that edition published in this collection, which keeps the printer's errors as printed; the English gives the intended sense. The Introduction and the Tenth Book were translated first, from the e-book text of the same edition, and then checked against the transcription. The Third and Ninth Books, with their notes, were translated from the transcription with Claude (an AI model) in fifteen parts, working from one brief and one glossary of terms; each part was checked by script against the Danish, printed page by printed page (page numbers, notes, paragraphs, emphasis, Greek), and where the Danish was corrected afterwards the English was mended to follow it. The remaining Books will follow by the same method; until then they are available in English only in B. E. Meyer's translation of 1900, made from the German edition.

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Introduction

It will be natural that, when a new presentation of an important period in the history of philosophy appears, the question should be asked what conception of philosophy the author takes as his ground, and what significance and value he ascribes to the history of philosophy. My answer is that the studies whose provisional conclusion is marked by the publication of this book had precisely the aim of orienting me with respect to what philosophy really is. As we come to know a person through his biography, so we must be able to come to know a science through its history. And it will be all the more natural to try this road, since experience constantly shows that in the domain of philosophy conflicting ways of seeing assert themselves, so that there is no presentation of philosophy to which one could point as giving an exhaustive concept of it. It will then be natural here — as in the domain of religion — to apply the comparative method. The history of philosophy treats the attempts which individual thinkers have made to discuss the ultimate problems of knowledge and of life. The most important result will be to get clear about which problems philosophy occupies itself with, how those problems are posed at different times, and by what the different ways of posing them, and of attempting to solve them, are determined in detail. If such a result can be approximately reached, the study of the history of philosophy will be of no slight importance for continued philosophizing. *[XI]*

For my own part I have been strengthened, by the study of the history of modern philosophy, in the assumption I had arrived at by another road: that philosophical inquiry turns upon four principal problems. These four problems are now to be briefly characterized. A more searching discussion and comparison of them I have given in the treatise *Filosofiske Problemer* (University Programme 1902). *[XII]*

1. *The problem of knowledge* (the *logical* problem). However different the several sciences may be in respect of subject and of method, they all nevertheless work with human thinking. Every time they form a concept, utter a judgment or draw an inference, they presuppose the general forms and principles of thinking. Here, then, there shows

itself the possibility of a peculiar discipline, one which investigates the forms in which thinking moves and the principles it must by its nature take as its ground, whatever the objects it may concern. This discipline, *formal logic*, treats however only one part of the problem of knowledge. Those forms and principles do not lead beyond thinking itself, but only make it possible for thinking to be in agreement with itself, to be consistent. Every time they are applied to given phenomena, which our thinking has not itself constructed but must take as they come, the question arises with what right that application takes place — with what right we assume that not only our thinking, but existence itself, which expresses itself in the given phenomena, is consistent, agrees with itself. Herewith shows itself the possibility of a discipline which investigates the conditions of a knowledge of existence, and the limits of such knowledge. This discipline is *epistemology*.

2. *The problem of existence* (the *cosmological* problem) arises when it is asked how the existence of which we are members must be supposed to be constituted, when we draw the ultimate consequences of all that we know, or can conjecture with the help of the most probable hypotheses. We call this problem the cosmological (from *kosmos*, world, and *logos*, doctrine), because it leads to a discussion of the possibilities that present themselves to thought when it seeks to work up what is given in experience into a general world-view, or when by speculation's bold road it attempts to construct one. The
[XIII] different philosophical systems are attempts in this direction — attempts whose value depends on how significant and how comprehensive are the experiences they take as their ground, and how much consistency and power of combination is displayed in their constructions.

3. *The problem of valuation* (the *ethical-religious* problem) arises through this, that we do not merely stand to existence in an intuiting and understanding relation, but also have our feeling set in motion by it, so that we utter judgments which ascribe value to it or deny it value. Of particular importance are the judgments we utter upon human actions, our own and others'. All such appraisal rests — like all knowledge and understanding — upon certain presuppositions, which it is a matter of exhibiting and determining. This task belongs under the ethical problem. When the valuation concerns not merely human actions and institutions but existence, life as a whole, the religious problem arises, which leads especially to a discussion of the relation between the ethical ideals and actual existence, so that it becomes a combination of the cosmological and the ethical problem.

4. *The problem of consciousness* (the *psychological* problem). For all three of the problems indicated it shows itself that their treatment presupposes an empirical acquaintance with human conscious life. Psychology presents the *actual* development

of human knowledge, which must be known before one can discuss the conditions of knowledge's *validity*. And since the relation between the spiritual and the material is a principal point within the problem of existence, psychology is presupposed in cosmology as well. As regards the problem of valuation, finally, psychology discusses partly the constitution of the very feelings that lead us to make valuations, partly the possibilities which actual conscious life presents for a further development in the direction valuation demands. In consequence of this close relation to the philosophical problems, psychology is itself to be reckoned a part of philosophy; and one will therefore also, conversely, quite naturally come in psychology to touch upon philosophical problems. And even if one were to declare all three of the preceding problems insoluble, or arisen through misunderstanding, the psychological problem — the question of the nature and the laws *[XIV]* of conscious life — would still remain as philosophy's last stronghold.

— If it is asked what factors will, in the nature of the case, come to influence the treatment and the answering of these problems, the first to be named must be the personality of the one who philosophizes. The problems named have this in common, that they lie at the boundary of our knowledge, where the exact methods fail; it cannot then be otherwise than that the inquirer's personality must come to determine his train of thought, without his needing to be conscious of it. The "personal equation" must be of greater importance in philosophy than in other scientific domains. The historical, comparative method therefore becomes of particular importance here; by its help the personal element can most easily be exhibited. That personal element is not always something to be pushed aside; its presence is often a condition of a problem's being able to come to be at all. There are thoughts which can come to be only upon a determinate psychological soil. — It will, in the second place, depend upon which observations are taken as the ground. Here the development of natural science acquires especially great importance for modern philosophy. As will appear, the decisive problems of modern philosophy are determined by the very fact that modern natural science came into being. But to this there come as well — especially as regards the problem of valuation — historical conditions, intellectual movements in other domains. — Finally, the posing of problems and their solution are determined by the consistency with which the starting points once seized are held fast and carried through.

Both the characterization and the criticism of every philosophical attempt will always lead back to these three points. Among them I lay most weight, in the presentation that follows, upon the first two. An inconsistency in a great thinker is often precisely a natural consequence of *several* trains of thought having opened themselves to his genius,

without his having been able to pursue them far enough to discover their mutual conflict. And it may be of the greatest importance that these different trains of thought be brought out. Best of all, of course, is when profundity and consistency are united.

[XV] An attempt at posing or at solving a problem may have a twofold interest. In the first place it may be regarded as a symptom, as the historical expression of intellectual currents. From this side the history of philosophy is a part of general cultural history. In the second place it may be examined with respect to the real, definitive clarity that has been reached by it. This consideration is more specifically philosophical and technical. The two sides from which the philosophical phenomena can be regarded will of course, in the case of different phenomena, stand in very different relations to one another. Sometimes the cultural-historical interest may even stand in inverse relation to the purely philosophical.

If I am briefly to indicate wherein my presentation of the history of modern philosophy differs from its predecessor in our literature — Brøchner's work of 1873–74 — I would name first the particular weight I lay upon the personal factor, upon the relation to empirical science, and upon the cultural-historical significance of the philosophical phenomena. Next I would point to my endeavor to give the posing of problems precedence over their solution. Solutions may die, and the problems live on; otherwise philosophy would not have had so long a life as it has had. Finally it goes without saying that a renewed study of the sources, as well as the use of the abundant literature of recent years, has set many phenomena in a new light. But in spite of all this it has been a dear thought to me, in this work, to be able to continue Brøchner's labor in the field of the history of philosophy; and I can only hope that my book may not be unworthy to be consecrated to the memory of that noble inquirer, whose admiration for the heroes of thought was at once so deep and so understanding.

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Third Book

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The Great Systems.

After the time of the new ideas and discoveries there comes a time of attempts to order and systematize, to lead the many thoughts and facts back to simple and fixed fundamental thoughts. These attempts were made with great confidence that the right foundation had been found. In place of analysis, construction was put. This had the great importance for thinking that it could now become fully clear what lay in the points of view which the Renaissance and the new natural science had brought forward. They were established with a dogmatic assurance of which the minds of the preceding time had not thought, because they did not give themselves so clear an account of the individual principles and their range. What had earlier hovered more or less obscurely before the mind was now formulated with express consciousness. And out of the presuppositions so formulated systems were built up, each of which laid claim to take the place of the old scholastic system, which was only now wholly overthrown. From scholasticism, however, was inherited the endeavor to give knowledge an absolute conclusion, to bring thought to rest in a principle which itself should contain no problem. Men wished to build as high as the building that had been torn down had reached. The natural need, which always stirs more or less strongly in the human mind, to bring coherence into all representations that are taken to be valid, expressed itself here so vividly and energetically, and at the same time in so brilliant a manner, as in no other period of the history of modern philosophy. *[203]* This was connected with the fact that tasks had been set which had been set neither before nor later with the same novelty and sharpness. It was the new picture of the world and the new science that had to be worked together with the rest of the spiritual content that stood firm for consciousness. It therefore had to be the problem of existence that came to stand in the first rank. Bruno had treated it on the basis of the new picture of the world. But now the new mechanical explanation of nature had been added, and the great question of the relation between the material and the mental was thereby set

before thinking. The most important of the hypotheses that are possible concerning this relation were put forward by the systems of the seventeenth century with a clarity and energy that give these attempts of thought a lasting value. But other problems weave themselves together with this one. The question of the relation between God and the world even presses forward for a while in front of it. With it, in turn, the question of the unity or multiplicity of existence is connected. And finally it becomes an important problem whether the mechanical explanation of nature makes it possible to ascribe any real significance to the concept of purpose.

The problem of knowledge and the problem of valuation recede before the problem of existence, although they constantly exert their influence as more or less consciously motivating and impelling forces. In Descartes, the first in the series of the great systematizers, there is still, in the presentation as well, a clear endeavor to find the way to constructive thinking through analysis. In Hobbes and Spinoza construction overshadows analysis. In Leibniz the analytic tendency begins again to work its way forward, as a transition to the predominant importance the problem of knowledge and the problem of valuation were to acquire in the eighteenth century.

The conditions of cultural history in the seventeenth century present an analogy to the concluding, systematic tendency that at this time formed the main current in philosophy. There is one fundamental mood that pervades the different domains, politics, church and thinking. It is the time of the absolute sovereignty of the state. The state
[204] has freed itself from ecclesiastical tutelage and now demands absolute submission of individuals and of the smaller circles. The principle of sovereignty is even carried through, in its very sharpest forms, by the two thinkers who mark the high point of construction. The individual now seeks shelter and rest after the storms of the Renaissance and the age of the Reformation. Intellectual contemplation and mysticism draw many to them. A tendency to quietism makes itself felt. But in the philosophical systems these tendencies are satisfied through a thinking elaboration of the new ideas and discoveries. They thereby acquired the significance of experiments for testing the carrying power of important trains of thought.

1. René Descartes

a. Biography and Characterization

This man, the founder of modern philosophy, was born in Touraine on 31 March 1596, of a noble family. The sickly boy early showed unusual gifts, and his father used to

call him "my philosopher" on account of the many questions he had to ask. So that he might receive a careful upbringing, he was sent to the Jesuit college La Flèche, recently founded by Henry IV. He always remembered his Jesuit teachers later with gratitude, and it was a great sorrow to him that the Jesuits came forward as opponents of his philosophy. At La Flèche he learned natural philosophy and philosophy according to the scholastic system, but was most occupied with mathematics, and very early he seems to have busied himself with ideas that led him to his great mathematical discovery, the founding of analytic geometry, the application of algebra to geometry. He has himself described his development in the years of his youth in his epoch-making work *Discours de la méthode,* since it is at the same time the history of the genesis of his philosophy. On leaving school he felt dissatisfied, despite the much he had learned. Many facts he knew, and many fine thoughts had been imparted to him; above all he admired the stringent method of mathematics. But those facts and thoughts stood before him as disconnected fragments, and mathematics was to him only useless cobweb-spinning. He then laid *[205]* his studies on the shelf and threw himself into the worldly life of Paris. His theoretical interest, however, he could not wholly deny: among his papers was found a treatise on the art of fencing that dates from this time. Soon he grew weary of the thoughtless life and suddenly vanished from his friends, withdrawing to a solitary part of Paris in order to study in peace. It now became more and more his ideal to lead a retired life devoted to reflection and study. His motto became: "Happy he who lived in concealment!" (Bene vixit, qvi bene latuit!). His friends found him after the lapse of two years, and he was torn out of his solitude. Then he resolved to come to know "the great book of the world." Perhaps practical life, which puts all thoughts to the test, had taught men truths that learned speculation could not discover! And besides, he wished to try himself under the vicissitudes of life. He then went as a volunteer, at his own expense, into military service under Maurice of Orange, though he used every quieter time for studies, especially mathematical ones. From Holland he went to Germany, where the Thirty Years' War was about to break out. He joined the army the Elector of Bavaria was gathering against the rebellious Bohemians. While he lay in winter quarters (1619—20) at Neuburg on the Danube, a scientific breakthrough took place in him, as he found the general method that later guided him both in his philosophical and in his mathematical studies. In a memorandum from that time, which was found among his papers, he has even given a definite date for the emergence of the decisive thought: — "the 10th of November 1619, when I was filled with enthusiasm and found the foundation of a wonderful science!"[42]) He lived shut up in his room and gave himself up to thoughts that led him to his general

doctrine of method. It occurred to him that, as in general a work done by many in common will be more imperfect than one carried out by a single person, so our imperfect knowledge derives from the many different teachers who each instill their opinions in us, from the influence of the different drives, from the different conflicting judgments we hear from scholars and from practical people. Let us then, in order to remedy this
[206] imperfection, begin afresh, disregard what has been handed down, and slowly erect our building on one single foundation! The right method consists in accepting only what is thought clearly and distinctly, in resolving every difficulty into its individual parts, and in starting from what is simplest and most easily understood, so as then step by step to come to the more composite questions. It was the analytic method that stood before his inner sight in its whole generality. In the mathematical domain this method led him to the idea of a science that was more general than the individual mathematical sciences, since it investigates relations, proportions in general, no matter whether they obtain between figures or numbers or other things. It was a general theory of magnitude or of functions, of which analytic geometry was a special application. — His thoughts worked so eagerly with these ideas that he fell into an exalted state. He had remarkable dreams, and the next day he promised the Holy Virgin to make a pilgrimage to Loreto, so that she might give his thoughts success. (He did not fulfill this vow, however, until a convenient opportunity offered itself). This vow of pilgrimage is a peculiar introduction to modern philosophy — a counterpart to the supernatural voice in which Herbert of Cherbury a few years later heard a sanction of his "natural religion"! — Yet he still thought it was too early to proceed to the working out of his philosophy. After having taken part in the capture of Prague and in a campaign in Hungary, he returned home to France and took over some estates that had fallen to him by inheritance. His family would gladly have seen him married and placed in an official position; but his mind was not set on that. He formed the resolution to devote his life to science, and in order to be able to pursue it in peace, he moved to Holland (1629). Before that he had repeatedly had occasion to develop his philosophical ideas, especially about method, in literary circles in Paris. Two remarkable unfinished treatises, which were first published long after his death, "Rules for the Direction of Thinking" and "The Search for Truth by the Help of the Natural
[207] Light," probably date from this time. They give a significant presentation of the analytic method. In the first period in Holland he (as is seen from Discours de la méthode, parts 3 and 4) occupied himself with the speculative ideas he later presented at length in his "Meditations." He laid the foundation of his theology and his psychology and found a way that, through doubt itself, leads him to the starting point of all coherent philosophical

knowledge.

When Descartes settled in Holland, it was surely not only because he sought peace and because — as he said — he could philosophize better in a cooler climate. A contributing reason was that he expected to find greater freedom there for his inquiry. The reactionary movement that brought Bruno to the stake and Galileo under mental torture had spread to France. When some young investigators in Paris in the year 1624 wished to defend atomism against Aristotelian physics in a public disputation, the theological faculty declared the doctrine that everything consisted of atoms heretical, since it conflicted with the Catholic doctrine of the Eucharist. The disputation was forbidden at the last moment, after nearly a thousand people had assembled to attend it. The authors were arrested and sent away from the city. And on 6 September 1624 the Parlement forbade — on pain of death! — the putting forward of fundamental propositions that conflicted with the old and tried authors, and the holding of disputations not approved by the theological faculty! — This certainly could not but make the atmosphere in Paris less agreeable for a philosopher. In Holland he could expect to find freer air. Besides the purely philosophical speculations mentioned, he occupied himself in his new place of residence with studies in natural science and worked out a treatise that was to be called "The Universe" (le Monde) and to give a presentation of the development and structure of the world according to purely mechanical laws. He supposed that God had created matter as a chaos and thereafter ordered it according to the same laws that he now constantly follows in the preservation of the world. In this way he reconciled his faith in the dogma of creation with the idea of a development according to natural, still demonstrable laws, in the application of which to the genesis of our world-system he stands as the predecessor of Kant and Laplace. But then came the news of Galileo's *[208]* condemnation and of the Inquisition's condemnation of the Copernican doctrine, which Descartes had presupposed in his hypothesis — and he then laid the work aside. He would not, he wrote to his friend Father Mersenne, teach anything that conflicted with the faith of the Church, and besides it was, after all, his motto that one lived best in concealment (bene vixit, qvi bene latuit); he wanted above all to have peace and be free from fear and vexation; and therefore under the present circumstances he would confine himself to studying for himself alone. — Descartes's fine motto thereby received an ugly application, and this whole affair has rightly been called a stain on his character. From his letters it is seen that he is in full agreement with Galileo in his results. And even if one now has every reason to suppose that Descartes was a sincere Catholic, there is nevertheless hardly any question that fear, and perhaps most of all the need of peace,

was the predominant motive for his reticence. Later (in "Principia philosophiæ") he presented his hypothesis of cosmic development — but in disguised form. He threw, as his first biographer says, sand in the eyes of the Inquisition.

Descartes had not yet had anything printed. And yet his philosophical and scientific ideas were known in wider circles, partly in Paris, partly also in Holland. The Cartesian philosophy, as has rightly been said, was taught before it could be studied in print. Several of his disciples brought it forward at the Dutch universities, where it occasioned hard struggles, which also caused its founder much disquiet. Only in the year 1637, at the urgent requests of his friends, did he have four treatises published (*Essays philosophiques.* Leyden 1637), which were to give characteristic examples of his inquiry and its results. In the first treatise (Discours de la méthode), the only one that has direct philosophical significance, he gave here the history of his ideas and the main features of a new epistemology and metaphysics. In the second and third treatises (Dioptrique, Météores) he gave examples of a strictly mechanical explanation of nature, and
[209] in the fourth (Géometrie) he founded analytic geometry. — The complete presentation of his philosophy he gave some years later in his *Meditationes* (1641) and in *Principia Philosophiæ* (1644). He had had the Meditations sent in manuscript to several of the thinkers of the time, e.g. the famous Jansenist Antoine Arnauld, Gassendi, Hobbes, and their objections, with Descartes's replies, were printed together with the work itself, which thereby acquires an interesting dialogical character. The discussion with Gassendi continued for a long time and took on a rather bitter character. While the opposition on the side of Gassendi and Hobbes was purely philosophical and therefore always very instructive, the new philosophy met resistance for quite other reasons on the side of the Jesuits and of orthodox Protestantism. At Utrecht, Gröningen and Leyden hard struggles were fought, since the theologians held to the scholastic philosophy as a bulwark for the faith. At last prohibitions against new opinions were issued. The Dutch, says Descartes in a letter, honor the beard, voice and bearing of the theologians more than their integrity. He considered the Protestant theologians worse than the Catholic ones. He had come under a double fire. The Protestant theologians accused him of skepticism, atheism, the dissolution of universities and of church and state, but besides warned against his philosophy as papism; and the Catholic theologians accused him not only of heretical opinions, e.g. the assumption of the motion of the earth (which he had nevertheless sought to conceal), but also of inclining to Protestantism and attending Protestant services.

The last work of Descartes that appeared in his lifetime was the interesting treatise on the emotions *(Les passions de l'âme),* which appeared in 1649. Its origin is owed

to Princess Elisabeth of the Palatinate (a daughter of the unfortunate Bohemian king Frederick of the Palatinate), with whom he was in lively correspondence. In his letters to her he developed his ethical ideas. With another gifted princess too he came into correspondence, namely Queen Christina of Sweden. At her invitation he traveled to Stockholm in order personally to introduce her to his philosophy. This stay "in the land of bears, ice and rocks" (as he calls it in a letter), as well as the life of the court, cannot have been favorable to his health. The year after his arrival he contracted an illness that *[210]* caused his death (1650).

The main features of Descartes's character appear clearly in his life. A love of study and thinking, which was faithfully preserved through his whole life, and which in particular moments, when new ideas dawned on him, could pass over into affect and enthusiasm, was the most important and finest trait of his nature. His favorable and independent circumstances made it possible for him to carry through the plan of life he had laid down, which aimed above all at preserving peace and equanimity for the sake of his studies. He did not lack courage, as he showed on his travels; but he was too anxious and deferential toward authorities. The least attractive trait in him is his lack of ability to acknowledge the merits of others. He had a great feeling for what was new in his standpoint and firmly denied having learned from his predecessors, maintaining that even the assumptions he had in common with them received in his hands a quite different justification. He names Plato, Aristotle, Epicurus, and among the moderns, among others, Telesio, Campanella, Bruno as authors he has known, but — despite all agreement — has not learned from, since his principles were quite different. Only Kepler does he acknowledge having learned from; in particular, the German investigator's mechanical view of nature, his emphasis on the importance of the concept of quantity and his partial discovery of the law of inertia were of great influence on Descartes's philosophy of nature. But this trait too is connected with his complete absorption in his own thoughts. As a thinker, sharp distinction, the clear leading back to simple points of view, was his chief quality. He has thereby done a deed of lasting significance: definitively put an end to arbitrariness and fantasticality as far as method is concerned. If errors are committed in his thinking, it is in any case easy to see where the error lies. Descartes found an essential difference between geometry and metaphysics in this, that in the former there is no difficulty about the first principles, whereas what matters is to draw consequences from them, while in metaphysics, on the contrary, the great dispute stands precisely about the principles themselves, or about the right determination of the fundamental concepts, since it requires great effort of thought to work oneself free of the prejudices *[211]*

we are inclined to cherish. This he himself gives (Resp. ad sec. obj.) as the reason why in his Meditations he has followed the analytic method. But his ultimate purpose was nevertheless a construction, a synthesis, that could present the whole system of our knowledge in its logical grounding. And he often makes the transition from analysis to construction all too quickly, takes the fundamental concepts as finished before they are. Moreover, it was his conviction that concerning the special relations of nature we can only arrive at hypotheses if we do not wish to remain standing at the individual data of experience. In putting forward such hypotheses he was very bold, and his historical significance rests in part precisely on the enthusiasm and the resistance he thereby aroused.

b. The Method and Presuppositions of Knowledge

Since in all knowledge, whatever object it concerns, we use our understanding, it is, Descartes thinks, of importance to investigate this itself closely. The understanding works alike everywhere; therefore the science of it becomes the universal science, and the true method can therefore be only one. It consists essentially in starting from what is simple and clear, and from that seeking to understand what is composite and obscure. First of all one must by experience (*induction* or enumeration) bring together everything that is needed to throw light on the question to be treated. This material is then ordered in such a way that the simplest, immediately evident relations are laid at the foundation. To find something immediately evident is the first condition for being able to think. Every transition of thought takes place by immediate intuition, *intuition.* It is by intuition that I learn that a triangle is bounded by three lines, that a sphere has only one surface, that when I think, I must exist. By such immediate intuiting the first principles are apprehended. There are propositions that are so simple that we cannot think them without holding them to be true. To these belong: that the same thing cannot both be
[212] and not be, that nothing cannot be the cause of something, and that there cannot be more in the effect than there is in the cause. It is not, however, Descartes's meaning that such propositions immediately are *given.* He expressly maintains that intuition relies on individual cases; but in the individual example a general truth can be found contained, and we then know it immediately as self-evident (simplici mentis intuitu), even if it is not yet formulated as a general proposition. The great question of how general principles can be derived from individual cases, so that intuition can pass over into a fundamental proposition, Descartes does not go into more closely. He only admonishes us to be very cautious in putting forward principles[43]).

From the propositions that are clear in themselves we derive further consequences by *deduction*. It consists in a series of intuitions, in that by a continuous movement of thought one goes from link to link and so connects what was not immediately evident in its connection. If one here comes upon something that cannot be comprehended, one must stop, since it would be fruitless to go further. Either one lacks sufficient experiences, or one stands before questions which, by the nature of the human mind, cannot be answered. If one succeeds in carrying the deduction through, we have led the unknown back to the known, and the first, simple truths have been extended to hold for a larger domain. The items of knowledge accessible to us are mutually connected by a wonderful bond and can be derived from one another by way of inference, so that one will be able to find them if one only advances gradually from the very simplest items of knowledge. — Descartes, as is seen from the way in which deduction is conditioned by intuitions of simple relations of experience, is no opponent of the method of experience, just as he was always eager to gather experiences and make experiments. But he maintains that scientific knowledge is reached only by strict thinking, and that experience has its proper significance in giving occasion for it. He does not, on the other hand, discuss searchingly whether it is possible to reach a complete collection of everything that can throw light on the subject (complete induction or enumeration), and he underestimates besides *[213]* the necessity that the results of deduction be verified. Science arose for him through inferences from data of experience and consisted in logically deduced hypotheses. —

But how far must I now go back to find the very simplest? Where do I find an unshakable foundation for my knowledge? These questions Descartes answers in Discours de la méthode (Chap. 4) and — more fully — in the Meditations. About every content of knowledge, every object of knowledge, we can doubt. All assumptions and sense perceptions may rest on illusion. Let us suppose that an evil being had created us and incessantly instilled errors into us! Even then there would still be one thing we could not doubt: our own thinking, our own consciousness. All doubt, all error is after all a thinking, an activity of consciousness. However many incorrect, non-existent things I represent to myself, the very faculty of representation (vis imaginandi) still constantly proves its reality. In so far as I think (or: have consciousness), I exist: Je pense, donc je suis (Cogito, ergo sum). It is, as already remarked, an immediate intuition that cannot be further grounded, any more than it can be further defined what thinking or existence is. It has an immediate clarity and distinctness that is the model and measure for every other insight. — Descartes declares it a misunderstanding to take that proposition as an inference, but he has undeniably himself given occasion for this misunderstanding by

saying *donc* (ergo)[43 b]).

That simple intuition, which cannot be shaken by any doubt, is not yet any knowledge. How do we get further? How can we ground the objective validity of our knowledge, the reality of what we represent and think? — To answer these questions Descartes uses the causal principle, or rather the proposition that effect and cause must correspond to each other. This proposition belongs, as we have seen, according to Descartes to those that can be discovered by intuition. The natural light teaches us that there cannot be more in the effect than there is in the cause. From this it follows that nothing can arise
[214] from nothing, and that the perfect cannot arise from the imperfect. If we apply this to our representations, then it is clear that some of these can be explained as having arisen from external causes, others as produced by ourselves. But in neither of these ways can the representation of God as the infinite being, the sum of all perfection and reality, be explained. Since I myself am a finite being (of which my doubt and my desire convince me), I cannot have produced such a representation. It cannot have arisen through the combination of individual observed perfections; for then it would not contain the unity and indissolubility that is the distinctive mark of the representation of God. Every external cause is also finite. There remains, then, only that God himself is its author. It is innate in us in the sense that our faculty of thinking, as soon as it awakens to clarity, can by its own power grasp this representation as true. — By an innate representation (idea innata), then, Descartes understands not a representation that is *given* from the first, but one that we have the capacity to develop. The idea of God, says Descartes, is innate in the same sense as the representation of myself is. "The innate representations proceed from the faculty of thinking itself"[44]). — But the word "innate" was fateful; it gave occasion to many misunderstandings.

When the reality of the representation of God has thus (be it noted: by virtue of the causal axiom) been secured, a firm foundation, according to Descartes, has been won for the reality of our knowledge. Everything I know belongs to the infinite reality that this representation expresses, is one of the perfections the concept of God contains, and thereby stands firm for me. Only a finite being can err; the possibility of erring falls away in the infinite being, in whose perfection I share through my clear and distinct representations. Popularly, Descartes expresses this by saying that God, who is the ultimate cause of all our sensations and representations, cannot deceive, since he is a perfect being. But, when one looks more closely, he indicates a less external connection between the representation of God and the reality of our knowledge. Since I can perceive my imperfection only with the help of a more or less distinct representation of perfection,

and since the representation of God is the representation of the highest perfection, which *[215]* comprehends everything, this representation serves me as an ideal by which I can measure and correct my imperfect knowledge[45]). When Descartes draws the concept of God into epistemology, it is not really God in the religious sense he means; the idea of God is for him one with the idea of the continuous and all-embracing connection in existence, in which everything that is to have reality must be able to find its place. Thus it is said in the sixth Meditation: "By nature as a whole (natura generaliter spectata) I mean nothing other than either God himself or else the connection (coordinatio) of created things established by God." What I am to acknowledge as true must be able to find its place in this connection, which, then, could just as well be called nature as God. It is by its connection with everything else that the individual thing acquires reality for me. The criterion for distinguishing between dream and the waking state is that the experiences of waking life I can connect with all my experiences and memories without a break in the connection (perceptionem earum absqve ulla interruptione cum tota reliqva vita connecto). — This is the deeper meaning of Descartes's theological grounding of the validity of knowledge. But interesting as this line of thought is, the theological turn could nevertheless have been avoided altogether, since already the causal principle, which is after all in its turn a presupposition of the reality of the representation of God, is sufficient to bridge the gap between the knowing subject and existence. The causal principle and the other first fundamental propositions Descartes (in Responsiones secundæ) expressly exempts from the knowledge that is first to be grounded by the representation of God. On a closer development of what lies in the causal principle, it would appear that it leads precisely to the assumption of a connection within which the individual phenomena have their definite place. But the time for a closer discussion of the causal relation and the causal principle came only a century after Descartes's time. *[216]*

c. Theological Speculation

Descartes begins analytically and critically, but very quickly leaps over into theological and spiritualistic speculation. In his "Je pense, donc je suis" he finds the starting point for both; for the present we keep to his theological ideas, which are of interest to us in that they provide a stricter development and grounding of the content of "natural religion."

Descartes denies that the existence of God can be proved from the world given in experience. The series of causes could after all go on to infinity, and we must not confuse the impotence of our understanding to follow such an infinite series with the necessity of assuming a first cause. Moreover, the concept of God is presupposed as known

beforehand when I am to prove the right to call such a first cause God. His proof is more direct. It consists, as we have already seen, in the impossibility that the representation of an infinite being could be formed by a finite being. This representation is indeed even a presupposition of the representation of myself as a finite, limited being; for the finite presupposes a limitation of an infinite; the infinite is the positive, by whose limitation the finite comes into being. The concept of God is then formed by removing the limitation (thus by the negation of a negation found to hand). With this last determination Descartes comes considerably closer to the view that the concept of God is formed as we form all our ideals: by extension and by the removal of barriers (besides by the combination of features from different contexts). Gassendi pointed out against Descartes that we in fact form ideal representations by extension and combination; but Descartes answered that this very capacity to extend the representation of the relatively perfect given in experience presupposes that we have our origin from God. Instead of following Gassendi to the study of the psychology of the formation of ideals (a subject that even to this day has been very little treated), Descartes keeps to the *finished* representation of God: since it concerns something infinite, it supposedly cannot have been produced by a finite
[217] being. This inference itself would after all only be correct if it were the representation itself, and not merely its object, that had to be infinite. —

Yet Descartes has still another proof, the ontological. While there he went from the representation of God back to the cause of this representation, here he seeks to show that when we clearly and distinctly think the concept of an infinite and all-perfect being, we must also assume the existence of such a being: for by denying it existence we would deny it a perfection and so contradict ourselves. Against this reasoning (which is found in the fifth Meditation) Gassendi already objected that existence is certainly the condition for something's being able to have perfections or imperfections, but is not in itself any perfection. Descartes held to the point that we do after all ascribe existence to things in the same way as we ascribe properties to them. —

Descartes is clear that the concept of God whose validity he thinks he has proved by the two proofs cited is not the popular one. Most people do not regard God as an exalted and infinite being, the sole author of everything, but think by the word God of a finite being that can be worshipped by men. It is no wonder that there are those who deny the existence of such a God. His own higher concept of God Descartes develops further by determining God as the absolute substance. That God is a substance means that he is a being which is able to exist by itself (per se), needing no other thing in order to exist. Descartes does indeed also use the concept of substance of finite beings

(thus of soul and matter, which we shall go into more closely in what follows); but he declares that it cannot be used in one and the same sense (univoce) of the infinite being and of the finite beings. Since the finite beings all depend on the infinite, they can only improperly be called substances. Especially in his letters Descartes carries the strict concept of substance through. "One cannot," he writes, "prove the existence of God without regarding him as the most perfect being of all; but he would not be so if anything happened in the world that did not proceed from him. Natural philosophy already teaches that not the least thought can come up in the human mind without *[218]* God's willing it and having willed from eternity that it should come up." This is also one of the reasons why Descartes rejects atomism: an atom would be something that subsisted by itself; but only one being, the highest of all, can be independent of every other being! — In his writings Descartes expresses himself more vaguely, since a wider and more popular definition of the concept of substance slips in beside the strict one. According to the laxer usage, substance means the same as thing or being. Substance is that which has certain properties. From the properties we infer a substance. In this sense (as bearer of properties) a finite being can very well be a substance. It is clear that it must lead to unclarity when the two meanings of the word are not kept sharply apart. But Descartes conceives soul and matter as substances also in the sense of absolutely independent beings. And just as he thus applies the strict concept of substance to finite beings, so too he conversely applies the loose concept of substance to the infinite being, since the content of this concept is after all all possible reality and perfection — thus all possible properties! It was therefore a criticism that cut deep into Descartes's system when Hobbes and Gassendi objected against him that we have no positive concept at all of substance as bearer of properties, since we know only properties (accidents, attributes — or, as we would now say, phenomena). Descartes conceded that we do not apprehend substance immediately, but infer it from the properties; but he maintains that we can think it, even if we cannot form any image of it[46]). Here too lay a problem that came to play a great role in the subsequent philosophy. —

It was with the help of the causal principle that Descartes arrived at the assumption of God's existence. But this same proposition might seem to lead beyond the concept of God: for must not God then also have a cause? From this difficulty Descartes seeks to free himself by distinguishing between that which has its cause outside itself and that which has its cause in itself. God must be, in relation to himself, what the cause is *[219]* in relation to the effect, thus be the cause of himself (causa sui). As the infinite being, God has an inexhaustible power in himself, so that he needs nothing other than himself

in order to exist. In this way the concepts of substance and cause of itself are closely connected. Antoine Arnauld, the great Jansenist theologian, who later became an ardent Cartesian, objected (with a line of thought that already occurs in Thomas Aqvinas) that the concept of cause of itself contained a self-contradiction and in any case could not be applied to God. For, he said, no one can give what he does not have, and for anyone to be able to produce himself, he would have to exist already beforehand. And if God is cause, he is also effect; the cause precedes the effect in time; but in God there can be no distinguishing between past and future or between possibility and actuality. Descartes answers that when we prove the existence of God by means of the causal principle, we must also apply this principle to God himself, which is possible only by conceiving him as the cause of himself. But he adds that the expression cause is used of God only by analogy, on account of the imperfection of human knowledge, and that one may well call God the cause of himself, but not the effect of himself, since thereby something in him would come to stand lower than something else! The relation of time has, moreover, no validity at all for God, since it is due only to human apprehension, is a modus cogitandi. — There seems then undeniably not to remain much of the concept of cause in its theological application. The Cartesian philosophy of religion could not but give rise to a need to discuss the justification and significance of the application of our fundamental concepts in theology.

The proposition that the relation of time has no validity in God must particularly be remembered when Descartes teaches that the eternal truths (such as, e.g., that a thing cannot both be and not be, — that the sum of the angles in a triangle is equal to two right angles) depend on the absolute will of God. God's reason and his willing are not distinguished. God has not first perceived the eternal truths and then laid them down.
[220] His willing is eternal and unchangeable like his essence. Here too the great difference shows itself between Descartes's concept of God and the current one. There is really not a single psychological expression that can be transferred to God. But this consequence Descartes did not yet draw.

One difficulty remains, even if one grants Descartes to be right in all his assumptions. Through our doubt we perceived our imperfection, and its concept presupposed the concept of the perfect. Thus we rose up to the concept of God. But after we have risen to this height, imperfection nevertheless continues to subsist — and how is it at all possible that imperfection could have come into being, when it is an ideal and perfect being that lies at the foundation of everything? Descartes answers that imperfection necessarily clings to everything finite, and everything created is finite. It is, besides, only a negation

that brings about imperfection; and this negation makes the world more perfect than it would otherwise be, since the multiplicity in the world thereby becomes greater. Even if the eye is the most beautiful thing in the body, the other parts of the body still have no reason to complain, since the body could not consist of nothing but eyes. Descartes indicates in these remarks a theodicy similar to the one Leibniz later presented at length.

d. Philosophy of Nature

By virtue of the criterion of truth Descartes had found, he concludes that our sense-sensations must derive from something that is different from our consciousness. But this something need not — according to Descartes — by any means be so constituted as the senses present it to us. The significance of sense perception is first and foremost practical; it is to inform us of what is beneficial or harmful to us. And even if we regard it as the task of sense perception to help us to a knowledge of things, it is not necessary that our sensations *resemble* things, provided only that they correspond to them as the word corresponds to the thought. If we look to the way in which sense perception takes place, we find that all impressions on the senses are contacts; it is thus only the surface of things that we come into relation with. If we wish to think how things are constituted *[221]* apart from the way in which they affect us, then there remain only extension, divisibility and mobility as properties we cannot — even in our imagination — think away. They are the simplest and clearest representations we can have of matter, and it turns out that by keeping to them — thus to the purely geometrical properties — we can reach a simple and clear understanding of everything that goes on in the material world. It is the phenomenon of motion that contains the explanation of all other phenomena. On the other hand, nothing is explained by assuming "forms" or qualities in matter. The qualities belong to our consciousness; but what really goes on in the material world is only motions outside our organism or within it. In science we must think of the material world as it would be if no human being perceived it. — It is the principle of simplicity that played so great a role in the struggle for the new picture of the world, and that is now used to ground the leading back of all the properties of matter to extension and motion. A consequence of this leading back is that the qualities are relegated to the subjective domain. Since Descartes can hardly have known Galileo's "Saggiatore," he has probably come to the principle of the subjectivity of the sense qualities by his own path, through the attempt to think through the conditions for a purely objective and strictly necessary explanation of material processes. From notes in Descartes's hand from the years 1619—20 it is seen that he did not yet at that time adhere to the

mechanical explanation of nature. The transition to it must have taken place in him in the years 1620—29, in connection with his studies in natural science, especially in the domain of optics and physiology. Besides Kepler's influence, there has probably also been at work here the influence of a circle of younger French investigators who adhered to atomistic views and were especially influenced by Sebastian Basso, the real renewer of the atomistic doctrine in modern times; Descartes himself names Basso among his predecessors. It was two young men of this circle who wished to hold the
[222] disputation in Paris mentioned earlier and thereby gave occasion to the terrorist decree against new opinions. Since Sebastian Basso in turn seems to have been influenced by Giordano Bruno's writings, a historical connection appears here between Descartes and the philosophy of the Renaissance. — Harvey's discovery of the circulation of the blood too has (as we shall see later) intervened strongly in Descartes's line of thought.

In the attempt to explain all material phenomena by means of extension, divisibility and mobility as the fundamental properties of matter, Descartes proceeds deductively. He goes from cause to effect. He is fully conscious that by this path only a hypothesis can be reached, since there is always the possibility that the given phenomena could have been produced by other causes than those we start from. Here, then, experience and experiment must come in as confirming, and Descartes declares that for this so great a quantity of experiments is needed, and such costly ones, that they far exceed his means and can be brought about only by the cooperation of many. Descartes, then, has in no less a degree than Bacon an eye for the significance of experience, but he lays greater weight than the latter on the guiding principles and hypotheses. From Galileo he likewise differs in his interest in principles and deductions. He took the deductive procedure to be the properly scientific one and lacked Galileo's ability to throw light on great and comprehensive relations through investigations of particulars — and this although he himself, in his doctrine of method, sets the intuition bound up with individual observation as the starting point of thinking. But his task was a different one from Galileo's. His mission was to bring to full clarity and consciousness that the time was past when an explanation of nature could be reached by assuming mystical forces or supernatural intervention. He set up the ideal for all natural science: the derivation of phenomena from their causes with the same transparent necessity with which a mathematical conclusion follows from the presuppositions laid down. The principle of simplicity, to which he appeals, is really at the same time the principle of transparency. In the ideal thus set up he was not mistaken. But he ascribed to the deductive application
[223] of general principles, and to the hypotheses grounded on it, greater significance than

they have in reality, and his philosophy of nature thereby acquired a dogmatic character, which, especially in the hands of the Cartesians, stood out markedly and provocatively. In many a mechanical scholasticism took the place of the scholasticism of substantial forms and qualities. —

From the fundamental properties of matter it follows that there can be no absolute atoms, since no limit can be set to divisibility — that there can be no empty space, for space means extension, and extension presupposes an extended being, thus matter — that the material world is infinite, for it is impossible for us to set limits to extension, and where there is extension, there is also matter. Further, it follows that when material changes occur, these must all be explained according to the laws of motion, which thus become the highest laws of nature. These laws too Descartes seeks to derive by deduction, and the foundation for this deduction he finds in the concept of God, which in his philosophy of nature comes to play a role analogous to the one it plays in his epistemology. From the unchangeableness of God, which is part of his perfection, Descartes derives that the sum of motion produced at the creation is preserved during the preservation of the world (which according to Descartes is a constantly continued creation). Motion can be distributed differently among the different parts of the world, but no motion is lost and none begins anew. — There lies in the theological grounding Descartes gives of the old proposition of the constancy of motion an involuntary admission that the fundamental properties of matter from which he starts in his philosophy of nature are too simple and abstract. For it turns out that it is really the divine force constantly at work in the world that is constant; the constancy of motion is derived from the constancy of this force. Had Descartes seen this clearly, he would also have found a more correct proposition than the one he sets up. For it is not motion but force, energy, that is constant. With his proposition Descartes gets into difficulty at every transition from motion to rest or the reverse. But even in its imperfect form his proposition is an interesting predecessor of the modern principle of the conservation of energy. — Special laws of motion too Descartes *[224]* derives from God's unchangeableness. So first and foremost the law of inertia, which in him for the first time is consciously and systematically set up. It would, Descartes thinks, conflict with the divine unchangeableness if any thing that can be regarded as simple and undivided could lose the state it is in without an external cause. Thereby the possibility is rejected that such a thing could of itself (sua sponte) and without external influence pass over from motion to rest or the reverse. Descartes probably arrived at the principle of inertia under the influence of Kepler, who set up one half of it, but independently of Galileo. Already in the treatise "Le Monde," which was worked out before Galileo's

Dialogues appeared, he had in fact set it up. — In setting up the principle of inertia Descartes makes a reservation, leaving open the possibility that souls and angels could act upon matter. This reservation is made in the interest of his spiritualistic psychology, which we shall consider later.

Descartes's concept of matter bears the stamp of simplicity and clarity and is indeed also grounded by the principle of simplicity. It is a natural outcome of the tendency that also found expression in Kepler and Galileo and sought to lead all qualities back to quantitative relations, whereby exact natural science became possible. But just as he is inclined to overlook the hypothetical character of his deductive results, so he is also inclined to regard his definitions, set up by abstraction and by virtue of the principle of simplicity, as exhaustive. He thinks he knows the essence of matter completely by the determinations extension, divisibility and mobility, without, however, being able to give any guarantee for it. When he speaks of matter as *the extended substance,* he does not notice that it is an abstraction he has made into a being subsisting by itself. He has not proved the justification for regarding the geometrical properties of things as absolute. For it does not necessarily follow from the fact that these properties are the most important for the knowledge of nature. Properties such as extension, divisibility and mobility may very well be due to the action of things upon us (like colors, taste, smell,
[225] etc.), without their therefore *resembling* the things themselves. And besides there is — as already shown — a determination that is still more fundamental than the three properties Descartes keeps to, namely force, energy. — It became the business of subsequent inquiry to discuss the questions indicated here. Descartes's philosophy of nature signified for the time being, in all its one-sidedness, an immense advance in clarity. We shall now briefly consider some of the most important applications he makes of the general principles he has set up. It is always the fundamental thoughts we keep to; in detail his explanations are often very unfortunate and arbitrary. Spinning hypotheses was his fault; his greatness was the happy grasp of general points of view.

When everything in material nature is to be explained from extension, divisibility and mobility as the only properties of matter, it follows of itself that final causes cannot be applied. But Descartes gives a theological rather than an epistemological grounding for the exclusion of the teleological explanation of nature. Since God is an infinite being, there must be much in his working that we cannot grasp; it would be presumptuous to wish to search out his purposes. There is, after all, in the unbounded world infinitely much that does not affect us at all; what meaning can one then attach to its having been created for our sake? The only purpose that can be thought for everything that happens

must be God himself. It is thus because we dare not set limits to the essence of the world and of God that the teleological view is rejected.

If we are now to understand nature in detail, then according to Descartes we must start from the simplest, most easily seen-through phenomena, from the processes we constantly see going on before our eyes, and by means of them explain what goes on in the small and in the hidden, and what has gone on in the past. The principle of simplicity thus passes over, for Descartes, into the principle of actuality, which requires the distant and unknown to be explained by the near and known. Best of all, we must take as our foundation insight into the construction and working of our machines, since we have the most distinct representations of these. Therefore, says Descartes, I describe the world as *[226]* if it were a machine (terram totumqve hunc mundum *instar machinæ* descripsi). The present state of the universe we explain to ourselves by supposing that the parts of matter from the very first (for matter and motion were created at the same time) were in whirling motion about certain centers. In these centers the smaller parts had to gather, which arose during the whirling motion through the friction of the larger parts against one another. Thus arose the various larger and smaller globes. Some of these globes gradually lost their independence and were swept along by the vortices that go on about larger globes. So it has gone with the earth. Descartes thinks that according to his theory he can call the earth immovable, since it does not change place within the medium that whirls it along with itself around the sun, and since, besides, outside the part of the universe we can observe, there may be stars in relation to which the earth is immovable! By such a subtle application of the relativity of motion Descartes seeks — as it turned out, in vain — to escape standing as a Copernican heretic. Gravity is explained as a consequence of the pressure of the vortex currents on the parts in space, whereby they are pressed toward the globes or in any case held in their vicinity.

Descartes is the first in modern times who has attempted to give a mechanical theory of the evolution of the world-system. He does indeed make a polite reservation toward theology, conceding that the world was created perfect and finished; he only wishes to show the *possibility* that the present state of the world can have developed according to natural laws from a more imperfect initial state. If one asks how this initial state is then to be thought, he answers that it will not matter much whether we think of it in the one way or the other: for by virtue of the natural laws matter will run through all the states it is capable of assuming, so that after the exhaustion of more or fewer imperfect states the present state will in any case be reached. This declaration (Princ. phil. III, 47), which contains the program for all mechanical theory of evolution, is of greater interest

[227] than Descartes's attempt to depict the cosmogonic process in detail. — A chief defect of his hypothesis was that it made no attempt to explain the connection that Kepler's laws had established. These laws continued to stand as an indifferent fact until Newton's brilliant thought discovered the physical connection from which they could be derived, and which could be derived from them. —

Just as Descartes seeks to explain the connection and development of the world according to the general laws of nature, so he also seeks to explain the organism and organic life according to purely mechanical laws. Not only astronomy, but also physiology was in his view to be a purely mechanical science. Just as in his cosmogonic hypothesis he abstracts from theology, so he abstracts in his organic mechanics from psychology and thinks of the human body as formed of material parts and working according to the laws of heat and motion, without any soul (whether "vegetative" or "sensitive" or "rational") intervening. This view, which is a consequence of Descartes's general principles of the philosophy of nature (since the organism, as a material being, must fall under the general laws of matter), received for him a confirmation from experience in William Harvey's discovery of the circulation of the blood (1628). Harvey stands in the first rank among the founders of modern natural science; he is for physiology what Galileo was for physics. He gave the mystical forces their deathblow in the domain of physiology by showing that the motion of the blood is not due to its own force or that of the soul, but is determined by the contraction of the heart, which presses it out into the body. The general laws of motion thus hold in the organism as outside it. Descartes was one of the first eminent men to adhere to Harvey's doctrine, and his adherence, which was declared in Discours de la méthode (chap. 5), was a powerful support for the new doctrine, which had so much resistance to overcome, since it stood in sharp conflict with the old view of organic life[47]). Descartes depicts in various writings (especially in Traité de l'homme) how the human body can be conceived as a mere machine. Here too his general way of looking at things has brought clarity about, even if his special explanations were
[228] unfortunate. One of the greatest anatomists of the time immediately following, Niels Steensen, acknowledged that the method Descartes introduced serves to show the insufficiency of what had earlier been taught about the significance of the various organs, and to put the questions about it in a clearer way. Descartes extended the mechanical view that Harvey had carried through for the circulation of the blood to the domain of the physiology of the nerves. In the nerves he assumes, in agreement with ancient physiology, currents of "animal spirits" (esprits animaux), which are formed by the fine parts of the blood, after being heated in the heart, being carried up to the brain and filling

its cavities, while the rest of the bloodstream continues its way through the vessels. From the brain they stream out through the nerves to the muscles. These currents can now, according to Descartes, be set in motion by impressions that do not become conscious to us; this happens in involuntary movements, as when in falling we put out our hands, or when we continue our walk without thinking of it. Such involuntary movements can take place purely mechanically, even against our will (mente invita et *tanqvam in machina).* Descartes expresses what goes on here by saying that the "animal spirits" are reflected (esprits réfléchis), and he gives a clear description and explanation of what is now called reflex movement. In animals there is, he thinks, no reason to assume anything other than that all functions and actions go on in the involuntary and machine-like way. There is no reason to ascribe a soul to them. When the lamb flees at the sight of the wolf, it is because the rays of light that from the wolf's body strike the lamb's eye, through the "reflected" currents of "animal spirits," set its muscles in motion. And when the swallows come in spring, it is to be explained in analogy with a clock's striking its strokes at a definite time. One sees, after all, that animals cannot adapt themselves to new and more complicated conditions, any more than a machine can. In any case: when we can explain the behavior of animals purely mechanically, there is no reason for assuming an animal soul. And when man can make wonderful automata, should not nature be able to produce still more wonderful ones? — The assumption that animals are mere machines recommends itself to Descartes also for the reason that, in his view, *[229]* one would otherwise have to ascribe immortality to them — and should an oyster or a sponge be immortal? It is otherwise with man, in whom, on account of the consciousness that makes itself known in each of us, we must assume a soul, a thinking substance in interaction with the material substance, with the capacity to intervene in directing the motions of the "animal spirits." Immediately, according to Descartes, the soul stands in connection only with a single part of the brain, the pineal gland (glandula pinealis), which drew Descartes's attention because it is not, like so many other parts of the brain, a double organ, and because it seemed to him to be placed about in the middle of the brain and above the passage through which the "animal spirits" from the anterior cavities of the brain are in connection with the posterior ones. The "animal spirits" strike against the pineal gland and thereby affect the soul, awaken sensation, feeling and drive in it, and the soul answers in turn — likewise by striking (!) against the pineal gland and thereby directing the animal spirits in a certain direction. Often (as in self-control) thrust and counter-thrust go in opposite directions, and it then depends on who can strike hardest![48])

Descartes has the merit of having described and analyzed reflex movement and thereby thrown clear light on the involuntary activity in us. He has also the merit of having maintained that mental activity is bound to the brain, and immediately only to it. And finally he has — despite the imperfect anatomical presuppositions at his disposal — the merit of having presented the spiritualistic psychology with great vividness and in its full consequence. His psychological theory we shall now consider somewhat more closely.

e. Psychology

Only in man does Descartes assume a soul, and he conceives it as a being distinct from the body, a substance of its own, the bearer of the mental properties, just as the body is the bearer of the material properties. Since, as we have seen, he carries his mechanical
[230] view of nature through into the domain of organic life as well, he must assume that in the organism, as little as in the rest of material nature, can any motion arise that is not a continuation of earlier motion, or any motion vanish that is not continued in a following motion: for the principle of the constancy of motion is indeed a chief proposition of his philosophy of nature. But when he now lets the soul "push" the pineal gland in order to set the "animal spirits" in motion, and in turn itself receive "pushes" through the action of the "animal spirits" on the pineal gland, then motion must indeed arise or vanish at this point without compensation! Even if one could attach any meaning to the soul's pushing or being pushed, Descartes's spiritualistic psychology nevertheless comes here into open contradiction with his philosophy of nature. In reality he thinks of the soul as a material being when he lets it push and be pushed; this is what always happens to spiritualism when it is thought through with full clarity. Descartes's gifted correspondent, Princess Elisabeth of the Palatinate, remarked quite rightly that it was easier to conceive the soul as material than to understand how, if it is not material, it can move matter and be moved by it. In one single place Descartes seems to want to restrict the soul's action to bringing about a change of direction in the "animal spirits", which in themselves could just as well move in the one direction as in the other; but even this mere directing would then conflict with the principle of inertia and could only be set up as an arbitrary exception to it.

When Descartes in general sets up the soul as a peculiar being, distinct from the body, it is because consciousness and extension are such wholly different properties that he can conceive them only as borne by different beings. Je pense, donc je suis! was his starting point, and by thinking he meant, as he expressly explains, consciousness in general

(omnia, qvæ nobis consciis in nobis fiunt). He could therefore, he remarks, just as well say: I feel, therefore I am, or: I will, therefore I am. It is the fundamental fact of consciousness that he wants to bring out. And he has deserved well of psychology by having set up consciousness (mens) as the starting point, whereas the concept of soul (anima) is — *[231]* and especially was — an ambiguous and mystical concept. He thereby paved the way for modern empirical psychology. It is not impossible that in this he was influenced by Vives, who distinguished so clearly between mental phenomena as what is really given and the essence of the soul in and for itself; Vives's psychological work is cited in several places by Descartes. But Descartes thinks that in thinking or consciousness he has a perfectly clear and exhaustive concept of the essence of the soul, just as in extension he thinks he has a perfectly clear and exhaustive concept of the essence of matter. Now if, says Descartes, I can think the concept of the soul clearly and completely without needing to presuppose the concept of matter, and can think the concept of matter clearly and completely without needing to think of the concept of the soul, is this not proof that we have before us here two wholly different beings or substances, independent of one another? (It is — as Descartes himself expressly remarks in one place — not the individual body that is substance in the strict sense, but matter considered as a whole: corpus generaliter sumptum). — We have earlier mentioned the double meaning in which the concept of substance appears in Descartes. It does not merely denote a bearer of properties, a something that "has" the peculiarities that experience presents. Even in this vaguer meaning of the word Descartes infers too much when he makes the "something" of thinking different from the "something" of extension; it could indeed be the same "something" that had both properties! But in the stricter meaning substance is a wholly self-subsistent and independent being. It lies, says Descartes, in the nature of substances that they *mutually exclude one another.* And this he assumes of soul and matter, because he can think of each of them by itself alone without needing to think the other party along with it. Man, then, is to consist of two beings that mutually exclude one another! Thereby the interaction between them becomes incomprehensible. When the one substance affects the other, they cannot be wholly independent of one another; something then takes place in the one that cannot be understood without my thinking of the other. The relation becomes incomprehensible, and Descartes himself admits it. *[232]* Those who never philosophize, he says in a letter, have no doubt that soul and body interact and make up one being. But philosophically it is impossible to think at once the difference and the connection between soul and body. No inference and no comparison, he says in another place, can teach us how consciousness, which is incorporeal, can set

the body in motion; but a sure and evident experience teaches it to us daily; it is one of those things that must be understood through themselves, and that are obscured when we compare them with other things. — Descartes speaks here as though he had seen the soul push the pineal gland and receive a push in return! The knot of the problem is precisely that immediate experience cannot decide the matter, but that inference and comparison are our only way to an intelligible hypothesis free of contradiction.

It is a common assumption that medieval dualism recurs in Descartes's theory of soul and body. But what really led Descartes to his theory is his endeavor to form clear and distinct concepts. From the independence of the concepts of one another he then infers the independence of the corresponding beings of one another. It is the clear and sharp characterization of the difference between mental and material phenomena that is Descartes's great merit; but it is also this that leads him astray. As a dogmatist he trusts in the completeness of his concepts and in the agreement of existence with his concepts. The psychological dualism that we have met in Telesio, Bacon and Campanella Descartes carries through in extreme form — and he has thereby contributed much toward leading beyond it. He has formulated the popular metaphysics of soul and body so sharply that its difficulties and contradictions stand out clearly. — Gassendi and Arnauld rightly objected against him that, because consciousness and extension are the chief properties of the soul and of the body, it does not follow that their essence is thereby exhausted. They have thereby struck precisely the decisive point in Descartes's theory. —

The consequence of Descartes's sharp distinguishing between soul and body is felt in
[233] his special psychology. He investigates carefully which mental activities and phenomena derive from the soul itself, and which are due to the body's influence on the soul.

Sense perception is due to the body; it arises through the "animal spirits" acting on the soul by way of the pineal gland. Yet Descartes emphasizes that popularly more is counted as belonging to sense perception than should be. For a distinction must be made between the motion in the brain produced by the impression (which can also be found in animals), — the action of this brain-motion on the soul, whereby the sensation (e.g. of color) arises, — and thirdly the judgments that are involuntarily passed on external things on the occasion of the motions of the bodily organs. Such an involuntary judgment underlies the conception of things as external and as situated at determinate places in space. It is the motions of the eye and of the head that, through the corresponding motions in the "animal spirits", occasion the soul to pass such a judgment. This third moment in sense perception is due to the soul itself.

Likewise there are two kinds of memory: one of material things, which rests on after-

effects or traces of the impressions in the brain, and one of mental things, which rests on lasting traces in consciousness itself. — Thinking proper (intellectio) and representation (imaginatio) are distinguished from one another in that in thinking proper the soul acts alone, but in representation it makes use of bodily images. Representation, like sense perception and material memory, belongs to the soul only insofar as the soul is united with the body; the soul in its purity (anima pura) can be thought without representation and sense perception. — The difference between drive and will likewise rests on the drive's being called forth by the body, while the will belongs to the soul itself. Descartes ascribes to us an immediate consciousness of the causal freedom of the will, although he admits that the indifference of the will is rather an imperfection in our knowledge than a perfection in our will. The will has a greater scope than the intellect; we can choose and hold fast to something that we do not understand. The will is unlimited, the intellect limited: by this Descartes explains the possibility of error. — The emotions (passions) are due to the body's influence on the soul, but the inner feelings (émotions intérieures) arise *[234]* in the soul in consequence of its own thoughts and judgments. The inner, purely mental feeling is indeed, so long as the soul is united with the body, accompanied by an emotion, a sensuous feeling: but this is then due to the soul's action on the brain, inasmuch as, e.g. in intellectual love, the mood of the soul affects the pineal gland. In emotion joy precedes love, in inner feeling the reverse. Although by this dualistic conception Descartes really blocks his own way to a coherent history of the development of the feelings, he has nevertheless deserved well of the psychology of the feelings by his attempt to trace the various feelings back to certain elementary forms from which they arise by combination. As elementary feelings he assumes wonder, love, hatred, desire, joy and sadness. Of these and their combinations he gives a purely descriptive and analytical account, constantly making use of his physiological theory. In explaining the physiological conditions of the emotions he lays the chief weight on the interaction between brain and heart. Only in the case of admiration, which he conceives as a purely intellectual feeling, does he not think that any influence of the brain on the heart takes place. He emphasizes the connection of the emotions with the instincts: they dispose the soul to will what is useful and to shun what is harmful; even without the soul's intervention, the motion in the "animal spirits" that calls forth the emotion leads to the performance of purposive actions. But the purposiveness that shows itself here is not absolute; therefore experience and reason must intervene to correct it. Of importance for the development of the feelings is the connection between a representation in the soul and a certain state of the "animal spirits"; only through such a representation, not quite directly, can the will intervene

determiningly in the bodily motion. The struggle between reason and sensuousness is carried on, according to Descartes, not within the soul itself, but is a struggle between the soul or the will on the one side and the body or the "animal spirits" on the other, and
[235] the struggle is over which can push the pineal gland the harder! (Passions de l'âme. I, 47). — Here too we find the spiritualistic theory carried through by Descartes with all desirable consistency.

f. Ethics

When Descartes set up universal doubt as the starting point of his philosophy, he provisionally excepted the moral domain, although he saw that his philosophizing must have consequences for it too. So that his conduct of life should not be left without guidance while he analyzed everything theoretically, he set up (in the Discours de la méthode, chap. 3) the following provisional rules for his conduct: 1. To keep to the religion and manners of his fatherland. 2. To carry out consistently what he has once resolved. 3. To strive to conquer himself rather than fortune, to subordinate his wishes to the order of the world, and to remember that our own thoughts are the only thing that is wholly in our power.

Later, when he had developed and published his philosophical views, he was wary of venturing into the moral domain. The theologians, he says, who were so angry with him on account of his physics, would certainly not leave him in peace if he wrote on ethics. Apart from some statements in the work Passions de l'âme, it is in his letters to Princess Elisabeth, to Queen Christina and to Chanut (French envoy in Sweden) that we find his ethical ideas developed. They are only a further unfolding of the provisional rules of the Discours de la méthode; they betray the influence of Stoic philosophy, particularly of Seneca.

What means, asks Descartes, can philosophy teach us for attaining the highest happiness, which vulgar souls vainly expect from the course of things, but which we can only get from ourselves? Three means present themselves: clear knowledge of what is right, firm willing of it, and the suppression of every wish and every want that is directed at what is not in our power. Only through the intervention of the will does knowledge come to rule in us, so that the motion of the "animal spirits" follows it. Whereas the
[236] sensuous feeling or emotion confuses our thought and makes it overvalue external goods, it now becomes possible to hold fast to the inner enjoyments of the soul, which are eternal like the soul itself, because they are built on the firm ground of the knowledge of truth. The most important knowledge is the knowledge of God as the author of all

things, as the all-perfect being of whom our mind is an emanation, and the knowledge of the soul as a being distinct from the body. Next comes the knowledge of the infinity of the world, which roots out the belief that everything should exist for the sake of the earth and the earth for our sake. Knowledge teaches us to regard ourselves as parts of a whole (family, society, state) and to prefer the interest of the whole to our own interests. But above all it is a matter of distinguishing between what is in our power and what is not. Only the use of our will is wholly in our power. In the feeling of this consists *magnanimity* (générosité), which is the key to all virtues. The magnanimous man feels at once his own power to accomplish great things and his own limitation. Instead of the notion of a capricious fortune, which is a mere phantasm, it is a matter of holding fast to the conviction that everything rests on the eternal and unchangeable will of Providence. Thereby the passions are purified, and room is made for the spiritual love (amor intellectualis) of God, the sweetest and most beneficial of all the feelings we can have in this life (although it must be left to the theologians to decide whether it, such as the natural man can have it, is sufficient to bestow eternal blessedness on man). By sufficient exertion of heart and thought this feeling can also become the strongest of all. — As in Descartes's epistemology and in his philosophy of nature, so in his ethics too thinking culminates in the idea of the infinite being as the ultimate foundation. A mixture of rationalism and mysticism is characteristic of Descartes, as of the seventeenth century as a whole. This side of Descartes's philosophy was particularly developed by his immediate successors. *[237]*

2. Cartesianism

The Cartesian philosophy could not but exercise great influence on the age. By its demand for methodical doubt and analysis, for disregarding all inherited presuppositions, it could not but awaken free inquiry and lead to independent thinking. On this point its most important significance doubtless lies. It has thereby exercised influence on many more than those who became Cartesians in the narrower sense of the word. By its demand for a purely mechanical explanation of nature it intervened as a reforming force in natural science, especially in physics and medicine, even though several of its special hypotheses soon proved unfortunate. It stated the principle of mechanism with an energy that has been of lasting significance. By its spiritualistic psychology and its theological speculations it appealed to the current conception, for which the resolute and absolute distinction between soul and body was very clear and appealing, and which

also found in the Cartesian proofs a new foundation for agreement between theology and science, in place of the one that was falling with the scholastic philosophy. In the last part of the century we accordingly see the greatest minds of the French Church, Bossuet and Fénélon, as zealous Cartesians. In eminent Jansenists, such as Arnauld and Nicole, it likewise found adherents, although the greatest figure of Jansenism, Pascal, here took up a special position, which will be characterized later. Under Arnauld's direction an excellent account of logic was worked out in this circle *(La logique ou l'art de penser.* 1662), which is still of interest and value. In the learned Order of the Oratory too Cartesianism found great support. In wider circles it was spread by popular lectures in Paris and the provinces. From Madame de Sevigné's letters one sees what interest it aroused in the high aristocracy, among women too. It gained no small spread in Holland through the appearance of several younger university teachers. But both in France and in Holland it also met with violent opposition. The Jesuits and the orthodox Protestant clergy sought in every way to crush it. The new philosophy ascribed too much to free human reason
[238] and gave doubt a dangerous significance. It taught the subjectivity of the sense qualities, which was not compatible with the dogma of the Eucharist (especially in its Catholic form), and besides it assumed the motion of the earth and the infinity of the universe. The cause of theology seemed to be bound up with the old scholastic doctrine. At several Dutch universities Cartesianism was forbidden; in any case the Cartesians were kept away from theological chairs and from offices in the ministry. In Rome Descartes's works were put in 1663 on the list of forbidden books, and in France repeated royal decrees forbade the teaching of Cartesianism at the universities. All this did not prevent it from becoming the ruling philosophy, whose dominion was first broken by the tendency proceeding from Locke and Newton, a tendency which it had, moreover, itself prepared.

Among the more independent Cartesians the difficulties and contradictions in Descartes's doctrine could not, in the nature of the case, but lead to attempts to develop the master's thoughts further. In the dogmatic part of the Cartesian system there were especially two points that contained stings. The one was the relation between the absolute substance, which played so great a role in Descartes's theory of knowledge, doctrine of nature and ethics, and the finite substances, the finite world as a whole. How can an absolute being have a finite world outside itself, and what reality can such a finite world have? This problem could not but press forward particularly in an age that had a bent for mysticism and for the recognition of absolute powers; it is indeed the age of absolutism in politics, and it became so by a natural analogy in philosophy too. Soon the Cartesian philosophy glides over into mysticism. The other point was the relation between soul and body

as substances that mutually excluded one another and yet were to stand in interaction with one another. The master himself had indeed even declared that this could only be understood so long as one did not philosophize. There were some of the disciples who went on philosophizing on this point, and who found here an incomprehensibility that for them recurred everywhere in experience where different things were to act on one another. By this road Cartesianism was led on to restricting knowledge to establishing the succession and external connection of phenomena, while the inner connection be- *[239]* tween them was declared incomprehensible. Thereby the treatment of the problem of causality by the later epistemology was prepared. — Both points work together in the so-called *occasionalism,* yet in such a way that the mystical element, the tendency to gather all reality and all causality in the absolute substance, is the predominant one. We consider in all brevity the most important representatives of this tendency.

Louis De La Forge, a French physician, stated clearly (in his *Tractatus de mente humana,* 1669, in French as early as 1661) that just as difficult as it is to see how the mind moves the body, just so difficult is it to comprehend how one body can move another body. Perhaps, however (cf. *L. Stein* in Archiv für Gesch. d. Philos. I. p. 56), another Cartesian before him, Géraud De Cordemoy, a Parisian advocate who later became tutor to the Dauphin, had stated this thought; in that case he must be regarded as the real founder of occasionalism. And while de la Forge declares that the relation between the soul and the body is intelligible only on the assumption that God has originally put them in connection with one another, Cordemoy goes further and teaches that it is just as impossible for the soul to get new representations as for the body to get new motions, if God does not intervene. The finite beings, souls as well as bodies, are only occasions for God to intervene. Thus, e.g., our will is the occasion for God to move our arm. How the tendency to concentrate all activity in an absolute being cooperates in the development of occasionalism is seen from the following statement of the Cartesian Sylvain Régis (in his *Cours de philosophie)*:"In order to produce actions properly so called one must act of oneself and by oneself, i.e. by one's own power, and it is certain that only God can act thus. From this it follows that only God is a truly efficient cause."

These thoughts came to consistent development in the Netherlandish Cartesian Arnold Geulincx (the name is pronounced Gölinks). He was born in Antwerp in 1623 of Catholic parents, studied at Louvain and became professor there. As a Cartesian he was in ill favor, and they contrived to drive him from his post. He then went to Leyden, *[240]* where he went over to Protestantism. In Leyden too Cartesianism was struggling with the old scholasticism, and Geulincx met with much opposition. After a series of straitened

years, in which he worked as a teacher of philosophy, he died of the plague in 1669, shortly after having obtained a professorship at the university. The most important of Geulincx's works that appeared in his lifetime is the *Ethica* (1665) (which, however, appeared complete only later, in 1675). The occasionalist doctrine appears partly as the background of his ethics, partly — more fully — in the Metaphysics published after his death (1695). — His collected works have recently (1891—93) been published by *Land*, who has also written an interesting monograph on him. *(Arnold Geulincx und seine Philosophie.* Haag 1895). In the following characterization we shall keep especially to the thoughts that are of significance for Geulincx's ethical standpoint.

It is a peculiar mixture of rationalism and mysticism that characterizes Geulincx's philosophy. And at the same time it shows itself clearly in him how the problem of causality grows out of the Cartesian doctrine of substance.

Geulincx sets his ethics in sharp opposition to the prevailing, Aristotelian ethics, which built on man's striving after happiness and laid great weight on practice and unconscious habit. The ethical consists, according to Geulincx, in obedience to the law of reason that God has implanted in us. Not in obedience to God's will; for this is fulfilled whatever we may do, and besides: everything, be it ever so exalted and holy, is subject to the discussion of reason. Ethics presupposes the knowledge of nature. For man must come to know his position in the world in order to be able to act rightly. Self-examination (inspectio sui) is the foundation of ethics. It teaches me that only my thought and my will belong to me, whereas my body is a part of the material world. And in the material world I have produced nothing and can produce nothing. For by what right dare I assert that I produce something, when I do not know at all how it comes to be? And besides: my acting cannot reach beyond the limit of my being, cannot flow out to other things. My
[241] will has significance only for my own soul. Just as little can things act on one another or on me. How can the acting of a thing, which is its own inner state, acquire significance for other things? It does not itself know, after all, how the change outside it is produced; its inner state does not, after all, extend wholly over into the other things! — If you think that you yourself move your limbs, although you do not know how it happens, you might just as well think that it is you who composed the Iliad, or who makes the sun go across the sky, and the child in the cradle could with the same right believe that its will directly sets the cradle going when the mother fulfills its wish to be rocked. If my willing and acting are to have significance beyond my own being, the Deity, the only true and real cause, must intervene. The same one who put motion into matter and gave it laws also formed my will and united it with the material body in such a way

that will and motion come to correspond to one another, just as two clocks that go alike and strike at the same time, not because the one acts on the other, but because they are produced by the same one. Better than this image, which had already been used by Cordemoy and has become classic in the discussion of the relation between soul and body, is an image that Geulincx uses in his *Annotata majora in principia Cartesii.* When I can procure food and clothes for money, it is not the metal's own natural power that brings this about, but its value rests on human institution (ex hominum instituto, qvo valor iste ei consignatus est); in a similar way it is not the material motion itself that is able to produce sensations and representations in me, but this happens by virtue of divine institution (nulla vi sua, sed instituto qvodam decretoqve divino). — So strongly does Geulincx emphasize the Deity as the *only acting* being that *Land* is probably right that the expression occasionalism does not quite fit his conception, since what *occasions* does after all to some degree *cooperate.* He stands in any case nearer to the standpoint of Spinoza and Leibniz than the other "occasionalists".

By the sharp emphasis on the difference and self-containedness of the individual beings the problem of causality, which concerns the possibility of a transition and *[242]* connection between them, clearly emerges. And since the divine will is conceived as the unifying power that embraces all the individual beings and acts in them, it is also intimated that the problem becomes wholly insoluble if a unity in existence is not assumed behind all differences, a continuity that extends through them all and beyond them all. Since, finally, it is said that not motion itself, but an element that is put into it or attached to it, is able to produce the mental states, a deeper solution of the problem of soul and body than the Cartesian one is intimated. Admittedly all this appears in Geulincx in a theological or mythological form that prevents full clarity and consistency. But his philosophy offers great interest through its energetic emphasis on the point that the acting of a thing and its being cannot be separated.

Geulincx himself draws from his doctrine the ethical consequence that we stand as spectators before the world, or rather before God (for the world cannot itself produce a picture of the world in me; only God can do that, and as a spectator of what takes place in the world I am therefore a constant miracle: ego ipse spectator mundi maximum sum et juge miraculum!). We are absolutely dependent on the will of the Deity. The supreme principle in ethics then becomes that where I can accomplish nothing, I shall also will nothing: ubi nihil vales, ibi nihil velis! The chief virtue in Geulincx's ethics therefore becomes humility. A characteristic contrast to the magnanimity of Descartes and of Telesio and Campanella! The vigorous self-assertion of the Renaissance had

given place to its opposite. I have, says Geulincx, no right to anything, not even to myself. It cannot then be my task to further my happiness, but only to do my duty. Yet happiness is perhaps furthered in just this way in the best manner; for, thinks Geulincx, our misfortunes really come from our all too eager striving after happiness. — Into the details of this ethics of thoroughgoing resignation we cannot go either. In many of Geulincx's ethical remarks a fine and heartfelt mood comes to light. —

[243] Occasionalism gets its extreme form in Nicolas Malebranche. The mystical side of Cartesianism decidedly presses forward into the foreground here. Malebranche, the son of a high official in Paris, entered the Oratory in the year 1660, at the age of 22. His weak health had already, while he was a child, nourished in him the desire for a quiet, contemplative life, and the last part of his life he spent for the most part in his cell. By chance he came one day upon one of Descartes's writings (Traité de l'homme), and the clear development of thought interested and inspired him to such a degree that for emotion he could scarcely finish reading it. From that moment his life was dedicated to philosophy. His chief work is the *Recherche de la vérité,* of which the first volume appeared in 1674. Of his other works, particular mention must be made of the *Entretiens sur la Métaphysique* (1687), which many regarded as his finest work. As an author Malebranche is clear, elegant and witty, but also somewhat verbose. The religious and the philosophical interest go together in Malebranche, and he thought he had found the way to a complete reconciliation between them. He did not see that his thinking led him to results similar to those that Spinoza, whom he abhorred (le misérable Spinoza, as he calls him), had already reached earlier. Just as little as Spinoza does Malebranche seem to have known the earlier occasionalists (Cordemoy and Geulincx), a proof that a natural tendency of development was stirring here. Malebranche kept his freshness of mind into his high old age and died in 1715.

Malebranche begins in his Recherche de la vérité by investigating the causes of errors and the means of freeing oneself from them. A chief source of error is the belief that the senses, which are given us for practical purposes, should be able to teach us to know the essence of things, and that we rightly transfer the sensuous qualities that we sense to the things themselves. Material things are in themselves only extended, not colored, hard, soft etc. The imagination leads us astray, since it stands under the influence of sense perception and is also dependent on our feelings. Malebranche undertakes an interesting investigation of the psychological and physiological nature of the feelings
[244] and here states views (particularly on the role that the influence of the nervous system on the blood vessels plays in emotion) that to this very day have attracted attention. —

The errors both of sense perception and of the imagination derive from the action of the body. But the understanding too, which after all belongs to the soul in itself, must be subjected to a critique. The chief question is, whence our representations (ideas) come. Only our own mental states and activities do we apprehend immediately; other things we know only through the representations or ideas we have of them. Now these ideas can be produced neither by the things nor by ourselves: for no finite being can be an absolute cause. To be a cause is something divine, and it is therefore paganism to ascribe causality to finite things. There can be only one cause, the Deity; the natural causes are only occasional causes (causes occasionelles — hence the name of this whole tendency of philosophy). When a ball is set in motion because another pushes it, this other ball is not the cause. There is no necessary connection between the motions of the two balls[49]). The collision of the balls is only the *occasion* for the author of all material motion to carry out the decree of his will by imparting to the pushed ball a part of the motion of the pushing one. The moving force of bodies is only his will, which upholds the bodies. And it is likewise with the human mind: without God's action it can neither sense nor will. One cannot help oneself by assuming that God has given bodies and souls the capacity to act, imparted to them a part of his power. For God cannot do that: God cannot create gods! Just as the true religion teaches that there is only one God, so the true philosophy teaches that there is only one cause.

Our knowledge, then, can only be explained by our being dependent on God with regard to all our thoughts. God stands in an immediate relation to every mind; he is "le lieu des esprits". Through this connection with God, the author of all things, our mind beholds the created things: the representations of them must indeed be in God, since he could not otherwise have created them. Strictly speaking, all the particular representations we have are only limitations of the representation of God, the infinite *[245]* being. Every act of thought is a closer determination of an indeterminate representation of what is, which never lets go of us. In a similar way all our striving is at bottom a striving toward God; only in union with him is true happiness, and it is he who awakens the need of happiness in us; but we often stop at a finite, limited good and regard it as the highest good — just as we regard the finite beings as true causes. Malebranche's ethics thus forms a complete analogy to his theory of knowledge.

A difficulty in Malebranche's doctrine that we behold everything in God is this: why it is then really necessary to assume an actual material world. Our knowledge of nature, according to Malebranche, finds its full explanation in God's action on us, and the actual material world is not needed at all: God's representation of it (the intelligible extension,

as Malebranche calls it) is enough. According to one report Berkeley, on a visit to Malebranche shortly before the latter's death, is said to have made this objection to him. The same objection also came out in a correspondence that Malebranche's former pupil, the mathematician Mairan, carried on with him. Mairan puts to Malebranche the alternative: either material extension does not exist (since, according to Malebranche, it is not needed to explain our knowledge of it), or else material extension, as Spinoza taught, belongs to the very essence of God (it then being not merely the intelligible extension, not merely the representation of extension, but actual extension itself that we behold in God). The old mystical philosopher was ill at ease at these objections, and the correspondence with Mairan closes with his appealing to religious faith as the ultimate foundation of all certainty. —

While Cartesianism in the series of thinkers we have discussed here led over into a theological idealism, in another circle of thinkers it helped to produce new forms of skepticism. It contained several elements that could lead in this direction. In the first place the principle of methodical doubt, the whole critical tendency with which
[246] Descartes's philosophy began. Not everyone could pass over from doubt and analysis to dogmatic settlement with the same speed as the founder of Cartesianism. In the second place the very dogmatism of Descartes could not but challenge criticism, especially as the dogmatic results themselves raised new problems, or old problems in a new form. In the third place Descartes himself maintained an opposition between positive religion and scientific knowledge, an opposition that, when more sharply stressed, could lead to denying any decisive value to the philosophical results Descartes might have reached, since they did not, after all, lead to the highest truth. These motives worked in varying degree and form on a series of men at the end of the seventeenth century who, for all their differences, have this in common, that at least for a time they stood under the influence of the Cartesian philosophy and later went over to a more or less pronounced skepticism in philosophical respects. Only a quite brief characterization of them can be given here.

Joseph Glanvil (1636— 1680) worked his way in his youth out of Puritanism and the scholastic philosophy that reigned in Oxford during his time as a student. He became a great admirer of Bacon and Descartes, especially of the latter, and cherished great hopes for the future of the science of experience. In his chief work *(Scepsis scientifica* 1665), which he dedicated to the Royal Society, he spoke up for unprejudiced inquiry; he counted himself a skeptic only in the original meaning of this word, in which it denotes one who seeks. He joins Descartes entirely in the demand for a mechanical explanation of

nature (the mechanical hypothesis) as opposed to the scholastic doctrine of qualities and forms. Nature is for him the great machine (the great automaton). But he stresses more strongly than Descartes that all our knowledge of nature is hypothetical; he traces out with great zeal all the grounds that indicate that our knowledge will always be afflicted with imperfection; and he shows how precisely the advance of experience brings new phenomena that may necessitate a change in the hypotheses we formed earlier. Like the occasionalists, but independently of them, he emphasizes very strongly the difficulties *[247]* of conceiving an interaction of soul and body, and like Geulincx and Malebranche, but before them, he comes to see the difficulties that lie in any causal relation whatever. "No cause", he says (Scepsis scientif. cap. 23), "do we know by immediate intuition, but only through its effects. When we infer that one thing is the cause of another, we build only on its constantly going together with it; for *the causal relation itself cannot be sensed* (causality it self is unsensible). But an inference from concomitance to causal relation is not compelling." Glanvil here anticipates Hume. He lays particular weight on such cases in which cause and effect are highly unlike one another, and where we therefore can have no clear and necessary insight into the relation between them. When, for all his critical and empirical sense, he defended the belief in witches, it was because he held it dogmatic to deny witchcraft, since one could not know a priori what causal relations took place in the world. His skepticism thus led him not to put full trust in the "mechanical hypothesis". He is here, like the kindred spirits who are to be named in what immediately follows, evidence that the new thinking had not succeeded in driving out the medieval notions.

High above all the men in this group, and high up into the sphere of eminent minds as a whole, towers the figure of Blaise Pascal. Pascal too (1623—1662) was influenced by Descartes and always kept some Cartesian views in his circle of ideas. But his violently and deeply stirred inner life led him to turn his back on philosophy. And yet there is always a certain kinship between Pascal's religious thinking and Descartes's philosophizing: both proclaim conviction free of authority. In his *Fragment d'un traité du vide* Pascal speaks out against the power of authority in the scientific domain, and in the treatise *De l'esprit géometrique* he gives clear indications of a doctrine of method, with express acknowledgment of Descartes's Discours de la méthode. Later, after his religious breakthrough, he fights on the side of the Jansenists against the validity of papal authority in the face of facts. From Rome he appeals to heaven. And on the occasion *[248]* of Galileo's condemnation he utters (in the 18th Provincial Letter) the often repeated saying that if the earth really goes round, no decree will be able to prevent it; it will go

round with all men, whether they believe it or not. Like Bruno and Descartes he assumes the infinity of the universe and uses this thought in his religious agitation to make the soul dizzy with the feeling of being a vanishing element. He agrees with Descartes in his conception of soul and body. The body is to him only a machine, an automaton. The sharp opposition between thought and extension he uses to impress upon us man's spiritual worth: as a material being man is admittedly to be reckoned as nothing over against the enormous masses; he is only a reed — but a *thinking* reed! This is a genuinely Cartesian thought. But even though this thought could fill Pascal with enthusiasm, it did not for him contain the highest. Above the world of matter and of mind there stood for him the supernatural world of love, which comes into being for man through God's immediate revelation. Religion is the heart's immediate experience of God (dieu sensible au coeur). What philosophical proof can lead to is at best an abstract God, perhaps the God of truth, but not the God of love, the one true God. Deism and atheism come for Pascal to much the same thing. What he yearns for is to have his heart filled, and no inquiry could help him to that. But besides this direct need to attach his feeling to an infinite object, another psychological motive is also at work in Pascal, which conditions the decisive difference between him and a figure such as Malebranche. It is the anguish with which he feels himself existing in a world without limits, without stability, a world concerning which knowledge can arrive at no firm results. Skeptics and dogmatists contend; neither of the two parties can be proved right — and yet one of them must be right; the very contention between them shows that the skeptics are at bottom right. It is no use wanting to ground the first principles of knowledge on human nature: for where is human nature to be found? It changes constantly, or else it is ruled by habit.
[249] In practical respects man is driven on by restlessness and vanity, or he follows custom and usage. Man is full of oppositions and contradictions. Doubt and uncertainty reign within him. "He is to be judge of all things and is yet a wretched worm; he is to be the bearer of truth and is yet a sewer of uncertainty and error; he is at once the glory of the universe and its disgrace!" — In the face of all these contradictions Pascal knows only one rescue: to take refuge in the historically revealed Christianity. There his own loving and trembling soul found rest. In order to lead others there he had devised an account that rested on the biblical history and was to show how all oppositions and contradictions in the nature and condition of man fell away through the revelation given in the biblical writings. It is certainly a piece of good fortune for Pascal's memory that the positive part of his apology for Christianity was not carried out, so that his thoughts *(Pensées)* on human nature and its condition, which were only to make up the

preparatory part of the apology, can now act by their own power. It is the strong feeling of the weight and value of life that animates Pascal. In opposition to the rationalistic tendency, which in Descartes's philosophy received such good support, Pascal appeals to the living personality, to the heart, to conscience as the ultimate decisive motive in the view of life. A century before Rousseau he raises the question of the value of scientific knowledge for personal life. In the face of his need of love and his anguish of soul the answer becomes that "the whole of philosophy is not worth an hour's trouble!" And he had the plan of writing against those who go too deeply into the sciences (qui approfondissent trop les Sciences). That he himself found the solution in the Christian, particularly the Catholic, more particularly the Gallican, and most particularly of all the Jansenist doctrine, he would not regard as the result of habit or tradition; he would not see that here he still left a place open for the skeptical arguments that he himself had brought into the field. And when he was asked what one was then to do — what one was to resort to in order to bring the doubting thoughts to rest, he answered as a good Cartesian: We are automaton as well as we are mind. Both must be brought to believe, *[250]* the latter by conviction, the former by habit. And when conviction will not come, then begin with habit: act as if you believed; take holy water and have masses said; it will make a beast of you and make you believe (cela vous fera croire et vous abêtira)! That is cutting the knot — with the help of the philosophy that he at once uses and combats. —

Pierre Daniel Huet, Bishop of Avranches (b. 1630, d. 1721), likewise wished to abolish free inquiry in favor of religious faith, which in him appears more pronouncedly than in Pascal as the faith of the Church. (Cf. *Barach: Huet als Philosoph.* Wien 1862). He also still holds to essential parts of the scholastic philosophy. For a time he had been a Cartesian, and he felt great enthusiasm for the clear and simple method of the new philosophy. But he later came to doubt whether there could be any rational criterion of truth at all: we must then again have a criterion that this is the right criterion of truth — and so on to infinity! Descartes urges us to begin with the simple and from it to pass over to the composite; but there is nothing so easy and simple that it could stand as indubitable! Descartes thinks that we cannot doubt thought itself; why not? We know thought, after all, only in the same way as everything else: by directing our attention to it — that is, by a new thought. An absolute beginning is therefore not reached. Descartes's "Cogito, ergo sum" is an inference whose legitimacy can be called in doubt, and his proof of the existence of God falls away when one sees that existence is not a property. Only the authority of tradition and of the Church could furnish a foundation for knowledge, and Huet, who was a learned humanist, offended at the Cartesians' disdain for the ancients,

therefore seeks to show that the religions and philosophical systems of antiquity are due to a tradition stemming from the Jewish people.

Pierre Bayle (1647—1706) had come to know Cartesianism during his student years in Geneva, and in him it now displaced the scholastic philosophy into which he had been introduced at a Jesuit college. He admired especially the Cartesian demand for
[251] clear and distinct concepts and for leading complicated questions back to simple and immediately evident principles. To several of the doctrines of Cartesianism he later (during his university work at Sedan and in Rotterdam) took up a critical attitude. He was quite as much a scholar and man of letters as a philosopher; he took an interest in literary phenomena and speculative opinions in their motley multiplicity, and the need of clear and distinct concepts led him to sharp pointing up of the difference between standpoints, to decided emphasis on the sting of problems, and to the unmasking of illusory solutions. In his masterly characterization of Bayle Ludwig Feuerbach shows that Bayle was by no means a systematic doubter, but that it was the difficulty and multiplicity of the individual subjects that gave birth to doubt in him. Philosophy was for him essentially criticism, its significance more negative than positive. Its results decidedly conflict with those of theology, which Bayle shows in the case of a single dogma (that of the Fall) by setting up a series of theological and philosophical fundamental propositions that prove to be wholly incompatible with one another. By this setting up (in his *Réponse aux questions d'un provincial)* Bayle opened the most important debate on the problem of evil since Jakob Böhme's "Aurora". Already in his *Dictionnaire historique et critique* he had maintained (especially in the article on the Manichaeans) that the assumption of two principles in the world, a good and an evil one, would be very difficult to refute if it were defended by unprejudiced and acute thinkers. If Augustine had not left Manichaeism, he would, thought Bayle, have been able to make something of it that could have confounded the orthodox. By the way of reason this doctrine could not, in his view, be refuted. According to Bayle a choice must be made between reason and faith, and faith shows itself all the more glorious the more contrary to reason the dogma is that is to be believed. Bayle himself had certainly made his choice and was sincere in his sharp distinction. But this choice has not prevented him from acknowledging a humanity independent of all dogmatism. He lays (especially in *Pensées diverses à l'occasion de la comète)* great
[252] weight on the natural instincts that keep the individual and the species alive without conscious reasoning. And as regards in general the conduct of men in public and private life, he maintains that it is determined far more by their natural temperament and their character than by their belief or unbelief. He can therefore (which aroused great offense)

very well conceive of a state of atheists. Ludvig Holberg later remarked, precisely on the occasion of this line of thought of Bayle's, that it made "the whole disputation" on whether freethinkers could live morally "in vain" (Ep. 210). This line of thought stands in natural connection with Bayle's zealous struggle for toleration, and no less with his maintaining the independence of ethics over against theology. The Cartesian method is applied by him here in an interesting way: just as the proposition that the whole is greater than the part is so clear and distinct that no revelation can be accepted if it conflicts with it, — so too the simplest ethical judgments, by which we distinguish between good and evil, are so clear that no revelation can abolish them, but the validity and value of every revelation must conversely be tested with them as the standard. If one rejects the natural light here, one has no standard at all. On this point a possibility comes out in Bayle that would make the conflict between faith and knowledge lose its significance: namely this, that the natural instincts and the immediate sense for the true and the good were sufficient to carry life! It became the task of the eighteenth century to develop this possibility, which the Renaissance had already seized in its discovery of man, but which the victorious reaction had not allowed to get beyond the narrow circle of the thinkers. The thought also still needed a fuller development. —

3. Pierre Gassendi

formed a counterweight to Descartes, especially in the French world; yet more by his method than by his results. He wished to be a philosopher of experience and to plead the cause of sense perception against Descartes's intuitive and deductive method. While Descartes, with greater speed than his own method allowed, rose to the highest ideas and *[253]* tended to ascribe to them an immediate grounding in the human mind, Gassendi maintained the empirical origin of our concepts. The very polemic *(Disqvisitio metaphysica.* Amsterodami 1644) that he published against Descartes is among the most interesting documents in the debate between the empirical and the deductive school. When Gassendi nevertheless ends in spiritualistic results, this is due to his clerical position, which made it necessary for him to combine theology and empirical philosophy. He was born near Digne in the south of France in 1592, became a priest in 1617 and later professor of philosophy at Aix, a position he later exchanged for a professorship of mathematics in Paris, in which he worked until his death in 1655. In his youth Gassendi published a sharp critique of the scholastic philosophy; after that the severe decree of 1624 against the new doctrines and the condemnation of Galileo caused him to hold back. He occupied

himself with physical and astronomical studies, but did not venture to declare himself for the Copernican doctrine. In his later writings he developed the atomistic doctrine partly in the form of a renewal of the Epicurean doctrine. He defends the personality and philosophy of Epicurus against the current attacks. These works became of great significance, since the atomistic theory, which Bruno and Basso had prepared, was excellently suited for use by modern natural science. Gassendi was an admirer of Galileo, and the fundamental propositions that he sets up for the knowledge of nature betray Galileo's influence *(Opera omnia.* Lugd. 1658). Starting from the premise that nothing can become nothing or arise from nothing, and that something must therefore subsist through all changes, he shows that if change is to be explicable at all, the something that subsists through all changes, i.e. matter, must be a multiplicity of solid and indivisible elements, i.e. atoms. Mathematically, division can be continued to infinity, but physically divisibility has a determinate limit. All changes in nature are to be explained by the motions of the atoms: all causes are causes of motion. Between the atoms there is an
[254] empty space; otherwise their motion would not be possible. Space, like time, is a thing of its own kind, to be called neither a being nor a property. Only by contact can motion arise and cease; therefore all causes must be material. The soul moves the body only insofar as it (as vegetative and sentient soul) is itself material. Gassendi differs from Descartes by maintaining that solidity, not merely extension and mobility, must be ascribed to matter, and by setting a limit to physical divisibility. He teaches moreover (under the evident influence of Galileo) that it is not motion itself, but the impulse of motion (impetus), that is constant; the impulse of motion does not fall away because motion passes over into rest. These improvements of the new mechanical view of nature mark a significant advance. On the other hand Gassendi recalls the predecessors of Descartes and Galileo, especially Telesio, by assuming sensation in all atoms and using this assumption, as well as the notion of a world-soul, in his philosophy of nature. A similar half-heartedness appears in his attempt to combine spiritualistic notions with his atomistic and mechanical doctrines: the atoms are created by God; and to the proposition that all causes are material God and the supernaturally implanted soul or part of the soul (anima rationalis or intellectus) form an exception. This compromise was even more inconsistent than the Cartesian one, since the soul itself was split into two parts, a material and an immaterial. But it brought the advantage that the learned, and especially the theological, world grew used to not regarding atomism as an absolutely godless doctrine. Natural science could then make use of the atomistic hypothesis undisturbed, as its needs required.

4. Thomas Hobbes

a. Biography and Characterization

When in the year 1588 the Spanish Armada was about to put to sea in order to attack England, the terrifying rumors of war caused the wife of the clergyman in Malmesbury *[255]* to give birth prematurely to a little son. She bore, says the latter later in a versified autobiography, twins, namely me and fear. He derived from this his timorous, peace-loving temperament, which nevertheless — as his life shows — did not exclude energetic thinking and a delight in literary feuds. The boy, Thomas Hobbes, received his first instruction in his native town and at the age of 15 came to Oxford, where he learned the scholastic logic and physics, which interested him very little. He left the university early and became tutor to a young nobleman of the family of Cavendish. To this family he remained attached all his life. His position gave occasion to repeated journeys round Europe. He gathered experience of the world and cultivated literature with zeal, especially the classical historians and poets. He followed the events in his native country with an attentive eye, and a translation of Thucydides which he published in 1629 was not prepared without regard to the political storm clouds that already in the first years of the reign of Charles I threatened on the horizon. For a time, however, his interests were drawn in an entirely different direction. By chance he came upon Euclid's geometry and thereby discovered that there existed a rigorous deductive science. He had not suspected this until then, for mathematics did not yet belong to the regular subjects of instruction in England, and was indeed even regarded as devilry. He now obtained a model for his thinking. A mathematician by profession he did not, to be sure, become; his later attempts in that direction turned out most unfortunately and involved him in long literary feuds in which he decidedly drew the shortest straw. But the foundation of his philosophical tendency had now been laid. The great weight he lays on deductive thinking places him close to Descartes, while it places him in decided opposition to Bacon, with whom he had stood in a personal relation, having in Bacon's last years been of assistance to him as secretary and in the translation of his books into Latin. From Bacon he received no decisive philosophical impulses, even if one can perhaps show on individual points that, after his own peculiar philosophizing had been set going by impulses from other quarters, he took up Baconian ideas and wove them into his system. *[256]* The problem that set his thinking going he is said, according to what his first biographer relates, to have received in a conversation that probably took place in a scientific circle in Paris, and in which the question was raised in passing what sensation is. As this

question was not answered, Hobbes set himself to think about it. It then dawned on him that if material things and all their parts were always at rest or in uniform motion, all difference (discrimen) between things, and consequently all sensation, would fall away. From this he then concluded that difference of motion (diversitas motuum) is the cause of everything. Now he thought, by his own account, of nothing but motion day and night, waking and sleeping. It had dawned on him that the deductive method, which he had not long before come to know in its most perfect form, was excellently suited to application on the basis of the fundamental proposition that everything is motion. Probably he also at the same time (c. 1630) arrived by his own path at the conviction of the subjectivity of the sense qualities[50]). That Galilei had already in 1623 stated this principle, Hobbes did not know. Only after it had dawned on him does he seem to have become aware of Galilei's writings, and later on a journey to Italy he came to know the man of whom he himself later (in the dedication of De Corpore) said that "he first opened to us the first gate to the whole of physics, namely the nature of motion." Thereby his already begun line of thought received a more definite form and a firmer basis. Galilei is said also to have drawn his attention to the possibility of treating ethics deductively in analogy with physics. If this account is correct, then Hobbes's whole system had thereby found its provisional conclusion for him. For a long time yet, however, only in his mind and thought and in the confidential exchange of ideas that took place within a circle of men whom the monk Mersenne, Descartes's and Gassendi's friend, gathered about him in Paris, and to which Hobbes also belonged. "From the moment," says Hobbes in his autobiography, "that I had communicated my ideas to Mersenne, and he in turn
[257] had made others acquainted with them, I too was counted among the philosophers. His cell was better than all schools." It was also Mersenne who later sent Descartes's Meditationes round to various thinkers, among others Gassendi and Hobbes, and thereby called forth one of the most remarkable philosophical discussions of the 17th century. Hobbes's critique of Cartesianism is an interesting document for the illumination of his own philosophy. Hobbes, to whom the late Lord Cavendish had secured economic independence, now intended, when in the year 1637 he came back to England after several years' absence, to work up his ideas into a threefold system: Corpus — Homo — Civis, a doctrine of the laws of matter, of man and of society, as far as possible by the deductive method on the foundation of the general laws of motion. But the execution of this great plan was prevented for a long time by the outbreak of the English Civil War. His temperament made him look with anxiety on the political strife growing in passion. It seemed to him to shake the foundation of all ordered social life. He feared that the

elementary drives would stir with unbound and unconditioned violence. His experience of the world had taught him how much the impulse of self-preservation and egoism can work not only for good but also for evil, and that a strong authority is needed to keep the stream of these forces within its proper bed. He was ultra-conservative precisely because in his theory he was ultra-radical and went back to the purely elementary presuppositions. But surely his moving in the circles of the high nobility also contributed much to determining his political sympathies. He therefore misjudged the significance of the English Revolution, although the grounds he gave for his condemnation were purely naturalistic and thereby diametrically opposed to the grounds that held the adherents of the Stuarts together. For Hobbes the principle of authority was a derived principle; for them it was an absolute, divine, supernatural principle: this was the great opposition that came out as soon as Hobbes worked out his political theory. And it is the grounds given, not the results, in Hobbes's political theory that have given it its lasting and in a certain sense unique significance in modern ethics and political theory. In the year 1640 he composed a work that contained his psychological and ethical-political theory *[258]* in its main features. This work, which circulated in copies and was later published in two parts, Human Nature and De Corpore politico, has recently (1888) been published by TØNNIES from the oldest manuscripts and with the original title *Elements of Law.* It is one of his freshest and most instructive works and ought to form the foundation of every study of Hobbes. It was composed with the troubled circumstances of the time in view and as a word of warning. Hobbes believed that he would have been exposed to personal danger from the side of the opposition if the King had not dissolved Parliament. When the unrest increased, and when the new Parliament began to proceed severely against the King's men, he fled (at the end of 1640) over to France in order there to be able to continue his studies in peace. But the grave circumstances of the time gave him no peace to turn his thoughts away from political problems, and he then worked up the last part of "Elements of Law" into an independent work *(De cive),* which appeared in 1642 in a small, in 1647 in a larger edition. It was intended to be the third part of the whole system. From the earlier treatise it differs by stressing far more strongly the opposition between the state of nature and life in the state, and by insisting on the necessity of the sole authority of the power of the state in religious matters. Hobbes was an opponent on principle of all hierarchy, the Protestant as well as the Catholic. This made his stay in a Catholic country and in a circle of exiled clerical royalists insecure. There came a formal breach when he had once more laid his studies in natural philosophy aside in order to write a political work. This was the famous work *"Leviathan* or the

Matter, Form and Power of a Commonwealth ecclesiastical and civil", which appeared in London in 1651. It was named after the mighty sea monster mentioned in the Book of Job, of which it is said: "Upon earth there is none like him; he is made without fear." With this mighty one Hobbes compares the absolute power of the state. He here carries through the doctrine of sovereignty founded by Bodin and Althusius to its extreme consequences — but on a naturalistic basis and in an antihierarchical spirit. That it was not especially the absoluteness of the royal power but of the power of the state that he
[259] asserted was interpreted as if with this work he were angling for Cromwell's favor. And the episcopal royalists were indignant at the statement (in a concluding section which was left out in the later Latin edition of 1670) that just as Elizabeth had overthrown the Catholic hierarchy, the Presbyterians had overthrown the Episcopalians, only to be overthrown themselves by the Independents — "and so we are brought back to the independence (independency) of the first Christians." The consequence was that Hobbes lost his position as teacher of mathematics to the young King Charles II and was denied access to the court. In England conditions were now calmer; there was again a firm and ordered power of the state, an exerciser of sovereignty, and Hobbes thought that it was now safest for him to return home. After a toilsome journey in wintertime he came home to England at the end of 1651, and he could now enjoy the benefit of complete freedom of the press to work in peace on the publication of his works, which he had never lost sight of. In 1655 appeared *De corpore,* which comprises logic, the doctrine of fundamental concepts (philosophia prima), the doctrine of motions and magnitudes, and the doctrine of physical phenomena. In 1658 appeared *De homine,* which for the most part is an optics (to illuminate the nature of the sense of sight) and otherwise gives only brief outlines of the psychology of language and of the feelings. This second part of the system can measure itself neither in content nor in form with the first part of Elements of Law or with the opening chapters of Leviathan. The third part of the system was the earlier published work De cive, and the whole system was thus now complete. Then came the Restoration, which Hobbes greeted with joy. He came into favor with his former pupil Charles II, who often conversed with him. He lived with the Cavendish family as an old student, much occupied with mathematical and theological polemics. His antihierarchism was made into atheism, and a Hobbist counted as the same as what was later called a freethinker. Although he ascribed to the power of the state the sole right to determine what should be taught, he himself interpreted the Bible in a critical and rationalistic spirit. For him the original faith of Christianity was only this, that Jesus was the Messiah, not any speculative doctrine whatsoever. The eternity of the

punishments of hell he rejected. He assumed no incorporeal spirits. But for the most part *[260]* he referred to the incomprehensibility of the doctrines of faith and advised accepting them without thinking about them, just as one does best to swallow bitter pills whole without chewing them. Hobbes preserved his physical and mental energy into his high old age; at 87 years of age he translated the whole of Homer into iambic meter. He died at 91 years of age (1679).

Hobbes is an acute and energetic thinker. Thinking, he says in the preface to De corpore, will be esteemed the highest enjoyment if one but once experiences the joy that is caused by the meeting between our mind and the glorious universe (pulcherrimi mundi cum anima congressus). He made the most thoroughly considered attempt in modern times to make the knowledge of natural science the foundation of all our knowledge of existence. It is the most profound materialistic system that has been set up in modern times. But Hobbes's writings, which are distinguished by their vigorous and clear exposition, contain besides many interesting remarks in logical and psychological respects. He may be called the founder of English psychology, and thereby of the properly English school in philosophy. Above all, however, it was his ethical and political points of view that became of far-reaching significance. By his bold, though one-sided naturalism he set people's thoughts in motion, partly by provoking them. He broke with scholasticism in the domain of the human sciences, just as Copernicus had broken with it in the domain of astronomy, Galilei in that of physics and Harvey in that of physiology. By the side of these men Hobbes, with justified self-esteem, places himself as founder of the science of society: this (he says in the preface to De corpore) is no older than his book De cive. By his naturalistic grounding of ethics and politics he roused a movement which has aptly been compared with the one Darwin roused in the 19th century.

b. First Presuppositions

Science stands for Hobbes in opposition partly to sensation and memory, partly to theology. It does not, like sensation and memory, stop at the individual phenomena, *[261]* but seeks to know them from their causes; now it goes from cause to effect, now from effect to cause. From this it follows that science can only concern what has an origin; thereby theology is excluded, the doctrine of the eternal God, who has not come into being. — So at least I define science, says Hobbes. Perhaps someone wishes another definition; that is his affair. Definition is a free and arbitrary matter, a mere explanation of what we mean by a name, and the use of names is arbitrary. Science begins with the definition of the underlying concepts. One arrives at these concepts by analysis

(resolutio) of what is given to sensation (a sensibus ad inventionem principiorum). The validity of the principles, the fundamental concepts, itself cannot be the object of proof, only of definition. They are known in themselves — otherwise they are not principles. Since they are fixed by arbitrary naming, they are produced by ourselves. It is art or construction, not science, that fixes the principles: here we ourselves produce the truth: ratiocinationis prima principia, nempe definitiones, *vera esse facimus nosmet ipsi* per consensionem circa rerum appellationes (De corpore. cap. 25, 1., cf. cap. 3, 8—9).

This sharp emphasis on the purely arbitrary in the setting up of the first presuppositions Hobbes makes properly only in the work De corpore; in Elements of Law it does not appear so clearly. It is due to the ever more predominant deductive method: when all true science is deduction, the presuppositions of deduction, which cannot themselves be deduced, must come to stand as purely arbitrary. But then it cannot be understood what Hobbes means by saying that the principles must be "known by themselves" (per se nota), if this is to mean more than that one is conscious of what is meant by the words employed. And this immediate consciousness itself is then a presupposition which properly ought to be set up by itself. It is a kind of intuition or, as Hobbes himself in one place[51]) calls it, a reflection. Besides, Hobbes declares, as we know, that we arrive
[262] at the principles by resolution or analysis (resolutio, analysis) of the given; it is then not altogether arbitrary what principles we set up, and there is more in the setting up than mere naming. Indeed, Hobbes says expressly that analysis is an inference from the given to the principles or definitions (analysis est ratiocinatio a supposito ad principia, id est, ad definitiones. De corp. cap. 20, 6): thus the first principles of inference are themselves set up by inference, in so far as they are set up by analysis. It turns out, then, that the situation in setting up the principles is more complicated than the pure theory of arbitrariness assumes.

Hobbes has also in another way limited the arbitrariness in setting up the principles, or in naming. One cannot, for instance, give one and the same thing two mutually opposed names. Of this principle Hobbes declares (De corp. cap. 2, 8) that it is the principle of all inference, and so of all science. But it is surely not itself set up arbitrarily; it precisely restricts arbitrariness: if I have given a thing a name, I cannot add to it *any whatever* other name. The same holds everywhere we construct: only the first step can be purely arbitrary; the following ones are determined by the first. — Apart from the wavering and one-sidedness that shows itself here in Hobbes, it is his great merit to have brought out so sharply the active and constructive element in knowledge. Neither in Descartes nor in Bacon did this come sufficiently forward, and the decisive point where

one turns from induction and analysis to deductive thinking therefore did not receive sufficient illumination with them. —

When we have now arrived by the way of analysis at the first principles, the task is to derive the phenomena from them. As a general principle Hobbes sets up that there is a single cause of all the properties that the phenomena present, namely *motion.* This assertion, he thinks, needs no proof; it will be granted by anyone who thinks it through without prejudice (De corp. cap. 6, 5). To this proposition, that all change is motion, a series of other propositions attaches itself, which Hobbes sets up on occasion without examining their closer relation to one another or to the one first set up. Thus the general causal principle (which is set up in the work on freedom and necessity), the principle *[263]* of inertia, the proposition that the cause of motion must always lie in the motion of a contiguous body, and the proposition of the conservation of matter. In a single place, however, Hobbes proves the proposition that all change is motion by means of the causal principle and the principle of inertia, and for these two propositions too he sets up proofs on occasion, which, however, on closer examination turn out to presuppose precisely what was to be proved[52]).

From the proposition that all change is motion, and the propositions more or less closely connected with it, Hobbes now wishes to build up his system by the deductive way. After logic has first examined the method, and philosophia prima the most important fundamental concepts, geometry discusses the mathematical laws of motion, mechanics the effects of the motion of one body on another body, physics the effects of the motions that take place in the small parts of bodies, the doctrine of man and the state (what would now be called the human sciences) the motions that take place in men's minds and determine their conduct toward one another. If the system could be developed purely deductively, we should in these various disciplines merely have progressive special applications of the general laws of motion, starting from the first self-created presuppositions, through the series of steps Corpus — Homo — Civis. And this is evidently the ideal that stands before Hobbes. But just as we have seen that the doctrine of the self-creating activity in the fixing of the presuppositions is, by his own account, subject to considerable restrictions, so it turns out also that at three places in the system there is a break in the strict deductive connection, since the relations are too complicated for a single, progressive method to manage them. These three points lie at the transition from geometry to mechanics, from mechanics to physics, and from physics to psychology.

In the first place, the system would form a strict unity if mechanics could without more ado be derived from pure geometry, without wholly new presuppositions being

taken up. But in mechanics the cause, or the concept of force, comes forward, while
[264] geometry presupposes only extension and the general logical principles. That Hobbes lays no weight on the difference between geometry and mechanics is connected with his not distinguishing between ground and cause. Yet he admits the difference in so far as he concludes the geometrical section of De corpore with the remark: "In this section we have considered motion and magnitude in themselves, and in an abstract manner. The next section deals with the phenomena of nature, i.e. with the motion and magnitude of real bodies." (De corp. XXIV, 9).

Although Hobbes lets science go partly from effects to their causes, partly from causes to their effects, he is nevertheless clear that only when one goes from cause to effect does knowledge have strict necessity. When we begin with a given phenomenon and regard it as effect, the knowledge of the cause can always be only hypothetical: what one can find out can only be the manner in which the phenomenon *could* have been produced; but whether it really *has been* produced in this manner cannot be decided purely deductively. When we start from the given and seek its real causes, we do not ourselves produce the principles, but must try to find the principles which have been employed by the Author of nature in producing it. Herein physics differs from geometry and mechanics (De corp. cap. 25, 1. De homine, cap. 10, 5). — We must therefore, at the transition from mechanics to physics, halt our deduction in order to examine what phenomena are given, and form our hypotheses by analysis of them. The question is whether something analogous does not already hold for the presuppositions of logic, of "the first philosophy," of geometry and of mechanics, so that these too are not apodictic but of a hypothetical character. This seems to follow from the fact that analysis is necessary in order to find them[53]).

The third point where the successive advance in the system gives place to more complicated relations is at the beginning of the human sciences. For, says Hobbes, the causes of the mental motions can be found not only deductively, that is, from the propositions of mechanics and physics; here everyone has, by self-observation (per uniuscujusqve proprium animum examinantis experientiam), access to knowledge of
[265] the principles on which political theory builds: the human feelings and drives and their tendencies. One can therefore come to political theory not only by the long series of deductions from geometry through mechanics and physics, but also directly, inductively, from the facts of self-observation, which are set up as new starting points for deduction. (De corp. cap. 6, 7). It is the independence of empirical psychology over against the science of material nature that Hobbes here acknowledges. He had all the more occasion to call attention to this point since he had himself, as we know, published psychological

and political-philosophical writings (Elements of Law, De cive and Leviathan) before the work which in the system comes before the human sciences (De corpore). He ought here to have gone a step further. For what could be grasped deductively from physics can in any case only be the physiological processes that are connected with the mental phenomena. As phenomena of consciousness the mental phenomena can be known only by the inductive way of self-observation, and the break in the deductive process is therefore greater than Hobbes sees.

The last two points are, moreover, on closer examination connected. For the reason why at the transition from mechanics to physics we must break off deduction is properly this, that physical phenomena appear to us as qualities, but these as such are due to the sentient subject. What is given in physical phenomena is thus subjective phenomena, whose objective cause is to be sought, which cause must necessarily — on Hobbes's presuppositions — be motion. (De corp. cap. 25, 3). It is thus the mental phenomena that at both points break the unity of the method and of the system.

c. The Object of Science

In agreement with his first fundamental proposition, that all change was motion, Hobbes declares that the object of science is the corporeal. By body (corpus) is understood everything that is independent of our thinking and fills a part of space. As independent of us it is called substance. Body and substance are one and the same. An incorporeal substance *[266]* is for Hobbes a contradiction. But substances and bodies we do not perceive immediately, only by inference. It is only the properties (the accidents) that we observe; of substance as such we have properly no representation, although we infer that there is something that underlies the properties. The essential properties of body are extension and motion. All other properties are only subjective phenomena in the sentient consciousness. Space and time too are, according to Hobbes, subjective, since he understands by space and time such representations as the extension and motion of bodies call forth in us, and which we could retain even if all bodies disappeared.

One might interpret this to mean that Hobbes renounces knowledge of the essence of things and keeps only to their properties, in so far as these can be apprehended with the help of sensation. And Hobbes has even a very lively understanding of the fact that the very circumstance that something can appear to us and be apprehended by us is for the philosopher the most remarkable of all facts. "Of all phenomena, the very fact that something appears to us as phenomenon (id ipsum *τὸ φαίνεσθαι)* is the most wonderful of all, so that, since the phenomena are principles for the knowledge of everything else,

sensation is the principle for the knowledge of the principles themselves, and all science is derived from it, while its own causes can only be found by beginning with it." (De corp. cap. 25, 1). Here the sentient consciousness is set as the first starting point for all knowledge. It is a statement that recalls Descartes's "Cogito, ergo sum," and which might seem to conflict with the assumption that Hobbes's system is materialistic. It is also only in so far as Hobbes proceeds strictly deductively from his original postulates or definitions that he can be called a materialist. His materialism falls away every time the strict deductive procedure, which would derive everything from the one proposition that change is motion, is broken. That Hobbes wants such a derivation is certain, though it is also certain that he cannot achieve it, and that he sees that he cannot achieve it. Whether
[267] one will call him a materialist or not therefore easily becomes a dispute about words. Edv. Larsen understands (in his monograph on Hobbes, Copenhagen 1891. p. 186) the matter thus: Hobbes is a materialist only in his doctrine of method, because our knowledge in his opinion could only give an account of motion, but he does not wish to give a materialistic metaphysics. Hereby, I believe, a distinction between method and system is put into Hobbes which he does not know. He is too much of a dogmatist to distinguish in this way. He trusts that things are such as they must be according to our definitions. He is a materialist in the same sense as Descartes is a spiritualist. Both become entangled in contradictions through their assumptions; but of these contradictions they were not conscious.

We therefore also see Hobbes, just after he has brought out with great emphasis what problem lies in the very fact that something becomes a fact for us, pass with great composure to asking whence, then, sensation, the principle of all knowledge, comes, and answer this question in consistency with his first fundamental proposition: since all change is motion, sensation too, which is due to a change, must be motion. Sensation, he says, is *nothing but* a motion in the small parts of the sentient body. (De corp. cap. 25, 2, cf. Leviathan, cap. 6). And what he here says of sensation he says, in the objections to Descartes, of consciousness as a whole (mens *nihil aliud erit* præterqvam motus in partibus qvibusdam corporis organici). Thus for Hobbes psychology properly belongs entirely under the doctrine of motion. And even if he distinguishes between motion itself and its appearance (apparition) to us, as when he insists on the subjectivity of the sense qualities, he nevertheless maintains that *what really exists*, when we have sensations or feelings, is only motion. The feeling of pleasure is, e.g., nothing but a certain motion in the heart (pleasure is nothing *really* but motion about the heart, Elements of Law, VII. 1). Appearance, apprehension, feeling, consciousness become for Hobbes, where he

speaks strictly deductively, only a seeming (nothing really, El. of Law, I. 2, 5). — But a seeming, we must add, that presents a great problem: for motion, according to Hobbes, *[268]* can produce only motion, and it then stands as an inexplicable marvel that at certain points, besides motion, there comes into being the seeming, the "apparition", which we call consciousness. It is Hobbes's great significance that he has so clearly established the limitation of the materialistic hypothesis. His energetic deductive thinking has here led him to the decisive point. No later materialistic attempt has been carried through with the clarity and freedom from prejudice of his. He has thereby given a contribution of lasting value to the elucidation of the place of mental phenomena in existence. What is cast in one piece always carries its reward with it. —

As regards Hobbes's special doctrine of nature, it need only be remarked that he no more than Descartes assumes an empty space, but conceives the smallest solid particles as moving in an ethereal fluid. Motion vanishes for him no more than for Gassendi in the transition to rest. He conceives space and time as composed of small parts, each smaller than any part experience can show us (minus qvam datur); even in such vanishingly small parts of space and time there is motion, but only as tendency to motion (conatus, endeavour). And this tendency to motion does not vanish because an equally great opposite tendency to motion holds it in check; it then becomes tension (nixus). By the help of these concepts, set up (De corpore, 15, 2; 22, 1), it becomes possible for Hobbes, who here (like Gassendi with his corresponding ideas) stands under the influence of Galilei, to maintain the continuity of motion in a more perfect manner than Descartes. And he has (as is seen from a treatise first printed in the appendix to Tönnies's edition of Elements of Law) applied these thoughts in a polemic against Descartes (El. of Law p. 215).

d. The Limits of Knowledge. Faith and Knowledge

Every representation we can form is, according to Hobbes, finite and limited. A knowledge of the infinite is therefore excluded. When one hears the words "eternal and infinite," one must be prepared to hear absurdities. "Infinite" can only be used in a negative sense, *[269]* of that for which we can set no limits. This word signifies nothing in the thing itself, but a lack of power (impotentia) in our mind. We express by it our own limitation, not a positive property of any being. — For Descartes, conversely, the infinite was the positive concept: the finite arose by limitation of the infinite. These are two different views of life that express themselves in this opposition.

Just as little as there can be a knowledge of the infinite, just as little can there be a

knowledge of the world as a whole. Only a few questions can be raised about the whole of the world — and none of them can be answered. Science cannot say anything about the world's magnitude, duration or origin, not even whether it has an origin at all. The knowledge of the first things, the first fruits of wisdom (primitiæ sapientiæ), is reserved for the theologians, just as in Israel the first fruits of the harvest were offered to the priests. So Hobbes expresses himself in De corpore (cap. 26, 1). He further argues here that we cannot close the causal series in such a way that we could know for certain that it could not be continued further back. Even if we assumed a first cause, this cannot be something unmoved, for nothing can be set in motion by something that is not itself in motion; what we stop at must then be something eternally moved. But there is in itself no contradiction whatever in assuming that the world has no beginning. Therefore, says Hobbes, I cannot praise those who boast that they have proved by natural reasons that the world has an origin. — In a remarkable opposition to this stands the manner in which Hobbes expresses himself in his political-philosophical writings, (El. of Law I, 11, 2; De cive, cap. 14—16; Leviathan, cap. 11—12). He here acknowledges, to be sure, the incomprehensibility of God, but he finds no difficulty in inferring back from the world to God: by continually going back, we must necessarily come to an eternal cause which in turn has no cause! Yet it is not only theoretical but especially practical reasons that lead to natural religion, since man, on account of his weakness and dependence, renders
[270] God obedience. — This opposition is all the more striking as it cannot be explained by the political-philosophical writings being older than De corpore. For in the objections to Descartes, which were written at about the same time as De cive, it is expressly maintained that the creation of the world cannot be proved. The matter can hardly be explained in any other way than that Hobbes in his political-philosophical writings, from practical motives, attributed greater probative force to the grounds of reason than he strictly could. — But he always maintains that we can form no representation of God. It goes with us here as with the blind man who warms himself at the fire: of the cause of the warmth he can form no representation. We dare not transfer our properties and states to God — neither pain nor need nor understanding nor will; for they all presuppose a limited being in order to have any meaning. Only negative, superlative and indefinite expressions are permissible, and they are then used to express our obedience and admiration, not what God in himself is. The word incorporeal too, applied to God, is only an honorific predicate. For since everything that is, is body, God must also be so, which also agrees with the opinions of the old Fathers of the Church. — Natural religion is not sufficient, for reason is imperfect and the affects are strong. A revelation

is then necessary. But the discussion of it belongs in political theory. For since it goes beyond human comprehension, it can be grounded only on authority, and authority a private man can obtain only by working miracles; since now the time of miracles is past, the state ought to decide all religious questions. Religion is not philosophy but a law; it demands not discussion but obedience.

e. Psychology

Hobbes opens the series of eminent investigators in the domain of psychology who are the pride of English philosophy. With his usual vigorous grasp and his eye for the great, elementary lines he has given valuable contributions to the understanding of mental life. It was properly the endeavor to obtain an empirical foundation for ethics and political theory that led him into psychology; but this gains independent interest in his exposition. *[271]*
— We first bring forward Hobbes's physiological theory, in so far as it concerns the conception of the life of consciousness. — What goes on in the external world is, as we have seen, only motion. When this motion propagates itself inward through sense organs and nerves to the brain and from this to the heart, it meets here with a certain resistance and reaction, since the inner organs are themselves constantly in a certain motion. It is on account of this reaction, which is like a striving outward, that we place the object of sense-sensation outside us. How the relation more precisely stands between these inner processes (action and reaction) and sense-sensation, Hobbes, as we have already seen, does not go into. — As is seen, Hobbes stands somewhat closer to ancient and scholastic physiology than Descartes, since he (though only in De corpore, cap. 25, 4, not in Elements of Law, I, 2, 8) regards the heart as the place where sense-sensation arises. This assumption was probably called forth by the effects on feeling attached to sense-sensation, in which the heart plays an essential role. In the heart is the source of life, and according to the manner in which the impression from without alters the vital motion, which is one with the motion of the blood, namely either facilitates or impedes it, there arises pleasure or pain respectively. Thereby is called forth in turn an involuntary striving to hold fast the motion that brings pleasure with it, or to check the motion that brings pain with it. Already in the movements of the fetus this original need (conatus primus) makes itself known.

What really takes place in the world is, as we have seen, according to Hobbes only motion — motion outside us and in us. The subjectivity of the sense qualities is therefore in Hobbes a consequence of his metaphysics of motion. But not only by the deductive but also by the inductive way he seeks to establish the subjectivity of the qualities. He

points to the fact that we see images of things in places where they in fact are not, as in reflection, in dreams and in sense illusions, — that different persons can apprehend the properties of things differently, — that we sometimes see things double, — that by a
[272] blow on the eye or pressure on the optic nerve light arises, which nevertheless in fact is not an external light. From this he concludes that the content of our sense-sensations is always only "appearance" of motions in brain and nerves, nothing real.

Sensation arises when there is a reaction from the interior of the organism against the impression (a reactione cordis versus exteriora). But, says Hobbes, one cannot turn the proposition round and teach that everything that reacts has sensation. Then all bodies would be able to have sensation! Even if a glimmer of sensation always arose in such a reaction (phantasma), it would at once vanish if there were no capacity *to hold it fast and recall it,* and it would not be sensation proper (sensio). To sensation proper belongs a judgment about the objects, built on comparison of the sense impressions. (Nam per sensionem vulgo intelligimus aliqvam de objectis per phantasmata judicationem, phantasmata scilicet comparando et distingvendo, De corp. XXV, 5). Sensation therefore presupposes memory and comparison. The individual sensations must be capable of being distinguished from one another. It is therefore a necessary condition of sensation that the sense impressions *vary.* Under entirely uniform influence all sensation falls away. Just as little as I have any sensation of the bones in my arms, although they are surrounded by very sensitive membranes, just as little would a man be able to have sensation if he stared motionless at one and the same unchanging thing. To sense the same thing constantly and not to sense at all come to the same. — This was, as we have seen, the point with which Hobbes's whole philosophizing began: that without change there would be no sensation. Through the proposition that all change is motion he then passed over into his philosophy of motion, from which (as he certainly does not himself notice) he has difficulty in coming back to sensation. No wonder that he so strongly stresses that the very fact that something is apprehended by us (or can be phenomenon for us) is the most remarkable of all phenomena; about it his whole philosophy turns, since it constructs partly the causes of sensation, partly the effects of sensation. —

[273] Just as several different sense impressions obscure one another, so the memory images, which are nothing but sense-sensations weakened by the removal of the object, will be further weakened by the new sense impressions that come. In dreams, where such new impressions do not come, the memory images come forward as the expression of something present, often together with features and properties that are taken from other representations. — The memory images arise according to certain laws. They follow the

same order as the original sense perceptions. The motions in the brain are connected in such a way that when the first is repeated, the others follow after on account of cohesion, just as water on a smooth table is drawn after the finger that leads a part of it in a certain direction. Thus the representation of the apostle Andrew can bring us to think of the apostle Peter, his name can bring us to think of a stone, etc. This is the law that is now called the law of contiguity. But Hobbes also lays great weight on what might be called the law of interest. The course of representations is regulated not only by the connection between the original sense-sensations, but also by the influence of feeling and drive. A certain order and connection comes into our representations from the fact that we always have a purpose we seek to reach, and which makes us seek for means to reach it. It is the constant regard to our purposes (freqvens ad finem respectio) that brings system into our thoughts. In dreams (besides the external impressions) this consciousness of the purpose is lacking, and thereby the capriciousness of dream representations is better understood. — For the origin of language, the involuntary association between an object that sets feeling in motion and a vocal outburst or an action in which the feeling finds vent is for Hobbes not a sufficient explanation. Language arises, as already mentioned, according to Hobbes by arbitrary naming. Man sets a mark in order to be able to recognize a thing again, as when seafarers set a mark on a rock they pass by, in order to be able to recognize the place again. Language is a mark (nota) for the individual before it is a sign of communication (signum) for him over against others. And only when signs of communication have thus been introduced — does a society become possible. — There lay in Hobbes's own doctrine of association conditions for a more natural and more *[274]* correct psychology of language than this theory, which ascribes too much to conscious reasoning and arbitrariness. — Arbitrary naming plays, as we have seen, a great role in his epistemology. It makes possible the construction of the first fundamental propositions or definitions. And when such construction has once taken place, all thinking, according to Hobbes, proceeds as an addition or subtraction of names. He seeks to make all thinking a kind of algebra of names. In a judgment, e.g., it is expressed that one name denotes the same thing as another name. Since Hobbes has not seen the significance of the relation of similarity in the involuntary association of representations, he consistently stops at this wholly external conception of thinking, which at the same time suits so well his deductive tendency. —

Pleasure and pain are, as we have seen, closely bound up with the facilitation or checking of the vital process. The impulse of self-preservation is the fundamental need in man. Pleasure and pain arise from its momentary satisfaction. When experiences

have been made and there is a capacity to form representations of the future, the more composite feelings arise. These can, according to Hobbes, be traced back to the feeling of power or of impotence. When I feel pleasure in something that in the future may cause pleasure, this presupposes that I have power to procure it for myself. This power may be my own mental or bodily strength, or that of my friends, on whose assistance I rely, or an authority by whom I feel myself protected; perhaps even God the Almighty alone disposes of the power that makes it possible for me to feel pleasure in the representation of the future. The feeling of pain arises, on the other hand, from the representation of impotence to procure future goods or avert future evils. All men have a constant and restless craving for power, which only death brings to an end. It is not always that one expects greater joy than that which one possesses; but one wishes constantly to secure better what one has. The special feelings arise during the struggle or race between men. There arises triumph in those who come before others, humility in those who fall behind, hope when one is in breath, despair when one is tired, anger at a sudden obstacle, the
[275] feeling of pride when a strong obstacle is broken through, weeping when one suddenly falls, laughter when one suddenly sees others fall, pity when we see one whom we wish well fall behind, indignation when we see one whom we wish ill get ahead, love when we help another during the race, the feeling of happiness when we constantly get ahead of those before us, the feeling of unhappiness when we constantly fall behind, — and only with death does the race end! As long as we live, we set ourselves purposes and feel craving. All feeling of pleasure is bound up with a craving toward a goal. If we conceive a highest goal, and that it has been reached, then not only all craving but also all feeling would fall away. Unhindered progress to ever more distant goals would be the greatest of all goods. The enjoyment itself of what has been reached is a craving: the soul of the one who enjoys moves through the parts of what is enjoyed. For life is motion, and if this cannot proceed straight forward, it becomes circular motion.

In his closer explanation of pity and love Hobbes traces these feelings back to the impulse of self-preservation and the feeling of power. Pity arises from our coming, at the sight of others' misfortune, to think of a possible misfortune for ourselves; love consists in the feeling of one's own power that one has in being able to help others. The joy of knowledge, too, is a kind of feeling of power, as well as the joy of wealth and rank; only science gives very little power, since so few understand it. Yet Hobbes also assumes an immediate joy of knowledge, which is an inconsistency. — Cruelty, he explains, is contempt for another's pain and sorrow; it springs from the assurance of being oneself in safety; on the other hand, Hobbes takes it to be impossible that anyone, without

something being gained for him thereby, directly rejoices in the suffering of others. Hobbes's explanation of all feelings from the impulse of self-preservation becomes artificial and one-sided in detail. But he nevertheless always has firm and natural ground under his feet in regarding the feelings in their appearance and motivation during the struggle to subsist and to get ahead. There is not the refinement in him that there is in his contemporary La Rochefoucauld, who in his "Maxims," in which he set down his "experiences" of the Fronde and of salon life, constantly operates with the trick of putting a conscious ulterior motive into every feeling. Hobbes has far more sense for *[276]* the involuntary. He would hardly have been able to write down an assertion such as the one La Rochefoucauld has put forward, that one weeps only either to get a name for tenderness of heart, or to be pitied, or to avoid the shame of not weeping! —

As regards the psychology of the will, there are found in Hobbes good remarks on the relation between drive, deliberation and willing, directed against the scholastic psychology, which makes "the will" a peculiar, abstract faculty, without connection with the more elementary expressions of activity. Besides, Hobbes has given a brilliant plea in defense of determinism in the little work "Of Freedom and Necessity," which was called forth by a discussion with an English bishop. The bishop held that if the will were not free of causes, the laws would be unjust when they forbid actions; deliberation would be useless, admonition, praise, blame and punishment in vain, and books, weapons, instruments and medicines of no use! Hobbes answered: the law acts as a deterrent, and so as a motive; punishment is not meant to torment the criminal for what cannot be otherwise, but to make him better and to deter others; without deliberation the action often turns out badly, does not become what it otherwise can become, and for the same reason instruments and medicines etc. often cannot be dispensed with; praise and blame are conditioned by the goodness or corruptness of the action and have nothing to do with necessity or non-necessity.

f. Ethics and Political Theory

With the same energy with which Hobbes founds his whole conception of existence on the laws of mechanical motion, he founds his whole conception of ethical and social life on the impulse of self-preservation. If one asks about the relation between Hobbes's ethics and political theory on the one side and the rest of his system on the other, Hobbes has already conceded that one can arrive at ethics and political theory by two roads, partly through deductions from the doctrine of bodies, partly through immediate psychological *[277]* experience. But the deductive road he has not attempted at all, which is connected

with the fact that his ethico-political writings were composed before his writings on natural philosophy. He makes no attempt to derive the impulse of self-preservation from mechanics; he does not see self-assertion as a general tendency in all that exists, as Telesio and Bruno did before him, and Spinoza later. He contents himself with establishing it as a fact. There is, however, yet a third road by which, according to Hobbes, ethics and political theory could be introduced: by arbitrary definition, arbitrary fixing of the first principles. For in ethics and political theory we are dealing with our own life; its valuation and ordering rest on our will. Therefore, according to Hobbes, ethics and political theory can be constructive sciences like geometry and mechanics. For all moral and political laws presuppose a contract among men to lead a social life on certain conditions. From this contract the particular moral and political rules can be deduced. As with the general scientific presuppositions, however, regard for experience plays an essential role here too in the setting up of the principles; this setting up is not unmotivated. On the contrary, it is not the least interesting part of Hobbes's ethico-political theory that sets out to show how men come to construct a social life for themselves.

It is experience of bitter need and danger that leads them to it. It is wrong to suppose (with Aristotle and Grotius) that man has an original need of and capacity for social life. On the contrary. Since each individual's need and feeling of power lead him to take possession of as much as possible of what nature offers, he comes into collision with the others, who have a like need and feeling of power. Fear will perhaps even lead them to try to forestall one another in using violence, so as not to be taken by surprise themselves. That this is how matters stand is shown by the accounts of our barbarian forefathers and of savages; and it is seen in the precautions for safety that we all take, and in the relations of states to one another. The state of nature, i.e. the condition of man as it is apart from the life of the state, is a *war of all against all* (bellum omnium
[278] contra omnes). Here no other rule holds than the individual's need and power. Where power is slight, cunning takes its place. — This does not mean that all men are evil by nature. It is not human nature itself that is evil; but ruinous actions can spring from it. And even if there were fewer evil men than good, the good would still have to take precautions against the evil.

There would be no way out of this dismal condition if reason did not have its root in human nature just as much as passion does. Right reason leads men to seek better means for their preservation than they have so long as they fight each for himself. They discover that the evil is common, and that there are common means to employ against it. Thereby *the first and most fundamental moral law of nature* comes into being: peace is

to be sought, and if it cannot be attained, aid for war is to be sought! But the condition for this is that the individual give up the unconditional right that he has in the state of nature. It is the impulse of self-preservation that dictates this surrender. From this there further follows: keeping one's word, gratitude, obligingness, forbearance and equity are to be practiced, but arrogance and encroachment avoided; what cannot be divided is to be used in common; mutual strife is to be settled by arbitration; each individual is to keep his mental faculties unimpaired (for otherwise right reason and its commands cannot come to prevail). All these consequences can be summed up in the old saying: Do not do to others what you would not have done to yourself. — Should anyone think that these commands of reason could have no power against the passions, Hobbes concedes that fear and hope, anger and ambition, greed and vanity can indeed prevent their being acknowledged; but — he adds — there is no one who does not at times have a calm mind (sedatus animus), and then he will acknowledge them in their necessary connection with the impulse of self-preservation. All duties and virtues follow from the one proposition that peace is to be procured; they are means to peace. Therefore the natural moral law is also the will of the Deity; reason is implanted by God, and Christ, the apostles and the prophets proclaim the same commands that can be derived from right reason. *[279]*

It is not impossible that Hobbes was influenced in this line of thought by Herbert of Cherbury, who, as we know, saw the significance of the natural law in its preventing mutual destruction, and saw in the right setting out of the "common notions" a means to peace. It is known from Hobbes's biography (Vitæ Hobbianæ Auctarium) that Hobbes stood personally close to Lord Herbert. —

When Hobbes appeals to right reason, he is fully aware that he is not assuming, with Plato and the Scholastics, a special mystical faculty that could furnish a common standard in all decisions about what is good and evil. He expressly denies (Elements of Law, II, 10, 8) that any such ready-made recta ratio is to be found in the world: Right reason is not existent. What stands firm is only the end after which all living beings strive, and when reason awakens to clarity, it will also find the necessary means of reaching this end. Hobbes gives reason its basis and content in self-preservation and its means, and is fully clear that reason works only when the mind is calm (De cive III, 26: sedato animo). In constructing the presuppositions of ethics and political theory he takes such a calm state of mind as his basis. And it is experience that has taught him that only through the union of the impulse of self-preservation with the composure that makes deliberation possible can there be any question of ethical laws for human actions. An idealization of the given is thus necessary, for passionate excitement, shortsightedness

and inconsistency must all be kept away if reason is to be "right." The ideal and consistent demands that peace requires must, moreover, be upheld by a constant will. It is not merely a matter of outwardly observing the commands, but of inward adherence to them for their own sake. The natural moral law binds the individual before the tribunal of his own conscience (in foro conscientiæ). — Hobbes has undeniably not noticed to how high a degree he is idealizing here. If the impulse of self-preservation in its elementary form is always to be the only motive, then such a conscience as Hobbes here conceives,
[280] a feeling or will that holds fast to the necessary conditions of peace for their *own* sake, becomes impossible. It goes with him here as when he assumes a desire for knowledge as an immediate feeling of pleasure; this too cannot be directly derived from the impulse of self-preservation. Hobbes lacks the idea of a displacement of motives, by which what first stands as a means can pass over into becoming a purpose. Such a metamorphosis is quite conceivable on the basis of the impulse of self-preservation: it might well be that one uses the means best when one does not use them *as* means!

It is of little use, now, that the individual acknowledges the natural law before the tribunal of his conscience. An authority must be brought into being before which all bow. The individual must hand over his natural right to such an authority on the presupposition that all others also hand over theirs to the same authority, and this authority, the power of the state, then concentrates in itself all the right that in the state of nature was distributed among the many. This contract, by which the life of the state replaces the state of nature, contains at once an agreement of the individuals among themselves and a common submission to an authority (whether this be a single person or an assembly of several persons). In this Hobbes's doctrine differs from that of Althusius, who distinguished between the social contract and the constituting of the authority; for Hobbes the two coincide. But Hobbes, no more than the earlier teachers of natural law, assumes that this contract is entered into with full consciousness. It can be understood or supposed (a supposed covenant, pactum subauditum), and need not be express (pactum expressum). Examples of understood contracts are found in the relation between the victor and the vanquished whose life he has spared: had he not trusted in his obedience, he would not have spared his life. Likewise a tacit compact is entered into between the infant and the one who brings it up and protects it, as a rule the mother. That the contract has taken place Hobbes *infers* from the nature of the relation. When the contract is understood or tacit, it comes close to being a mere point of view, used for illumination
[281] and valuation. But it is then characteristic of Hobbes that he uses precisely this point of view. The contract presupposes (where it is express) the individuals standing over

against one another as independent beings, each with his impulse of self-preservation, and society is now to be the product of their interaction. By this point of view it becomes possible for Hobbes to deduce the life of the state from individual self-preservation. He imagines men sprung up out of the earth like mushrooms, each in his own place, and afterwards joining together. This is a thought experiment that is necessary for the constructive method. But Hobbes does not inquire expressly how experience is related to this thought, as indeed in general the relation between experience and the arbitrary presuppositions of constructive thinking remains unclear with him.

The contract theory is meant to show that it is in every individual's well-understood interest to obey. The authority that is to be obeyed must be unconditional. The power of the state must be as strong as possible, so that each individual over against it is utterly insignificant. It must have the authority to punish, the right to wage war and make peace, disposal over all property, and the decision as to which opinions and doctrines may be proclaimed. To absolute authority on the one side corresponds absolute obedience on the other. Only what would be worse than death — e.g. to kill oneself or one's nearest — the power of the state cannot command. When this enormous power is abused, there is a breach of the moral law of nature, by which the holders of the power of the state are also bound before the tribunal of their conscience, — but not a breach of the laws of the state, which cannot bind the very authority whose will upholds them. If one finds these hard conditions, Hobbes answers that with the power to protect there follows unavoidably the power to oppress. Human life simply cannot be free of inconveniences. The freedom of the state of nature is bound up with insecurity and strife, the constraint of the life of the state with security and peace. It is impossible to limit the power of the state. For power can be limited only by power, and to limit power is therefore the same as to divide it. If one restricts the sovereign's power, then sovereignty in reality lies with the one or those who can punish or dismiss the so-called sovereign. This holds whether one *[282]* wishes to distinguish between ecclesiastical and secular power, or between executive and appropriating power. — Hobbes regards monarchy as the best form of government. Democracy is in reality an aristocracy of orators. Monarchy has the advantage that there is only one who can commit abuses, but above all that party strife is avoided and secrets are more easily kept. It is (by virtue of the original contract) in reality the people that governs; its will is carried out through the monarch, to whom in the original contract everyone's natural right has been transferred.

In all religious and moral questions it is for the sovereign to decide. The sovereign is to determine how God is to be worshiped: otherwise what is worship for one would be

blasphemy for another, and a constant source of strife and dissolution would be present. Just as little can individuals decide what is good and right in practice; that would be a constant seed of rebellion. It is clear, too, that only when the definitions are common can there be certainty that the conclusions will be common. When Hobbes calls ethics and politics constructive sciences because "we ourselves" fix the first principles, then in practice this "we ourselves" becomes one with the sovereign. Here the strangeness of Hobbes's whole standpoint comes clearly to light. His free, naturalistic inquiry has led him to see the necessity of an absolute authority whose reason must be acknowledged as recta ratio. But he then nevertheless himself gives a grounding of this authority, grounds its recta ratio by his own recta ratio, and in this grounding makes use of the freedom of thought and of expression whose extirpation he demands. The fruits of knowledge that he himself has plucked he uses to impress on us the necessity of a prohibition against eating of the tree of knowledge. He used the freedom of the press under the rule of the Independents to publish the first edition of Leviathan (1651), and later, under the Restoration, he took refuge in the Dutch freedom of the press to bring out the second edition of Leviathan (1670); Charles II had expressly refused him permission for a new edition of the heretical work. Like so many of the best men of the seventeenth
[283] century, Hobbes unites in himself the Renaissance's vigorous self-confidence and feeling of freedom with the conviction of the absoluteness of the great sustaining powers. In his doctrine he marks the strong reaction against the Renaissance and the Reformation. The rebellious opinion concerning freedom of conscience and concerning the right of individual knowledge in the moral and political domain he derived partly from the study of the Roman and Greek authors (especially Aristotle) at the universities, partly from the free access to the reading and interpretation of the Bible that came with the Reformation. Against this he set up the strictest caesaropapism: Christ's kingdom begins only with his coming again to judgment; until then God speaks only through his vice-gods: the holders of the power of the state. He declares himself against Catholics, Anglicans and Puritans alike. If the pope is to be the shepherd of the faithful, he becomes their real sovereign, and the state is dissolved. If each individual believer is to be considered a partaker of the divine wisdom, there will be a strife of all against all. If the bishops are to have any authority beyond what follows from royal appointment (the Anglicans think that they have "something more, they know not what, of divine right"), there will also be strife between church and state — and an ecclesiastical tyranny besides. The Protestant clergy thought that the papacy had been overthrown so that they could step into its place. But — asks Hobbes — what then have we gained by deliverance from the pope's

tyranny, if these insignificant men (these petty men) are to replace him? Especially since they are people whose mental qualities are not favorable to a peaceful social life (qvorum ingenium paci et societati aptum non est)! Hobbes thus begins with polemic against free discussion, but ends with polemic against the small authorities who had pushed themselves forward in the place of the great one. It is really they whom he means to strike. And in this relation lies the explanation of his strange position.

Hobbes shares the tendency to seek a center, the concentration and the resignation that are characteristic of the seventeenth century, and that in philosophy find expression not only in his political theory but appear in analogous features in the mysticism of Fenelon and Malebranche and in Spinoza's pantheism. And he has the great merit in political theory of having established that in a state there can be only one center of *[284]* gravity. But he mistakes the center of gravity: all forces other than gravity vanish for him, and consistently, then, all masses gather at the center. Power and obedience become all in all. The spontaneous content of life, manners and custom, public opinion, the quiet and hidden ways by which even the holders of power are dependent on what moves among the many, and on the other hand the whole unfolding of life which it is the task of power to protect and to subordinate itself to as its serving foundation — for all this Hobbes has no eye. His eye is held captive by that one point. And his thinking is in the lapidary style; the finer nuances escape him. All the greater is his sense for the great, elementary conditions of life of human society. He belongs to the same side of the English mind from which Malthus and Darwin, both of whom can be said to continue Hobbes's thought, have sprung.

Nevertheless there are, as already touched on, hints in him that lead out to a wider horizon.

The duty of the ruler is summed up in the proposition that the welfare of the people is the highest law. For the state is not instituted for the sake of the rulers, but for the sake of the citizens. The despotism for which Hobbes speaks is of the same kind as that which appears in the enlightened, reforming statesmen of the eighteenth century (Frederick II, Joseph II). He is not only an ardent opponent of hierarchy in all its forms, but also of every kind of class rule, such as may be found even in "free" countries. By the absolute center of gravity that he establishes he means to abolish the small centers of gravity, so that all the elements can come to move in natural orbits. And the absolute authority itself is grounded by way of reason. Hobbes makes use of free discussion. He hoped that his "Leviathan" would "fall into the hands of a prince who would consider it himself, without the help of any interested or envious interpreter, and then use his sovereign authority to

protect the public teaching of the doctrine it contains, so that the theoretical truth might be converted into practical utility"[54]). But he hopes not only for the enlightenment of
[285] princes, but also for that of the people. Against the hierarchy he relies not only on the jealousy of the power of the state, but also on the advancing intellectual culture and criticism of the people: Paulatim eruditur vulgus! (De homine, XV, 13). Self-contradiction between the propositions of the doctrine, or between the doctrine and the life of its preachers, will sooner or later be discovered. A wide horizon opens here, wider even than the one that stood before the heroes of the Enlightenment in the eighteenth century; for Voltaire and Diderot expressly disclaimed the thought of an enlightenment of the vulgus. And this wide horizon does not, properly speaking, come about through an inconsistency, when one goes to the bottom of Hobbes's presuppositions. For by the "contract" through which individuals give up their self-will, this will is not really meant to vanish, but only to be taken up into a higher connection. What is lacking in Hobbes is the means of showing how the isolated will can undergo a metamorphosis by being thus worked into a larger whole. But by his first and his last word he points beyond the imperfection of his own doctrine, an imperfection that is outweighed by the natural force and the acuteness in the foundation and construction of the doctrine.

g. Opposing Tendencies in English Philosophy

The history of English philosophy in the seventeenth century is not exhausted with Bacon, Herbert of Cherbury and Thomas Hobbes and with the tendency they represent. Besides this tendency, which takes its foundation in what is given in external or internal nature, and works on this foundation by means of the inductive or the deductive method, there is also another tendency, which historically points back to Neoplatonism, and which assumes a foundation raised above all experience for the highest ideas, especially for the ethical ideas. To this belongs Robert Greville, Lord Brooke, who in his thirty-fifth year fell at Lichfield (1643), fighting for the cause of Parliament. In his work on "The Nature of Truth," which appeared two years before his death, and for which he perhaps received the impulse from Lord Herbert's work, he protested against knowledge's being
[286] merely a human faculty, empty in itself, that would first have to go looking for its content: only light is receptive to light, and knowledge is a ray of the divine nature itself. True knowledge raises us above sense perception — of which the Copernican system is an example. All plurality, all relations of space and time are mere appearance; thinking finds rest only in the knowledge of the one cause of all things. The knowledge of the unity of all things in God frees us from envy, shows me the happiness of others as my

own, disperses the darkness of the soul and heals all pain. Experience, on the other hand, can show us no real causes; it can only show us that the one phenomenon precedes the other[55]).

The Platonic tendency to which Lord Brooke belonged found a home about the middle of the century at the University of Cambridge, while Oxford continued to hold to Scholasticism. A year after Leviathan had appeared, there appeared (1652) Nathaniel Culverwel's "Discourse of the Light of Nature." Culverwel had died in 1651; he was Master of Arts and Fellow at Cambridge. His writings are Platonizing like Brooke's, but not so mystical, and he lays greater weight on psychological experience. But he assumes eternal and universally valid truths in both the practical and the theoretical domain. These truths are one with God's essence and are upheld in the world by God's will. Thus for him, as for Brooke, philosophy and theology merge into one, whereas Bacon and Hobbes drew a sharp boundary between them[56]). In a more systematic form the same tendency, whose distinguishing mark is a close union of Platonic philosophy and Christian theology, appears in Ralph Cudworth († 1688) and Henry More († 1687), both attached to the University of Cambridge. Whereas for Bacon and Hobbes it was a principal matter that truth must work its way forward in the world, come into being in man's consciousness through strenuous labor, these men returned to the Platonic doctrine of the eternal truth, which does not come into being, and to which man's consciousness need only open itself in order to have a share in it. This tendency turned both against Puritanism, which was hostile to philosophical thinking, and against Hobbes (and, in More's case, also against Spinoza), who set philosophy in opposition to theology. It is characteristic that Henry More was led into philosophical studies through *[287]* doubt about the truth of Calvinism; he had been brought up strictly Calvinist, and the fact that he was able to work his way out of the doctrine that had been impressed on him by the energy of his own thinking was to him a testimony to the independence of thought over against everything external, an independence that he then asserted not only against the narrow Puritan theology but also against the realist tendency of Bacon and Hobbes. More was in touch with Descartes and took up several of his ideas, only not the sharp distinction between mind and matter; for him the mental was extended, though not an object of touch. In his work *De anima* (Rotterdam 1678. cap. 1) he defines a mind as a substance that can indeed be penetrated but not divided (substantia penetrabilis et indiscerpibilis), whereas body is a substance that can be divided but not penetrated (substantia impenetrabilis et discerpibilis). What is characteristic of body is not that it is extended, but that it is impenetrable. Everything that exists has extension in length,

breadth and depth, the mind as well as the body. *(De anima.* cap. 8: Responsiones ad Hobbesiana). In this way he mediates between Hobbes and Descartes and thinks he has refuted Hobbes's objection to spiritualism, that we cannot form any concept of an immaterial substance. Extension is to be common to both kinds of substance, and More goes even a step further, in that he, like several of the philosophers of the Renaissance, regards space as a divine attribute: that God has extension follows from his omnipresence. *(De anima.* cap. 2, § 4—5). The difference between God and the individual soul is only that God is unlimited, the soul limited. In this bold mysticism More is influenced by the Kabbalistic doctrine. His mystical conception of space recurs in Newton, who on the whole was influenced by More. — From Descartes More differs not only in assuming that all substance is extended, but also in taking activity (in the case of mind original activity, in the case of matter communicated activity) into the concept of substance. *(De anima.*
[288] cap. 1 § 3). Hereby he prepares (like Gassendi and Hobbes) Leibniz's philosophy of nature. — We cannot go further into this tendency here; it is more a learned mysticism than new philosophical thinking. It has had great significance for the spiritual development of England, but has not brought new thoughts into the history of philosophy.

A greater interest attaches in this respect to an author who met Hobbes on his own ground, the inquiry into the psychological foundation of ethics and social life. Richard Cumberland (b. 1632, studied at Cambridge, died as Bishop of Petersborough 1718) published in the year 1672 a work remarkable in the history of ethics: "Philosophical Inquiry into the Natural Laws" *(De legibus naturæ disqvisitio philosophica),* which is expressly directed against Hobbes. He declares that he means to continue the work of Hugo Grotius by another road than the one the famous Netherlander followed. Instead of inquiring how the moral laws of nature make themselves known in experience and history, he means to inquire into their origin and their causes. In this inquiry he means to start from experience and to reason his way forward from the nearer to the more remote causes. He is sure that by this road he will at last come back to a divine source of those laws; but he will not use the easy method of the Platonists, which lets the innate ideas bear immediate witness to this lofty origin: "I have," he says, "not been so fortunate as to arrive at the knowledge of the natural laws in so convenient a way." And in any case he will not build his ethics on such a hypothesis, although he acknowledges the sound and good influence of "our Platonists" in moral respects. Against Hobbes, on the other hand, he points to the spontaneous tendencies in human nature, whereas Hobbes had taken the arbitrary adaptation of wills as his basis. From birth onward there takes place an unfolding of the human faculties and powers, a spontaneous use

of the organs, which is bound up with an immediate satisfaction. Sense impressions are met by a spontaneous inclination to receive them, to turn toward them. In this way the first experiences are had and the first mental operations performed: remembering, comparing and counting. In this immediate need (impetus), which is not produced by *[289]* the impressions, but only released by them, the natural and self-evident truths of reason (naturalia rationis dictata) have their ground. And such an immediate need also underlies the first striving to produce goods and shun evils — and that without the individual's own goods and evils being at once separated from the common ones: on the contrary, it is spontaneously acknowledged that the good that is common to all rational beings stands above that which falls to a single being only, and that the single being attains its peculiar happiness precisely by striving after what is good for the whole to which it belongs. Clear consciousness and arbitrary willing can only develop further what the spontaneous tendencies already lead to; art can work on the foundation of nature, but cannot take its place. — While Cumberland thus stresses the original natural instinct, in opposition to Hobbes's construction of society out of clearly conscious, independent individuals, he maintains on the other hand that the natural is not only what can be traced in the first manifestations of life, but also what appears as ripe fruit when growth and development are complete. Human nature is therefore to be sought not only in the elementary drives, but just as much in what the highest spiritual development displays. Even if man, then, at first confines his interest to himself, it is not on that account unnatural for him to attach his interest to what brings happiness to all. The state of nature is not given up because there develops a magnanimity (generositas) for which the good is what works toward the preservation and perfecting of the whole human species, nay perhaps even of the whole universe, so far as we have a concept of it. — What Cumberland urges, then, is partly the possibility of a development from the elementary drives to magnanimity, without the external and mechanical notion of contract needing to form the middle term, partly the harmony between individual and common goods. In both respects he put forward thoughts that were more fully grounded and developed within English philosophy in the eighteenth century, and that already, in Cumberland's own time, were quietly taking shape in Spinoza's Ethics and Leibniz's philosophy of law. *[290]* Cumberland's exposition is still unclear and imperfect, but he has pointed to decisive points. Even if his work (to judge by its title page and preface) was called forth chiefly as a plea against Hobbes, it nevertheless has an independent significance in the history of ethics.

5. Benedict Spinoza

a. Biography and Characterization

Spinoza is the central thinker of the seventeenth century. All the threads of thought run together in him: mysticism and naturalism, theoretical and practical interest, which in the century's other thinkers stand in greater or lesser opposition to one another, and — where they appear in the same personality — awaken inner discord, are here sought to be carried through consistently, and the reconciliation between them is found precisely in their being carried through consistently. Whereas so many know no other way of connecting the threads of thought than to cut each of them off separately and then tie them together in an external fashion, the greatness of Spinoza's thinking is that he will set no arbitrary limits, but trusts in the harmony of the trains of thought in their innermost essence. Through the quiet labor of thought to which he devoted his life, he sought to shape a work that could take the new world-view and the new science into itself and let them unfold all their consequences, and yet maintain the reality and independence of the mental in existence, — a work in which the multiplicity and individuality of phenomena could be fully acknowledged, and yet the unity, the all-embracing order of the world, be seen as the power that in its innermost bears all and works in all. And this work was for him not a purely theoretical labor; it was to him the satisfaction of a personal need of clarity and self-understanding, and it helped him to the cheerful fundamental mood that is won by working one's way through the unrest and darkness of existence to the understanding of eternal laws. His chief work (*Ethica*) has thereby become a work of art, not merely a purely speculative work. It has aptly been
[291] compared, in its five books, with a drama in five acts. And this comparison can be carried through in detail. In the first book the most general propositions are given, which unroll before our eyes a great, infinite background for human life. In the second book there is added to these a series of propositions borrowed from physics, by which the strictly mechanical view of nature is laid down, and the book then passes on to an inquiry into human knowledge. A provisional conclusion is offered by the end of the second book — and, on a not unfounded conjecture, the first two books originally formed a whole by themselves. There appears a prospect of attaining what Spinoza had set as his goal: the knowledge of the connection in which our mind stands with the whole of nature. In this knowledge he saw (in an autobiographical fragment) the only good that was closely bound up with his personality, and of which no external accident could rob him. But so swift a conclusion of the drama is prevented by a cloud appearing over the horizon: it

is not only thoughtlessness and false combinations of representations that hinder true knowledge; it is first and foremost the feelings and passions by which we are tossed about. Here a new and serious resistance is to be overcome — and in order to be overcome it must first and foremost be known. In the third book, then, Spinoza gives a masterly natural history of the feelings. He demonstrates the connection of the feelings with the impulse of self-preservation, and shows how they are transformed under the influence of representations. His chief endeavor is to find a fixed causal connection in the mental as well as in the material domain. He wants to find the law of the swell of waves by which we are tossed about. In this very law he finds deliverance. In the fourth book the light begins to break through again. No passion, it is true, can be directly extirpated; but it can be checked by another passion that is stronger. Once this has been seen, the task is to make the knowledge of the good, that is to say, of the conditions of the preservation of our being, a power in the mind, and this comes about by this knowledge itself awakening pleasure in us, both through the end it shows us and through its being itself a mental activity, an expression of our power. This knowledge will unite us with individuals akin to us in essence, since it turns out that we are subject to common conditions, and that *[292]* only common striving leads each of us to the goal. With this the drama approaches its conclusion a second time. But there still remains a question: that of the individual's complete working-through of his personality by uniting the scientific knowledge, whose possibility the first two books showed, with the practical course of development, the discipline through the struggle for existence, that the third and fourth books depicted. In the fifth book it is shown how the clear understanding of our passions in their natural connection raises us above them, and unites with all our other knowledge of nature into an immediate beholding of our being as one single one of the forms in which the eternal Deity, working in the world of mind as in that of matter, unfolds its being. As we thus see ourselves — and all things — "under the aspect of eternity," all the unrest and obscurity of time and finitude vanish, and with the mental freedom whose development the fourth book depicted there is now united the inward feeling of being one with the eternal and infinite being. — Thus within one frame Spinoza finds room for philosophy of religion, physics, theory of knowledge, psychology and ethics, just as the carrying through of realism by the strict application of the law of causality does not hinder, but for him precisely makes possible, the mystical unity with the Deity. If we add that this whole great and manifold connection of thought is set out in the form of propositions and proofs after the mathematical pattern (more geometrico), we have the picture of a work unique of its kind. No wonder that it was not understood in its own time; nay, that

the time of understanding began only a hundred years after its appearance, so far as the idealist side of its content is concerned, and came only in our own day, so far as the realist side of its content is concerned. For the history of philosophy the task with regard to a work such as this is first and foremost to inquire how it came into being, and on what historical presuppositions it rests. It stands before us like a cut crystal; if possible, we should discover something of the process of crystallization. Here Spinoza's personality and personal circumstances and his philosophical development may perhaps be able to
[293] help us. There can hardly be any doubt that the manifold content that is here gathered into so peculiar a frame is held together first and foremost by the personality of the author. After that, an analysis of the system itself will show us how the presuppositions and elements that the author has taken into himself are used by him, and how far he has reached his goal. —

Baruch Spinoza was born on 24 November 1632 in Amsterdam, the city in which his parents, Spanish Jews who had fled from the Inquisition, had taken refuge. The gifted boy made his first studies at the Jewish academy in Amsterdam, where he was introduced to the Talmud and to medieval Jewish philosophy. Thereby the foundation was laid for an essential side of his thinking, the endeavor to hold fast to and carry through the thought of the Deity as the one infinite being, as this thought comes forward in the higher popular religions, especially in Judaism. In the course of his studies, Maimonides (a Jewish philosopher of the twelfth century) and Leo Hebraeus (a Jewish mystic of the time of the Renaissance) seem in particular to have been of significance for him. Their books were in his library. The oriental or mystical tendency always lay at the basis of his thought and gave him the quality of being cast in one piece, the determined direction toward one point, that distinguishes his thinking. Early on he harbored doubts concerning the Mosaic theology, and he therefore became an object of suspicion on the part of the Jewish theologians, which in turn led to his withdrawing from the synagogue. He felt the need of a wider intellectual horizon and began to study the humanities and natural science. In Latin he was taught by the physician van Ende, who had a name as a freethinker. The most important source for Spinoza's biography, the pastor Colerus, who — in spite of his great abhorrence of Spinoza's doctrine — collected information about his life with great love of truth, thinks that in this "school of Satan" Spinoza no doubt learned other things than Latin. According to Colerus's account, Spinoza is said to have been in love with van Ende's gifted daughter, who helped her father with the teaching, but to have been rejected for a fellow pupil who knew better how to make his
[294] way as a suitor. There is, however, some difficulty in reconciling this account with what

is known of the ages of the two persons. — Colerus says that Spinoza at this time left theology and gave himself wholly to "physics." It is probable that among the authors he studied in order to become acquainted with the newer view of nature, Giordano Bruno was one of the most important. Bruno, it is true, is never named by Spinoza, and his books were not in Spinoza's library, but the latter's first work (especially a little dialogue that is incorporated in the work *On God, Man and his Happiness*) clearly recalls Bruno. Here he found a philosophical conception that made it possible for him to unite what was for him the essential element in the religious ideas with a scientific understanding of nature. Bruno's doctrine of the infinity and divinity of nature made it possible for him to unite the concept of God and the concept of nature in an inward way. This influence of the philosophy of the Renaissance on Spinoza has become probable only since the discovery (some thirty years ago) of Spinoza's first work. Previously one held to Colerus's remark that Spinoza, after he had given up theology for "physics," was long in two minds about which teacher he should follow, but that then Descartes's works had fallen into his hands and had appealed to him by their endeavor to build everything on clear and distinct grounds. It was thought that Spinoza began in philosophy as a Cartesian and only gradually turned critically against Descartes. Against this stands the fact that the very first work we know from Spinoza's hand criticizes Cartesianism on some of the most decisive points. Spinoza was never a Cartesian, but he learned much from Descartes and used several of his concepts, and in part followed him in his terminology. But in this period of transition, too, before his own system took more definite shape for him, he must have studied the more recent Scholastic authors, who at that time — before the philosophy of Bacon and Descartes had made its way — dominated philosophical teaching in Germany and the Netherlands. Many expressions and propositions in Spinoza's later as well as in his earlier works point to such a Scholastic influence, just as Spinoza himself in several places refers to the Scholastics. It appears — when one takes all this together — *[295]* that they are very comprehensive and diverse circles of thought, and no small literature, that Spinoza traversed in the course of his philosophical development. Closed and cast in one piece as his chief work stands before us, it nevertheless has its roots branching out to many sides within the philosophical tradition[57]). This does not diminish his originality; it is the mark of genius to be able to take in and work over as no one else can. The peculiarity and originality of the building do not suffer from the stones' having been fetched from many places.

Under the influence of all these motives of thought, which streamed in upon him from many sides, Spinoza in his views moved further and further away from the synagogue.

His open apostasy was foreseen, and in order to avoid it attempts were made to hold him back by offering him a pension. But neither this nor a fanatic's attempt to assassinate him kept him from following the way of his thought. He was then solemnly expelled by the Jewish clergy (1655). They even succeeded — since the Protestant congregation too regarded him as a dangerous man — in having him banished from Amsterdam for a time. He took up provisional residence with a friend in the country near Amsterdam. He supported himself by grinding optical lenses, a skill he owed to his rabbinical upbringing, since every future rabbi had to learn a trade. His friends in the city fetched his lenses and sold them for him. It seems that already at this time a circle of younger men had gathered around him and his thoughts. At this time the work on God and man (the "short treatise") and perhaps the first draft of the later "theological-political treatise" came into being. Spinoza turned with indignation from the fanaticism of the positive churches, and with ever greater inwardness and self-understanding found the task of his life in the independent working out of a circle of thoughts that could spread clarity over the nature of man and his position in existence. Experience had (as he says at the beginning of the unfinished treatise "on the Improvement of the Intellect") taught him that neither riches,
[296] sensual enjoyment nor honor could be a true good for man, but that the only thing that could fill the mind with ever new satisfaction was the constant striving after knowledge, by which the soul is bound to what endures while all else changes. Spinoza's thinking had — however speculative and abstract the forms it took, and however remote from ordinary human life it might seem to lie — a decidedly personal and practical motive. Full mental clarity was to him a necessity of life.

In the year 1661 he moved to Rhynsburg, a small town near Leyden. Here he began the working out of his famous chief work. It is mentioned in letters to his friends (especially in the letters to Oldenburg and to Vries). Copies of the worked-out part of the work existed early, and Spinoza's young friends in Amsterdam, mostly medical men, studied the book together and in cases of doubt turned to the master for explanation. His life was not so solitary as it has often been depicted. His correspondence, of which we have had fuller knowledge only since the find of manuscripts that was made some forty years ago, and in which the "short treatise" was also discovered, shows him in touch with a not inconsiderable circle of men of different station and intellectual tendency. The study of this correspondence is of considerable interest for the understanding of Spinoza and his thoughts as well as of the spiritual condition of the time; in my popular account of Spinoza I have treated this subject in a separate chapter, "Spinoza og hans Korrespondenter," to which I must refer here. In the nearby university town of Leyden

he had several friends. Thus he came here into a confidential relation with the Danish naturalist Niels Stensen, who was studying in Leyden at this time. Later, when Stensen had gone over to Catholicism, he addressed to Spinoza in an open letter (Epistola ad novæ philosophiæ reformatorem), in which he calls him his former very intimate friend (admodum familiaris), an appeal to follow him into the bosom of the one saving Church. Niels Stensen's biographer, the keeper of the national archives A. D. Jørgensen, explains Spinoza's silence in the face of his former friend's appeal in the following way: "The restless spirit of love of Christianity drove Steno to an attempt to make his friend a *[297]* partaker of its happiness; philosophy's infinite indifference to the individual personality with its desires and its cares let Spinoza see that this man was irretrievably lost to the knowledge of the truth, and that the right answer to his address was therefore silence." I believe that the excellent historian (who was the first to bring to light the interesting relation between Steno and Spinoza) here does both Spinoza and philosophy a great injustice. I have great reverence for the figure of Steno, for his religious as well as for his scientific personality. But I nevertheless believe that in all propaganda, however noble the name it calls itself by, other motives are at work besides the spirit of love (if this is always present at all); and in any case this spirit of love did not prevent Steno and many others from looking forward to a blessedness from which, they believed they knew, very many would be excluded. It is in any case unjust to deny Spinoza the spirit of love because he will not force his views on others. His loving faith was directed to man, and he found what is human behind very different forms of faith. An unwilling witness, the pastor Colerus, relates of him how sympathetic he was toward others, and how he comforted and cheered the members of his household when they were visited by sorrow or sickness. When his landlady in the Hague asked him whether she could be saved in the religion she had, he answered her that her religion was good and that she should not seek any other; she would surely be saved by it if she led a pious and quiet life. He sympathized most with the freer tendencies within Protestantism. He had a different view of the significance of Protestantism from that of his kindred spirit Bruno, and quite naturally so, since he lived in the freest country in Europe, which admittedly did not mean much at that time. When later a young man, Albert de Burgh, who had lived in Spinoza's household and who had gone over to Catholicism, repeated Steno's attempt at conversion, Spinoza answered him in a letter in which he maintains the validity of free knowledge and sets justice and love as what is essential in all true religion. It must not be forgotten that what not least offended Steno in Spinoza was that the latter had spoken *[298]* up for freedom of religion. To work for freedom of religion was for Steno to do harm to

one's soul! Should there not lie in Spinoza's conviction that a true faith cannot be forced on anyone either by spiritual or by bodily violence, and in his caution about shaking the faith of others when only there was seriousness in it, a greater and truer interest in the individual personality than the one that finds outlet in passionate propaganda?[58]) —

After a couple of years' stay in Rhynsburg Spinoza moved to Voorburg near the Hague (1663) and later (1670) to the Hague itself. Here too he had many friends, in part men in high positions, such as the brothers de Witt. His way of life was frugal, and he showed great unselfishness toward relatives and friends in money matters. A bright and cheerful mood marked his way of taking life. "Why," he asks in a passage of his Ethics, "should one rather still hunger and thirst than drive away melancholy? In my conviction no deity, and in general no one who is not envious, can take pleasure in my impotence and my misfortune, or count tears, sobbing and fear as merit; but on the contrary, the greater the joy we feel, the greater the perfection to which we pass, that is to say, the more we partake of the divine nature. Therefore the wise man will refresh and strengthen himself with food and drink, as well as with fragrance, flowers, music, tasteful dress, bodily exercise and plays." In spite of his mystical vein Spinoza was no ascetic. In his life as in his philosophy an energetic seeking to be absorbed in one thought went hand in hand with a sense for the manifold reality whose inner law, in his conception, this thought precisely expresses. Both his life and his philosophy are distorted if one does not hold fast to the different currents that actually moved in him, even if he had perhaps not (as he believed) found the redeeming word that could let all these currents come into their full right and into reconciliation with one another.

The first work Spinoza published was an exposition of the Cartesian philosophy (1663), originally intended for a young man (the Albert de Burgh already mentioned)
[299] whom he did not yet want to initiate into his own philosophy. It is characteristic that he thus regards Cartesianism as an introduction to his own doctrine. On the "Ethics" he worked and polished constantly. He was very cautious about communicating it to others; before he allowed his disciples to show the manuscript to strangers, he gathered careful information about their character and circumstances. Thus he refused his friend Tschirnhausen permission to show the Ethics to Leibniz; only after a longer personal conversation with Leibniz did he himself show him his chief work. According to a hypothesis put forward by TØNNIES[59]), the first two books of the Ethics were worked out by themselves, whereas the realist psychology of the third book, which betrays the influence of Hobbes, who also strongly influenced Spinoza in political theory, seems to be due to a later working out. This explains several discrepancies between the conception

in the second and in the third book. In relation to the "short treatise" the "Ethics" marks the advance that Spinoza no longer conceives the relation between mind and matter as a causal relation, but as a relation of identity. Spinoza made several attempts at publishing this chief work of his, but gave it up again when even the rumor of its publication aroused great agitation. He was in ill repute after the publication of his theological-political treatise (1670), although he had not yet developed his real system there. The book defended freedom of religion as a human right, with particular reference to the fact that the diversity of personalities must also bring with it diversity in faith. It further undertook a purely historical inquiry into the biblical writings (especially those of the Old Testament, whose late origin is maintained) and a psychological characterization of their authors. A definite distinction was made between religion, whose purpose is practical, in that it aims at bending the minds of men to obedience and morality, and science, which has a purely theoretical purpose; and on the basis of this distinction the complete freedom of science was maintained. Finally there was polemic against the assumption of miracles, since the laws of nature are the laws of God's own essence. — This work, which contained so many thoughts of the future, was regarded as the epitome of all ungodliness. The experiences Spinoza had in this way led him not only to hold back *[300]* his "Ethics," but also to decline an invitation to take up a professorship at Heidelberg; he feared that he would — in spite of all promises — not have sufficient freedom of teaching. — A disease of the chest that was hereditary in his family, and that had been consuming him for a long course of years, brought on his death (21 February 1677). About his death, as about his life, wild rumors were spread, which the truth-loving pastor Colerus refuted after careful inquiry — although he himself is astonished that a freethinker could live so beautifully and die so calmly, and although he is offended that Spinoza's barber, in a bill submitted to the estate, calls his deceased customer "the late blessed Mr. Spinoza!" —

In Spinoza's personal development we thus see the different elements of his system come forward from different sides. It will now be the task, through an analysis of the finished system, to see how he has united these elements, and how far the union has really succeeded.

b. Theory of Knowledge

If in setting out Spinoza's system one wished to follow his own "Ethics," one would have to begin with certain definitions and fundamental propositions and from them in turn derive a series of propositions. The question would then remain unanswered how Spinoza came to his definitions and fundamental propositions, which contain the whole

system within them, — and yet it is really this question that is of the greatest interest historically. If one begins with the definitions and fundamental propositions on bare ground, as Spinoza himself does in his chief work, the most natural supposition is that these first principles are either set up quite arbitrarily (as Hobbes thought) or else are the object of an immediate intuition. Neither was Spinoza's meaning. By following the fashion of the time, which demanded a strictly deductive exposition, he has therefore given occasion to misunderstandings of his system. From the unfortunately unfinished treatise "on the Improvement of the Intellect" it is seen that he gave serious thought to
[301] scientific method and had in mind to write a kind of inductive logic. In a certain sense it can be said that the principles on which Spinoza builds the deductive exposition in the "Ethics" came about for him by reflection on, and construction of, the presuppositions on which our empirical knowledge of existence rests.

If by experience one understands merely an apprehension of things as they present themselves to us spontaneously and by chance, it gives us no true knowledge. It stops at the given as it appears at a definite time and place, and an assumption then settles fast in consciousness merely because no contrary facts have so far shown themselves, without there being any guarantee that such are not to be found. This kind of experience Spinoza calls experientia vaga; it corresponds to Bacon's inductio per enumerationem simplicem. True knowledge must be a knowledge of reason, which does not stop at the single given phenomenon, but sees it as the consequence of a general law, as a single expression of a general order of things that extends far beyond it. Knowledge of reason considers the phenomena given in experience in their definite connection with one another, and finds in this connection the explanation of each single phenomenon. Knowledge of reason (ratio) starts from the given. Spinoza urges the necessity "that we always derive our concepts from real beings, proceeding as far as possible along the series of causes." First, then, an account of the facts must be procured, and after that the fundamental concepts and fundamental propositions must be derived from them. In his "theological-political treatise" Spinoza draws an analogy between the interpretation of a book (the Bible) and the explanation of nature. First it must be established what the given content is. Next one must make sure of certain general relations and laws that hold good at every point. In the case of physical nature these are the laws of motion; in the case of mental nature Spinoza would most likely name the laws of the association of representations. It is, in a word, the idea of the lawfulness that reigns in all things, which reveals itself in the single changing phenomena and makes each one of them explicable to us. In this lawfulness
[302] there is revealed for Spinoza the true, eternal essence of things. Not to the advantage of

clarity, he distinguishes between "the fixed and eternal things, on which the changeable single things depend in so inward and essential a way that without them they can neither be nor be conceived," and "the laws that are written in those fixed and eternal things as in law books." What he calls the fixed and eternal things can be nothing other than the order of nature itself in the material and in the mental domain, that is, the very fact that definite laws hold for the phenomena. These laws reveal themselves equally completely in nature as a whole and in its single parts. It is to them that one must go back if one wishes to reach a completion of knowledge. The series of phenomena is infinite, inexhaustible; there we could go on passing from link to link. But true knowledge concerns what connects the links, and what is therefore present everywhere, — those fixed and eternal things, which thought is able to grasp, while the series of phenomena extends to infinity, without completion. — Spinoza distinguishes expressly between the real connection that those fixed and eternal things (entia realia) express, and the mere general concepts (abstracta et universalia), which have no objective reality.

The truth and validity of our knowledge rest, according to Spinoza, on no external criterion, not on agreement with anything outside it. It is the clarity and distinctness that go with perfect consistency that give the assurance of truth. Error always rests on something limited and isolated having been made into a whole. Error vanishes when one simply proceeds with strict consistency from the erroneous presuppositions in which one is entangled. Intellectio fictionem terminat. In strict logical consistency, then, our thinking has a norm of truth that it can apply everywhere, even where it is in error. Truth, says Spinoza, illuminates both itself and error (veritas norma sui et falsi est), just as light makes known both itself and darkness. The starting point of all our inquiry naturally becomes, then, the investigation of the rule of truth that is given in the very nature of our intellect. This is the foundation on which we build in our inquiry after *[303]* what is eternal and necessary in things, after what underlies all things (res omnium prima).

Spinoza here hints that it is only by means of our subjective reason that we can find "the eternal things," "the first of all things," in nature. Had he pursued this hint further, the problem of knowledge in all its sharpness would have presented itself to him: by what right do we assume that existence itself follows the norm that holds for the relations of our thoughts to one another? But this problem does not arise for him. It stands fast for him that our rational presuppositions are also those of things, nay, that what is eternal in things corresponds to them; an objective order of the world corresponds to the subjective order of thought. There is the same relation between the objective objects of our thoughts

(which Spinoza denotes by the Scholastic expression essentiæ formales) as between our thoughts among themselves. To what is for us the first thought, from which all other thoughts are derived, there corresponds what is in existence the first thing, the origin and source of all else. When Spinoza has the laws that explain the single phenomena be "written in the fixed and eternal things as in their law books," he means thereby (we could say from our standpoint) to express that the connection that makes things intelligible is just as much a reality as the single detached "things" each by itself, nay, that it far rather than they deserves the name of reality, since it is eternal and all-embracing. And here too we must see the origin and meaning of what he in his deductive exposition calls substance. For by substance he understands that which exists in itself and is conceived through itself, and through which all else exists and is conceived. Spinoza's substance, the fundamental concept of his whole philosophy, is thus really causality represented as a thing or a being. For causality, the causal relation, is according to Spinoza understood through itself and is what makes everything else intelligible to us. And since the causal relation is the presupposition of a knowledge of real existence, which makes itself known to us through experience, we see that Spinoza's whole philosophy can be said to be a construction of the presuppositions that hold for all scientific experience, in such a
[304] way that these presuppositions are made one with the innermost essence of things. But causality, or the causal relation, which Spinoza thus thinks of as a real power on which the single things depend and in which they exist, corresponds in turn for him to the necessity with which we derive conclusions from premises, or in other words: Spinoza does not distinguish between ground and cause; the causal relation is for him not a relation in time, where the cause goes before and the effect follows after, but an eternal relation of ground; the relation between "the fixed and eternal things" (or "the first of all things") and the single phenomena is not that the former were first and then the latter arose as their effects. For true knowledge, according to Spinoza, the relation in time drops away. Knowledge of reason is, according to Spinoza, an apprehension of things "under the aspect of eternity" (sub specie æterni): it keeps to that in things in which the eternal necessity makes itself known in them, and sees them only as examples of it. The causal principle itself Spinoza regards as a necessary presupposition. In the Ethics he sets up as a fundamental proposition that a thing must have its cause either in itself or in something else. Now since substance, that which exists through itself and is conceived through itself, cannot have its cause in something else, it must have its cause in itself and thus exist with necessity. If one denies this proposition, one must also deny the proposition that there must necessarily be a cause of every thing (either in it or outside

it). And that it is a proposition of reason is evident also from the fact that for two things to be related as effect and cause means that the concept of the one thing can be derived from that of the other; the knowledge of the effect is only the knowledge of a particular property of the cause; there is something common in their concepts. Only when in this way the relation in time, and in general *the difference between cause and effect, drops away,* so that the effect is seen as the logical consequence of the cause (which would admittedly be the ideal completion of knowledge), can substance or "the eternal things" be called the cause of phenomena. And only on the presupposition of this confusion of ground and cause does it become intelligible how Spinoza could hope to build up a purely rational knowledge of existence, or, as one can also express it, to know existence *[305]* as being rational in its innermost essence. The counterpart to Spinoza is formed here by the conception that Brooke and Glanvil (as well as the occasionalists) had already hinted at and that Hume later carried through: that cause and effect are *wholly different* things, an assumption that must lead to doubt about the intelligibility of existence. —

Knowledge of reason (ratio) does indeed lead us to see what is eternal and necessary in things, but it is nevertheless, according to Spinoza, not the highest form of knowledge. In it the universal law and the single phenomena still stand in opposition to one another, and thinking here is still discursive, and wins its result by comparison and inference. The highest form of knowledge, according to Spinoza, is the intuitive, where the single phenomenon is seen immediately as if shone through by the general order of things, as when I see immediately that two lines are parallel to each other when each of them is parallel to one and the same third, or as when in the knowledge I have of something I at the same time see immediately what belongs to knowing something. The difference between the universal and the individual has here dropped away entirely; their unity is seen at one glance. It is Spinoza's highest aim to understand as much as possible in this immediate, intuitive way; he declares, however, that it is very little that he has so far been able to understand in this way. In the last book of the "Ethics" he seeks to understand the single soul in its inward unity with the eternal being: an intuition that forms the conclusion of all his theoretical and practical striving. The chief condition for rising to this intuition is that one tear oneself loose from external, accidental experience and find the necessary connection with the help of knowledge of reason. — Yet it is not easy to hold fast to a sharp difference between knowledge of reason and intuition; for both show us things "under the aspect of eternity," and moreover: in his description of intuitive knowledge Spinoza says that it *"proceeds* from an adequate concept of the essence of certain divine attributes to an adequate knowledge of the essence of things":

[306] here, then, the discursive has not entirely dropped away. — What Spinoza aims at is nothing less than the highest goal of all knowledge: to unite individuality and continuity, the single and the whole connection, in as inward a way as possible. He succeeds in this only by postulating an intuition that can recall now the artistic apprehension, now mystical beholding, according as the weight is laid on the individual or on the universal moment. —

It is a great pity that Spinoza did not get his theory of knowledge finished. But the fragment we have nevertheless shows that the fundamental definitions and axioms do not come in him as if shot out of a cannon, but are constructed on the basis of a reflection on the presuppositions of all understanding of existence. What is dogmatic in Spinoza appears in the confidence with which he sets up these presuppositions and treats them as realities, not in his having altogether overlooked the existence of such presuppositions.

c. The Fundamental Concepts of the System

α. When Spinoza begins his deductive exposition by defining what he understands by cause of itself and by substance, and by laying down as a fundamental proposition that what does not have its cause in something else must have it in itself, he has really thereby only postulated the possibility of a conclusive knowledge of existence. Existence could not be known if the causal principle did not hold, and causal knowledge could not be completed if there were not something which had its cause in itself and did not again point the inquiring thought beyond itself, as every single phenomenon does. When Spinoza defines substance as "that which subsists in itself and is understood through itself, that is to say, whose concept does not need the concept of any other thing from which it should be formed" (per substantiam intelligo id, qvod in se est et per se concipitur, hoc est id, cujus conceptus non indiget conceptu alterius rei, a qvo formari debeat), he has nevertheless, in his view, not thereby given a purely subjective determination of
[307] a concept, but formulated an objective fact, a reality. For him there is no doubt that substance exists. Its existence is given; but when we reflect on what substance is, we see that its existence is also necessary, since there is nothing that can exclude it, inasmuch as it has its cause in itself. Likewise it follows from the essence of substance that it cannot be produced or pass away; that it cannot be divided; that it cannot be limited; that there cannot be more than one substance. Everything else to which we ascribe existence can then only be properties (attributa) of the one substance or single phenomena (modi, ways) in which it reveals itself. That there appear in existence different properties (such as consciousness and extension) which we cannot reduce to a common form does not

justify (as Descartes thought) the assumption of different substances: they must be conceived as different properties or attributes of one and the same substance. Just as little does the fact that experience shows us a multiplicity of finite things or phenomena, separated in time and space, justify us in assuming a multiplicity of substances. The existence of finite things or phenomena is conditioned through and through, and true subsistence (substantiality) can be ascribed only to the universal order of things, the comprehensive being which bears them all.

β. With the concept of substance, the concept of God and the concept of nature coincide for Spinoza. Through the concept of substance he points back to Cartesianism and scholasticism, through the concept of nature also to Bruno and the Renaissance, and through the concept of God to the religious ideas which first of all set his thinking in motion. God is called substance as the infinite being which reveals itself under an infinity of attributes. Everything that is said of substance must then hold of God too, so that everything that is, is in God and can neither be nor be conceived without God. Both matter (extension) and mind (consciousness) are then divine attributes and do not denote beings outside God, and just as little are the single things or phenomena (modi) outside God. Just as little as God has anything outside himself, just so little are there differences or oppositions in God's essence; here there is no difference between possibility and actuality, no difference of time. Everything that follows from his essence follows with *[308]* eternal necessity; God's freedom consists precisely in this necessity determined by his essence. When God is called the cause of things, it must be well noted that cause and effect are here not two different things, but the effect a revelation of the essence of the cause. God's work is not different from himself, as the work of a human master is. God is the cause of things in the same sense in which he is his own cause; he is an indwelling (immanent), not a transcending (transscendent) cause; his work remains in him, and he in it; he cannot go beyond the compass of his essence. A distinction must indeed be drawn between God and the sum of all the single phenomena (modi): the phenomena make up the produced nature (natura naturata); God is the producing nature (natura naturans). But an external separation does not exist. According to Spinoza it is a purely *abstract* way of regarding things when we regard the single phenomena isolated from "the whole order of nature," that is, from substance, the producing nature. We thereby throw away the condition of understanding them. To see existence *abstractly and superficially,* as it appears to the senses and the imagination, through the isolated appearance of the phenomena, is for Spinoza the opposite of seeing it *as substance* and with the aid of the intellect. Every single phenomenon is only a limited form of the one infinite substance,

arising through a negation, a canceling of all the other forms under which it appears. Every particular determination is a negation[60]).

γ. Substance, God or nature, appears under different properties or attributes. "By attribute I understand," says Spinoza, "that which the intellect perceives of substance as making up (constituting) its essence." Of the infinitely many attributes which the infinite being that we call God or nature has, we know only two, mind and matter. The essence of God or nature appears to us only under these two forms, as working partly in the world of material phenomena, partly in that of mental phenomena. Spinoza's relation to
[309] experience appears clearly here. Not only is it a matter of experience that we know only two attributes, but the very definition of the concept of attribute rests on experience, namely on the fact (which must have dawned on Spinoza in the interval between the "short treatise" and the definitive working out of the "Ethics") that the mental and the material are each understood according to its own laws. The difference between them is then irreducible; we cannot derive the mental from the material or the material from the mental, but within the mental world by itself and within the material world by itself we can demonstrate connection and causal relation. The fact that the mental and the material are irreducible differences in existence is what Spinoza expresses by calling them attributes and by defining the attribute as that which expresses the essence of substance: for substance is, after all, that which subsists in itself and is understood through itself. One must therefore not leap over from one attribute to another. "So long," says Spinoza, "as things are regarded as mental phenomena (modi cogitandi), we must (debemus) explain the order of nature, or the causal connection, through the attribute of mind alone; and so long as we regard them as material phenomena (modi extensionis), the whole order of nature must (debet) be explained through the attribute of extension alone." There is thus only one substance, one causal connection, but it can be regarded in two ways. It has hardly been Spinoza's meaning, however, that the attributes should be only points of view which *we* apply to substance[61]). He says expressly, after all, that an attribute is what the intellect perceives as *constituting* the essence of substance, and a doubt whether what our intellect perceives as constituting also really constitutes cannot be shown in Spinoza. When he says, then, that we *must* explain the mental by the mental and the material by the material, this is a necessity which, like all necessity for him, is an expression of the eternal essence of existence.

If it is asked what it really comes from that one cannot explain the mental side of existence by the material, or conversely, the answer lies contained in the ideal of causal explanation which Spinoza has set up, and according to which cause and effect must

be homogeneous (4th and 5th axioms of the first book of the Ethics). In a letter (Ep. 4) *[310]* Spinoza says still more clearly: "When two things have nothing in common with each other, the one cannot be the cause of the other; for since there would be nothing in the effect which was also in the cause, everything that was in the effect would have arisen out of nothing." When this fundamental proposition is consistently held fast, one has the key to Spinoza's whole system. For from it follows the doctrine of God or substance as the *indwelling* cause of all phenomena (modi), and so the abolition of the theological dualism of God and world. And from it follows further the impossibility that the cause can be material and the effect mental, or conversely, and so the doctrine of mind and matter as expressions of one and the same existence, one and the same causal connection, and with that the abolition of the spiritualist dualism of soul and body. Just as it was the *general* epistemological consideration of the conditions for understanding what is given in experience that led Spinoza to set up the concept of substance (as the concept of hypostatized causality), so it was the *special* consideration of the possibility of understanding the mental and the material that led to the doctrine of attributes. This doctrine is essentially conditioned by the conviction of the possibility of conceiving the material side of existence as a continuous whole, and of the impossibility of deriving the mental side of existence from it. Spinoza makes a stand against Descartes and against Hobbes alike. His theory of attributes is an attempt to solve the difficulties from which Cartesianism and Hobbesianism (spiritualism and materialism) suffer. He has set a new, original hypothesis over against spiritualism and materialism.

Now Spinoza has not strictly held fast to this meaning of the doctrine of attributes. With the problem of the relation between mind and matter he has in several places confused the problem of the relation between knowledge and its object. It is easy to see that these are two different problems; for both mind and matter (existence from its mental as well as from its bodily side) are objects of knowledge, and the problem of knowledge comes up with regard to all the sides from which existence can be seen. But since Spinoza, *[311]* as a dogmatic philosopher, confidently relied on the agreement between our reason and the essence of things, he had to assume — as already set forth — that to every relation of thought there corresponded a relation in existence. And this *epistemological* doctrine of identity he then confused with the *psychophysical* doctrine of identity which his doctrine of attributes expresses. When he lays down the proposition: "The order and connection of representations is the same as the order and connection of things" (Eth. II, 7), this proposition is ambiguous, since it can be understood both epistemologically and psychophysically. Spinoza himself understood it rather epistemologically. This is

seen from his example: to my concept of the circle corresponds the circle existing in reality (Eth. II, 7 Schol). Psychologically it would have had to run: to my representation of the circle corresponds the state my brain is in when I have this representation. But when Spinoza, in immediate connection with this, maintains that mental phenomena are to be explained by mental phenomena, material phenomena by material ones, it is an entirely different problem that he has to do with. In the third book of the Ethics, where the point of view is psychological, not epistemological, he accordingly applies the doctrine of attributes purely psychophysically. — This confusion is naturally connected with the fact that the problem of knowledge in Spinoza does not appear as an entirely independent problem. Only through a critical revision of our knowledge — with regard to its capacity to know the mental as well as the material side of existence — does the difference between the two problems come clearly to light[62]).

Just as, according to Spinoza, there is an inward relation (an indwelling relation, a relation of immanence) between substance (God or nature) and the single phenomena, so for him the great fundamental forms and fundamental properties that existence presents must also be substance's own attributes. It was a bold consequence of this when Spinoza declared matter or extension a divine attribute. Matter, on the usual theological conception, is only something derived, a product of mind. Just as little as Bruno and
[312] Böhme could Spinoza rest content with this conception, which conflicts above all with his fundamental thought of the indwelling causal relation. With regard to Spinoza's doctrine of matter as a divine attribute it must be well noted that by matter he here means what is substantial in material phenomena, that in them which subsists through all changes and in spite of all divisions, and which can neither suffer, arise nor pass away. The material side of existence is, just as much as the mental side, a revelation of the Deity or of substance; to its "infinite power of thought" (potentia infinita cogitandi) corresponds the infinite material extension, of which the single phenomena are parts. What lies at the ground of existence, from whichever side it is regarded, is the causal connection and the unity to which the causal connection bears witness.

δ. The single phenomena (modi) are, as will be remembered from what has gone before, the particular ways in which substance comes forth within the different attributes. Since the attributes hold for substance, they hold also for the single modi: every single phenomenon can therefore be regarded from infinitely many sides, just as well as the whole of existence, of which it is a member or part, only that we know, after all, no more than two of these sides. The definition of modus is: a determination of substance, or: that which has its subsistence in something else, through which it is

also understood. As affectio substantiæ a modus is understood only through substance and has its subsistence in it — just as, one might say, the single wave subsists only as part of the sea and is understood only by being seen as part of the sea. In every single phenomenon there stirs an impulse of self-preservation which is a part of the divine power working in all things. But in each single modus this power stirs only in a limited way; one cannot, then, understand the way in which the single modus acts or suffers without seeing it in its interaction with all other modi. The state of the single modus is therefore understood only through something other than itself. Here an external (transscendent) causal relation comes into play. But however externally the single phenomena stand over against one another, they are nevertheless, all together and each by itself, only determinations (affectiones, modificationes, determinationes) of the one infinite substance. In the produced nature (natura naturata) the producing *[313]* nature (natura naturans) is at work at every point.

Within natura naturata Spinoza distinguishes between necessary and infinite modi, which follow immediately from God's eternal nature, and finite modi, which need other modi for their coming into being. Within the attribute of extension he names (Ep. 64) as infinite modi motion and rest; here he seems in particular to be thinking of the conservation of motion, or of the law that through all the changes of matter the same relation between motion and rest is preserved, — the law which makes it clear that every single body is only a part of the whole universe. Within the attribute of thought he names the "infinite intellect" (intellectus infinitus), which is an immediate effect of the infinite divine power of thought, and which — since it is placed in analogy with the constancy of motion — must be taken to mean the mental energy that constantly remains the same in the world, however much the single mental phenomena may change. (Perhaps it is chiefly the conservation of motion and "the infinite intellect" that Spinoza, in the treatise on the improvement of the intellect, has in mind with the expression "fixed and eternal things.")

ε. For a closer characterization of the system set forth here, a series of problems and difficulties to which the system leads shall only be indicated further. — God or substance is spoken of as acting, although the form of time is to hold only for the modi, natura naturata, not for substance, natura naturans. This is connected with the confusion of ground and cause mentioned above. — There subsist two causal relations: one among the modi mutually, an external, transscendent one, and one between substance and the modi, an immanent one. How are the two to be reconciled? — The unity of substance and the multiplicity of the attributes and modi are not brought into harmony; no answer is given

to how the one substance can appear under different points of view, or how an infinity of effects can follow from the one substance. — And when for Spinoza the concepts substance, nature and God cover one another, he introduces with the designation God a
[314] determination of value which is grounded only by his defining perfection as reality; the question whether this definition can be held fast is not examined more closely, and at a later point in his exposition he even sets up — as we shall see — a concept of perfection according to which not all reality is perfect. — Thus even the boldest speculative system leaves unanswered questions enough behind it. What is great in Spinoza is his resolute carrying through of the thought that existence must be rational; from this he concludes that its essence must be identity, absolute unity. Consistently, he would have had, instead of several systematic fundamental concepts, only a single one: substance. To this concept one may perhaps ascend — but once one has reached it, one can in any case not get down again. — These objections to Spinoza's system do not, however, strike his fundamental endeavor, the fundamental direction of his thinking, but are made possible for the most part by the dogmatic procedure he employs.

d. Philosophy of Religion

It will be of interest to stop for a moment to see in what relation Spinoza's philosophy stands to the current religious views. When he named his highest concept (substance) God, he had himself at once given occasion for such a comparison. Already in his correspondence we see this point discussed. On a particular occasion he writes (Ep. 73): "I hold an opinion of God and nature far different from the one which the modern Christians are accustomed to defend. For I maintain that God is (as one says) the indwelling, not the external, cause of all things. That is to say, I assert with Paul that in God all things are and are moved . . . But when some think that the 'theological-political treatise' aims to show that God and nature, where by nature one understands a mass or corporeal matter, are one and the same, they are entirely mistaken." — From the usual theological conception Spinoza's philosophy separates itself especially on two points: in not conceiving God and the world as two different beings, and in not ascribing human
[315] properties to God. In the first period one made short work by calling his conception atheism and materialism. Henry More, e.g., says in a writing in which he sets out to refute Spinoza's "gross atheism" (crassissimus atheismus) that God and matter are one in his system. Therefore, says More, stones, dirt, lead, dung are God, since these things are, after all, matter! So Spinoza is then a "sordid and befouled atheist" (sordidus et lutulentus atheus)[63]). A small sample of the polemics of the seventeenth century! — People were

at a loss with the new system on account of its peculiarity; it could not easily be fitted into the current rubrics of heresy, and in its sharply cut forms it nevertheless provoked offense on several points. Later the expression pantheism came up (first, it seems, in Toland). Here one had a word that hit the mark better. And yet there is no full agreement about its use. According to some it denotes a view that conceives God's relation to the world as immanent; according to others what is essential to pantheism is that God is not thought of as a personal being[64]). By the first of these marks there is no doubt that Spinoza's system is pantheistic; by the second it is less clear how one is to designate it.

In one of his letters (Ep. 54) Spinoza says that, so as not to confuse the divine nature with the human, he does not ascribe to God human properties such as will, intellect, attention, hearing, etc. The more precise grounding of this is found in several places in the "Ethics." — It is clear that the applicability of the psychological determinations must fall away with the falling away of the relation of time, and time holds for Spinoza only for natura naturata, not for natura naturans. With the relation of time all change falls away too; God can pass neither to greater nor to lesser perfection and can therefore feel neither joy nor sorrow, love nor hate. Further, God as substance has nothing outside himself, no external object toward which feeling or will could direct itself. Our consciousness, our intellect as well as our feeling and our will, is determined by such external objects, and when they fall away, consciousness too falls away for us. — If one wants to use words such as intellect and will of God, one must make it clear to oneself that their meaning *[316]* must undergo such a change that at bottom it is only the word that is common to the two uses, much as one uses the word dog both of the constellation the Dog (Sirius) and of the barking animal on earth. The whole impulse to transfer human properties has therefore no objective validity. If triangles could think, they would conceive God as triangular; if circles could think, they would conceive God as round.

On the other hand Spinoza recognizes the mental as an independent side of existence. God is substantia cogitans as well as substantia extensa; cogitatio (thinking, mind) is one of God's attributes, and so expresses God's essence! Intellect and will belong only to the produced nature, not to the producing; but there must be a ground in the latter for their being able to arise, and if the "infinite intellect" has eternal subsistence through all the change of mental phenomena, this too must have its ground in the essence of the Deity. One might perhaps express Spinoza's conception thus: God is mind, but not personality, because only a limited being can be a personality. (Spinoza mentions the expression personality in a single place only, but declares it obscure.) It must be noted, however, that Spinoza, by declaring cogitatio a divine attribute and by speaking (as will

be mentioned later) of God's infinite love of himself (of which man's love of God is a part), himself undertakes just such a sublimation of the psychological properties as he denies on other points. Against cogitatio as a divine attribute objections could be raised similar to those which Spinoza himself brings against the current conception. — Those to whom it is absolutely a matter of bringing a philosophical system under one "ism" or another will to that extent be at a loss with Spinoza, especially since every thinking
[317] theism must undertake sublimations of the same kind as Spinoza's.

e. Philosophy of Nature and Psychology

By setting up extension as one of the attributes of existence and by demanding (by virtue of the very definition of the attribute) that in the explanation of a phenomenon one is to keep to the attribute under which one regards it, Spinoza has already established a purely mechanical explanation of nature as the only scientific one. As falling under the attribute of extension, all bodily phenomena are to be explained only by bodily causes, by causes that act from other points of space than the one at which they themselves are. It is therefore very easy for Spinoza to give a proof of the principle of inertia: this really lies already in the setting up of extension as an attribute. Just as dogmatically as Descartes and Hobbes, Spinoza sets up the fundamental propositions of mechanics as eternal truths — although he must concede that he cannot derive motion from extension itself[1]. There can be no differences of essence between bodies; these differ only in respect of motion and rest, speed or slowness. The laws of motion are eternal laws of nature and of reason. — In the doctrine of attributes is contained also the rejection of every assumption of purposes in nature. Since the one attribute cannot determine the other, no material phenomenon as such can have its ground in a thought, which the teleological
[318] explanation presupposes. In a longer note to the first book of the "Ethics" Spinoza seeks to show that the teleological conception springs from man's practical need and wish, from his impulse of self-preservation in connection with his original ignorance of the real causes; he assumes that everything is arranged for the good of men and that the gods work for this; where no reasonable purpose can be found, they content themselves with finding the explanation in God's will, this refuge of ignorance (asylum ignorantiæ),

[1] *Spinoza* makes this concession in the last letter we have from his hand (of 15 July 1676), an answer to the philosopher and mathematician Tschirnhausen, who had asked with great acuteness how one could derive from the general concept of extension the figure and motion of bodily things. Tschirnhausen pointed out that this difficulty does not exist in Descartes, who assumed that God created matter in motion; but Spinoza, for whom extension is a divine attribute, then lacks the concept of force which in Descartes was conceived in theological form. Spinoza declares the definition of matter as extension insufficient and announces a closer discussion of the question, which he did not manage to give.

which is always at their service. If mathematics had not pointed out a rational way to knowledge, the teleological conception would forever have hindered the knowledge of nature. The teleological conception, however, not only hinders the knowledge of nature but also presupposes unworthy representations of the Deity: if God acts according to purposes, he is, after all, not perfect, but becomes so only when he has reached his purposes! Men make bold to apply moral and aesthetic concepts to nature, find order and confusion, beauty and ugliness, good and evil in it, supposing that it is designed to appeal to their feeling. They do not see that the perfection of nature is one with its might, and that this might is infinite, so that it is able to produce all possible degrees and forms of existence, those which to us stand lowest just as well as those which to us stand highest.

In a series of lemmas in the second book of the Ethics Spinoza indicates how he conceives a mechanical view of nature carried through. — According to the law of inertia every body is determined to motion or rest by another body, this by a third, and so on. The way in which a body is determined depends partly on its own nature, partly on the nature of the body that acts on it. — When bodies of the same or different size are so held together by other bodies that they touch one another, or when they communicate their motions to one another in a certain ratio, they are said to be mutually united and all together form one body or one individual, which is distinguished from other individuals by the definite union between the single bodies of which it consists. Such an individual continues to be the same even if some parts are taken in and others given off, and even if the size of the parts and the direction of the motions change, provided only that the *[319]* whole form of the composition continues to be the same. Several such individuals can be united into an individual of the second order, and several individuals of the second order into an individual of the third order, and so on, until at last the whole of nature stands before us as one single great individual, whose single parts vary infinitely without the whole individual undergoing any change.

In all its stages and degrees nature is animated. This too follows from the doctrine of attributes: since each attribute expresses the whole substance, to every form under the one attribute there must correspond a certain form under the other attribute. To the different stages of individuality in material nature, then, there correspond different stages of mental life in mental nature. Just as the human body consists of many composite individuals of different constitution, so the thought which makes up the nature of the human mind is also composed of many different thoughts. The mind and the body are, after all, one and the same substance, or rather one and the same part (modus) of

substance, and what under the attribute of extension appears as more or less composite forms of motion appears under the attribute of thought as more or less composite forms of thought. When I have a drive or make a decision, there takes place simultaneously under the attribute of extension a bodily determination according to the general laws of motion and rest, and this is to be understood in such a way that at bottom it is one and the same thing that happens. By this assumption, as Spinoza shows, one is spared from maintaining an influence of the soul on the body or conversely, something of which no one can form a clear concept. To say that the soul sets the body in motion is at bottom to say that one does not know how this motion comes about. This assumption has the further advantage that one does not set arbitrary limits to the activity of material nature. No one, after all, has yet found the limit of what can be effected by the laws of material nature itself, so that one has no right to invoke the intervention of the soul because the
[320] body seems to act in a marvelous way. Nor does the appeal to the "freedom" of the will help. This so-called "freedom" (real freedom, according to Spinoza, is that which consists in following the necessity of one's own nature) is an illusion which arises from one's not knowing the causes of one's own acting. The chatterer thinks he acts freely when he lets his tongue run, just as the stone, if it could think, would think that it fell to the ground of its "free" will. Besides, the decision is, after all, dependent on memory, and of this we are not always masters! —

Although Spinoza rejects every explanation of mental phenomena by material causes, he nevertheless uses — which is also fully justified on his presuppositions — the connection between the material and the mental to throw light on mental phenomena where these cannot be understood immediately. Every sense-sensation corresponds to a bodily state. Now this state has indeed its cause in the influence of other bodies, but this influence is modified by the constitution of our own body: from which it follows that the sense-sensation corresponds not only to the nature of the external bodies but first and foremost to the nature of our own body. Nevertheless we involuntarily regard every sense-sensation as the expression of something real, until other sensations or representations arise in us which exclude the assumed reality. According to Spinoza, then, it is through the mutual struggle of sensations and representations that our conception of reality is cleared and determined. — Sensations call up representations of what resembles, or has occurred together with, the object of the sensation[65]). The laws of the connection of representations are for Spinoza mental laws of nature, the counterpart of the laws of motion within the attribute of matter. Spinoza distinguishes sharply between the association of representations and thinking proper. This difference has been discussed in

the section on his theory of knowledge. Sense-sensation and the association of representations express man's state as a member of the series of phenomena, of the infinite natura naturata. The higher stages of knowledge, rational knowledge and intuitive knowledge, are reached by going back to that which stirs in everything as common law and essence, that is, to the infinite substance, and seeing everything taken up into it, or seeing it *[321]* sub specie æterni. — Hobbes's psychology has here the advantage over Spinoza's that it shows the possibility of how thinking proper, by concentration on a purpose or a task, can develop out of the involuntary association of representations. Spinoza still, in scholastic fashion, posits thinking proper as an entirely peculiar activity, different in essence from ordinary representing. All the more remarkable, then, that he nevertheless confuses the epistemological and the psychophysical points of view in the doctrine of attributes — although on the other hand this confusion would hardly have occurred if he had seen that the laws of the association of representations are in reality not suspended in thinking proper. —

The treatment Spinoza gives of the psychology of the feelings belongs to the finest he has accomplished, and that has been accomplished in psychology at all. In the first place he has the merit of maintaining the necessity and the justification of a purely natural-historical investigation of the human feelings and passions. He will (as an inquirer) neither mock them, laugh at them nor weep over them, but seek to understand them, and will therefore regard them with the same equanimity as if it were material phenomena of nature or geometrical figures that he had before him. Next, he has brought forward the most important points of view that come into application in the explanation of the feelings.

As regards the position and significance of feeling in mental life, Spinoza held different views at the different stages of his philosophical development. In his first work, "the short treatise" (II, 16, 5), his conception is entirely intellectualistic. It is knowledge, the representations, that determine all the phenomena of mental life. The will is merely a consequence of the representation, an affirming or rejecting. And here, moreover, the representations themselves are still determined by the influence of the objects, since in this writing he assumes an influence of the one attribute on the other. Already in his writing on the Cartesian philosophy he indicates that the representations are themselves expressions of our activity, so that one cannot distinguish between the arising of the representation and its affirmation or rejection. And with the definitive doctrine of attributes this indication is carried further, but in such a way that intellect and will *[322]* are now made completely one. To think is an activity of the mind, and all activity of

the mind is thinking: so, roughly, one might express Spinoza's conception here; it is formulated in the corollary to the 49th proposition of the 2nd book of the "Ethics": Voluntas et intellectus unum et idem sunt. But in the 3rd book of the "Ethics," where the natural history of the feelings is given, an entirely new conception of the will appears, since the will is here determined as conscious impulse of self-preservation (Eth. III, 9 Schol). It is now so far from being one with, let alone a consequence of, knowledge, that knowledge is precisely dependent on the will: "we do not seek, will, desire and wish for anything because we think that it is good, but conversely, we think that something is good because we seek, will, desire and wish for it." Here, then, need and will are one, and the representations dependent on them. In his treatise on the history of Spinoza's development, in which Tönnies has thrown light on this point in a brilliant way, he conjectures with great probability that it is Hobbes's influence that has produced this last change in Spinoza's conception. Hobbes puts drive and will in close connection with each other and with the impulse of self-preservation, and he also stresses the influence which the thought of a purpose has on the connection of representations. In Spinoza's work as it lies before us there is an evident contradiction between the definition of will II, 49 Coroll. and the definition given III, 9 Schol., so that it is highly probable that two drafts have been worked together without all traces of different conceptions having been removed.

From the impulse of self-preservation as the expression of the nature of each single being Spinoza now proceeds in his exposition of the psychology of the feelings in the 3rd book of the "Ethics." Even if he thereby changed something in his psychology, he changed nothing in his general philosophy. In every single being the infinite nature, the Deity itself, is at work. The striving of each single being to preserve itself is then only a part of the infinite divine working in the whole of existence. Already in the "short treatise" (I, 5) he saw (like Bruno) in this impulse of self-preservation, living in all beings, the revelation of divine providence. And this need is one with the very nature
[323] of things, takes on a different character according to the nature of the single beings: to exist is to strive to preserve oneself. This striving is called drive when it is bound up with consciousness: by drive, then, is understood the very essence of man, insofar as it is thought of as determined to activity by reason of a change in its state. When this drive is regarded in particular from the mental side, it is called will. Pleasure arises when this striving is furthered, pain when it is disappointed. Be it noted: it is the very transition to a more favorable ("more perfect") state that is bound up with pleasure, and the very transition to a less favorable ("imperfect") state that is bound up with pain. The states

themselves, as at rest, are not bound up with feelings. Nor, however, is any express comparison made between the states: the feeling corresponds immediately to the very change of the states.

Just as Spinoza here states what we now call the biological significance of feeling, its connection with the advance or decline of life, and just as he also has a clear eye for the significance of the relation of contrast for the arising of the feelings ("the law of relation in the domain of the feelings"), so he has also put the third main point in the psychology of the feelings, the significance of the representations and the combinations of representations for the development of the feelings into special kinds, in an excellent light. — From drive, pleasure and pain he derives, with the help of the laws of the combination of representations, the special kinds of feeling. We love what causes pleasure, but hate what causes pain. We love further what furthers what we love, just as we hate what harms what we love. And when we represent to ourselves a being that resembles ourselves as having a certain feeling, we involuntarily get the same feeling ourselves. From this compassion can arise — but also envy and wounded ambition, namely when we wish to possess alone what the other enjoys, or when we wish the other to be disposed entirely as we are disposed. — According to these general principles Spinoza now carries out his consideration in detail. He has founded the theory of evolution in the domain of the feelings, by showing how our feeling, under the influence of the representations and by mutual composition and mixture, can take on forms that are highly different from the *[324]* elementary initial forms. It is the conditions of growth of feeling that he has formulated. — How far it is possible to keep exclusively to the one starting point he has seized on, the individual impulse of self-preservation, is a question that cannot be discussed here.

f. Ethics and Political Theory

It is characteristic of Spinoza that there is a practical motive for his thinking, and that his thinking nevertheless rises high above the world of finitude and experience in which alone practical motives and practical striving are possible. Spinoza rejects the validity of every valuation of existence; existence follows its eternal laws, and our ethical judgments, which rest on a comparison between the more or the less perfect in relation to the ideal of human life that we form for ourselves, cannot find application to the eternal essence of nature itself, for which time, measure and number do not hold. Spinoza insists on the subjectivity of the ethical qualities as energetically as the subjectivity of the sense qualities had been insisted on from Galilei's time onward. Insofar as Spinoza holds fast to his conception of existence "under the aspect of eternity," he can have no real ethics,

and needs none either. Of ethics there can be question only in a world where the relation of time holds, where there is becoming and development, change and opposition, and where models can be formed and purposes set up. As Spinoza has aptly expressed it: the judgments of good and evil rest on comparison — but in the eternal unity with itself of the infinite, divine nature all comparison falls away, because all difference and opposition has fallen away. The investigation of Spinoza's theory of knowledge showed us that all *riddles* fall away under "the aspect of eternity"; we now see that all *tasks* fall away too.

But just as Spinoza distinguishes between natura naturans and natura naturata, he also distinguishes between the purely theoretical standpoint and the practical standpoint.
[325] The practical standpoint holds only within natura naturata. Spinoza sets up an ethics with the same right as that with which he maintains the significance of the modi of the world of experience, the finite world. There is no greater inconsistency in his setting up an ethics than there is in his setting up, at all, other concepts than the concept of absolute substance.

The possibility of an ethics is closely connected in Spinoza with his psychological doctrine of the feelings. The foundation of ethics is the impulse of self-preservation which lies at the ground of the manifold feelings and passions. For out of the impulse of self-preservation there can develop a need not to be tossed about by all the feelings and passions which external causes call up in us. As a part of nature, as a modus among other modi, man is dependent and is tossed about like the waves of the sea without knowing goal or direction. We naturally strive, then, to assert our being, after full, inward activity. The fundamental ethical virtue is *strength of mind* (fortitudo), an inward might which makes man free and independent. Virtue is the very essence or nature of man, insofar as he has the might to produce effects that can be understood through the very laws of man's nature. For man to be able to attain this character, he must make it clear to himself that no feeling and no drive can be checked in us except by a stronger feeling or drive. (This proposition was already maintained by BACO: De Augm. Scient. VII, 3). Spinoza builds his proof of this proposition on the law of inertia, which, by virtue of the identity of mind and matter, has its counterpart within the mental side of existence. Ethical development becomes possible through the fact that feelings can be called up which gradually displace the feelings by which they themselves were originally called up. When, by comparing different forms and stages of human conduct of life, we form for ourselves an ideal of human nature (idea hominis, tanqvam naturæ humanæ exemplar), we call everything that brings us nearer to this ideal good, everything that hinders us from approaching it evil, and there arises a drive to approach the ideal as much as

possible. The knowledge of good and evil is — since it concerns what is truly useful and harmful to us — nothing other than conscious joy at approaching the ideal or conscious *[326]* sorrow at moving away from it, and this knowledge can, precisely because it is one with a feeling, become a might in the soul. Man now soon sees that he cannot attain full strength of mind and freedom of mind by his own help. The whole of outward and inward life must be ordered and made secure. Above all it is a matter of developing a true knowledge; this too is reached only by united strength, just as, on the other hand, there is nothing that better unites men than the striving after such goods as can be common to all. The passions that bring strife among men always concern something which the individual alone can possess. The more each individual attains freedom and strength of mind, the less strife will there be among men, and the more will they act in common, because they all seek what is truly useful to human nature. Nothing is so useful to men as men. Best of all would it be if all could become as one soul and one body, seeking the common utility. Those who strive after their own true utility therefore wish nothing for themselves which they do not wish for other men, and thereby they become just, faithful and honorable in their conduct. The strength (fortitudo) in which virtue consists therefore appears not only as *courage* (animositas), by which man immediately asserts his personality, but also as *magnanimity* (generositas), in that man strives to help others and to bind them to himself in friendship. — By distinguishing between courage and magnanimity Spinoza makes an advance in relation to Hobbes and the thinkers of the Renaissance, in that he gets an eye for the different elements and possibilities, of which some can lead more in an individual, others more in a social direction. Courage is a predominantly individual virtue, magnanimity a social virtue. But Spinoza is hindered, already by his starting point in the pure impulse of self-preservation, from entering on a closer investigation of the relation between self-assertion and devotion.

Characteristic of Spinoza is the valuation which, after setting up the model of human development, he makes of the different feelings. The feeling of pleasure is in itself good, the feeling of pain in itself evil. Therefore hatred, fear, contempt, pity, remorse *[327]* and humility are evil — insofar as they are not necessary intermediate links for the arising of feelings of pleasure. This is grounded by the fact that the feeling of pleasure corresponds to an advance in perfection, the feeling of pain to a decline. He who is led by true knowledge, the spiritually free man, has his gaze directed toward the positive development of life in himself and in others. He thinks of nothing less than of death; his wisdom is a meditation on life (meditatio vitæ), not on death. He strives, as far as he can, to outweigh the hatred, anger and contempt of others by love or magnanimity. He

who fights hatred with hatred lives a wretched life; but when hatred is conquered by love, the conquered yield with joy, not because their strength diminishes but because it grows. There is no way in which a man can better display his strength of mind than by developing others to live according to their own free knowledge[66]).

While Spinoza's ethics leads in this way from the impulse of self-preservation as its foundation out to practical action and the founding of society, there also stirs in him another tendency, which from the same starting point leads in a mystical and individualistic direction. From his earlier psychological standpoint there still constantly presses forward the presupposition that man's real essence consists in pure knowledge: the more this can develop, the more does man's real nature come forth, the more is he truly self-active, and the true impulse of self-preservation therefore leads to the development of knowledge. It was, after all, also the knowledge of the connection in which our mind stands with the whole of nature that stood for Spinoza as the only good which could not be taken from us by external causes. Therefore Spinoza can, almost in the same breath (cf. Eth. IV, 22 Coroll. with 26 Dem.), set up the two propositions that the striving after self-preservation is the first and only foundation of virtue, and that the striving after knowledge (conatus intelligendi) is the first and only foundation of virtue. Alongside the realistic psychology, according to which Spinoza conceives feelings and passions as positive powers in our being, there runs another conception, according to
[328] which feeling is defined as imperfect or confused representation (idea inadæqvata sive confusa, see the concluding explanation in the 3rd book of the "Ethics"). According to this other conception feeling falls away with the full and clear development of knowledge. The more we understand ourselves and our states as determined by the infinite nature, by the Deity which stirs in us as well as in everything that acts on us, the more do we see ourselves not as single, isolated, powerless beings, but as taken up into and as one with the Deity. We feel pleasure at this thought, since it is the fruit of our mind's highest activity, and pleasure at the thought of the being that is the cause of the joy of knowledge we feel. Thereby there arises a spiritual love of God (amor intellectualis dei), which gives us the highest peace. God does not stand here as an external object; our own innermost power is a part of the infinite power that stirs in everything, and our love of God is one with the infinite love with which God loves himself: God's love of men and men's love of God are one and the same. We see ourselves and everything under the aspect of eternity. — Spinoza the mystic comes into conflict with Spinoza the psychologist; for the intellectual love (the human and the divine) is to be raised above becoming and opposition, but Spinoza has himself, after all, explained feeling

as corresponding to a *transition* — either an advance or a decline! Only by explaining feeling as unclear thinking and abolishing the law of relation can Spinoza hold fast to the mystical conclusion of the ethics.

In the fifth book of the "Ethics," where this mysticism prevails, is found also Spinoza's doctrine of immortality. In that intellectual love of God, grounded on the understanding of our unity with the absolute being, our true essence expresses itself: this essence, then, is not dependent on time, but is a part of the infinite intellect (intellectus infinitus), which continues to subsist although the single mental phenomena arise and pass away, just as motion subsists through all material changes. As an immediate form of the divine working we are immortal. But this immortality — which is not a *continued* subsistence but an *eternal* subsistence, since the relation of time has no significance at all here — seems able to belong only to those in whom the highest knowledge, and with it the full *[329]* and clear self-activity, is developed[67]). Whether this subsistence as an eternal element of thought (modus cogitandi æternus) is personal or not cannot be seen from Spinoza's exposition; his notion of immortality is in any case very different from the popular one. And at the close he remarks that the assumption of the eternity of our mind is not a condition of ethics, pointing out that he himself has developed his ethics without using this assumption. Blessedness is not the reward of virtue but is one with virtue itself. For only when we can feel joy in our mind's highest working can we master our passions. —

The deep and sound thoughts in Spinoza's ethics would have come to a fuller unfolding if he had carried out his psychological theory in detail. As it is, intellectualism gains the upper hand in him, especially when he is to depict the highest forms of spiritual life. His ethics thereby becomes too much an ethics for philosophers. The significance of immediate devotion and of artistic intuition does not come into its own, although in his "intellectual love," grounded on intuition, there are hints in these directions. But no one will be able, with an unbiased sense, to immerse himself in Spinoza's wonderful work without confirming Goethe's words about the boundless unselfishness that shines forth from every proposition, about the lofty resignation that once and for all bows to the great laws of existence instead of scraping through life with the help of petty consolations, and about the air of peace that runs through the whole book. —

But peace, according to Spinoza, must be won through strife. It is won by the individual through the struggle with the passions, in which a new, higher passion is developed, and it is won by men in union through the savagery of the state of nature, where everyone takes as much right as he has might, being replaced by the life of the state through the establishment of an authority that can subdue the conflicting drives

and bring harmony between the impulses of self-preservation of the individuals. In his political theory, as in his psychology, Spinoza is decidedly influenced by Hobbes. Only
[330] he does not have Hobbes's blind and one-sided fear of the multitude. He is consistent enough to hold fast that human nature is the same in all, and that safeguards are needed just as much against those who rule as against those who are to be ruled. One can just as little presuppose the best motives in the rulers as in the multitude. And if it is thought that the multitude has no understanding of public affairs, Spinoza answers that this comes from its being kept in ignorance of them. Spinoza is one of the first democratic philosophers of law in modern times; but since his "political treatise" was not completed, we do not know his doctrine of democracy in detail. It is characteristic that while Hobbes maintains that a democracy is in fact always an aristocracy — of orators, Spinoza maintains that a monarchy is always an aristocracy — of officials. In reality, of course, the political functions must always be exercised by a few — but it depends on where one finds the ultimate source of authority. According to Spinoza it depends on the state's being built on the united might of the many, in that they are united into one will, just as the strength of the individual also grows by being united with that of the others. But this spiritual unity, which is the condition of the unity of might, can be present only where the state strives after what sound reason points out as useful to all. One can put up with much for the sake of peace; but if slavery, barbarism and isolation are to be called peace, then peace is the most wretched of all things for men. The state is not to make men into beasts or machines, but precisely to bring it about that their mental and bodily activity can unfold freely, and that they can use their free reason.

Only on the free adherence of individuals does the might of the state rest, and the individual therefore does not give up his natural right on entering the state, but has it precisely secured and developed. The state would act against its own purpose by suppressing the individuals' freedom of thought and of speech; it would first and foremost make the noblest and best men in the land its opponents.

As Spinoza maintains the freedom of expression, so he also maintains religious freedom. It is the main subject of his "theological-political treatise." He wants to show
[331] here that religion has nothing to do either with the state or with science, and that the state, by its nature and its purpose, must let the inner spiritual life develop with full freedom. The state can have power only over outward action, not over disposition. It can lay down an outward cult, and its dominion in this respect is to be preferred to a hierarchy. But even when it uses its might in this way, it does best to let religious worship consist merely in the exercise of love and justice. The mental nature of men

(ingenium) is so various that what awakens reverence in the one causes laughter in the other. It is therefore best to let everyone keep his free conviction and right to interpret the religious documents according to his own intellectual tendency (ex suo ingenio). — From science religion separates itself by its practical character. It aims at piety, not at truth. The prophets were not men who were especially highly gifted in intellectual respects. Imagination was stronger in them than understanding. Their task was to impart to the people, through images and symbols, representations that could bring it to live in that obedience to God which shows itself in love of one's neighbor. The multitude, which cannot follow the inferences of reason, must have historical models for guidance. Only thereby does it come to obey not out of mere fear but out of its own will and devotion. What theoretical opinions men have of God — whether they think he is fire or spirit or light or thought — is a matter of indifference, so long as they have in him a model that can lead them on the right way. — While according to Spinoza the prophets do not see the truth in its pure form, but only in images, the divine wisdom was revealed immediately in Christ, the only one who has had the capacity to apprehend something which is not contained in, and cannot be derived from, the first grounds of our knowledge. In a letter (Ep. 73) Spinoza declares that he does not understand what the Church means by the doctrine that God has taken on human nature; it seems to him as absurd as that the circle should take on the nature of the square. But he assumes that God's eternal wisdom, which reveals itself in all things and especially in the human mind, has revealed itself most of all in Christ. — Spinoza demands that the biblical writings be subjected to a purely historical investigation. The Bible is to be explained by itself, *[332]* just as nature is explained by itself. Only when one has established what the Bible really teaches can one raise the question whether this teaching is true or not. Just as Spinoza has maintained the independence of the investigation of nature and of ethics, so he here maintains the independence of the historical method. His historical eye shows itself in the demand that in the interpretation of the writings one is to take account of what one knows of the individualities of the authors. One is also to investigate the origin of the writings. The investigation Spinoza undertakes leads him to assume that the five books of Moses were composed in the time of the Exile. Even if more recent research does not always go so far, it nevertheless has a forerunner in Spinoza.

It was through his philosophy of religion that Spinoza first had an influence. Before Lessing and Goethe, and after them the speculative philosophers, pleaded his cause at the end of the eighteenth century, his views had an influence of no small extent in the Netherlands, both in such a way that individuals quietly found nourishment for their

life of thought and feeling in his ideas, and in such a way that religious sects were even formed, now of a more mystical, now of a more rationalist character, whose conception and application of Christianity derived, through more or fewer intermediate links and modifications, from the writings of the solitary philosopher. Right up to the end of the eighteenth century Dutch orthodoxy had frequent occasion to denounce Spinozist thoughts as heresy; his books were repeatedly forbidden and confiscated. In its more mystical form religious Spinozism united with the influence of Jakob Böhme's writings, which from the end of the seventeenth century were eagerly read in the Netherlands. — Remarkably enough, the three elements which were bound together into a peculiar unity in Spinoza's thinking have later each in its own time won enthusiastic adherence: first the religious, in the Dutch sects, then the idealist, in the German Renaissance at the end of the eighteenth century, and finally the realist, among philosophers and natural scientists in our own day. And even thinkers who officially would not acknowledge him
[333] were nevertheless deeply and strongly influenced by him. This holds in particular of the man who has often been set up, by himself and by others, as Spinoza's antipode: Leibniz.

6. Gottfried Wilhelm Leibniz

a. Biography and Characterization

In Descartes, Hobbes and Spinoza we find three great systems developed which, for all their differences, have this in common, that they assert a purely mechanical connection of nature as regards the material side of existence. In these systems the consequences had been drawn from the principles whose establishment had made the rise of exact natural science possible. They were attempts to show how existence looks when these principles hold absolutely for all material phenomena (whether one then holds that there are other phenomena than the material ones, or not). To many of the old school these new principles and systems were the epitome of arbitrariness and ungodliness. A reaction makes itself felt in various forms at the end of the seventeenth century. In philosophical respects the reaction of greatest interest is the one which will not give up what had really been won in scientific respects, and which, not in an external way, by compromising or by patching incompatible elements together, but in an internal way, through an investigation of the presuppositions that lay at the basis of the mechanical view of nature itself, sought to lead beyond it and to reconcile it with the ancient and medieval conception on which men had turned their backs with such great energy. In this sense Leibniz signifies a reaction against those three great thinkers. But perhaps never

has a reaction been carried through in so profound a way, even if Leibniz, consciously and unconsciously, accommodated himself in his manner of expression to conservative views far more than his standpoint entitled him to. What from the first offended Leibniz in the modern ideas, for all his admiration for them and for those who bore them, was that they threatened the aesthetic and the religious contemplation of nature. The legitimacy of the concept of purpose was rejected in the world-view, and the world seemed to become *[334]* a mere machine. The aim of his life was to bring about a reconciliation between ideas from opposite regions of the world of the spirit, and in working to reach this aim he developed a wealth of thoughts and points of view, of discoveries and hints, which is perhaps unique in the history of thought. In almost every scientific field he intervened creatively and stimulatingly, and there is an inner agreement between, a common type for, his thoughts in the widely different fields in which he worked.

Gottfried Wilhelm Leibniz was born in Leipzig on 21 June 1646. His father, who was a jurist and professor of moral philosophy, died early, and the boy, who was often left to himself, was very fond of spending his time in his father's library, where he read all the books he could manage, at first chiefly novels and historical works, later scholastic works. Despite all the obstacles of foreign languages and of the abstruse character of the content, he nevertheless sought by his own efforts to understand what he got hold of. Those around him, he relates in an autobiographical sketch, feared first that he would become a poet, and next that he would become a scholastic: "they did not know that my mind could not be filled exclusively by one kind of thing". The struggle between the mechanical and the teleological view of nature began early in his mind. He relates that as a quite young man he wandered about in a wood near Leipzig and pondered whether he should hold to the scholastic philosophy or join the new doctrines. The need of an all-round education and of acquainting himself with the exact sciences led him away from Leipzig. On the occasion of a slight he gave up taking the doctor's degree in his native city. He took the degree in law at Altorf and then went to Mainz, where, despite his youth, he was taken into the Elector's service. Plans for the improvement of the science of law and of the law-books occupied him greatly in these years alongside his philosophical and scientific studies. He wanted to root out the scholastic barbarism in jurisprudence, just as at the same time he became more and more convinced of the decisive victory of modern natural science over the scholastic doctrine of nature. There was even, by his own later *[335]* statement (in the Nouveaux Essais), a time when he inclined very strongly to "the side of the Spinozists". But the ruling thought in him was nevertheless this: to maintain the assumption that existence is determined by purposes without giving up mechanical

natural science. With zeal he defends the new scientific conception against the men of the older time, his teacher Jakob Thomasius and the famous jurist and physician Herman Conring. But it was his endeavor to show that the mechanical connection of nature, far from excluding, actually presupposes a divine thought of providence. In a small treatise *(Confessio naturæ contra Atheistas)* and in his letters to Jakob Thomasius from the years 1668—69 this double endeavor appears clearly. Leibniz is here still a dualist; there is still for him a gulf between the two principles he wants to connect. In studying Kepler, Galilei and Descartes he feels, by his own account, "as if transported into another world"; but the old world is to be connected with the new. A journey to Paris on a mission for his government gave him the opportunity for new studies. Alongside his practical business and alongside the remarkable proposal he laid before Louis XIV for a French conquest of Egypt (in order to divert the lust for conquest from Germany), he immersed himself in mathematical studies in Paris under the guidance of Huyghens and was led to the invention of the differential calculus, at the same time as he zealously studied the Cartesian philosophy. His standpoint around the year 1670 bears the stamp of the influence of Hobbes's writings. In a treatise on natural philosophy *(Hypothesis physica nova.* 1671) he asserts, in a manner similar to Hobbes and Gassendi, the continuity of motion by means of the thought of tendencies of motion (conatus) in the smallest moments and the smallest parts. This thought stands in connection with the discovery in the following years of the significance of infinitely small quantities in mathematics. From this time on he combats Descartes's doctrine of the conservation of motion and begins to put the conservation of force in its place. He finds the difference between mind and matter in this: in the material domain a tendency which does not pass over into motion
[336] can exist only in the single moment, but in mental life the tendency can, by means of memory and thinking, reach beyond the moment. He has expressed this pregnantly in the proposition that a body is a momentary mind or a mind without memory. In a letter from the year 1671 we read: "As the activity of bodies consists in motion, so the activity of souls consists in striving (conatus) or, so to speak, in a minimal or punctual motion, just as the soul properly is only in a point in space, while the body occupies a place". We have in these hints what one might call Leibniz's second philosophy. He seeks to go beyond dualism and asserts continuity by putting mental striving into the differentials of matter, but is still not clear about the relation between the material and the mental. An obscure point of union is found in the concept of striving; beyond this indefinite hint Leibniz does not here get. Only this stands firm for him, that the solution can be found only by going back from what is given in the phenomena to its elements

and presuppositions.

Just as Leibniz worked in the fields of various sciences, it was also his endeavor to come into personal contact with eminent men of various intellectual tendencies. In Paris he associated with Huyghens, to whose mathematical and scientific ideas Leibniz owes much: through him he was surely led to the principle of the conservation of force, which plays so great a part in his thinking. Even earlier he had addressed himself to Hobbes in writing, though no correspondence seems to have come about. During his stay in Paris his attention was directed to Spinoza, whose "Theological-Political Treatise" had appeared a few years before, while the "Ethics" was as yet accessible only to a narrow circle. The mathematician and philosopher Tschirnhausen, who belonged to this circle and was one of Spinoza's most acute correspondents, was together with Leibniz in Paris and gave him a survey of Spinoza's views. Through a common friend Tschirnhausen informed Spinoza that he had met in Paris "a man well versed in the various sciences and free from the usual theological prejudices, by the name of Leibniz, with whom he had formed a friendship, since he was a man who, like himself, worked at developing his *[337]* mind and regarded this as the most important thing". Tschirnhausen considers his new friend worthy to know the "Ethics" and asks the master for permission to show it to him, reminding him that Leibniz had already earlier written Spinoza a letter concerning the "Theological-Political Treatise". Spinoza answers: "I believe I know this Leibniz through his letters. But I do not know why he, who is a court councillor in Frankfurt (i.e. Mainz), has gone to France. So far as I have been able to see from his letters, he is a man of a free mind and well versed in every science. Yet I believe it would be unwise to entrust my writings to him so quickly. I should first like to know what he is doing in France, and to hear what my friend Tschirnhausen thinks of him when he has associated with him longer and come to know his character better". Spinoza evidently did not have real confidence in Leibniz, whose many-sidedness and pliancy were certainly not always favorable to his intellectual character and in any case gave his spiritual life and his activity a stamp that was entirely opposite to Spinoza's withdrawn and concentrated existence as a thinker. Spinoza's mistrust was justified. For however near Leibniz stands to Spinoza in his final results, and when he is consistent, he nevertheless in his later writings seizes every opportunity to distance himself from him, and that often in expressions one would not have expected of a man who was "free from the usual theological prejudices". At this time, however, he was very eager to come into contact with Spinoza. After a four years' stay in Paris (1672—76), where he — a proof of how different minds he knew how to value and learn from, — associated with the great Jansenist theologian and zealous

Cartesian Antoine Arnauld, he went by way of England to Holland, where he stayed for two months, during which he repeatedly visited Spinoza and discussed philosophical subjects with him. Only through the most recent literary finds[68]) has this personal relation between the two great thinkers been discovered, a relation which became so intimate that Spinoza at last overcame his misgivings and showed him the "Ethics". Leibniz himself does mention in a single place (in the "Theodicy") his visit to Spinoza,
[338] but gives the impression that the conversation turned on the events of the day! It was not commendable to have been so near to Spinoza. One sees accordingly that several of the letters in the edition of Spinoza's works which his friends saw to immediately after his death have been abridged, and the names of the writers of the letters often omitted, and Leibniz, who had looked forward to the publication with great longing, was very dismayed to find one of his own letters (an entirely innocent one, to be sure) printed with his name given. That Leibniz had written Spinoza several letters, no one had previously suspected. — Spinoza's line of thought exerted a strong influence on Leibniz, especially in leading him definitively beyond the dualistic theory. One finds Spinoza's thought again in him when at this time he objects against the Cartesians that the difference between mind and matter did not exclude their being properties of the same being. We have marginal notes and excerpts of Leibniz's dating from his first reading of Spinoza's "Ethics". There are three points on which he at once raises objections, and all three are characteristic of his line of thought. The first is the fifth proposition of the first book, which asserts that there can be only one substance; the second is the exclusion of intellect and will from "the producing nature" (Eth. I, 31); the third is the appendix to the first book with the famous polemic against final causes. — As little as Leibniz ever was a Cartesian or a Hobbist, so little was he ever a Spinozist. But he received mighty impulses from his predecessors. As regards Spinoza it must especially be noted that Leibniz embraces the identity hypothesis put forward by Spinoza and carries it further. Hardly any greater misunderstanding of Leibniz could be conceived than to conclude, from his assertion of the plurality of substances, that he disputed the identity hypothesis. His endeavor aimed precisely at uniting both assumptions; how far he succeeded is a question in itself. —

At the end of the year 1676 Leibniz took up a position with the Duke of Hanover (Braunschweig-Lüneburg) as court councillor and librarian. A mass of administrative,
[339] historical and political work laid claim to him, and his restless zeal to keep up in every field brought with it an enormous correspondence. In the library in Hanover are kept the correspondences he carried on with over a thousand different persons. But just

as it was the main thought of his philosophy that existence consists of a multitude of individual beings, each peculiar in itself, in mutual harmony and continuity, so too there is a connection between his endeavors in the various fields. He had an eye for small relations, for the vanishing differences which nevertheless, by their joining together and totalization, can prove to be realities. His mathematics, his physics and his philosophy show us this trait. But no less weight did he lay on continuity, the inner connection between all things. It was the connection between past and present that interested him in his historical studies, and it was a similar interest that led him to procure information about China through the missionaries of the Jesuit order. He had a sense for the practically useful as well as for science and Church. He saw to the development of the mines in the duchy, worked for the founding of academies, proposed that the monasteries be transformed into scientific institutions, negotiated with Bossuet about a union between the Catholic and Protestant Churches and later with Protestant theologians about a union between the Lutheran and the Reformed Churches. To Peter the Great, whom he highly admired and with whom he had several meetings, he put proposals for reforms and for measures to promote philology and natural science. As a philosopher of language he had so sound an eye and asserted with such energy the significance of experience for the study of language that Max Müller has been able to say of him that, had he been understood and supported by the learned men of his time, the science of language would have been founded a century earlier as an inductive science. In the field of medicine too he put forward significant ideas. — No wonder that he can declare in a letter: "How extraordinarily distracted I am cannot be told. I make researches in the archives, bring out old papers and collect unprinted documents. I receive and answer letters in great number. But I have so much that is new in mathematics, so many thoughts in philosophy, *[340]* so many other literary observations which I do not like to let be lost, that I often do not know what I should do first." Nor, accordingly, did Leibniz succeed in collecting himself for a work typical of his personality and his line of thought. He was not a "man of one book" like Spinoza. His ideas he expressed in an infinity of letters and small treatises, which are not yet all printed. The most important philosophical works brought to light so far are found in Gerhardt's edition (7 vols.); but much still lies in the library in Hanover, and a new collected edition of his works is being planned, arranged by the international association of the scientific societies of the various countries. An incredible capacity for work and mobility of thought this man had. And just as existence in his system is mirrored in countless ways in the individual souls, so too he had a need to make his system intuitive from the very different standpoints with which he came into contact. It

was during his correspondence with the Jansenist Arnauld that he developed his system to full consciousness in himself. But he had nothing against giving it a scholastic form in order to make it acceptable to his friends among the Jesuits. He cast it in a more popular form when he philosophized with the Electress of Hanover or with the Queen of Prussia, or when he was to give Prince Eugene an exposition of his doctrine (the later so-called "Monadologie").

But we have not yet followed the development of Leibniz's philosophical views to their conclusion, to the point where he could say that he believed he had found "a new expression for the inside of things" (une nouvelle face de l'intérieur des choses). We have seen how he held fast to the legitimacy of mechanical natural science, but wanted to reconcile the teleological way of looking at things with it, first in a more dualistic manner, later (following Hobbes and Spinoza) in a more monistic way. It was his conviction that "the moderns had carried the reform too far" by making everything into machines, but it is no less his conviction that hitherto men had only "prostituted themselves" by trying
[341] to refute the strict mechanical connection even in the material processes that proved to be most closely bound up with mental phenomena.

In a letter from the year 1697 Leibniz wrote: "Most of my opinions have been matured at last after a deliberation of twenty years; for I began at a very young age to think . . . Yet I have changed and changed again my views according to the new knowledge I acquired, and only about twelve years ago did I feel satisfied." This statement points to the year 1685. In that year Leibniz worked out a small treatise *(Petit discours de métaphysique),* which at the beginning of the following year he sent to Arnauld. It states that existence consists of individual substances; in each of these the world is expressed from one of the infinitely many possible sides from which it can be seen in the divine thought; each one is a divine thought — realized by constant "emanation". Extension and motion are only phenomena, just as much as the sense qualities. Only force, not motion, persists; and force is one with substance. This treatise had been prepared by an earlier one (of 1680), in which it is first developed that the innermost nature of motion — motion, to which the modern systems reduced everything in the material nature! — was not yet sufficiently cleared up (intima motus nondum patent), and then it is asserted that to subsist (to exist, to be substance) and to act are one and the same, and that all force or striving is an acting[69]). The proposition that every substance acts, and that everything that acts is substance (omnem substantiam agere, at omne agens substantiam appellari), is perhaps the most important in the whole Leibnizian philosophy. By it an end was made of the mysticism or mythology that conceives the essence of things as a resting

essence, an unchangeable subsistence in itself. Leibniz himself found in it the possibility of his union of teleology and mechanism, in that the forces which make up the innermost nature of individual beings and phenomena, and are that which persists through all changes, are assumed to flow out from the divine being and to act in the direction of valuable purposes. The divine purposes are then fulfilled in nature without a breach of the lawful connection, but precisely through it. The advance in the "Discours" of *[342]* 1685 over the treatise of 1680 consists chiefly in this, that the *individual* character of the substances is asserted, and that they are conceived in analogy with the souls of men and animals. Each individual being is a little world, develops according to its inner laws, by its own force, a peculiar radiation of the divine force, and this development takes place, on account of the common origin, in harmony with the development of all other beings. — With this Leibniz's system was complete. It first came before the public in a treatise which was printed in the Journal des Savants in 1695 *(Système nouveau de la nature et de la communication des substances).* And the name monad he used only later (1696). He hardly borrowed the name from Bruno; on the whole no direct influence of Bruno on Leibniz can be shown historically, however much kinship there is in several of their ideas. Rather (according to what has been conjectured in the most recent time) he took the name from the mystical chemist van Helmont the Younger, with whom, just shortly before it is first found in him, he had come into personal contact. —

Leibniz fulfilled his own definition of substance as that which constantly acts. Even after his system was formed, he went on working to shape its fundamental thoughts to the greatest possible clarity. His extensive correspondence and his wide personal acquaintance gave him ample opportunity for this. Thus during a stay in Vienna he worked out (1714) for Prince Eugene an exposition of the doctrine of monads, which after his death was printed under the title *Monadologie.* At the same time he followed attentively what was stirring in his time. With two eminent thinkers he entered into a discussion, which he conducted with great chivalry. John Locke's attempt to derive all knowledge from experience he criticized in his *Nouveaux Essais* in a way that is brilliant and still to this day instructive, asserting in particular the significance of the inner dispositions, of the conditions lying in the knowing beings' own nature. As Locke died before Leibniz had his work finished, he would not publish it, and so it came not to exert the influence on the philosophical discussion that it would surely otherwise have *[343]* had. Only in 1765 did it appear. The other contemporary against whose views Leibniz felt called upon to take up his pen was Pierre Bayle. Bayle had given a critique of the attempts of various theologians and philosophers to demonstrate a harmony between

the assumption of an almighty divine government and the moral and physical evil in the world, and had asserted that philosophically the assumption of two opposite principles, a good and an evil one, could be defended far more easily given the world's actual constitution. Against this Leibniz sought in his *Théodicée,* which appeared in 1710, to show the agreement between theology and philosophy and to ground the claim that the actual world is the best possible world. It is Leibniz's weakest work, diffuse and loose in form, vague and in a bad sense popular in its line of thought. Nowhere, moreover, does he accommodate himself more to notions to which, by strict consistency, he could give no place in his system. It was unfortunate that it was above all this work through which Leibniz was known in the eighteenth century. The "Nouveaux Essais" lay forgotten in the library in Hanover, and the brilliant and inspired small treatises in journals in which he had expressed his real fundamental thoughts were accessible only to few.

There can be no doubt that Leibniz meant his attempt to unite and reconcile religion and philosophy honestly. It was a thought which from his youth on runs through his whole life. But a properly religious nature he was not. This is shown by the ease with which he could disregard confessional oppositions. It was natural rather than positive religion that lay close to his heart. And it was his sense for the individual and characteristic that made it easy and at the same time interesting for him to familiarize himself with the various ecclesiastical systems, just as it was his faith in the harmony among all individual peculiarities and endeavors that led him to the wish for a reconciliation between the contending Churches, a wish which was not shared by men like Bossuet and Spener, who could see no legitimate peculiarity in confessions other than their
[344] own. Arnauld urged him rather to be converted to the true faith than to speculate about the connection of things. Characteristic of Leibniz is a story he liked to tell about two brothers, one of whom had become a Catholic and then sought to convince the other, while the latter just as eagerly sought to win him back for Protestantism; the result was that they both mutually convinced each other, — and the end of the whole thing was that God, on account of their great zeal, had mercy on them both! What interested Leibniz was the inner force and striving in the monads, not merely the particular way in which force manifests itself in the individual monad, and which is often regarded by the latter as the only right one. Leibniz's faith concerned the universal harmony, which for him was not first a result of the future, but was already present in minds, if only they could become conscious of it. In every book Leibniz read — and he read many books — he found something that was interesting and instructive to him, and since he knew how differently things can look from different points of view, he was inclined to

defend and excuse even where he could not approve. With people of all stations in life he knew how to talk about their interests. His recognition of the monads extended to the lower forms. He avoided killing animals, and if he had used an insect for microscopic examination, he put it carefully back on the leaf from which he had taken it. Mild and cheerful, not subject to strong emotions, constantly occupied with thinking and study — so he went through the world. — His last years were lonely. Because of his many labors he had, as he himself said, not had time to marry — before it was too late. Whereas in an earlier time he had been in favor at the Hanoverian court, and had advanced to "Geheime-Justizienrath", he seems in later years — after the death of his princely patrons — to have been in a kind of disgrace. The clergy and the people suspected his orthodoxy because he did not go to church; his name was interpreted in Low German as Lövenix, i.e. Glaube-nichts. When he died (1716), his bier was followed by only a few. — His motto was: "As often as an hour is wasted, a part of life is lost"; it was set on his coffin together *[345]* with a symbol he had been fond of: a spiral with the superscription: Inclinata resurget (it bends only in order to rise).—

b. The Doctrine of Monads

Now that we have gained a survey of the historical development of Leibniz's philosophy, the task is to give a systematic exposition of it. The writings which must chiefly be taken as the basis here are *Discours de métaphysique* (1685), *Système nouveau* (1695), *Lettre à Basnage* (1698), *De ipsa natura* (1698) and *Monadologie* (1714). But a mass of letters and smaller treatises must be compared with these.

It is by an analysis of the fundamental propositions of natural science that Leibniz grounds his philosophy. He acknowledges these fundamental propositions as unconditionally as Hobbes and Spinoza, but he finds them to be derived from presuppositions lying deeper. Physics has its ground, for him, in metaphysics. The first and foremost thing, then, is to see how, according to Leibniz, the transition from physics to metaphysics is to be made.

When the new science aimed at reducing everything in nature to motion, Leibniz entirely approves of this endeavor. But as a philosopher he asks: what reality has motion itself, and what cause has it? — All motion is relative. Whether something is assumed to be moved will depend on the standpoint from which it is seen. The real reality is not motion itself, but the force which is its cause, and which persists even if the motion ceases. The Cartesians, who taught that the sum of motion in the world is the same, but also that motion passes over into rest and conversely, are forced to assume constant

breaches of continuity, since both the transition from motion to rest and the converse became inexplicable. Only by means of the *concept of force* (or tendency, conatus) can the connection be maintained. There is then only a relative difference between motion and rest: they are different phenomenal forms of force. It is force that persists. There is no
[346] absolute rest. Not in the smallest moment one may think of is a moved body in one and the same place in space. If one asks what force then is, Leibniz answers: *that in the present state which brings with it a change in the future* (ce qu'il y a dans l'état présent, qui porte avec soi un changement pour l'avenir) (Lettre à Basnage). This is what is real in existence. Leibniz also arrives at this concept through a critique of the concept of the atom as the atomism of the time conceived it. The assumption of absolutely hard atoms would conflict with the conservation of force and with continuity, since in every collision force would be lost. The atoms must therefore (as Huyghens already taught) be elastic — and must accordingly consist of parts; absolute atoms are consequently a self-contradiction. There is, when we examine the matter thoroughly, activity everywhere. "The atoms are," says Leibniz in a letter, "only the effect of a weakness in our imagination, which likes to rest and therefore hurries to come to an end with the divisions and analyses". (Letter to Hartsoeker 1710). In his polemic against the Newtonian Clarke, Leibniz distinguishes between a body's losing motion, considered as a whole, in a collision with another body, and force's being lost: the motion which the body considered as a whole (in relation to other bodies) loses passes over to the parts, which by the impact are set in an inner vibration; the forces are not annihilated, but distributed among the small parts. The same thing happens here as when one changes large money into small coin. (Cinquième écrit de Mr. Leibniz § 99). The difference between the solid and the fluid is relative, just as motion is relative. From this point of view too only force has a real reality.

That force is the real reality Leibniz expresses, in the language of the time, by saying that force is substance and that all substance is force. But the concept of force is in turn closely connected with the *concept of law:* force is, after all, only that in a state which brings it about that a change occurs in the future. The presupposition of every application of the concept of force is then a lawful connection between the changing states. The principle of sufficient reason, which says that there is a ground for everything,
[347] is therefore the real fundamental proposition of Leibniz's philosophy, although he sets it up as an express fundamental proposition only in his later writings (the Theodicy and the Monadology). And with the concept of law another main concept of Leibniz's is in turn connected: *the concept of individuality.* For individuality means for him the law according to which changes in a being's states succeed one another. Each individual has, so to

speak, its formula. And just as in a mathematical series one can infer the following terms when one has found the law for the succession of the first terms, so it is also with the inner changes of an individual being. In the law for their succession consists individuality itself: *la loi du changement fait l'individualite de chaque substance particulière* (Lettre à Basnage). In the "Discours de métaphysique", where the concept of the individual substance is introduced for the first time, the relation between the substance and its changing states is compared with the relation between subject and predicate in a logical judgment: just as in a logical judgment it must be possible to find the predicate contained in the subject as soon as one knows the subject completely, so he who knows the nature of the individual will be able to derive from it all its changing states. There is continuity between these, since the following one always has its ground in the preceding one. Each of these substances is therefore a little world for itself, develops from its own inner drive and has its own inner life. Only in a miraculous way can such an individual substance arise or pass away. And each substance has its own inner law, different from others.

If it is asked why the real, the persisting (the substance) must be individual, Leibniz answers: because only *unities,* individual beings, can exist. He reproached the Cartesians not only for ascribing absolute reality to motion, but also for ascribing absolute reality to extension and making it substance. Already in his notes on reading Spinoza's "Ethics" he writes that he is not sure that bodies are substances; it is otherwise with souls. This remark indicates the direction in which his thinking went. In his later works the following line of thought recurs constantly under manifoldly varied turns of phrase: extension presupposes something which is extended, that is, repeated several times. Extension *[348]* means multiplicity, composition, aggregate. But the nature of the composite depends on what composes it (the components). Not the aggregate, but the ones of which it consists, have reality. If there were not absolutely simple substances, there would therefore be no reality at all. The assumption of the reality of extended matter brings with it, moreover, the difficulty that its divisibility, by virtue of the law of continuity, goes on to infinity. The true unities therefore cannot be material, extended; if, on the other hand, no such absolute unities existed, matter would be a mere seeming without any real significance. Only the absolutely simple and indivisible can be substance, and it cannot be material.

Before we ask what, then, it is, one more important determination must be brought forward. Leibniz started from the fundamental propositions of natural science. The most important of these was for him the principle of the conservation of force[2]. It was for him

[2]As already remarked, this law derives from Huyghens, who in his doctrine of the pendulum further developed hints which Galilei had given, and which we have also found in his disciples Gassendi and Hobbes. (See E. MACH: Die Geschichte und die Wurzel des Satzes von der Erhaltung der Arbeit. Prag

the foundation of the laws of nature (fundamentum naturæ legum). This very foundation of the laws of nature can, according to Leibniz (see especially De ipsa natura, Erdmann's edition p. 155), be understood only by a teleological argument: it is necessary that the same quantity of force be preserved in the world; but this necessity depends on the divine wisdom and order reigning in the world; the first principle or "sufficient reason"
[349] of mechanism is a final cause, *a teleological principle.* In a small treatise directed against the physicist Papin *(De legibus naturæ,* Dutens' edition, III, p. 255) Leibniz carries this out as follows: cause and effect must always correspond to each other, be equivalent. If there were more in the effect than in the cause, we should get a perpetuum mobile; if there were less in the effect than in the cause, the same cause could never be brought about again, nature would constantly go backward, perfection would be diminished. Both would conflict with the divine wisdom and immutability. When the equivalence of cause and effect obtains in nature, to the exclusion of the two other possibilities, then this, according to Leibniz, cannot be grounded purely mathematically or mechanically, but must be explained teleologically.

The law according to which the individual beings, which are the true realities, unfold their changing states is for Leibniz a law according to which they advance to more perfect states, not merely to new states. The relation between cause and effect is thus here identical with the relation between means and purpose. In its innermost ground existence is so disposed that the purely mechanical laws act in the direction of what the laws of justice and of love demand. In the law inherent in its own being (lex insita) both kinds of laws are united in the case of every individual. — Leibniz is, as one sees, so occupied with the law of the conservation of force that he overlooks that, teleologically seen, what matters is not that the same quantity of force is preserved, but that it is preserved in a form favorable and useful for the subsistence of life and for continued development. This is, moreover, a consideration which has come clearly to light only through the investigations of the most recent time. — But although Leibniz thus grounds the mechanical connection of nature teleologically, it is nevertheless not his intention to reintroduce final causes and to permit an explanation of natural phenomena by appeal to inner forces or to purposiveness. In particulars one must proceed strictly mechanically; it is only mechanism as a whole, or rather the first principles of mechanism themselves, that prove to become intelligible only through teleological consideration. The mechanical

1872. p. 10 f.). Leibniz maintained the principle of the conservation of force against the Newtonians, who denied it. See his interesting polemic against Clarke (Opera philos. ed. Erdmann. p. 746 ff.). On the relation between Descartes, Newton and Leibniz with regard to this principle see Rosenberger: Isaac Newton u. seine physikalischen Principien. Leipzig 1895. p. 409—412.

connection of nature is the fact from which Leibniz's whole philosophy proceeds, just as much as Hobbes's and Spinoza's. — *[350]*

The technical name for the absolute individual beings which for Leibniz make up the true reality is *monads.* It is a Greek word which means unities, and which thus designates the beings only from the formal side. If we now ask what these individual unities, with their inner states unfolded according to a definite law, really are, Leibniz answers that we must conceive them in analogy with our own souls. In the "Discours" this is not stated definitely. But in the following correspondence with Arnauld we read: "The substantial unity presupposes a complete (accompli), indivisible and naturally indestructible being, since its concept encompasses everything that is to happen to it. Such a thing cannot be found in figure or motion, . . . but only in a soul or a substantial form *in the likeness of (à l'exemple de)* what is called an I". (Letter to Arnauld, Nov. 1686). The objective analysis is thus here supplemented by a subjective analysis. In ourselves we find, by way of self-observation, inner states (sensations, feelings, thoughts) which change, and a need or drive which leads us to pass over from one state to another. By means of this subjective observation we must — by way of analogy — make the nature of the monads intelligible to ourselves. Our soul is only a single monad, but we have no reason to believe that we stand alone in nature. On the contrary. By virtue of the law of continuity, which excludes leaps in nature, we must assume that there are infinitely many degrees of the kind of existence we know from ourselves: c'est partout et toujours la même chose, aux degrés de perfection près. The law of analogy bids us everywhere to embrace the principle: "Just as with us" *(tout comme ici)*[70]*). The law of analogy,* according to which we conceive all other existence as differing only in degree from our own, is for Leibniz only a special form of the law of continuity and springs, like it, from the principle of sufficient reason. If one does not assume such a general ensoulment of lower and higher degrees, it becomes entirely unintelligible how sensation and consciousness can arise. Just as the origin of motion became inexplicable for the Cartesians because they did not go back to force, so the origin of consciousness was also a riddle for them, because they took no account of the obscurer degrees and forms of mental life, the unconscious mental life, the need and craving which first awaken to the clarity of consciousness at higher stages of life. In ourselves we constantly observe, after all, an alternation of clearly conscious, obscure and unconscious states. In analogy with this we must conceive monads of all degrees with respect to clarity and obscurity. There are slumbering, dreaming and more or less waking monads. The uniformity of nature leads to this assumption. The part of matter which makes up the human body cannot, after all, alone be endowed with the *[351]*

capacity to sense and to will. Something analogous (whether one wants to call it soul or not) must be found at lower stages. In a letter (of May 1704) Leibniz calls this application of analogy "mon grand principe des choses naturelles".

The way by which all mythologies and earlier speculative systems — without being conscious of it — had come to the assumption of the mental as ruling in existence, this way Leibniz goes in full consciousness. He points clearly to the point where all metaphysical idealism springs up. And it is again the presupposition of the complete intelligibility of existence that guides him here. From this presupposition follow the continuity of all existences and the analogy of all beings with our own being.

But as Leibniz has come to this point in the development of his thought, he stands at a turning point. He builds his philosophy on an analysis of material phenomena, guided by the endeavor to find what is real and persisting in them. Now that he has arrived at the assumption that, properly speaking, only representing beings exist, with differing obscurity and clarity in their representations, matter becomes for him a *mere* phenomenon. It is a seeming for the senses when bodies seem to exist. The senses do not deceive us, however, any more than they deceive us when we believe that the sun goes round the earth. The phenomena are real insofar as there is a regular connection
[352] between them, a law according to which they succeed one another. It is this law that is the mark of what is reality for us. But it does not force us to assume that anything exists other than monads and their representations. — At this point there would be the possibility of an entirely new line of thought: a testing of whether, from the purely subjective standpoint, from the assumption that the world exists only in the form of representation, one would arrive at the same result as by the objective way Leibniz has gone so far. This new way Leibniz did not take, and when in his last years he heard of Berkeley's attempt in this direction, it struck him as a mere paradox (see his letter to des Bosses of 15 March 1715). Leibniz himself evidently did not become clear about the great difference between being a phenomenon *of* something and being a phenomenon *for* someone. Both relations can lie in the concept of phenomenon; Leibniz begins with the first relation and ends in the second. And it is only after he has set up the concept of substance as force and as individual unity that he — in accordance with the law of analogy — finds its essence to consist in representing activity. Precisely because this last assumption is the end point of his train of thought, it is understandable that he does not draw all the consequences of it[71]). Leibniz continues to treat material phenomena as the expression of an objective order of things. By virtue of the law of continuity everything in the material world stands in reciprocal action; a change in one part of the world

extends to all parts. And from this it is concluded that it is the *whole* universe which is represented by each monad in a more obscure or clearer way. Every monad represents the whole universe, just as every single point in the physical universe feels everything that happens in the whole universe. The very assumption that there is a multiplicity of monads shows that Leibniz did not go the purely subjective way, for then he would have had to do with only one monad and its picture of the world. The assumption of other consciousnesses than my own already presupposes, after all, the application of the inference from analogy on which Leibniz builds his metaphysical idealism. —

An empirical confirmation of his hypothesis that there is force, life and soul behind all apparently dead mass he found in the discovery of the small organisms which Swammer- *[353]*
dam, Leeuwenhook and other naturalists found with the help of the microscope. There opened up here for him a world of infinitely small beings in the domain of organic life, just as he himself had helped to open up the world of infinitely small quantities, and just as in his philosophy he had asserted that behind the apparently homogeneous mass the true reality was found, consisting of individual beings with a nature akin to our own. Arising and passing away, birth and death were for him only phenomena, expressions of processes of contraction or unfolding, of obscuration or clarification in monads. The absolute leaps and differences fell away (no less than the homogeneous mass) the more clearly he became aware of the infinite individual nuances of life. With visible satisfaction Leibniz in all his later expositions refers to the works of the great contemporary naturalists just named.

With his concept of the monad Leibniz apparently enters into a sharp opposition to Spinoza. For the latter there was only one substance, and all individual beings stood to it as the waves to the sea. For Leibniz the true reality is a multiplicity of individual forces, each acting according to its inherent law, so that all its effects unfold from its own original character. Leibniz himself often sharply emphasized this opposition, most clearly in a letter to Bourguet (Dec. 1714): "I do not know how one can extract Spinozism from my doctrine. It is precisely by the monads that Spinozism is dissolved. For there are as many true substances, . . . as there are monads, whereas according to Spinoza there is only a single substance. He would be right if there were no monads; then everything outside God would be vanishing and be dissolved into simple modifications". But it is the same fundamental concept of substance from which both Leibniz and Spinoza proceed. All the determinations and states of the substance are to follow from its own inner nature, not to be due to influence from outside. Leibniz asserts this just as definitely in the case of each of his monads as Spinoza asserts it in the case of his one and only

[354] substance. The difficulty of this absolute concept of substance became for Spinoza this: how the multiplicity of the attributes and of individual beings could be explained from the assumption of a single substance. The difficulty of the same concept became for Leibniz this: how there can be agreement between the states of the different monads when each of them is determined only from within, and none of them has holes, windows or doors (to use Leibniz's own images) by which something from outside can come into them. And the difficulty became for Leibniz, besides, that of reconciling the doctrine of monads with his theological presuppositions.

Leibniz regards both these difficulties as solved by one thought. That there is agreement between the states of the different monads he takes as a fact: there is a universal harmony or concord (accord) between all beings in the world. Clearly enough, as already touched on, he starts here from the observation of the general physical reciprocal action in the world and from the empirical (and yet grounded on an inference from analogy) assumption that the observations and representations of the individuals agree in such a way that a common picture of the world can be constructed in its broad outlines. But since the real reality is a multiplicity of entirely self-subsistent and independent beings which (as Leibniz says in a letter) are to be regarded as little gods, since they have life in themselves, there must — he concludes — be a common origin of that harmony. "God alone", we read in the "Discours" (§ 32), "brings about the connection or coherence between the substances, and it is through him that the phenomena of the one meet with and agree with those of the other, so that there is reality in our representations". God has from the first created the individual monads in such a way that in the one regard is taken of the other; the agreement between them is due to the common origin. They arise from God by an outflowing or radiation (émanation, fulguration), as individualizations of the one divine force, just as each of our thoughts is a radiation of the force of our mind. God sees the universe from infinitely many different points of view, and each of these different points of view he realizes by creation as an individual substance. The
[355] states of the individual monads thus correspond to one another as the different views one gets of a town from different vantage points agree with one another (Discours § 14). The spontaneity in the monads and the conformity between the monads thus both find their explanation in the fact that it is the divine force and thought that make up their innermost essence. The harmony between them is present from the first, is "pre-established". Each of them is a peculiar concentration of the universe which stood before God's creating thought, is a universe en miniature, and the universe consists only of such concentrations. Leibniz is a metaphysical Copernicus: he shows how the world

can be seen from far more points of view than one is accustomed to think, and that the one point of view in itself has the same legitimacy and validity as the other. The world is for him not something that lies outside or between the individuals, but consists precisely of and in the individual beings.

Leibniz thus puts the whole problem back into the thought of the creating God. He passes over from philosophy to theology. But that does not free him from the difficulties: for how does there arise in the divine thought or imagination a world that can be seen from infinitely many points of view? And these possible points of view do not stand in the divine imagination as resting images; but just as Leibniz constantly asserts that all possibility is a little actuality, an incipient reality, so he assumes here too an original need or craving for existence (inclinatio ad existendum, exigentia existendi): the possibilities seek to press forward to full reality, which is granted only to those that fit together best into a harmonious world-whole. It is a mythological prelude which decides the struggle for existence before actual existence begins, so that the latter can then be a harmonious unfolding of inner germs in each being. In the exposition of the Theodicy we shall return to this point. It has been to the decided detriment of the clarity and significance of Leibniz's philosophizing that theological points of view play so great a part in him from first to last. And it does not help at all toward the solution of the problems to clothe them in a theological form. A certain feeling of devotion makes one rather stop asking questions and raising objections; but such a stopping is no solution. *[356]* It is easily seen that the problem, in the form in which it comes to present itself to Leibniz, is insoluble. If the monads are absolute substances, they cannot be produced. Only the appeal to a supernatural creation covers over the evident self-contradiction. Just as Leibniz, by virtue of his concept of substance, excludes real reciprocal action between the monads, so he would consistently have had to exclude their coming into being out of a substance different from them. Here Malebranche said the redeeming word: God cannot create gods! — Leibniz's difference from Spinoza — if one disregards that evident self-contradiction — in reality falls away, since the creation of the monads is not concluded at any point of time once for all, but constantly takes place: the radiation or outflowing of the monads goes on uninterruptedly (Discours § 14. 32. — Monadologie § 47: Dieu seul est l'unité primitive ou la substance simple originaire, dont toutes les Monades créées ou dérivatives sont des productions, et naissent, pour ainsi dire, par des fulgurations continuelles de la divinité de moment à moment). Leibniz lays weight on its being the source, the force to act, that thus uninterruptedly wells out from the Deity into the monads; but Spinoza too, after all, explains the striving which leads each

individual being to assert itself as a single manifestation of the power that acts in all individual beings. Consistently carried through, Leibniz's philosophy here differs from Spinoza's only in that the individual beings are drawn more into the foreground and their common origin stands as an obscure background (which is lit up more strongly only when Leibniz seeks connection with the theologians), whereas in Spinoza the individual beings are seen more as offshoots or ramifications of the infinite being which is his first and his last thought. The difference between the two systems can be reduced to the same difference as there is, according to Leibniz, between two views of the universe, produced by a turning of it. For both, existence is a pyramid, which Spinoza sees from the apex, Leibniz from the base[72]). — There falls, as already emphasized, a shadow both
[357] on Leibniz's intelligence and on his character when, despite this state of the case, he so often distances himself from Spinoza in more or less heresy-hunting expressions. —

Leibniz's theory of the relation between soul and body is a simple consequence of his doctrine of monads. What we call our body is a group of monads, each of which develops for itself according to inner laws and by inner force, just as our soul does. From the state of one monad one can, by virtue of the universal harmony, infer the states of other monads; it is this that is popularly expressed by saying that one being acts on another. The monad from whose state one can, by such an inference, derive those of the others is said to act. When the inner development of the individual monad is hindered, this is due to the obscurity, the chaotic state of the elements in its own interior. But since the nature of every monad, that is, the degree of obscurity or clarity and the force to pass over to new states with which it is endowed, depends from the first on a regard to the other monads together with which it is set into existence, there correspond to the inner hindrances definite states in other monads, which are therefore popularly called their causes. When Leibniz wants to express himself quite popularly, he uses the image of two clocks adjusted to each other once for all to designate the relation between soul and body. Popularly and accommodatingly he also calls the body a composite substance. But this manner of expression is too crude for his real conception. Matter is for him, after all, only a phenomenon, a world of external multiplicity whose states express for the senses the inner changes in the monads, which are the real reality. One thing "expresses" or "represents" another, in Leibniz's usage, when there is a constant and regular relation between what can be asserted of the two things (Letter to Arnauld, Oct. 1687). What is concentrated, united into unity, in the interior of the monads appears to sense perception as an extended multiplicity. In a letter of 4 Nov. 1696 (to the Electress of Hanover) Leibniz writes: "Souls are unities, bodies multiplicities . . . But although

the unities are indivisible, they nevertheless represent the multiplicities, much as all *[358]* lines from the periphery meet in the center. *In this union (réunion) consists the wonderful nature of consciousness (sentiment);* and it is also this that makes every soul like a world for itself". — The stringency with which Leibniz, in agreement with Hobbes and Spinoza, asserts the mechanical connection of nature makes it impossible for him to keep to the usual dualistic conception. The identity hypothesis stands, on his presuppositions and according to his whole system, as the only possibility. The difficulties to which this hypothesis is subject Leibniz has made still greater by the very absolute substantiality he ascribes to the monads. From the stamp of unity which consciousness has, he concludes with dogmatic haste that the conscious being must be an absolute, indestructible unity which can receive nothing from outside. Had he, instead of merely conceiving the bodily in analogy with the mental, *also* conceived the mental in analogy with the bodily, he would have seen the necessity of assuming that the individual consciousnesses, despite their wonderful stamp of unity, stand in reciprocal action with the rest of existence by virtue of the law of continuity on which he himself lays so great weight, but is able to apply only within each individual monad, not to the relation between them. Development comes for Leibniz to mean only unfolding. The individual organism, for him, really does not come into being at all; it exists from the first; its birth and growth consist only in its being freed from its envelope and in its dimensions being enlarged. It was the so-called encasement hypothesis (the one germ being thought to lie inside the other up through the series of generations) which Leibniz embraced like many naturalists of his time. And so in the mental domain too development became not a taking up of new content, but only a clarification of the content which was given from the first in a chaotic and obscure form. — Concentration cannot come into being according to Leibniz; new centers, new monads cannot form. Here he cuts the knot. The problem of individuality perhaps contains an insoluble knot for human thinking; but to cut it is not to solve it.

Yet this very cutting is — just as much as the distance he felt between himself and *[359]* Spinoza — only an outcome of the energy with which Leibniz thought through the concept of individuality. It appears in him with a brilliance and clarity as in no other thinker. And the peculiar turn he gave the concept of substance by applying it to the individual elements of existence (the monads) is characteristic of the transition from the seventeenth to the eighteenth century. In place of mystical resignation and submission to the absolute powers there now enters the free, peculiar striving of the individual for clarity, for the unfolding of its own being. It is a revolution in the world of thought — a prelude to the revolution in the external world. But one presupposition Leibniz's system

shares with the other great systems: the presupposition of the complete rationality of existence. It is the principle of sufficient reason that guides him in his theological speculation and in his doctrine of monads as in his mathematical and scientific works. This presupposition too passed as an inheritance to the new century, whose typical initiator he was.

c. Psychology and Theory of Knowledge

From the significance that the analogy with the life of consciousness has for Leibniz as the foundation of his metaphysical idealism, it follows that his real point of departure must be the same as Descartes's: immediate self-observation, inner experience. It gives us the first term of the analogy. Leibniz is also clear that here lies a point of departure for his thinking, and he has devoted great attention to it. It is the "Nouveaux Essais" that is the chief work here. The polemical side of this work could only be properly assessed once Locke's theory of knowledge, against which it is directed, had first been set out. But in this context the weight lies on Leibniz's own psychological and epistemological views. The work mentioned gains significance as a verification of Leibniz's general philosophical ideas, since in the introduction he declares that here he will disregard his
[360] doctrine of monads and take his stand on purely empirical ground.

The chief question that presents itself is whether the soul is empty from the outset, like an unwritten tablet (tabula rasa). Long before, in the "Discours" (§ 27), Leibniz had expressed himself on this question. Even then he found that this notion of an unwritten tablet was false. It is due to inexact observation. One overlooks the small, obscure stirrings in the soul and keeps only to what is clearly conscious, which arises only later, when outer experiences have long been at work. The Cartesians made themselves guilty of this error just as much as the empiricists. The less difference and opposition there is between our sensations, the less any single thing in us stands out in relation to the rest of the content, the more obscure, in a word, the life of consciousness is, the more easily one overlooks it. More exact observation will be wary of setting limits. There are all possible transitions between obscurity and clearness. The obscure changes in us that do not come to real consciousness Leibniz calls perceptions. Already in them a multiplicity is united in a single state. At this stage stand the lowest beings (in Leibniz's system: the monads at the lowest stage). The expression consciousness (sentiment)[3] Leibniz first uses where the sensation (the perception) is more distinct and is accompanied by

[3]Sentiment in Leibniz should be translated by consciousness. By conscience Leibniz understands something more than necessarily lies in the word consciousness, namely connaissance *reflexive,* thus rather "self-consciousness".

memory. The highest stage of mental life presents express attention (apperception) or reflection, which is directed toward the obscure sensations. (These three degrees are best described in the little treatise *Principes de la nature et de la grâce* § 4, cf. *Monadologie* § 14 ff.). But at every stage it is an activity, a spontaneity, that unfolds itself. We are active, even in the most obscure states. Just as there is no tablet that does not by its constitution have an influence on what is written on it, so our nature is preformed from the outset. In an involuntary, instinctive way we apply principles of which we can only later become conscious. What is contradictory we reject, even if we have never heard *[361]* tell of the principle of contradiction. There are both practical and theoretical instincts. The impulse of self-preservation and the need to help others work involuntarily. There is on the whole far more in us than we think of. Here, as in material nature, it is the small magnitudes that underlie what comes forth distinctly and to the senses. No more than motion can consciousness arise from nothing or all at once. Corresponding to the conservation of force (or properly — by virtue of the identity hypothesis — as one with it) Leibniz assumes the uninterrupted subsistence of mental life through states of consciousness and of unconsciousness, through clearness and obscurity. The apparent disappearance of mental life is only a conversion into a more obscure, more elementary form. The question whether there are innate ideas Leibniz thus answers to the effect that there are innate capacities, dispositions, which unfold as soon as experience gives occasion for it, and which underlie all theoretical and practical activity. As an example of the obscure elements we easily overlook, he points to what may be called sensations of life, the as a rule faint sensations that correspond to the organic functions, and that have an influence on the state we are in at every moment. Our outer senses are indeed more distinct; but since the qualities of sense are utterly different from the motions that correspond to them within matter, Leibniz assumes that our sense-sensations are only apparently simple, but are in reality just as compound as the corresponding motions. What comes forth clearly for consciousness is for him always something compound that presupposes more obscure elements, just as, when we hear the roar of the sea, we really gather the impressions from thousands of waves into one sensation. If we pay attention to it, we shall find that we are never without sensation. Our soul is constantly active. Constantly we receive small sensations, and constantly they release our activity. We are never altogether indifferent; the assumption of indifference is only an expression of our inattentiveness to the small differences. Involuntarily we turn to the right or to the left, even where there are no outer grounds for the choice. Our nature is always working in us to produce greater satisfaction (à se mettre mieux à son aise). A certain *[362]*

unrest always makes itself felt, a lack of full satisfaction, which drives us forward, a spur which as a rule we notice only when from unrest it turns into pain. There are always small hindrances to overcome, and demands are always made for exertion of strength. No feeling of pleasure is possible without such unrest, without minimal hindrances; indeed, the hidden spur maintains itself in joy itself and heightens it, so that a constant progress becomes possible. Only through these small sensations (petites perceptions) does the connection between the different periods of life of one and the same individual become clear; for the clearly conscious states often stand as contradictory. And as they at once explain the difference and the connection between the states of one and the same individual, so they also explain both the difference and the connection between different individuals. Even if one will call the soul an unwritten tablet from the outset, one must nevertheless grant that no two tablets are alike in constitution. If one looks at the small nuances and differences, one will find in the whole of nature no two altogether alike beings. There are not two leaves that are alike, let alone two human beings. Here too the assumption of indistinguishability is due only to our ignorance. But in the obscure sensations of which the individual does not become conscious, the rest of existence makes itself known in him and determines him, without his knowing it. Through them he is connected with the rest of existence. Even in sleep this connection is not suspended. There is only a difference of degree between sleep and waking: the turning away of attention is a partial sleep.

It is a great merit that Leibniz has earned in psychology by drawing attention to the significance of the vanishingly small elements. He has opened a new world for self-observation, admittedly a world where it feels the want of an instrument that could correspond to the microscope in the domain of outer observation. And he has indicated points of view by which it becomes possible to maintain the continuity of the mental side of existence to a far higher degree than the usual conception assumes, and indeed at bottom also than Leibniz himself assumed when he let the monads be without windows
[363] and thereby shut them off from unconscious influences from the rest of existence. The elements of existence (monads) differ from one another only by the degrees of clearness or obscurity of their inner states, and likewise the states of the same individual differ only by such differences of degree. It is a herald of the fact that we are entering the century of the Enlightenment. All the conditions are assumed to be present — only the light must be brought: on this optimistic faith Leibniz and his posterity built. Yet it must be noted in a purely psychological respect that Leibniz points toward a deeper view of mental life than the one rationalism and the age of the Enlightenment took as

their foundation. In the first place it is for Leibniz an *infinite* content that is given in obscure form, and full clarification of it is thus impossible for a finite being. Secondly, he assumes, as an independent element beside the sensations or representations, the need and the tendency (appétit, tendance) to pass over to new sensations. Striving or willing is to that extent for him what is innermost in man, and it is a striving that cannot cease. Toward the passions (passions) he takes a somewhat wavering stand. At times he regards them as obscure thoughts (pensées confuses); and on this view they must then fall away with sufficient clarification; — at times he declares that they are different both from representations and from feelings of pleasure and pain, but are tendencies that are aroused by representations and accompanied by feelings of pain; on this view they are thus closely connected with the never resting striving or willing[73]). By carrying the latter point of view through, Leibniz would have arrived at a more realistic conception of mental development than the one that sees it exhausted by the concept of enlightenment.
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Self-observation gives us, as we saw, the first facts from which we proceed in our thinking, *the first truths of fact. The first necessary or a priori truths* we have in identical judgments, that is to say, those in which subject and predicate can be shown to be one and the same concept. Common to the first truths of fact and to the first a priori truths is the immediate relation: in the one case between the understanding and its object, in the other between the subject and the predicate of the judgment. This immediate relation is *[364]* in both places proof that nothing more certain can be reached. Leibniz came early to this distinction between a priori truths and truths of fact. As early as 1678 (in a letter to Conring of 3 Jan.) he makes this distinction and demands that all a priori fundamental propositions be traced back to identical judgments. He therefore divides the axioms into identical and a priori. The significance of identical judgments rests on this, that when other judgments can be traced back to such, their validity is thereby given. Leibniz thus demands that proof be given for all fundamental propositions one lays down, in so far as they are not identical. He sympathizes with Locke's criticism of "the innate ideas," in so far as Locke thereby wished to combat indolence and prejudice. But Locke ought to have distinguished between necessary truths and truths of fact. Even if a truth is innate in the sense in which Leibniz uses this word, a proof of it must nevertheless, on his view, be given; "innate" truths must be *learned* like other truths. The clarification may be a hard labor, and full clearness is found only in identical judgments. Leibniz has great merit in logic for having laid down the principle of identity, whereas the Aristotelian and scholastic logic kept to the principle of contradiction. He sketched a logic in which

every judgment is formulated as a relation of identity, and has thereby anticipated a method that English logicians (Boole and Jevons) have taken up in our day. But these sketches were not published before 1840, so they have not exercised any influence. — Just as the principle of identity is our criterion of truth for pure thinking, so the lawful connection that the principle of sufficient reason demands is the criterion of truth in the domain of experience. And just as Leibniz is the first to lay down the principle of identity as the fundamental logical principle, so he is also the first to lay down the principle of causality as a special principle for actually given relations. He thereby brings forward the principle that bears all real knowledge, and since he at the same time distinguishes it sharply from the purely logical principles, he makes it natural to discuss its nature more
[365] closely. It is not to be a formal logical principle; nor is it to be drawn from experience; but it is nevertheless to be a principle of reason and to hold for all actual relations (and that both in the supernatural and in the natural world!): how is this possible? That was the problem Leibniz left to his successors. He himself was content to analyze it out of the confusion, partly with general logical, partly with special physical principles, in which it had occurred among his predecessors. Indeed, he does not even distinguish clearly between ground and cause, which is connected with the fact that the principle is at once to be a principle of reason and to hold for actual relations, — and in the last instance with the fact that the laws of facts are seen as testimony to the divine reason that is the origin of the world[74]).

His whole life long Leibniz occupied himself with plans for a kind of alphabet of thought (Alphabetum cogitationum humanarum), which was to be formed by a thoroughgoing analysis of our knowledge, so that one arrived at the fundamental concepts that condition it. By combination of these fundamental concepts he then thought one would be able to find new truths. And by a suitable system of signs one would at the same time be able to obtain a universal language, something corresponding, for all concepts, to what the mathematical language of signs is for magnitudes. It was thus to be at once a logic, an encyclopedia and a grammar, at once a means of gathering what had been learned and of making new discoveries. As the sharp distinction between a priori truths and truths of fact dawns on him, he nevertheless restricts his "Ars combinatoria" or "Characteristica universalis" to the a priori domain. The plan was, despite many attempts, never carried out, essentially, no doubt, because he gradually saw that such a general language of signs really presupposed that our knowledge was already complete. But the plan is characteristic of dogmatic philosophy.

d. Theodicy

It was Leibniz's task as a thinker to lead beyond the mechanical view of nature without breaking it. The way he took was to try to show that the forces at work in the mechanism *[366]* are determined by purpose, so that the whole graded series of mechanical causes and effects, seen from within, is a graded series of means and ends. What is innermost in the world is at every point striving, development, progress. In this thought, which besides bringing harmony between mechanism and teleology also brought harmony between the states of the different monads and between soul and body, he thought he also had a means of reconciling religion (both the positive and the "natural") with reason[4]. It was his conviction that his philosophy satisfied the demands the strictest orthodoxy could make. But the religious problem was just then sharpened by a contemporary writer for whom Leibniz cherished the greatest esteem. Pierre Bayle maintained, as we have mentioned in its place, a disharmony between religion and reason and found especially the doctrines of positive religion contrary to reason. He relied here especially on the problem of evil and found the Manichaean doctrine of two principles in the world, one evil and one good, more in agreement with experience than the orthodox doctrine, to which he nevertheless bowed in the obedience of faith. Leibniz's gifted disciple, the Queen of Prussia, was strongly influenced by Bayle's misgivings and asked Leibniz to refute them. Leibniz thought himself well prepared for it. From his first sketches on he had had an eye for the problem of a theodicy, i.e. the problem of how the assumption of the world's origin in an almighty, all-good and all-wise being can be reconciled with experience of the physical and the moral evil in the world. Early on he had planned *[367]* a work on this; Bayle's "Reponse à un Provincial" and the Queen's request led him to work out and publish his *Essais de Théodicée sur la bonté de Dieu, la liberté de l'homme et l'origine du mal,* which appeared in Amsterdam in 1710.

Against Bayle's assertion that the doctrines of religion are contrary to reason and yet must be believed, Leibniz sets the distinction between something's *being above* reason and its *being contrary to* reason. This distinction is connected with his division of truths into eternal truths (which are grounded in the principle of identity) and truths of fact (which belong under the principle of sufficient reason). Against the first kind of truths

[4] In a recently published statement by Leibniz he places his conception of the relation between *God and the world* in a characteristic way together with his conception of the relation between *soul and body* and between *teleology and mechanism:* "The general mechanical laws are expressions of the will of the Deity (decreta voluntatis divinæ), and the special mechanical laws in each body are expressions of will or forms of the soul in its striving after perfection . . . Everything in nature can be proved by final causes (causæ finales) just as well as by efficient causes (causæ efficientes)." (Bodemann: Die Leibniz-Handschriften der kgl. öffentl. Bibliothek zu Hannover. 1895. p. 89).

of reason nothing that is to be believed can be contrary. The existence of God, e.g., is possible only if the content of the concept of God, the properties ascribed to God, are not self-contradictory. But an assumption can be raised above what experience teaches, since the rational necessity we find, with the help of the principle of sufficient reason, in the connection of phenomena is always conditioned: the very fundamental propositions of natural science, according to Leibniz, find their explanation only in a thought of providence, a teleological principle. It is the very principle of the truths of fact, the principle of sufficient reason, that leads us beyond experience, since the harmony between the monads and the lawful connection in all that happens finds its explanation only in an absolute being that has produced the world by a rational choice. Each single thing, each single event by itself is contingent: we then obtain no completion of the series of grounds or causes, no full satisfaction of the principle of sufficient reason, unless we go outside the series (extra seriem, see Opera ed. Erdm. p. 83) to a cause that is the cause of itself. That the divine choice by which the given world came into being was a rational choice is inferred from the fact that several worlds were possible besides this actual world: this one can then only have come into being because it was the best. But the best world does not mean a world that has no dark sides and defects; it means merely
[368] a world in which the defects are vanishing in relation to the perfections. Every possible world that was to consist of finite beings would have to present imperfections, since finite nature brings limitation with it and therefore cannot fully take the divine nature into itself. From limitation (metaphysical evil) spring suffering (physical evil) and sin (moral evil). Evil thus has in a certain way its origin in God: but not in God's will, which always wills the good; rather in God's intellect, in which the possible pictures of the world dwell from eternity. God has created actuality, not the possibilities. The source of evil is the imperfection that clings to every possible world of limited beings; but by the divine choice of creation this imperfection is reduced to the least possible and made to serve as means and as foil for so much the greater perfection. — In a somewhat more mythological form and carried out in livelier colors, Leibniz's conception becomes really the same as Jakob Böhme's. And both approach Bayle, since they (as Leibniz himself remarks), like him, assume two principles, only that they think they can bring them to agree within one and the same nature of God, whereas Bayle found it necessary to assume two gods. Leibniz and Böhme let the possibilities press forward and struggle within the divine nature, a prelude to the struggle in actual existence.

Bayle had maintained that the Manichaean doctrine had its strength in the testimony of experience to misfortune and sin in the world, while an a priori consideration would

rather maintain the unity of the ground of the world. Leibniz must then follow him over into the domain of experience. Frequently, to be sure, he confines himself to saying: this world must be the best possible; otherwise God would not have chosen it! He overlooks the objection that could be made (and that Schopenhauer, Leibniz's antipode, put forward much later), that even if the chosen world was the best possible, it was not thereby said that it was good enough to be actualized! The need to exist, the craving for actualization, which Leibniz ascribes to the eternal possibilities (he can, after all, everywhere think of possibilities only as minimal actualities), might well be of evil! *[369]* Leibniz must then try to show that the actual world deserves to exist, since for a thorough and comprehensive consideration there remains a surplus of light and perfection. — He demands that one must not look merely at a single part of the world, but think of the whole. A sound that by itself would sound ill can make an excellent effect in a musical ensemble. Cover a painting except for a small spot: it will then seem a meaningless and unclear juxtaposition of colors; but in the whole painting it helps to determine the impression. A bitter ingredient in a dish serves to heighten the flavor, while by itself alone it would be unpalatable. — But how far, then, is the consideration to be extended? Existence, Leibniz answers, is infinite. The thought of providence, the purpose that underlies existence and conditions its stepping forth into actuality, we therefore cannot reach. Only a small, vanishing part of the world do we know — and it is perhaps the one where the most evils are found! Human purposes and ideals must not be made the only thing or what is decisive in the last instance. God pursues purposes that comprise the whole of existence, in which there are surely many other beings than human beings[75]). — And should this great, infinite work then be altered or taken back for the sake of our misfortunes and sins? All the more since everything hangs together in such a way that if, e.g., Tarquinius's deed of rape or Judas's betrayal were taken away, the whole historical development would become a different one. God has placed the single beings in the world; but of their actions they are themselves the cause; the source, not the stream, he has created. And although he knew what would stream from this source, he nevertheless created it, because the world in which it occurred was after all better than any other. Without Judas's betrayal the Savior's atoning death falls away: all the more reason, then, to say with the old church hymn: O felix culpa! — When the evil in the world is emphasized so strongly by many, it is because it draws attention to itself, and because habituation weakens the sense for the good we possess. Hindrances and resistance there must be, so that our activity may constantly be aroused anew and we may not become dulled. Even the blessed and the angels must meet resistance if they are not to become *[370]*

"stupid." Although existence in itself is continuous and harmonious, it is nevertheless purposive that apparent leaps should appear, breaks in the connection, disharmony: it exercises thought and increases the beauty of the whole. — Leibniz even undertakes to justify the dogma of eternal damnation: in existence as a whole he assumes such a fullness of light and perfection that even if the majority of human beings were eternally tormented, their pain would nevertheless be vanishing in relation to that infinite sum of happiness!! — Whether existence regarded as a whole advances in perfection, or whether the perfection it possesses merely varies in form through changing states, is a question Leibniz will not decide. But he is in no doubt that the fundamental mood of one who looks at existence with a widened and understanding gaze will be joy in the beauty and perfection of things and in the constant development, a joy that is widely different from Stoic patience in the face of necessity. —

Leibniz comes, without noticing it, to grant Bayle far more right than he wishes. Apart from the fact that in a certain sense he too assumes two principles, he must in fact concede to Bayle that when the consideration is confined to *experience,* it is very difficult to maintain teleological optimism. Leibniz's appeal to the infinity of existence and the limitation of our knowledge is really an abandonment of the proof: for whence does he know anything of how things stand in other regions of the universe? It is a faith he appeals to; but thereby he declares the question insoluble by way of reason. In a letter contemporary with the Theodicy (to Bourguet, printed in Gerhardt III, p. 550 f), he accordingly also says: "We can survey only a small part of the chain of things, and it is this part that presents the most evils and is therefore well suited to exercise our faith and our love of God." — Nor does the appeal from the single to the whole untie the knot: for precisely in Leibniz's own system, where there is life and soul at every single point in existence, it becomes impossible to ground the sacrifice of the single for the whole. The suffering in the single monad does not fall away because it increases the
[371] beauty of the picture of the world that the monads together form. The single monad will rightly be able to complain of the role assigned to it, and say that it would rather not be than be in hell, even if this hell could not be dispensed with as a kind of bass in the great orchestra of existence. Leibniz's own sense for the infinitely small, for individual differences, ought consistently here to have led him away from teleological optimism.

e. Philosophy of Law

Leibniz's ethical ideas and his ideas in the philosophy of law are closely connected with his general philosophy. His doctrine of monads culminates in the idea of existence as a

"City of God," ordered according to the fundamental law of justice, according to which the forces that underlie all mechanism act. The teleological point of view is common to his conception of the world and his ethics.

The foundation of all ethics and philosophy of law (both are comprised for him in the concept of natural law) he finds in the immediate, instinctive need of happiness. Each single feeling of pleasure corresponds to a progress, a "perfection," which either consists in the force being increased, or in a greater inner harmony being attained in the content the force comprises. "Perfection," he says (in a little treatise *Von der Glückseligkeit,* one of the few German works we have from his hand), "shows itself in the force to act, as indeed every being consists in a certain force, and the greater the force, the higher and freer is the being. Further: in all force there shows itself — and the more so, the greater it is — a multiplicity springing from a unity and contained in a unity (Viel aus Einem und in Einem), since there is one thing that determines a multiplicity outside itself and forms it beforehand in itself. Now unity in multiplicity is nothing other than agreement." Leibniz means by this to express that the feeling of pleasure is conditioned by the fullness of and the harmony between the forces that sustain life, whether one is conscious of these conditions or not. Happiness consists in a constant feeling of pleasure. The condition for the production of happiness is wisdom: the clarification of the understanding and *[372]* the exercise of the will. But involuntary striving already leads in this direction. Leibniz assumes, like Grotius, the "incomparable Grotius," as he calls him, an involuntary need to act not merely for one's own but also for the happiness of others. Love is joy in the happiness of others, in that one has made it one's own; even disinterested love is possible only through the happiness of others as it were shining back upon ourselves. As the chief ethical virtue Leibniz, following Aristotle, lays down justice, which he defines as the love of the wise man (caritas sapientis): love sets the end, and wisdom knows how to find the right means and the right distribution. Justice consists in giving each his own, in distributing goods in the right way, both in the ordering of public relations and in the imparting of what the individual has at his disposal. Justice in this sense (as legislative and distributive justice) aims at benefiting all and is different from strict right (jus strictum), whose purpose is the preservation of peace in society. (See the treatise *De rationibus juris et justitiæ.* 1693).

There is something prophetic in Leibniz's philosophy of law in spite of his partial use of ancient and scholastic elements. By setting universal utility or happiness as the end of law and morality he is a forerunner of so-called utilitarianism. His ideas recall Cumberland's, and he was himself gladly surprised to find a kindred line of thought in

his younger contemporary Shaftesbury.

Although ethics and the philosophy of law are built in Leibniz on a purely psychological foundation, they nevertheless cannot, in his view, do without a theological sanction. He agrees with Grotius that morality and law are not to be grounded on the arbitrary commands of the Deity. But like all the highest principles and laws, the laws of morality and of law too spring in the end from the divine thought and are upheld by the divine will. In disinterested love, to be sure, the striving after one's own happiness becomes one with the striving after the happiness of others. But it is found only in chosen natures (les âmes bien nées). One cannot, without presupposing divine reward and punishment in an
[373] other life, prove that the moral good *and* what is most useful to the individual himself *always* coincide. Therefore natural religion is necessary as a guarantee of morality. And it gives at the same time the highest point of view for human morality and doctrine of right, since the consideration is widened to the divine law that reigns in the whole of existence.

In Leibniz's philosophy of law, as in his whole system, the fundamental thought of harmony between individual beings comes forth. It was this thought that bore his inquiry in all domains, and that he bequeathed to the new century, for which his system is in so many respects typical.

7. Christian Wolff

What Leibniz had expressed in letters and treatises that were accessible and intelligible only to a few, Christian Wolff developed broadly, flatly and systematically in voluminous works. He has an extraordinarily great significance for having introduced rational thinking into wider circles and made "the principle of sufficient reason" valid in all domains. Whereas Leibniz wrote predominantly in French or Latin (although for a time he thought of writing his "Theodicy" in German, precisely on account of that language's philosophical "virginity"), Wolff used the German language and in part created the German philosophical terminology. A popular Wolffianism became an essential component of the German Enlightenment in the eighteenth century. Through Wolff a whole series of thoughts from the great systems of the seventeenth century found their way out into the educated public. Another circle of thoughts came from England through the movement awakened by John Locke, which the following main division will set out.

Christian Wolff, who was born in Breslau in 1679, studied mathematics and phi-

losophy besides theology and became professor of mathematics at Halle in 1706, but lectured chiefly on philosophy. He was strongly influenced by the writings of Descartes and Leibniz, but through Tschirnhausen also by Spinoza, whose theory of knowledge Tschirnhausen (without naming Spinoza's name) had carried out in broader form in his *[374]* *Medicina mentis.* Although Wolff himself did not like to be regarded as Leibniz's disciple, and although he modified Leibniz's thoughts in various ways, it was nevertheless with full right that the philosophy he introduced into the German universities, and which reigned there right up to Kant's time, was called the Leibniz-Wolffian philosophy. To be sure, the innermost kernel of Leibniz's line of thought, his metaphysical idealism and the doctrine of monads, did not lend itself to such a popular systematization as Wolff gave. But the tendency to unite natural science and theology by conceiving the world as a great mechanism that is designed to serve divine purposes and brought by "pre-established harmony" into agreement with mental life is pervasive in Wolff[76]) as in Leibniz. The world is conceived as a law-determined whole of individual beings with God as the ultimate ground. And the logical bearer of the whole system is the principle of sufficient reason.

It is interesting to see how Wolff is not merely the popularizer of dogmatic philosophy but also becomes its completer. It was a significant step in Leibniz that the principle of identity and the principle of sufficient reason were laid down as two different principles, the one holding for all truths of reason, the other for all truths of fact. Wolff now tries to derive the latter principle from the former. He wants, then, to make it self-evident that there is a ground or a cause for everything that happens, and he found it a defect in Leibniz that he had given no proof of "the principle of sufficient reason," which nevertheless plays so great a role in his as well as in the other great systems. Wolff's proof (which is found in *Vernünftige Gedanken von Gott, der Welt und der Seele des Menschen, auch allen Dingen überhaupt,* den Liebhabern der Wahrheit mitgetheilet von Christian Wolffen. 1719. § 30) is as follows: "Where there is no ground present, there is also nothing from which one can comprehend why something exists, and so it must arise from nothing. Now since it is impossible that something can arise from nothing, everything that exists must have its sufficient ground why it exists." — It is easy to see that this proof moves in a circle; for that something cannot arise from nothing means, after all, the same as that *[375]* everything has its ground or its cause, and that was precisely what was to be proved. But by this line of thought Wolff got dogmatism set completely into a system. All certainty was traced back to the principle of contradiction, and philosophy became a system of propositions that could all be grounded purely logically. The principle of contradiction

contains, as must indeed be noted so as not to do Wolff an injustice, the guarantee for all empirical assumptions as well as for all inferences. When I observe something, I cannot at the same moment not observe it; this demand every observation must satisfy. Wolff by no means wants to make all science formal logic. But he wants to show that rational principles reign everywhere. It was rationalism that made its entry.

Wolff applies it not merely to the conception of the world but also to theology. Just as from the principle of sufficient reason he derives propositions of natural science such as the principle of inertia and the law of continuity (nature makes no leap, because the transition from one state to another would then be incomprehensible), so he also wants to derive from this principle not merely the existence of God (the world must have its ground, which has its ground in itself), but also the conditions to which every positive revelation must be subject. There must — Wolff thinks — be definite marks by which revelation can be distinguished from empty imagination and false pretense. Revelation cannot be contrary to God's perfection and cannot contain contradictions. Although it is a miracle and to that extent is contrary to "contingent" (factual) truths, it must not be contrary to the necessary truths: God can make the sun stand still, but cannot change the relation between the diameter and the circumference of the circle. And Wolff maintains that a world in which miracles are rare is more perfect than a world in which they are frequent; miracles require only might, but the natural order requires in addition a wisdom that thinks of the whole, not merely of the single.

Wolff himself became a victim of "the principle of sufficient reason." The pietist
[376] theologians in Halle saw a danger in the rationalism he proclaimed, and when in a university oration he had praised the Chinese philosopher Confucius for his pure morality and declared it to agree with his own, the storm broke loose. With King Friedrich Wilhelm I people are said to have made use of Wolff's application of the principle of sufficient reason to human actions, representing to the soldier-loving king that when one of his grenadiers deserted, one could not, according to Wolff's doctrine, hold him responsible. According to what Wolff relates in his autobiography, the king is said to have been shown a passage in the writings of one of his theological opponents where this consequence of his determinism was drawn. There then came (1723) a royal cabinet order according to which Wolff, on account of his ungodly doctrines, was to be dismissed and within 48 hours be out of the King's realms and lands; if not, he would be hanged. Wolff went to Marburg, where he continued his activity until the new King of Prussia, Friedrich II, who was himself among his admirers, called him back to Halle, where he worked until his death (1754).

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Notes to the First Volume (42–76)

[42]) P. 205. *Discours de la méthode* (1637), 2. partie. — *La vie de Monsieur Descartes.* [Par A. Baillet]. Paris 1691, I, p. 51—71. — *Millet*: *Histoire de Descartes avant* 1637. Paris 1867, p. 100 ff.

[43]) P. 212. In the account given here I follow especially Descartes's earliest sketch of a doctrine of method in *Règles pour la direction de l'esprit,* a work which, when carefully studied, removes many current misunderstandings of Descartes. Cf. besides the third Meditation and the Reply to the second set of objections, in which Descartes, at the request of others, attempts a systematic exposition of his ideas. — To the question of how general propositions are formed from the single intuitions, Descartes really gives only this answer: "It is precisely the nature of our mind to form general propositions from the knowledge of the single." *(Resp. ed. sec. obj.).*

[43 b]) P. 213. Whereas Descartes in his doctrine of method *(Règles* etc.) begins by speaking of the clear and distinct intuition of simple relations (intuition) and names the consciousness of ourselves as thinking among the examples of such intuition, he later begins (especially at the beginning of the 3rd Meditation) with the consciousness of ourselves as thinking beings and derives from the simplicity and immediate clearness of this thought the significance of intuition. — The line of thought that underlies cogito ergo sum occurs already in *Augustinus (De vera religione* cap. 49), indeed already in ancient Indian philosophy.

[44]) P. 214. At the end of the 3rd Meditation it is said: "The possibility then alone remains *[516]* that the representation of God is innate in me in the same way as the representation of myself is innate in me." This is for Descartes more than a comparison. For I cannot — he thinks — become conscious of myself as a limited and imperfect being without an unconscious use of the representation of a being raised above all limitation (3rd Medit.). — To Hobbes's objection: "There is no innate representation; for what is innate must always be present," — Descartes answers *(Resp. tertiæ* X): "When we say that a representation is innate, we do not mean that it always appears before us; for in that sense there would be no innate representation at all; but only that we have in ourselves the capacity to develop it." — In *Notæ in programma quoddam* Descartes says: "Since I noticed that there are thoughts in me which do not proceed from external objects or from the determination of my will, but merely from my faculty of thinking, I called such *representations*, which are the *forms* of those thoughts, innate, in order to distinguish them from others, which come from outside or are produced by myself. In the same sense we say that magnanimity, or that certain diseases such as gout or stone, are innate in certain families, by which we do not

mean that the children suffer from these diseases in their mother's womb, but that they are born with a disposition or capacity to contract these diseases." — The part of Descartes's theological line of thought that infers from the representation of God as the representation of an infinite being to an infinite being as the cause of this representation is due perhaps to an influence of *Vives* (De *anima.* 1538 p. 127 f.), by whom Descartes was also influenced on several points in his psychology.

[45]) P. 215. Cf. *Natorp's* excellent account *(Descartes' Erkenntnisstheorie,* p. 55. ff., 76 f.). Natorp has shown that the use of the concept of God in Descartes's epistemology is not of so external a kind as it appears from the usual accounts. Yet Natorp does not sufficiently emphasize what I believe one must lay the greatest weight on: namely, that this whole epistemological use of the concept of God was altogether unnecessary, since Descartes after all uses the causal principle to establish the reality of the concept of God; when the causal principle is presupposed valid, no theological guarantee is needed for the validity of knowledge.

[46]) P. 218. The strict concept of substance is found in Descartes *Medit. III, Princ. phil.* I § 51. The lax concept of substance is found: *Resp. sec.* and *Princ. phil.* I § 52. — Descartes himself *(Princ. phil.* I, 51) draws attention to the double meaning of the word substance.

[47]) P. 227. On Harvey's discovery and its prehistory see P. Hedenius *Om upptäcktan af blodomloppet.* (Upsala Universitets Årsskrift 1892). See especially on the relation between Descartes and Harvey p. 216 f., 277. Descartes assumed that the heating of the blood in the heart caused it to be pressed out into the body; he had thus not appropriated Harvey's thought of the heart as a contracting muscle. — Harvey too, to be sure, held fast to the old doctrine of the
[517] heating of the blood in the heart, but did not apply it in this way. It was only *Niels Steensen* who definitely showed that the heart is a muscle, and that there was no foundation for the significance as the very source of life and the seat of the soul that had been ascribed to it from ancient times. (Cf. *Jul. Petersen*: *Bartholinerne.* Copenhagen 1898. p. 84 f.).

[48]) P. 229. Descartes's localization of the soul in the pineal gland was altogether arbitrary and was criticized by the anatomists of the time. *Thomas* Bartholin *(Anatomia.* Lugd. Batav. 1651, p. 356 f.) objects partly that the nerves do not end in the pineal gland, partly that it is so small that the representations would be mixed together [!] in it, partly, finally, that it is placed where the "excrements of the brain" accumulate! In the brilliant lecture that *Niels* Steensen gave in Paris, and which is printed in *Winslöv's Exposition anatomique* (4th part), he remarks that since one does not know what fills the cavities of the brain, the doctrine of the animal spirits may be just as good as that of the excrements of the brain. Against Descartes, whose general method he credits with great merit, he objects that the pineal gland does not stand in connection with any canal, and that it does not float freely, but hangs together with the rest of the brain. — What significance the pineal gland has is not yet known. (*Edinger*: *Vorlesungen über den Bau der nervösen Centralorgane.* 6. Aufl. p. 138). On Descartes's significance for the physiology of the nerves cf. Huxley: *Om den Hypothese, at Dyrene ere Automater.* (Tidsskrift for populære

Fremstillinger af Naturvidenskaben 1874). — A remarkable utterance of Descartes's is found in a letter to Mersenne (February 1649), namely that when he denies the animal soul, he denies merely *thinking* (cogitatio), not *life and sense perception*: vitam nulli animali denego, utpote qvam in solo cordis calore consistere statuo; nec denego etiam *sensum qvatenus ab organo corporeo dependet.* This does not agree with other statements, especially not with *Resp. ad sextas obj.* § 9, where 3 degrees of sense perception are distinguished: the first is merely a motion in the brain called forth by the impression (cerebri motus), and this is common to animals and human beings; the second is the immediate sensation of pain, pleasure, thirst, hunger, color, tone, taste, smell, warmth etc., which arises in consciousness (in mente) as the effect of that motion; the third is the judgments that, on the occasion of that motion, we pass on external things. The sense perception he speaks of in the letter to Mersenne must be the first of these degrees; but of it, it is not enough to say that it *depends* on the motions of the brain: it is nothing other than these and cannot be called sensation. What he calls thinking in the letter must correspond to the third degree. There is thus an unclarity with regard to the second degree; it is precisely here that the transition takes place from motion (quantitative processes) to consciousness (where different qualities of sense come to correspond to the different quantities).

[49]) P. 244. By his denial of necessary connection between phenomena Malebranche is a forerunner of Hume. Perhaps Hume even got motives for his theory of causation from Malebranche, whom he (as can be seen from his *Treatise,* I, 3, 14; 4, 5) studied eagerly. *Mario Novaro* has in *[518]* his work *Die Philosophie des Nicolaus Malebranche* (Berlin 1893, p. 44—50) drawn an instructive parallel between the theories of causation of Malebranche and of Hume. — Geulincx's philosophy too would present parallels to Hume; but Hume hardly knew it.

[50]) P. 256. *Vita Thomæ Hobbes.* [Autore Rd. B. i.e. Richard Blackbourne]. 1681, p. 14 f. Cf. Telesio's doctrine of sense perception (above p 93). — In his *Vita carmine expressa,* p. 119 Hobbes describes how, after long brooding over the nature of things, he found out that there was only one single real thing [motion] in the world, although it was "falsified" (sic) in many ways. — F. Tönnies, whose energetic research in most recent times has thrown so much light on Hobbes and his philosophy (see his *Anmerkungen über die Philosophie des Hobbes.* Vierteljahrsschrift für wissenschaftliche Philosophie. III—V), has found a manuscript that contains a treatise by Hobbes which, by its content, must have been composed between the moment when the significance of the deductive method and then of motion dawned on him and the moment when the subjectivity of the qualities of sense dawned on him. This treatise, which is of interest with regard to the historical development of the principle of the subjectivity of the qualities of sense, Tönnies has printed in his edition of Hobbes's *Elements of Law* (Oxford 1888). — Hobbes maintains that as early as 1630 [thus before he came to know Galilei] he had held the conviction of the subjectivity of the qualities of sense. Cf. *Tönnies's* monograph on Hobbes *(Hobbes' Leben und Lehre.* Stuttgart 1896. p. 15).

[51]) P. 261. Cf. Hobbes's treatise *Of Liberty and Necessity.* 3. ed. London 1685, p. 314, where it

is said that as regards the definitions of spontaneity, deliberation, will, inclination, drive, freedom, "no other proof can be given than every man's own experience when he reflects on himself (by reflection on himself) and remembers what he himself means when he says that an action is spontaneous" etc.

[52]) P. 262. In *De Corpore,* IX, 9 Hobbes derives the proposition that all change (qualitative change too) is motion from the causal principle and the principle of inertia. — His "proof" of the causal principle (Of Lib. and Necess., p. 315) is as follows: "If something had no cause, it could just as well begin to come into being at the one moment as at the other; — there must then be a cause why it comes into being precisely at this moment!" That here precisely what was to be proved is presupposed, Hume has already shown. — An analogy to this is formed by his proof of the principle of inertia (De corpore, VIII, 10): if a body at rest were surrounded by empty space, there would be no reason why it should move more in the one direction than in the other, and it would thus not move at all unless an external cause were added. Here it is obviously presupposed that there are no "inner" causes. — Possibly this proof of the principle of inertia stems from Galilei. See his "Dialoger over Verdenssystemerne." 1st Day. (Strauss's German translation p. 21).

[519] [53]) P. 264. Hobbes therefore really has no reason to distinguish between definition and hypothesis. He says (De Corp., dedic.) that in his logic, his Philosophia prima and his geometrical doctrine of motion he proceeds from definitions, but in his physics from "not unreasonable hypotheses." But those definitions too are hypotheses. — Hobbes requires, for the full understanding of a phenomenon, the demonstration of a continuous progression from cause to effect. This continua progressio (De corpore, IX, 6) recalls Bacon's latens processus continuatus (Nov. org. II, 5—7). I do not consider it improbable that Bacon's doctrine of the continuous process may have come before Hobbes's mind when he had been led by another way to the idea that all change was motion. He made the principle of his deduction what in Bacon was to be the last result of numerous inductions. — Likewise it is not improbable that the name Philosophia prima for the second part of De corpore is due to Bacon, who (De augm. III, 1) designates by this name the science of the general fundamental propositions that hold for all domains.

[54]) P. 284. A. D. *Jørgensen* gives, in my view, too lurid a picture of Hobbes's political theory, especially in his comparison between Hobbes and Bishop Vandal. *(Peter Schumacher Griffenfeldt,* I, København 1893, p. 218 f.). The moral limitation of the prince's physical right is found no less in Hobbes than in the Lutheran theologian, and in Hobbes moreover with a grounding that makes absolute sovereignty a means, not an end. Cf. e.g. *De cive* XIII, 3: "The state is instituted for the sake of the citizens" (Civitas civium causa instituta est). — In the older edition of *Leviathan* (ed. 1651, p. 193) the hope of influencing princes is expressed somewhat more strongly than in the later one (ed. 1670, p. 172).

[55]) P. 286. *Freudenthal* has (Archiv für Gesch. de Philos. VI, p. 191 ff., 380 ff.) brought forward Lord Brooke's work and shown the interest that attaches to it.

[56]) P. 286. On Culverwel see M. M. *Curtis*: *An Outline of Locke's Ethical Philosophy.* Leipzig

1890, p. 9—18.

[57]) P. 293. *Colerus (La vie de B. de Spinoza,* tirée des écrits de ce fameux philosophe et du témoignage de plusieurs personnes dignes de foi, qui l'ont connu particulièrement. A la Haye 1706, p. 14) regards Spinoza merely as Descartes's disciple in philosophy. *Leibniz (Opera philos.* ed. Erdmann, p. 139) says of him that he has only cultivated some seeds that Descartes had sown. This conception was long dominant. *Kuno Fischer* and *Falckenberg* hold fast to it in their works on the history of modern philosophy. — *Joël (Beiträge zur Geschichte der Philosophie.* Breslau 1876), in opposition to this, emphasized the great and lasting effect of Spinoza's studies of Jewish theology and philosophy of religion. — Chr. *Sigwart (Spinozas neuentdeckter Tractat.* Gotha 1868) and *Avenarius (Ueber die beiden ersten Phasen des Spinozischen Pantheismus.* Leipzig 1868) pointed — on the basis of the youthful treatise of Spinoza published by van *Vloten (Ad Benedicti de Spinoza opera quæ supersunt omnia supplementum,* Amstelodami 1862): *Kort Traktat om Gud,* [520] *Mennesket og dets Lykke* — to the probability that Spinoza had been influenced by Giordano Bruno. — Finally *Freudenthal (Spinoza und die Scholastik.* In the Festschrift for Zeller 1887) has shown to how high a degree Spinoza has used scholastic concepts and proofs (especially in *Cogitata metaphysica,* which forms an appendix to Spinoza's exposition of the Cartesian philosophy; but also in the Ethica). And the same scholar has later collected all the older biographies of Spinoza and in general all documents that can throw light on his life and his writings, in his book *Die Lebensgeschichte Spinoza's in Quellenschriften, Urkunden und nichtamtlichen Nachrichten.* Leipzig 1899. — In my work *Spinozas Liv og Lære* (København 1877) I have given a popular account of Spinoza and his philosophy. Later, repeated studies have led me on single points to a different conception than in that work. I take the opportunity to state that the picture used as title vignette in the work mentioned, and which was taken from van Vloten's "Supplementum," has since been shown to represent — not Spinoza, but one of his friends, the philosopher and mathematician Tschirnhausen. A fine and, by all that can be ascertained, genuine portrait of Spinoza is found in van Vloten and Land's new edition of Spinoza's works.

[58]) P. 298. I must also oppose A. D. *Jørgensen's* reflections (*Nic. Steensen,* p. 57) on Spinoza's relation to experience; but on this point the section on Spinoza's theory of knowledge will contain what is needed.

[59]) P. 299. *Studie zur Entwickelungsgeschichte des Spinoza.* (Vierteljahrsschrift für wiss. Philos. VII) p. 161 ff., 334.

[60]) P. 308. The first book of the "Ethics" develops the concept of substance and the concept of God in a series of definitions, axioms and propositions. — On the connection between the concept of substance, the concept of God and the concept of nature cf. Avenarius: "Ueber die beiden ersten Phasen des Spinozischen Pantheismus," where it is aptly shown that all three concepts, carried through consistently, meet in one point in Spinoza. Yet Spinoza hardly kept the three currents of thought so sharply apart from one another as they are kept in Avenarius's account. — On the relation between substance and modes one must note, besides the first book of the "Ethics,"

Ep. 12 (in v. Vloten and Land's edition), where a distinction is drawn between the abstract and the substantial conception, and Ep. 50, where it is declared that every determination is a negation. — The expressions natura naturans and natura naturata occur already in Renaissance philosophers, mystics (Master Echardt) and scholastics. According to *Siebeck* (Archiv für Gesch. d. Philos. III, p. 370 ff.) they are said to have their origin in the Greek *φύσις* (as *φύον* and as *φυόμενον*) and to have arisen through the translation into Latin of Greek philosophical writings, probably in the thirteenth century. — As regards the exposition in the Ethics, the geometrical-deductive form is
[521] — as already remarked — only a shell. In reality it is a series of analyses that Spinoza undertakes, and by God or substance he understands the comprehensive reality to which all phenomena belong, and within which thought can therefore point them out. Cf. on this *Harold Joachim*: *A study of the Ethics of Spinoza.* Oxford 1901. p. 116 f. — Cf. *Descartes's* declaration (in Medit. VI) that by nature in general (natura generaliter spectata) he understands either God himself or the connection of things established by God (coordinatio rerum).

[61]) P. 309. Cf., against this conception of Spinoza's concept of attribute maintained by the elder *Erdmann*: H. *Bröchner*: *Benedict Spinoza.* København 1857, p. 47 ff. — *Mourly Vold (Spinozas Erkendelsesteori.* Kristiania 1888, p. 242) wrongly names Bröchner among those who interpret the attributes "as predicates necessary for our understanding, but accidental for God." Bröchner polemicizes precisely against this conception.

[62]) P. 311. Already in my work "Spinozas Liv og Lære" (1877) p. 100 I have pointed out this confusion in Spinoza. It has also been brought forward by *Windelband*: Geschichte der neueren Philosophie, I (1878), p. 211 and by *Tönnies*: Studie zur Entwickelungsgesch. des Spinoza. (Vierteljahrssch. für wiss. Philos. 1883) p. 176 f.—*Mourly Vold* (Spinozas Erkendelsesteori, p. 202) denies the confusion in Spinoza and regards the doctrine of attributes as springing essentially from the endeavor to ground a "correspondence between idea and object." But when he concedes that the doctrine of attributes in Spinoza also has a psychophysical significance, he must consistently concede a confusion in Spinoza. For epistemology and psychophysics are two different things, which is already seen from the fact that the "objects" are not merely of a physical but also of a psychical kind; the problem of knowledge concerns, after all, both the knowledge of the psychical and of the physical. — The confusion in Spinoza is seen clearly when one compares Eth.II, 7, Schol., where the relation between the concept of the circle and the existing circle is used to illustrate the relation between the attributes, with II, 17, Schol., where a distinction is drawn between the representation of Peter (idea Petri) that corresponds to Peter's own body as its soul (mens Petri), and the representation of Peter (idea Petri) that Paul has. The first idea Petri has a psychophysical, the second an epistemological significance. Spinoza's highly unfortunate usage of designating the soul (mens) as idea corporis has favored the confusion. This usage stems from Aristotle by way of Maimonides (see *Hamelin*: *Sur une des origines du Spinozisme.* L'année philosophique. 1900. p. 24 f.), to whom Spinoza himself expressly refers as regards the epistemological doctrine of identity, see *Eth.* II, 7 Schol. — In German speculative philosophy (in

Schelling and Hegel) the confusion flourishes.

[63]) P. 315. H. Morus: *Demonstrationis duarum propositionum, qvæ præcipuæ apud Spinozium* [sic] *Atheismi sunt columnæ, brevis solidaqve confutatio.* (Opera philosophica. Londini 1679, vol. I, p. 619). — On the different conceptions of Spinoza's concept of God there is an instructive treatise by J. H. Löwe *(Ueber Spinozas Gottesbegriff und dessen Schicksale.* Appendix to Löwe's book: Die *[522]* Philosophie Fichtes. Stuttgart 1862), which, however, does not mention More's characteristic interpretation.

[64]) P. 315. Cf. on this question *Ed. Zeller's* remarks in Vierteljahrsschrift für wissenschaftliche Philosophie, I, p. 285 f.

[65]) P. 320. In Eth. II, 18 only the law now called the law of contiguity is mentioned. But the law of similarity is presupposed in Eth. III, 27 and is named in Tract. theol. polit. cap. 4 as an example of a mental law of nature.

[66]) P. 327. A fuller account of Spinoza's ethics than I can give here is found in my work "Spinozas Liv og Lære," p. 120—146.

[67]) P. 329. When it is said in Eth. V.38, Schol. and 40, Coroll. that it is the *intellectus,* not the *imaginatio,* that is eternal, it seems that only those individuals who rise above the stage of imagination (sensuous representation) are organs of the divine activity of thought, which can no more disappear than motion can. A statement in the "short treatise" (2nd part, preface, no. 15) also points this way, to the effect that the soul can become eternal when through knowledge and love it unites itself with the absolute substance. — The thought we meet here in Spinoza often comes forth in the history of thought. I cannot go into it more closely here, but must confine myself to some literary references. Plato's Republic, VII, p. 534 C (Hejse's trans. of the Republic, 2nd part, p. 188). — *Chrysippos* (Diog. Laërt. VII, 157). — Maimonides (see Joël: Zur Genesis der Lehre des Spinoza, p. 66 ff. and Hamelin in L'année philos. 1900. p. 26 f.). — *Salomon Maimon*: Lebensgeschichte. Von ihm selbst geschrieben. Berlin 1792. II, p. 178 f. — *Goethe*: Gespräche mit Eckermann. 4 Febr. and 1 Sept. 1829. — *Wilhelm* von *Humboldt* (see Haym: W. v. Humboldt, p. 637 f.). — J. G. *Fichte*: Die Thatsachen des Bewusstseins, Stuttgart und Tübingen 1817, p. 197 f. — C. H. *Weisse*: Die philosophische Geheimlehre von der Unsterblichkeit des menschlichen Individuums. Dresden 1834. — *Louis* Lambert (see Ravaisson: La philosophie en France au 19. siècle, p. 223). — *Ernest Renan*: Feuilles détachées. p. 392 f.

[68]) P. 337. See on this Ludwig *Stein*: *Leibniz und Spinoza.* Ein Beitrag zur Entwickelungsgeschichte der Leibnizischen Philosophie. Berlin 1890. (Especially ch. 4).

[69]) P. 341. The treatise of 1680 has no heading. *Erdmann*, who first brought it forward from the library in Hannover, gave it (in his edition of Leibniz's *Opera philosophica,* 1840) the title: De vera methodo philosophiæ et theologiæ and assigned it to 1690—91. That it must be of earlier origin is shown by both internal and external criteria, especially the comparison with "Petit discours de métaphysiqve," which was worked out in 1685 and sent to Arnauld the following year. Cf. on this *Selver*: *Der Entwickelungsgang der Leibniz'schen Monadenlehre bis 1695.* (Philosophische Studien

herausg. von W. Wundt, III), p. 433. *Ludwig Stein: Leibniz und Spinoza,* p. 117 ff. — Stein, in his excellent account of the history of Leibniz's development, lays too great weight on the influence
[523] the study of Plato had on Leibniz. It is admittedly very characteristic that Leibniz, in the years so important for his philosophical maturing, occupies himself so much with Plato and, among other things, quotes the well-known passage in the Phaedo on final causes no less than twelve times. But it is nevertheless symptom rather than cause. The conviction of the legitimacy of final causes stood firm for Leibniz from the outset; it sprang from his religious presuppositions, and all through his youth he had sought means of bringing it into harmony with his scientific theories. Plato could bring him nothing new here. — In particular I find it an unnatural explanation when Stein derives Leibniz's proposition of the identity of substance and force from Plato's doctrine (expressed in a single passage) of the "ideas" as active forces. Leibniz had, after all, been occupied with the concepts of force and striving ever since 1670, and it is from his mathematical-physical studies that he carries these concepts over into his philosophy and uses them to bring the concept of substance to life. — When several writers designate the advance Leibniz made in the year 1680 as a *substantialization of force,* I believe that it would be more correct to reverse the relation and speak of a *conversion of substance into force.* It was in reality a dissolution of Descartes's and Spinoza's concept of substance that was brought about by the treatise of 1680.

[70]) P. 350. Leibniz's "Tout comme ici" has the same source as Holberg's "Ligesaa hos os," namely the comedy Harlequin empereur de la lune (Gherardi's Théatre italien, vol. I). See Nouveaux essais, IV, 16, 12 (Erdmann, p. 391 f.): On diroit quasi qve c'est dans l'Empire de la lune de Harléquin tout comme ici. — Besides this passage one must note with regard to the law of analogy: letter to Arnauld, Nov. 1686 (Gerhardt's edition, II, p. 75 f.). De ipsa natura (Erdmann, p. 157). Nouv. Ess. I, 1 (Erdmann p. 205) and III, 6, 14 (Erdmann, p. 312). Statements in letters (Gerhardt, II, p. 270; IV, p. 343).

[71]) P. 352. *Eduard Dillmann (Neue Darstellung der Leibnizschen Monadenlehre.* Leipzig 1891) seems to me, in his account of the doctrine of monads, which is in many respects instructive and interesting, altogether to overlook the great significance that the law of analogy has for the subjective completion of the doctrine of monads. Dillmann regards the faculty of representation as given from the outset in the concept of the monad and takes the concept of phenomenon in Leibniz merely in a subjective sense (phenomenon for). This conception is already refuted by the fact, conceded by Dillmann himself, that Leibniz without further grounding assumes that the monad represents the universe from the standpoint of its body, and that he supports the assumption that the monad represents the whole universe by the fact that every body, by virtue of the law of continuity, is affected by every change in any part whatever of the world. (Cf. Dillmann, p. 301 ff., 342 ff.). — On Dillmann's hypothesis it would be impossible to understand why Leibniz takes so dismissive a stand toward Berkeley's ideas as soon as he hears of them.

[524] [72]) P. 356. On the different ways in which the relation between Spinoza and Leibniz has been conceived in the course of time, see the first chapter of L. *Stein's* "Leibniz und Spinoza." — E.

Dillmann finds a fundamental agreement between the two systems, but he thinks at the same time that Leibniz's doctrine of the relation between God and the world is unassailable, *and* he lays no weight on the passages where Leibniz distances himself from Spinoza. (Neue Darstellung, p. 472 ff.). Dillmann lets himself be led by his great and justified admiration for Leibniz into playing the apologist for obvious self-contradictions and evident accommodations. — An instructive investigation of the relation between the two systems has been undertaken by *Robert Latta* in his treatise *On the relation between the philosophy of Spinoza and that of Leibniz* (Mind. 1899).

[73]) P. 363. On the difference between perception and appétit see Monadologie § 15. — Passions as pensées confuses: *Réplique aux reflexions de Bayle* (Erdmann p. 188 a), as tendances ou des modifications de tendances: *Nouv. Ess.* II, 20, 9 (Erdmann, p. 249 a).

[74]) P. 365. Cf. a treatise (without heading) which is printed in Gerhardt's edition, VII, p. 300 ff., where it is first said that all factual data must have their ground, since the creative will that manifests itself in them is not arbitrary, but acts according to grounds (ratio), and then: "It is thus established that all truths, even the most contingent, have an a priori justification (probatio) or a ground why they are rather than are not. And this is precisely what the common saying maintains, that nothing happens without a cause (causa), or (seu) that nothing exists without a ground (ratio)."

[75]) P. 369. The law of the effect of contrast Leibniz already uses in a letter from the beginning of the seventies (Gerhardt's edition, I, p. 61) to defend optimism. Later he develops it especially (as *lex lætitiæ)* in *De rerum originatione* (Erdmann, p. 149 f.). This aesthetic-psychological argument is of ancient origin; it is found already in *Plotinos*, and is repeated (remarkably enough) by *Augustinus*. Modern, on the other hand, is the emphasis on the infinity of existence. Leibniz remarks that Augustine does not yet know la grandeur de la cité de Dieu *(Théodicée* § 19). Cf. *Causa dei asserta §* 57—58.

[76]) P. 374. In the preface to the first edition of *Vernünftige Gedanken etc.* Wolff says: "I had indeed at the outset resolved to leave the question of the connection between soul and body wholly undecided; but since, by the reasons adduced in chapter 2, contrary to my expectation, I was quite naturally led to Herr v. Leibniz's pre-established harmony, I have retained it and placed this profound discovery in such a light as it has not yet received." — In the preface to the 2nd edition he expresses himself more precisely on the relation of his doctrine to Leibniz's. Leibniz, he says, combined idealism [the unbroken connection of the mental phenomena] with materialism [the unbroken connection of the material phenomena], without examining each of them more closely by itself. That Wolff now thinks he has done, and his result is that "the dualists can unite the theory of the idealists and of the materialists." — This is in reality a correct *[525]* determination of the difference between Leibniz and Wolff on this point: Leibniz embraces the identity hypothesis (though with a spiritualist modification), and Wolff embraces dualism (though with the assumption, taken from the identity hypothesis, of the unbroken connection of the

mental and of the material phenomena each by itself). — Very many of the objections that are usually advanced against the identity hypothesis hold for the Wolffian theory, to which the name "duplicism," which does not fit the real identity hypothesis at all, fits very well.

Ninth Book

[291]

Positivism.

For the spiritual life of the nineteenth century two intellectual tendencies are characteristic: Romanticism and positivism. The first proceeds from the needs of the heart and from the ideals of thought, the second seeks its support in what is factually given; for by the word *positive* is meant first and foremost the same as *real.* The two tendencies thus seem to be as opposed as can well be. They proceed from opposite starting points. And yet there is an inner kinship between them. They are borne by the same interest and rest on the same presupposition. Both are animated by a need of full spiritual content and of absorption in great realities. Romanticism wishes to grasp reality just as much as positivism does; only the former believes it can grasp it by the subjective way, while the latter will build on objective facts. The common presupposition is that an ideal which stands wholly outside reality is a false one. Therefore men turn away from the eighteenth century's criticism by the understanding and immerse themselves with enthusiasm in the great course of development of nature and of history. The concept of development reigns both in Romanticism and in positivism. Men seek especially to find the continuous connection within history which the age of Enlightenment and Revolution had broken apart. A more thorough understanding of the past and of the conditions for the development of the spiritual life forms an essential part of the significance of both tendencies in the history of thought. Even though it must be said that the spiritual oppositions — the oppositions in the domain of faith and of views of life — have steadily grown sharper in *[292]* the course of the nineteenth century, it is nevertheless the great debt we owe to these two intellectual tendencies that they have taught us to enter into standpoints which are essentially different from our own. By means of the historical sense, which is a form of human sympathy in general, a bridge has been thrown over oppositions which could not be overcome by logical discussion. This marks a turning point in the history of the human life of the spirit. A new organ has been acquired. What Immanuel Kant set up as

the program of the human sciences: to find the moving forces which have produced the works of the past, to investigate their laws and their range, and thereby to set them free to do the work of the future, — this Romanticism and positivism have each in its own way sought to fulfill.

It would therefore be wrong to regard positivism merely as a reaction against the philosophy of Romanticism. It would be wrong even chronologically; for the origin of positivism lies in the time before the full development of Romanticism. It was not surfeit with the one tendency that led to the other, even though positivism found more favorable conditions for spreading in wider circles once the dominion of Romanticism had been broken.

Yet there is one very essential point in which the two tendencies from the very outset diverge from each other. In its enthusiasm for the unity of thought Romanticism overlooked the multiplicity of reality, and in its assurance of the truth of the ideal it overlooked the strict mechanical conformity to law to which everything must submit that is to attain subsistence in the world of reality. Since positivism takes its starting point in what is factually given, it opens itself to the differences and oppositions of reality and strives to find the laws according to which phenomena arise and develop in the real world. Its difficulties and problems therefore come to lie at other points than those of Romanticism. Positivism seeks to go up from what is factually given toward the unity of thought and the validity of the ideal, while Romanticism went the opposite way. The great thinkers of positivism assume just as confidently that it must be possible
[293] to climb from below upward as the Romantics assumed that one must be able to climb from above downward. Comparative consideration must decide which way is the better.

While the philosophy of Romanticism had its home in Germany, positivism has its home in France and England. National differences show themselves here in a characteristic manner. The English school, which in Hume had reached a certain conclusion, now revives and appears in new forms, which still preserve the realistic stamp that distinguished English thought right from the Middle Ages onward. Only when the word positivism is taken in a wider sense does it also include the main tendency in the English philosophy of the nineteenth century. Its most important spokesmen have protested against being called positivists — but they were then thinking of the narrower sense of the word, in which it designates Auguste Comte's philosophy. It does not diminish the significance of modern English philosophy that it is here presented as that which has completed and extended the way of looking at things which Auguste Comte — the second great figure of French philosophy beside Descartes — had shaped in its broad

outlines. And all the less so, as Comte himself stood in considerable debt to the older English school.

Both France and England have in our century made valuable contributions to the development of philosophical thought which cannot be brought under positivism, even where this word is taken in the wider sense. But in the general history of philosophy we must lay the chief weight on the most significant tendency, and therefore also let it furnish the title for the Book that treats French and English philosophy in the nineteenth century. *[294]*

A. Comte and French Philosophy

1. Philosophy in France in the First Decades of the Century

a. The Renewal of the Principle of Authority

The French Revolution had wished to throw off the bridge to the past. The age of the Church and of the old faith was to be over. But even before the Revolution's opponents had triumphed outwardly, it became evident that as a spiritual power the Church was not defeated. Eminent writers praised religion for its significance in the service of humanity and for the poetry it shed over life, and amid the mighty upheavals there were those who sought an absolutely fixed point in the general swell of the waves. With the outward shipwreck of the Revolution, moreover, the ideas of the eighteenth century, as the French philosophers and their English predecessors had maintained them, also seemed to have played out their part. They had — so history now seemed to testify — been unable to lead to any fixed social order, while the Church had maintained and renewed itself "in spite of the syllogism, the scaffold and the epigram", to use an utterance of the principle of authority's most important representative, Joseph De Maistre *(Le pape.* 14. éd. p. 477). In defiance of everything the eighteenth century had thought and fought for, a theological school now sought to lead all ordering of human life and all understanding of existence back to supernatural principles. This school has more interest for the history of literature than for that of philosophy, and I therefore confine myself here to some characteristic features from de Maistre's chief works, while, as regards the tendency as a whole, I refer to the account in Georg Brandes: *Reaktionen i Frankrig* (2nd ed. 1892).

When modern philosophy, as the preceding sections of this work have shown, supports itself on natural science, partly as a further application of its results, partly as a discussion of its presuppositions, de Maistre seeks to deprive it of this support. He

[295] disputes that there can be purely physical causal explanations. What is material cannot — he asserts — be a cause. A physical cause is a self-contradiction[71]). All material motion is derived from original impulses, which can only stem from spiritual beings. In the consciousness of the influence of our own will we have a proof that motion begins with an act of willing. Scientific explanations, such as, e.g., the origin of water by the combination of oxygen and hydrogen, the origin of ebb and flood by the influence of sun and moon, the influence of chemical processes on the formation of the strata of the earth, de Maistre therefore declares to be "dogmas" which may rightly be doubted. Yet he is quite willing to let the learned amuse themselves in the domain of natural science, provided only they do not apply consequences from it to social and religious affairs. It is not science but faith that is to govern men. The truths that sustain life God proclaims not through the academies but through the authorities in Church and State. It is prelates, noblemen and high officers of state who are to teach the nations what is true and false in the moral and the spiritual domain. History has now furnished proof of the impotence of human reason to guide men. How few are able to think correctly! And no one is able to think correctly about all things. Then it is best to begin with authority. The ground of all misfortunes is that speech has been set free. It began with the Reformation — one of the greatest crimes men have committed against God! And it was continued in the philosophy of the eighteenth century — one of the most shameful episodes in the history of the human mind! The philosophers of the eighteenth century preached error as religion. They were not in good faith, but had formed a conspiracy (la cabale) against what is holy. Voltaire — this blasphemous buffoon (bouffon sacrilège) — gave himself the appearance of pleading the cause of innocence, although the innocence of Calas is not proven. By setting Locke on a pedestal he has only shown his fanaticism and at the same time his lack of patriotism. — An end to the misfortunes will come only when the absolute infallibility of the Pope is acknowledged. Without it the universal
[296] character of the Church cannot be maintained, social peace cannot be preserved, and the sovereignty of princely power cannot be upheld. It is a mysterious principle that is here appealed to: but everything is mystery in the social and in the physical world. Reason condemns war — and yet war reigns throughout all nature as a mysterious means to the subsistence of life. The executioner is the horror of society — and yet at the same time a force that holds society up: take away this incomprehensible social factor, and chaos will take the place of order! Only tradition and authority, not human reason, can guide us in this world, which everywhere presents terrible mysteries. Nothing therefore is more ridiculous than the assumption that man should gradually have raised himself

from barbarism to science and civilization. —

These ideas, which de Maistre develops especially in his *Soirées de St. Petersbourg*, a series of dialogues which are assigned to 1809 but appeared only after his death, in the year 1821, turn upside down everything that thought had sought to build up ever since the days of the Renaissance. As so often, there is here a kinship between the opposites. Just as unhistorically and uncomprehendingly as the philosophers of the eighteenth century had stood toward the Church and the Middle Ages, just as unhistorically and uncomprehendingly did de Maistre and the men of his tendency stand toward the philosophy of the eighteenth century, — only with the difference that the wicked philosophers now took the place of the wicked priests as the source of all evil. The philosophy of the eighteenth century was regarded as a deliberate conspiracy against authority, as a purely arbitrary process. And this was in good agreement with the fact that authority itself was conceived as what is purely arbitrary, commanding from without or from above. Hereby there arose again a new likeness with the philosophy of the eighteenth century: for when all knowledge is to be built on authority, its self-activity must be subdued, its free unfolding of its inner possibilities denied. De Maistre therefore consistently comes — just as Condillac did — to assume the complete passivity of man. It is then no wonder that the possibility that man should have arrived at civilization by the way of natural development appears to him as utterly absurd. *[297]*

b. The Psychological School

The significance of the authoritarian school lay in its passionate bringing forward of the value of the historical powers, in opposition to the psychology of the Enlightenment and the Revolution, which saw in the heart and understanding of the single individual an independent force. But as the principle of authority was conceived here, it was an external, brutal principle; its inner connection with mental life was not investigated — and could not be investigated from this standpoint, since every independent value that was ascribed to the individual's own inner life would entail a limitation of the absolute authority. Philosophically, therefore, as already remarked, the authoritarian school stood (when certain mystical stirrings, of which it itself becomes afraid, are discounted) on Condillac's standpoint. All the greater is the interest of seeing how, out of Condillac's own school, through the use of psychological observation and reflection, a deeper psychology develops, which places itself in definite opposition both to the French philosophy of the Enlightenment and to the new authoritarian school.

When the Revolution broke out, Condillac had triumphed. His philosophy became the

official one, and his adherents occupied the philosophical section in the National Institute established by the Convention. Before this philosophical academy the physician PIERRE JEAN GEORGES CABANIS in the winter of 1797—98 read a series of memoirs on the relation between soul and body, which were printed in the Academy's transactions and later in expanded form appeared as a book (1802) under the title: *Rapports du physique et du moral de l'homme.* Cabanis speaks of Condillac with great appreciation, but modifies his doctrine on essential points. Condillac lays the weight exclusively on the external senses; according to him man receives from the surrounding world in a passive manner the whole content of his consciousness and all its forms. Cabanis reproaches him for not having taken account of that in consciousness which corresponds to the organism's own inner states. Impressions constantly come to the brain from the various inner organs, and besides, the state of the brain and nerves themselves always plays a part
[298] as well. There is an obscure feeling or sense perception which is immediately bound up with the subsistence of life, independently of external sense impressions, and which has therefore already been present before the individual is at birth brought into relation to the great surrounding world, from whose influences Condillac derives everything in consciousness. Cabanis introduced the feeling of life into modern psychology. With this the passivity toward the surrounding world was already limited; for in the feeling of life the individual has an original fund which gains influence on everything it later takes into itself, and gives it color and stamp. And Cabanis brings instinct into close connection with the feeling of life. In instinctive actions too he finds facts that are incompatible with Condillac's theory. Instinct presupposes a fund of original force, which is released by impressions from the inner functions of life, as is shown especially in the reproductive and parental instincts. Cabanis lays so great weight on the concept of instinct that he even hints at the thought that a universal instinct might be at work in all processes of nature, a thought in which the germ of Schopenhauer's philosophy of nature has rightly been found. It was not, however, Cabanis's intention to put forward a philosophical system. He wishes to proceed in a purely psychological and physiological way and not to answer the ultimate questions. Although in a single place he expresses himself in a strongly materialistic way (in that he lets thought be secreted in the brain, as bile is in the liver), it would nevertheless be wrong to assign his work to the materialistic literature proper. His standpoint is characterized by the following statement (Rapports, XI, 8. éd. Peisse p. 597): "It took a long time before men came to acknowledge only a single force in nature; perhaps an even longer time will be needed to acknowledge rightly that, since we cannot compare this force with anything else, we cannot form any true notion of its

properties." In a later work *(Lettre sur les causes premières)* he gave a fuller development of his thoughts, which departs still further from materialism than the famous chief work had done.

Napoleon did not regard Condillac's school with favorable eyes. It did not, after all, content itself with investigating our sensations and representations; beside its *ideology* (so DESTUTT DE TRACY, an ardent Condillacien, called the doctrine of the origin of *[299]* representations, cf. his work *Eléments d'Idéologie* 1801) it also busied itself with theories of morality and law in the spirit of the eighteenth century. Napoleon derived all the misfortunes of France from the "ideologists", and in any case he would rather think for France, just as he acted for it. He suppressed the Académie des sciences morales et politiques established by the Convention, and writings of a freer philosophical content could no longer appear in France. Destutt de Tracy had to have his commentary on Montesquieu printed anonymously in America in English translation. The ideologists withdrew into narrower circles. In particular there gathered around Cabanis and Destutt de Tracy in Auteuil a band of younger men who were interested in philosophical studies. From this circle came forth France's first psychologist of this century, MAINE DE BIRAN (b. 1766, d. 1824).

Biran held administrative posts under the Revolution, the Empire and the Restoration and was a member of legislative assemblies. It was not, however, the great outward events of the time that filled his mind. He was a good patriot, but lacked the inclination and the capacity for outward action. The great historical drama worked upon him by strengthening the emotions in his own mind, emotions which he early observed both with the interest of the theorist and with an inward longing for peace and harmony within. His constitution must have been of such a nature that the relation between the passive and the active, the involuntary and the voluntary, or, as he himself aptly designates it, between temperament and character, became for him a theoretical and practical problem all his life long. Of this his *Journal intime* bears witness, the most remarkable of the works he left behind (published by E. Naville in his book: *Maine de Biran. Sa vie et ses pensées.* 1857), which contains entries from the beginning, the middle and the end of his life as a thinker. How early and how definitely his peculiar problem presented itself to him can be seen from a statement from the period when he still adhered to Condillac's doctrine (27 May 1794). "What is properly the so-called mental activity? I always feel the state of the soul determined by this or that bodily state *[300]* . . . If I could some day undertake a connected piece of work, I might have a wish to investigate in what degree the soul is active, in what degree it can alter the external

impressions, increase or diminish their strength by the attention it directs upon them, and try how far it is master of this attention . . . It would be highly to be wished that a man practiced in observation would analyze the will, as Condillac has analyzed the understanding." At the same time he complains of the constant shifting and streaming in his inner states (cette révolution perpétuelle, cette roue toujours mobile de l'existence), which hinders fixed moods and gives rise to doubt and unrest even toward what he seeks with honest will to hold fast.

In his first works Biran kept to the doctrine of Condillac and Cabanis. But gradually he laid ever more weight on the activity which in his view comes immediately to our consciousness in the effort of will. "The self makes itself known to itself through the inner sense in the effort or the movement of will, which the soul perceives (aperçoit) within itself as a product of its activity, as an effect that has its cause in its will." *(Rapports du physique et du moral de l'homme,* Oeuvres philos. 1841. IV, p. 75). Here there is for Biran an original fact which puts an end to Condillac's doctrine of passivity. But he does not forget — constantly repeated inner experiences saw to that — the involuntarily arising sensations and moods. He reproached the moralists who write on the problem of happiness for giving themselves up to general reflections and postulates and believing that one can immediately command one's feelings and inclinations. Wholly independently of our conscious willing a multitude of changing phenomena stir within us, which the self finds before it when it becomes conscious of itself, and which must stem from another inner cause than the self. There is also, outside the self or consciousness and independently of the relation to the surrounding world, a series of inner phenomena which self-observation discovers, but which can subsist without it. (Journal intime. 24
[301] Oct. 1814). This theory of sensations that were to subsist outside the self aroused great attention and much opposition in the small philosophical society which Biran gathered about him in Paris, and in whose meetings men such as the physicist Ampère, the historian Guizot and the philosopher Royer Collard took part. Only Ampère joined Biran definitely. They were both agreed in setting the immediate consciousness (apperception immédiate) of the energy of will as the central psychological fact. But outward from it mental life fades off into the unconscious, and when the act of self-consciousness sets in, the self finds a whole series of elements given — in any case it hears their echo. So also on awakening from torpor or on reflection upon habitual states. In a hitherto unprinted *Mémoire sur les perceptions obscures* (of 1807), which has been brought to light by Alexis Bertrand (La psychologie de l'effort et les doctrines contemporaines. Chap. 2), Biran had sought to explain this opposition between the passive and the

active in mental life physiologically by assuming different cooperating nerve centers. He also uses the phenomena of somnambulism to illuminate the duality within us. In his zeal for exact self-observation and for supplementing self-observation by other sources of psychological knowledge Biran is a forerunner of modern psychology. From his psychological standpoint he declares himself definitely against the authoritarian school, which attributed no significance to man's free self-activity. Only through absorption in the nature and laws of mental life can one, Biran maintains, find the point where the origin of the moral and religious relation lies. The men of the authoritarian school, who wholly disregarded the individual nature of the soul and "ex abrupto" set up authority as the foundation of all science, lacked in his eyes love of truth and were led merely by the need to act, to work upon men. And they fell, he continues, into skepticism and materialism, in that they made the soul purely passive and let its whole direction be determined from without, by authority! (Journal intime. 28 July 1823).

In the immediate consciousness of self-activity Biran sees the primitive and fundamental principle of all our knowledge. In it not only does our own self proclaim itself to us; but since self-activity meets resistance, we obtain consciousness of the existence of *[302]* the material world. The self-activity we are aware of is in fact the soul's acting upon the organism, where there is always a certain resistance to overcome. Thus Biran gives a spiritualistic interpretation of the consciousness of our acting and of the checking of our acting that may take place; and thereby there is gained at the same time an explanation of the opposition between our willing proper and what is involuntary in our inner states. Yet he does not adhere to the Cartesian doctrine of the soul as a substance: we know the self only as active force; there could be no immediate consciousness of the substance. — The force or acting which we thus perceive in ourselves serves us as a type to which we refer all external phenomena. We come to know in our self-consciousness the concepts of force, cause, unity and identity, and only because we can draw these concepts from the primitive inner fact, concepts which are very different from concepts of kinds, can we apply them to the phenomena of experience. — Neither external sense perception nor authority, but the immediate consciousness of our self-activity is the foundation of all our knowledge. Concerning the reality of this foundation Biran harbors no doubt; modern psychology will hardly be able to grant, however, that he has overcome Hume's doubt about the possibility of an immediate apprehension of causality[72]). —

Biran's most important writings were published only long after his death. He worked for a time on a larger work which was to comprise all his psychological studies *(Essai sur les fondements de la psychologie);* but political circumstances prevented him from

finishing it for publication, and later he laid his draft aside in order to work out a new exposition *(Nouveaux essais d'anthropologie).* Both works were published in 1859 by Naville (Oeuvres inédites de M. de Biran). The reason why in his last years he rejected his older exposition was that he had caught sight of a deeper layer in the soul than those he had hitherto taken into account in his reflections. Hitherto he had described the currents which spread from the feeling of life over the whole inner life, and, in opposition to
[303] them, the feeling of energy in which our self seeks to work its way out of the passive and involuntary. In the feeling of energy he had in stoic fashion found his refuge amid the outer and inner emotions. But this refuge was not enough for him in the long run. It did not suffice against the unrest in the outer and in the inner world. The need awakens in him for an absolutely fixed point of support, for a devotion of another and higher kind than that to the moods of the feeling of life, a devotion to something that is independent of the soul's own imperfect activity, and which yet comes not from without but from within. He uses Kant's distinction between noumena and phenomena and seeks his firm standing ground in the inner relation to something that lies beyond all phenomena. It is the fundamental thoughts of religious mysticism in which he now lands by the way of his own independent experience, at the same time as the study of the Gospel of John, De imitatione and Fénelon replaces the study of Descartes and Leibniz, which in its time had in turn replaced the study of Condillac and Cabanis. Biran's religiosity takes on no confessional character. He holds fast that religious feeling cannot be produced from without; it must arise involuntarily, and just as involuntarily under its influence the images take shape in which the soul finds the notion of the eternal and infinite expressed. Religion is feeling rather than faith; faith is subordinate to feeling. What our spiritual activity can do here is to bring about receptivity for the higher life, the spiritual life proper, *la vie de l'esprit* in opposition to the active life in will and reason, *la vie humaine,* which again is opposed to the currents of the feeling of life, *la vie animale.* This higher life Biran also designates as "the mystical life of enthusiasm", "the highest stage the human soul can reach, in that, so far as it lies with it, it makes itself one with its highest object and thus returns to the source from which it has sprung". (Oeuvres inédites. III. p. 541. 521).

But Biran's diary shows that his passion for psychological observation and reflection did not fall away with his passing over to mysticism. Again and again he discusses the question how far the psychological causal explanation can reach with respect to the most inward and highest states. Now it is said that those who seek to explain everything
[304] by natural causes cannot be prevented from asking whether even the highest state of

blessed rest and contemplation, where the soul is in ecstasy and feels itself under divine influence, — whether even that might not be explained by the working of organic moods, so that heavenly bliss would be replaced by unrest and tossing as soon as the organic state changed. Now it is declared as settled that the idea or feeling which the soul has of the perfect, the great, the beautiful and the eternal cannot spring from the soul itself; moral and religious truths have another source than psychological truths. (Journal intime. 26 August 1818 and 19 September 1818). Right up to the last leaves of the diary there is discussed in particular the question how our own mental activity in thinking and in concentration of the mind can prepare the way for that higher devotion by virtue of the constant reciprocal action between the active and the passive in us: and how, it is then asked, can one distinguish between what as a consequence of such concentration wells up from the soul's own ground and what is due to the influence of divine forces? Beyond a "whether — or" (soit—soit) Biran does not get, however great his need to hold fast that when we seek to penetrate into truth, truth also seeks to penetrate into us. (Journal intime. October 1823).

Only few had knowledge of Maine de Biran's thought before his writings — for the most part long after his death — were made public. To these few belonged first and foremost the famous physicist André-Marie Ampére (b. 1775, d. 1836), who stood in lively oral and written communication with him and together with him found the starting point of philosophy in the immediate consciousness of the energy of the self. Ampère acknowledged his friend's priority on this point, but while Biran all his life continued to circle about this one point, Ampère sought to give a detailed psychological and epistemological theory. Philosophical studies occupied him to a high degree; beside his scientific and mathematical lectures he also lectured on philosophy. Ørsted's discovery of electromagnetism led him into the brilliant series of investigations which have especially made his name famous; but he did not forsake his old love, and in his later years he was *[305]* occupied with studies on the encyclopedia and classification of the sciences, which led to the publication of his *Essai sur la philosophie des sciences* (first volume 1834, second volume with introductions by Ste. Beuve and Littré 1843), in which several of his interesting psychological and philosophical ideas were expressed. Later his correspondence with Biran and some fragments of a treatise were published with an introduction by his son, under the title: *Philosophie des deux Ampère,* publiée par Barthélemy St. Hilaire (Paris 1866). (Of the son's philosophy, however, there is not much in the book).

Ampère stands as a philosopher to Biran roughly as he stands as a physicist to Ørsted. In Ørsted's discovery he divined a single example of a general law, which he succeeded

in developing mathematically and demonstrating experimentally. Electromagnetism was extended by him into the electrodynamic theory. The influence of the electric current on the magnetic needle led him to discover the influence of electric currents on one another and the influence of the earth on electric currents. Biran's observation of the relation between the active and the passive in mental life led Ampère to two series of investigations, partly to an investigation of how our sensations and representations involuntarily combine prior to and independently of conscious activity, partly to an investigation of how the scientific knowledge of existence becomes possible on the basis of the mind's conscious use of its faculties. By the first investigation he has given interesting contributions to the psychological doctrine of association, by the second to epistemology.

In psychological respects Ampère applies the same method as the English association psychologists. He does not — like Biran and most other French psychologists — stop at description, but seeks to explain the origin of compound phenomena of consciousness by the fusion and combination of simpler elements. A fusion (concrétion) takes place, e.g., between the sensation of color and the sensation of resistance, which enter into so close a combination that one is inclined to take such compound combinations for
[306] simple and primitive ones. It is only analysis that discovers that a fusion has taken place, and that our own involuntary activity and the limits it meets have thereby played an essential part. Another example of concrétion is had in recognition, where the present sensation of an object and the reproduction of an earlier sensation fuse together. By the same process Ampère further explains that when during the performance of an opera one has the text before one, one can immediately *hear* the words, while without the help of the text one would not be able to hear them. The words one reads and the sounds one hears fuse immediately together[73]). It was this kind of explanation which, according to Ste. Beuve, made Ampère's psychological lectures so attractive.

In the very fundamental fact on which he was agreed with Biran, the immediate consciousness of self-activity, Ampère makes a distinction which Biran did not grant. He distinguishes between self-consciousness (autopsie [i.e. heautopsie], later: émesthèse) and the muscular sensation, showing that this latter sensation we have also when another person moves our arm or our leg. In one's own movement self-consciousness is the phenomenon that contains the cause, the muscular sensation the phenomenon in which the effect appears. Precisely through the fact that the muscular sensation can also be produced by another, I discover that I myself am in some cases the cause of it. In this experience, then, the causal relation dawns upon me. Immediately I have it given only in

my own activity, and all other acting I therefore apprehend by analogy with this. Together with the immediate consciousness, at which Biran stopped, there works according to Ampère the faculty of apprehending relations, the cognitive faculty proper. It is by it that we apprehend the relation between cause and effect, between places in space and time, etc. And it is by it alone that we can know more than mere phenomena. However true it is, according to Ampère, that everything that appears to us is only phenomenon, he is nevertheless convinced that the relations (rapports, relations) which obtain between the phenomena among themselves without depending on the phenomena's qualitative character have absolute (noumenal) validity. The mere concepts of kinds, which are *[307]* obtained by comparison and abstraction, have no higher validity than the sense qualities which make up their content. But the concepts which express pure relations and modes of coordination, such as causality, number, time and space, hold absolutely and form the bridge by which our knowledge reaches from the world of phenomena to that of noumena. On this point Ampère declares himself definitely against Kant, whom he otherwise highly admires. No less definitely does he declare himself against Reid, who assumes an immediate observation of external reality, whereby according to Ampére he confuses a "concrétion" with an elementary observation. Noumenal reality (matter in itself as the cause of sense impressions, the soul in itself as the cause of our own acting, God as the cause of everything) we discover only by way of inference and in the form of hypothesis. What we immediately apprehend is only phenomena, but the relations, independent of the qualities, which we intuit between the phenomena hold for the noumena too. By relations Ampère means the same as what Boyle and Locke called the primary qualities.

On this question Biran at first inclined to Reid's view. When his philosophical friend had eagerly urged him to study Kant, he became more of a Kantian than Ampère cared for. Against the latter's theory of relations he maintains (in a conversation which is mentioned in Journal intime 30 Oct. 1816) that a very great difficulty is bound up with making the transition from the consciousness of our own activity (the fundamental fact common to Ampère and Biran) to the assumption of external causes. Here, according to Biran, is an abyss over which analysis and induction cannot carry us. From the fact that I am not the cause of a passive state in me it cannot be inferred that there is necessarily a cause which produces everything that I do not produce. We are ourselves causes, and therefore it is natural for us to apprehend other things as causes: but that is no proof! — When Ampère held that the absolute validity of relations was proved by the fact that all inferences we draw from our theories of relations are confirmed by experience, Biran

answers (in a hitherto unprinted letter, used by A. Bertrand: La psychol. de l'effort p. 188):
[308] "What experience can teach us whether the forms of the coordination of phenomena are absolutely in things or only in the mind that apprehends? Can this doubt of reflection ever be cleared up by an external experience? Do not both possibilities agree equally well with the phenomena?" — To this Ampère has hardly been able to give any satisfactory answer from his theory of relations[74]).

In his world-view Ampère is a Cartesian. He divides the sciences into cosmological and noological sciences, of which the first concern the material, the latter the mental phenomena. The noological sciences are to be studied after the cosmological, partly because these show us the use of the human faculties for the knowledge of the world and thereby throw light on these faculties' nature, partly because the knowledge of the world, especially of man's physical nature, which they give, is of importance for understanding man's intellectual and moral faculties. As regards Ampère's classification of the sciences in detail, it is very intricate and cannot in simplicity and clarity measure up to the one Auguste Comte set up at about the same time. But there are in his "Essai sur la philosophie des sciences" many interesting remarks, and if he had been able to carry out his wish to give not only a classification but also an exposition of the most important truths, methods, problems and hypotheses in the various domains, there is no doubt that so excellent a thinker and investigator would have produced an excellent work and an interesting corrective to Comte's system. — At about the same time as Ampère and Comte the woman mathematician Sophie Germain (b. 1776, d. 1831) was also busied with ideas on the course of development of the sciences in its connection with the standard of truth which lies in the very nature of human knowledge. Under the evident influence of Kant she shows that by reason of the nature of our consciousness we have a need of unity, order and connection, a need which guides us not only in scientific inquiry but also in the moral and aesthetic domain. Her philosophical work *Considérations générales sur l'état des sciences et des lettres aux différentes époques de leur*
[309] *culture* (published 1833, new ed. 1879 in "Oeuvres philosophiques de Sophie Germain" by H. Stupuy) aims to show that the same type guides us in science, morality and art. "There is only one single type of the true. In morality, in science, in literature, in art we always seek the unity, order and proportion of existence among the parts of one and the same whole." The causal principle is only a special form of this general principle. We find application for the causal principle when we do not see the object we are investigating in its whole. It shows itself to us as a fragment, and we then ask what unity encloses it. We see it as a part and seek the whole to which it belongs. This need of unity and wholeness

has led to bold systems, to arbitrary use of analogies and to the assumption of mystical causes. But it has also led to many happy ideas, and the progress of science depends on its being alive. The coherence in the history of science depends on it. But men gradually learn to put methods in the place of systems and to ask *how and how much* in the place of *why.* Instead of creating the world according to the whims of his imagination man learns to trace out the real connection of the world, and when it thus gradually succeeds in finding a great unity in everything, the imagination will regain in surer form what it has lost by its bold images having been displaced by critical inquiry. —

While in the philosophical closets of Biran and Ampère a thinking was developing which led beyond the reigning Condillacism, this was defeated in philosophical teaching by the appearance of Royer Collard and Victor Cousin. ROYER COLLARD'S lectures at the Sorbonne (1811—14) introduced Reid as a philosophical classic in France. He laid great weight on psychological inquiry in the direction of the Scottish school and emphasized the significance of instinctive apprehension and of immediate moral conviction as against the limited point of view and abstract analyses of ideology. This was at the expense of the sharpness of the problems; an appeal to sound common sense all too often took the place of an argument. But a new tendency had been inaugurated, and the eye was opened to sides of the spiritual life which the hitherto reigning school had left in shadow. Royer Collard's considerable personality gave his appearance as a philosopher a peculiar *[310]* weight. His successor Victor Cousin (b. 1792, d. 1867) was an enthusiastic speaker with great talent for awakening the interest of the young. Young himself, he shared the young's expectation that with the free constitution and the fall of military despotism there would come a brilliant time for French spiritual life. His teaching had a decidedly historical stamp. He brought the history of philosophy into academic teaching in France. His own philosophical points of view he got partly from the Scots and Royer Collard, partly from Biran and Ampère, in whose philosophical gatherings he took part. Biran saw well enough that Cousin "was hunting on his preserves"; but he expected to get a share of the game, since Cousin's teaching would produce receptivity for his own great psychological work when it came (which unfortunately it did only some thirty-odd years after Biran's death). Cousin combined Reid's theory of the immediate apprehension of absolute reality with Biran's doctrine of the consciousness of our own self-activity and with Ampère's doctrine of absolute relations. And with this he later also united Schelling's and Hegel's doctrine of absolute reason. On journeys in Germany he had come to know German speculative philosophy, after Md. de Staël's book *De l'Allemagne* (which was written in 1810, but could appear only after Napoleon's fall) had given a

general impression of German philosophy from Kant's time. Cousin's philosophy became an eclecticism which from the various systems drew out what supposedly had lasting value: the systems were each by itself incomplete, but none of them wholly false! The standard for the choice was at first psychological observation after the pattern of Reid and Biran, later the universal reason of Schelling and Hegel, and in the last part of Cousin's career — when after the Revolution of 1830 he had become official director of all philosophical teaching in France — the chief doctrines of natural religion and of Cartesian spiritualism. The boldness and the inspiring quality of his first appearance[75]) as a philosophical teacher were damped by his historical studies and his official position. Aptly has Renan, who moreover long after Cousin's death expressed in warm words
[311] what he owed him, cited him as an example of the fact that it is not good for philosophy to win an all too complete victory. It must be added that the victory was only outward. Napoleon had looked with pleasure on Royer Collard's appearance, and it was the July Monarchy that supported Cousin.

Cousin's most important work is, Du *vrai, du beau, du bien,* his lectures of 1818, which were published by Garnier twenty years later in their original form, but in later editions were mutilated by Cousin himself, whom the heresies of his youth now embarrassed. It was this work that made so great an impression on Renan. Besides this, one must seek his ideas in the prefaces to five volumes of *Fragments philosophiques,* in which he treats subjects from the history of philosophy. In the preface to the first of these volumes (1826) he presents his characteristic doctrine of "impersonal reason", in which he has combined Reid, Ampère and Schelling. It is by means of the psychological method, by deepening self-observation, that he believes he has reached a point to which not even Kant had penetrated, a point where the apparent subjectivity and relativity of the necessary principles vanishes, in that there is revealed an involuntary grasping of truth as the foundation of all logical reflection and of all setting up of necessary concepts. Only by becoming an object of reflection does reason become subjective; in itself it is a light which, independently of all personal differences, shines within every human being as a universal revelation.

While Cousin did indeed believe that he was supporting himself on psychological observation, but very soon rose above it on the wings of rhetoric and fancy, Théodore Jouffroy (1796 —1842) had a greater and more faithful interest in psychological inquiry. He did, to be sure, keep one-sidedly to self-observation and disdain other sources of psychological knowledge, in that in a strictly spiritualistic fashion he distinguished between psychology and physiology as two subjects which had nothing to do with each

other. But he was in earnest — a whole series of investigations shows it — about getting clear on the relation between self-observation on the one side and natural science and philosophical speculation on the other. He does not have Cousin's romantic virtuosity in throwing bridges over abysses. His philosophy sprang from his purely personal need of *[312]* clarity in knowledge, after he had through a hard struggle seen the theological notions lose their value for him. But although there is a strong personal interest in his thinking, it does not lead him to compromise or to content himself with solutions that do not fully satisfy his thought. Rather, he then stops at the recognition that the solution of the problem is unattainable. "There are", he says in one of his lectures *(Mélanges philosophiques.* 3 éd. p. 350 f.), "two ways in which the man who thinks can obtain peace in his soul and rest for his mind; the first is to possess, or believe one possesses, the truth concerning the questions that interest humanity; the second is to see clearly that this truth is inaccessible, and to know why it is so. . . Since the facts we can observe are limited, the inferences we can draw from these facts are so likewise. Science therefore has its horizon, beyond which it cannot see; it is its task gradually to determine this horizon. At this outermost limit of its domain it must part from poetry, which alone can go further. This separation is its duty toward humanity, toward which it has the task of discovering the truth, and which has only all too often suffered from having hoped for and sought the truth where it was, and for ever will be, inaccessible to it." — It is far more by the manly way in which Jouffroy faced the problems than by his treatment of them in detail that he stands as a fine and attractive figure in the history of French philosophy. In so far as he points to definite results, these go in the direction of the eclecticism founded by Cousin.

c. The Social School

When Cousin's disciple Damiron in the year 1828 published a work on French philosophy in the nineteenth century, he set up eclecticism as the higher unity or the right mean as against the Condillacian school on the one side and the theological school on the other. The peculiar intellectual work of Biran and Ampère, which went deeper than *[313]* eclecticism, had not come before the public. But in the second edition (which came out the same year) he found himself induced to speak of the ideas of Saint-Simon and Auguste Comte, although he admitted that he could not clearly see their significance. At that time only Comte's first writings were available. Saint-Simon's career, on the other hand, was concluded. The germ had been laid from which modern socialism and positivism were to develop.

Saint-Simon (whose full name was Claude Henri de Rouvroy, comte de Saint-Simon) was more a journalist and social reformer than a philosopher. But he gains significance for the history of philosophy through his living conviction that a new social order becomes possible only if a new world-view makes its way. And this new world-view can, of this he is no less convinced, only be built up on the foundation of the positive sciences. Saint-Simon himself had not received any scientific education. As a young man (he was born in 1760) he took part with bravery in the North American War of Independence. Thereafter he threw himself into industrial enterprises, renounced his noble rank during the Revolution, took to speculating in national lands and appeared as a "grand seigneur sansculotte". He was in prison during the Reign of Terror, but was freed at Robespierre's fall. His wealth he used (as long as he had it) not only to live in high style, but also to prepare the scientific and social reform of which he had early conceived the thought. He surrounded himself with men of science from the Polytechnic and the Medical School in order to train himself to write an encyclopedia of the sciences. His great plans he did not give up because he became poor; on the contrary, he worked with ever greater zeal. He had a rare capacity for drawing eminent men to himself and getting them to work with him and for him. Augustin Thierry and Auguste Comte were for a time his secretaries and collaborators and called themselves with pride his pupils, until their ways parted. While Saint-Simon in his earlier period had put the production of a new encyclopedia in the first line, because only on the foundation of such a one
[314] could a new catechism be worked out, after the upheavals of 1814—15 he believed he could proceed directly to the new organization of society, whose main feature was that industrial and economic life should stand in the first line and the government system proper take a subordinate place. In place of noblemen and jurists, great industrialists and men of science were now to stand as the heads of society. The final purpose was to be that the working class, the class that suffers most, should be helped forward in intellectual and economic respects. While in France after the Restoration men were absorbed in constitutional and parliamentary struggles, Saint-Simon declared that it is the social renewal that matters, that the external form of government is indifferent, and that a government is on the whole only a necessary evil. As the leading motive to carry the new order he at first thought of insight into the harmony of interests, much as Helvetius and later Bentham conceived it. But in his last years he appealed to the love of man (philanthropy) and wished to found a new Christianity, which consisted essentially in the commandment of love, while at the same time it differed from the old Christianity by letting earthly life come into its full right and not making it a means to a

supernatural existence. The individual right of property can in his view be founded only on the universal utility which this institution offers, not on the single person's claim in and for itself. What matters now is association in order to exploit the earth. It is things, not men, that are to be governed. As his disciples later expressed it: in place of the mutual exploitation of men by one another the exploitation of the globe is to come. The state is to gather the human forces for great works, such as the building of canals and roads, drainage, cultivation. Labor and brotherhood was Saint-Simon's watchword. Surrounded by a small circle of adherents he died in 1825.

As Saint-Simon by his political program stood in opposition to both of the contending parties of his day, the Liberals as well as the Legitimists, so in the domain of ideas he stood in opposition both to the theological school and to the philosophy of the eighteenth century. Significant here is the way in which he conceives and evaluates the Middle Ages already in writings from 1807 and 1813. The Middle Ages are for him a great period *[315]* of organization. The civilized world was held together by brotherhood and common faith. The priests were not, as Voltaire thought, charlatans, but the most advanced part of the nation. Since the medieval system has fallen before criticism and revolution, we live in a spiritual and social chaos. Negation and egoism flourish. What matters is to reach a new organic period, and the spiritual power that is to lead to it can be none other than positive science, which it is a matter of putting into a system. Saint-Simon's friend the physician BURDIN put forward, in a conversation with him, ideas on the development of the sciences which Saint Simon took up, and which contain interesting intimations of positivism. All sciences began on the foundation of a small number of experiences and therefore had in their first period the character of guesses, often fantastic guesses. Astronomy began as astrology, chemistry as alchemy. As experience advances, the sciences pass from a conjectural to a positive form. Mathematics, astronomy, physics and chemistry have already reached the positive form; physiology and psychology are on the point of doing so. Finally philosophy too will become a positive science; it is due to the imperfect state of the particular sciences that there is still so great a distance between general science and the special sciences. On this conception of the history of the sciences Saint-Simon built his hope of a new world-view on a purely scientific foundation. The expression *positive philosophy* he was the first to use. Further than the general thought he did not get. He was not the man to carry through a great, connected work. And yet he marks a decisive turning point. By his recognition of the Middle Ages as a distinct period of culture he betrays a historical sense which was rare at the time when he appeared. And he thereby succeeds in being able to conceive history as a continuous process of

development. While in the period of Enlightenment the Middle Ages had come to stand as an arbitrary interruption of development, they now stood as a period of spiritual and
[316] social organization after a preceding period of dissolution. In return the period after the Middle Ages came to stand as an age of dissolution. Criticism and liberalism are for him only means of getting the obsolete system out of the way, and are not themselves a new system. Such a system had first to be developed, and in Saint-Simon's conviction it could not be reached except on the foundation of empirical science. The time must come when humanity would build its faith and its morality on experience and science. Only by this way can there arise a new total conception of existence about which all can gather, so that life can again be carried on with united forces.

This doctrine of Saint-Simon's his school generalized, in that throughout history it distinguishes between critical and organic periods. In Kant and Fichte, indeed already in Rousseau, a similar opposition occurs. Such a conception was bound naturally to come forward in an age that had seen criticism overthrow an old order of things in the domain of faith and of society without being able to produce a new order and without being able to root out the need of one. However much there was of charlatanry in Saint-Simon, he nevertheless had on a decisive point a correct historical eye and felt the spiritual and social needs of the time. It thereby becomes understandable that he had a great historian and a great philosopher among his disciples.

In Saint-Simon there is found only the germ of socialism. The germ was developed in his school[76]), which reshaped his doctrine in such a way that he would not have acknowledged it. They wished to restrict the right of inheritance and to entrust to the state the distribution of the product of labor according to each person's capacity and work. The school took on more and more a fantastic and utopian character. A new hierarchy was set up with "Father" ENFANTIN at its head; but the school was dissolved when Enfantin came out with ordinances concerning the sexual relations of the faithful, which brought conviction and imprisonment for the leaders. Before these last extravagances the sober and critical elements had withdrawn. The enthusiasm that had bound the Saint-Simonians together — the enthusiasm for the goal of completing man's dominion over nature by the cooperation of human forces — was nevertheless not without fruit.
[317] A great part of France's railways, canals, factories and banks owe their coming into being to former Saint-Simonians. The cutting through of the isthmuses of Suez and Panama was a Saint-Simonian thought. It was pupils from the Polytechnic School who in particular had joined the new doctrine with enthusiasm. This school, which had been established by the Convention, was the home of the sciences which according to

Burdin and Saint-Simon had already become positive, and which had the calling to form the foundation of the faith of the future. Positive philosophy is the philosophy of the Polytechnic School, while eclecticism reigned at the École normale (where teachers were trained for the higher educational institutions). The real founder of positivism, Auguste Comte, himself a pupil of the Polytechnic School, speaks in a letter (Lettres à Valat. p. 229) of the hidden and involuntary strife between *Normaliens* and *Polytechniciens,* which in his eyes was a special form of the strife between the metaphysical and the positive school. It was the philosophy of the Polytechnicians that was to become France's most significant contribution to the history of philosophy in the nineteenth century.

2. Auguste Comte

a. Biography and Characterization

At the root of the rise of modern positivism lies the development of the empirical sciences in the last centuries, and in particular the magnificent advance in the domains of chemistry and physiology which the French naturalists began at the end of the eighteenth and the beginning of the nineteenth century. One by one all the domains of nature had been drawn under the scientific principles and methods which Kepler, Galilei and Newton had established. LAVOISIER and BICHAT had continued in chemistry and physiology what the founders of modern natural science had accomplished in the domains of astronomy and physics. The question now is: what significance has this whole development for our view of life and our world-view? Positivism's answer is: Should it be the case that the progress of science is a joy only to the specialists and to the *[318]* industrialists? Should it not be a change in the whole spiritual life of man, in human faith and conduct of life, that results from all the domains of nature being subject to the method of the empirical sciences? There is, after all, in the human mind an involuntary need to apply the same method and the same theory everywhere. Now when we understand nature by explaining its phenomena according to the laws that experience points out, are we then, in our faith and in our way of conducting and ordering life, to be able to build on an entirely different foundation? In fact we find that empirical science constantly exerts a determining influence on men's religious and speculative representations, however much these are asserted to have sprung from entirely different sources. A certain degree of positivism is found in every world-view. But now all the domains of nature have been drawn under *positive science,* i.e. under the knowledge that builds on facts and on the laws which experience establishes for these facts. The moment then seems to have come when that influence is no longer to be hidden and limited, but when men can consciously

and resolutely use in their view of life and world-view only such representations as positive science can recognize and confirm. There is still lacking, to be sure, a positive science of the spiritual life of man; it must then be the task of positivism to found one. If that succeeds, what then stands in the way of carrying out the program of positivism? — The task Auguste Comte set himself was a double one: to make the human sciences into a positive science, and then to give a systematic exposition of the chief facts, chief laws and chief methods of all the positive sciences. The foundation for the world-view of the future would thereby be given. The time of theology and speculation was over; the positive philosophy now stood as the only possible tendency. The questions that man cannot get answered with the help of positive science would then gradually fall away.

There nevertheless always remains a difference between positive science and positive
[319] philosophy. For it is not the single isolated facts, nor the special laws, that can condition a world-view. The human mind seeks, as Comte expressly emphasizes, unity in method and in doctrine. The positive philosophy first comes into being when the single positivities are worked together into a whole. The point, then, is to unite positivity and generality or totality. Positivism therefore cannot use the given as it immediately lies before us, but must submit it to a working-up, unite it into a whole. Here the difficulty of positivism comes to lie; for in this working-together there is necessarily need of hypotheses and presuppositions, and the question then is on what their validity rests.

Auguste Comte finds his philosophical predecessors in the eighteenth century. He refers especially to Diderot and Hume, to Kant and the Scottish school. But he admires de Maistre and regards the Middle Ages as the last *organic* period we have had. It was his conviction that only by carrying out positivism in faith and conduct of life would we regain the unity and harmony in our spiritual and social life which was lost through modern criticism and through the Revolution. Side by side with the enthusiasm for positive thinking there goes in Comte the fervent, at last mystical love of humanity, a need of feeling that demanded its satisfaction with such energy that it several times threatened to burst his consciousness. The founder of positivism felt himself spiritually akin to the mystics of the Middle Ages. And at last he even wanted to found a new religion, which was to be positive in a sense other than the one in which people are accustomed to speak of positive religion. —

Auguste Comte's life shows us how the various motives that led to his work developed in him. He was born on 19 January 1798 in Montpellier. His family was strictly Catholic. By his own account (Cours de philosophie positive. Tome VI. Préface personelle) he was scarcely fourteen years old when he "ran through all the essential degrees

of the revolutionary intellectual tendency and felt the need of a general political and religious regeneration." Perhaps it was precisely the rapid development that led him away from the representations of his childhood which brought it about that the need of settled dogmas and definite symbols became so vivid in his later years, after his best *[320]* strength had been spent on his main philosophical work. The period in which he was seeking was only short, and he did not reap sufficient experience of what value such a period can have for the freedom, inwardness and fullness of personal life. He came to share the age's aversion to the *critical* periods. Already as a boy he was refractory toward all compulsion and all regulation; but full of reverence before intellectual and moral superiority. (Cf. the characterization he gives of himself as a young man in a letter printed in Littré: *Auguste Comte et la philosophie positive,* 2. éd. p. 493). In the year 1814 he entered the Polytechnic School in Paris. This school for the exact sciences, founded by the Convention, he later regarded (cf. a letter to Stuart Mill of 22 July 1842) as the first germ of a true scientific corporation. It comprised the sciences that had decidedly reached the *positive* stage, and it was for him the foundation on which all higher education should build. There was, moreover, among the school's pupils a republican temper and a brotherhood which united them in a common enthusiasm over the significance their subjects could have for the development of culture. It was this that won Saint-Simon's ideas so great a following among the school's pupils. The thought of positivity and the thought of humanity sprouted forth together. The reactionary government used, in the year 1816, a demonstration against a teacher as the occasion for closing the school and sending the pupils home. But Auguste Comte could not bear to live away from the center of the movement. In spite of his parents' prohibition he came back to Paris the same year to continue his education. He studied biology and history to supplement his polytechnical training, and acquired an excellent encyclopedic foundation. He earned his living by teaching mathematics. Of great significance for his development was his coming into close relation with Saint-Simon. He writes of this in a letter from that time (1818): "Through collaboration and friendship with one of the men who see farthest in the domain of philosophical politics I have learned a host of things which I would have sought in vain in books, and my mind has, in the half year *[321]* I have been in contact with him, made greater progress than it would have made in three years had I been alone. This work has developed my conception of the political sciences and has besides indirectly raised my representations of the other sciences, so that I have noticed that I have got more philosophy in my head and a more correct, more elevated view of things." *(Lettres à Valat.* p. 37). What Comte owed to Saint-Simon was

partly the notion of the necessity of a new spiritual authority in place of the hierarchy of the Middle Ages, which in the opinion of both had been an admirable institution for its time, partly the interest in the social question, with which the sharp emphasis on the opposition between militarism and industrialism is connected. In a treatise of 1820 *(Sommaire appréciation de l'ensemble du passé moderne)* Comte dates the dissolution of the old social system back to the twelfth century, finding the first germs of this dissolution in the emancipation of the communes and in the introduction of the positive sciences into Europe by the Arabs. Thereby the ground was laid for the industrial system founded on labor gradually taking the place of the territorial system founded on conquest, and for positive science replacing theology. Even if Comte had already, by his own path, arrived at similar ideas, his relation to Saint-Simon nevertheless hastened his development, and it was with great injustice that Comte later regarded it as a great misfortune for himself that he had come to know Saint-Simon[77]). The two men were, however, too different in their nature and line of thought to be able to go on working together. The break came when Comte felt his independence. In principle they came to disagree about the place science was to take in relation to labor and to the new "priesthood" Saint-Simon had in mind in his later years. As we have seen, the latter, in the last part of his career, pushed the theoretical reformation aside in order to think directly of the practical one. In this Comte disagreed with him. The last work in which Comte still calls himself a pupil of Saint-Simon occasioned the break; Saint-Simon dissociated himself from it, and
[322] Comte later published it in his own name. This work, whose title was: *Plan des travaux scientifiques nécessaires pour réorganiser la société* (1822, published again in 1824 under the title: Politique positive), marks Comte's breakthrough as an independent thinker. The most important obstacle to the development of civilization he sees in the constant preponderance of the revolutionary tendency, especially in freedom of conscience and the sovereignty of the people, which are critical, not organic, principles. There is lacking the capacity to enter into a coherent circle of representations and to recognize a rationally grounded authority. A new social system cannot be constructed all at once, but must develop successively. That is sufficiently proved by the many failed constitutions in the most recent history of France. The fundamental condition for working together toward a common goal is a common way of thinking and temper. The old society had such a common foundation in theology. In order that the new society may get a corresponding foundation, there must first be brought about a comprehensive edifice of thought that can come forward with the same authority as that with which the results of the single sciences come forward in the special domains. Most important of all is that politics be

made a positive science. While in the mathematical, physical and biological domains one recognizes definite laws for all phenomena, one still believes that in the social domain one can intervene at one's own discretion, e.g. introduce a thought-out social system without further ado. This belief will fall away when the conformity to law of human relations is recognized. And such a conformity to law Comte seeks to point out in the close connection that exists at every point in history between the social organization and the whole of civilization, that is to say, between the order of society and spiritual development as this appears in science, art and industry. Civilization in its turn springs from man's instinctive need of perfection[78]). The various stages that civilization runs through Comte, relying on the history of the sciences, reduces to three: the theological, the metaphysical and the positive, and he seeks to show how definite stages in social development correspond to them. Since *the law of the three stages* is a chief point in *[323]* Comte's philosophy, the detailed account of it must be deferred to the exposition of that philosophy. — This writing of Comte's aroused great attention among politicians, historians and scientists alike. Guizot, the mathematician Poinsot, Alexander von Humboldt, the Duke of Broglie and others expressed their approval. It went so deep that it led beyond the contending parties, while at the same time it did justice to what was of value in their tendencies. But Comte's real plan had still not been carried out. Only now did he proceed to the chief work of his life: to give an encyclopedia of the content and methods of the positive sciences. He writes to his friend Valat (1824): "I will work all my life and with all my powers at the founding of the positive philosophy." And at about the same time he writes to another friend (Gustave d'Eichthal) that his real task is "an encyclopedic recasting of all our positive knowledge, which in reality ought to be conceived as a single mass." A few years later he set out his "positive philosophy" for the first time before a small but select audience (to which there belonged, among others, A. von Humboldt, Poinsot, the mathematician Fourier, the political economist Dunoyer, the physicians Broussais and Esquirol, the engineer Hippolyte Carnot).

Comte supported himself by literary work and by teaching mathematics. He had married a young, gifted Parisian woman. Very early the relationship seems to have become less happy, partly because Comte's parents were against the union (which in their eyes was already abominable through having been contracted in civil form), partly also, as is seen from a characteristic letter to Valat (16 Nov. 1825), because Comte had not found in his wife the qualities he valued most in a woman: "devotion of heart and gentleness of character, together with the kind of subordination which the feeling of her husband's intellectual superiority can inspire in her," qualities which "it is certainly very

difficult nowadays to find united in a woman of outstanding intellectual gifts." Comte wishes these qualities to his friend's bride in such a way that it shines through clearly
[324] that he misses them in his own wife. She had been too independent for him. She showed, however, great faithfulness and energy during an attack of mental illness which — no doubt chiefly as a result of overwork — struck Comte. He was placed in an asylum, and his parents sought to use the opportunity to bring him under their influence and wanted to place him in a monastery. Madame Comte, whose civil marriage the parents had wished to pass over in silence, asserted her right and demanded that her husband be placed in his own home, and her nursing then succeeded in bringing back his mental health, so that he could resume his interrupted lectures. (Comte's parents nevertheless had him — while he was still insane — married in church, and thereby relieved their hearts of the mortal sin of having given their consent to the civil marriage).

Now there followed for Comte a series of vigorous and happy years, in which he worked out and published his main work, *Cours de philosophie positive* (in 6 volumes, 1830—42). The content of each single volume he first thought through at length on solitary walks. As far as the material was concerned, he could rely on his great memory. When the meditations were concluded, he wrote them down in the course of a comparatively short time. Little weight was thereby laid on the form of the exposition. His style is heavy, unaesthetic, and suffers from repetitions and a heavy use of technical expressions; but it is distinguished by clarity, thoroughness and weight, and bears the stamp of energy and earnestness. Often (especially in the last three volumes of the work, which deal with the social sciences) the stream of thought becomes so mighty that it carries the reader along. Comte sought to work for the positive philosophy in a popular way through public lectures. From the twenties on, former pupils of the Polytechnic School had held popular lectures on natural science round about the country. In the year 1830 Comte and some of his comrades founded the Association polytechnique, which was to work in this direction. For a long series of years (until the coup d'état of 1851 put an end to it) Comte gave free public popular lectures, first on astronomy, later (after 1848) on positive
[325] philosophy as a whole. In a little writing, *Discours sur l'esprit positif* (1844), he has set out the general considerations with which he introduced his astronomical course; it gives the best introduction to Comte's philosophy. —

A fixed official position Comte never obtained. After 1830 he had hoped for a professorship in the history of the positive sciences, for which he was in the highest degree qualified; but an approach to Guizot, now so powerful, was without result. In his Mémoires Guizot even, by an incomprehensible lapse of memory, presents it as

though it were the first time he saw this man, with whom he had nevertheless, in the years of opposition in the twenties, had so many conversations and felt a community of thought. Nor did he obtain a professorship of mathematics at the Polytechnic School, although he had for a year temporarily filled this post in a manner that won general recognition. He had to be content with the more modest position of répétiteur and of examiner at the entrance examination, in which last capacity he made annual tours of the provinces. This last post too, to which one had to be elected annually by the teaching council of the Polytechnic School, he lost when in the preface to the last volume of the "Cours" he had made a sharp attack on mathematical specialism and its arrogance. It was now, he held, biologists and sociologists who were to stand in the front rank in intellectual respects: the supremacy of the mathematicians was over, since the more concrete domains had been drawn under positive scientific treatment. The provocative manner in which he expressed this caused him not to be re-elected examiner, and he was now again reduced to living by private teaching. English and French admirers and friends (among the English are to be mentioned Stuart Mill and the historian Grote, among the French the learned Littré) nevertheless made his economic circumstances easier for him by giving him support. — In his domestic circumstances too it came to a crisis, his life together with his wife coming to an end after a long period of growing discord. Yet his wife continued in the finest manner to show her interest both in his ideas and in his personal circumstances.

Amid all these conflicts and the emotions they caused, and after the great exertion *[326]* of strength he had spent in the twelve years he worked on his main work, it came to a new nervous crisis, not indeed so severe as the earlier one, but yet of such a kind that, in his own expression in a letter to Stuart Mill, there was "danger cérébral"[79]). Other circumstances worked together in this crisis and gave it its peculiar character. He had come to know a woman who became for him what Beatrice was for Dante: the one toward whom his need of feeling, which he had carried all his life without getting satisfaction for it in any intimate relationship, could find a complete outlet. For the first time he came to know the whole inwardness of mood and of devotion, and this young woman, whose name was Clotilde de Vaux, continued, when she died after the lapse of a year, to stand for him as the representative of humanity (just as Beatrice for Dante was that of theology), to whom he addressed his daily thoughts and the "solemn outpourings of noble feeling" (solennelle effusion des sentiments généreux), as he called his prayers. She became the genius who inspired him to his second main work, which was to systematize the feelings, just as the first had systematized the ideas. This work

appeared under the title *Politique positive ou traité de sociologie, instituant la religion de l'humanité* (4 volumes, 1851 — 54). In a letter to Stuart Mill (of 14 July 1845) Comte states that the new work, the plan for which he had already formed earlier, had received its decisive character during that crisis and the "méditation exceptionelle" it brought with it. For it dawned on him that the second part of his philosophical activity ought to differ from the first in this, that feeling should get as great a place in it as the understanding had in the former. Now that the purely theoretical work has been done, the point is its social application, and this consists essentially in a systematization of the human feelings, a consequence of the systematization of ideas and an indispensable basis for the systematization of institutions. — Had the new work aimed only at this, it would not have marked any decisive opposition to the earlier one, but a continuation and extension
[327] or deepening of it. In reality it went, as the title already shows, far further. It was to found a new religion. And whereas the earlier work started from the world or nature and on the basis of the knowledge of nature sought to arrive at an understanding of man, this objective method was now to be replaced by the subjective one, which started from man, regarded the whole of nature from the human point of view, and humanity itself as the highest being (le grand être). In subjective absorption in the thought of humanity, in fervent devotion to it, consisted the new religion, whose high priest Comte soon felt himself to be. He would not give up the positive philosophy; its content was to be the dogmas of the new religion, the religion of humanity; but he added cult and practice. In his *Catechisme positiviste ou sommaire exposition de la religion universelle* (1852) he has given a shorter exposition of what the "Politique positive" develops at great length.

This last turn in his line of thought had been prepared by what he called his "hygiene of the brain" (hygiène cérébrale), which consisted in his abstaining on principle from all study and absorbing himself solely in the working out of his works. Thereby he would keep disturbance away and see to it that the unity of the plan was held fast. Earlier he had pursued far-reaching studies, and his solid memory enabled him now to use what he had gathered at that time. Together with this isolation from everything new in science and literature, an isolation which brought with it the cessation of all real discussion and of critical testing of his own ideas, there went absorption in music, Italian and Spanish poetry, and the reading of De imitatione Christi. He demanded of every positivist that he read every day in a poetic masterpiece at least as much as corresponds to a canto of Dante. The writing De imitatione he regarded as a great poem on human nature; in his reading he everywhere put humanity in place of God, and so found in the old mystical book an aid to a fervent entering into the life of humanity. According to an

eyewitness's description of Comte in his last years, his whole being was pervaded by gentleness and goodness. He had, however, had the disappointment that his most famous disciple Littré, and several with him, withdrew from him when he passed from appearing *[328]* as a philosopher to feeling himself the high priest of the religion of humanity, just as he himself had withdrawn from Saint-Simon when the latter voiced a similar pretension. After the long and severe labor of thought, the mythological need of childhood had now, favored by the cerebral hygiene, broken through again, and on this last part of his way he was followed by only a few[80]). Yet the new religion of humanity has its congregations and its places of worship round about in France, England, Sweden and America. This religion without theology is a characteristic sign of the times. For Comte himself it was a resting place where his thought gathered the memory of the greatest and best it had found in humanity's knowledge and action, and from which it looked with hope toward the future that could be expected for the constantly progressing species. Love as principle, order as foundation and progress as goal! that was the motto of the religion of humanity. — On 5 September 1857 Auguste Comte died.

b. The Law of the Three Stages

Our knowledge runs through, according to Comte, *three stages of development,* which can be pointed out in the case of each single science. The more intricate and composite the subject matter of a science, the longer it will be before it has run through this scale. The abstract sciences come first to the definitive stage, the concrete ones later. In Comte's opinion the point of time has now come when the most concrete of all the sciences, the science of society (sociology), is about to enter the third stage. With that the circle is closed, and a retrospect becomes possible.

The first stage is *the theological.* Here a very limited number of observations lies at the basis, and imagination therefore plays the greatest part. The explanation of natural phenomena, the connecting bond between the given facts which the human mind by its constitution cannot do without, is found at this stage in the intervention of personal *[329]* beings. It is only with the help of the representations of gods and spirits that man can from the beginning make the world comprehensible to himself. Herein lies the great benefit these representations have rendered to the development of human knowledge. Had these representations not been available, and had the wish to find again the working of those beings in what happened round about in nature not asserted itself, the activity of knowing would not have got under way; the original torpor would have persisted. The understanding man believes he gains at this stage is, moreover, absolute; for beyond

those divine beings he thinks of nothing, and if he can explain a natural phenomenon by their intervention, he has nothing more to search for. Therefore man at first has no doubt at all about the possibility of reaching an absolute knowledge. — In practical respects the theological representations have exerted an influence no less important. They have furnished a firm and common ground for moral and social life. This stage of development is the time of authority. Men live in a common trust in unshakable powers. In politics the age of kingship corresponds to this.

The theological stage comprises a series of steps. In fetishism a mental life like the human one is ascribed directly to the objects of nature. In polytheism, the step most characteristic of the theological stage, material objects are deprived of their immediate life, and the source of their motions and changes is sought in various beings, mostly invisible, who form a higher world. The transition from fetishism to polytheism is the most important step taken within the theological stage. The god is removed from the natural phenomenon to which he corresponds, and thereby the notion of matter as living can be replaced by the notion of matter as subject to the law of inertia. It is owing to the beginning capacity for observation and induction that this transition is made. In monotheism the distance and the opposition between the principle from which the explanation is taken and the phenomena that are to be explained are still greater. It is
[330] therefore vague and brings with it less spiritual community than polytheism. It forms the transition to the second great stage: *the metaphysical.* Here the explanation is found not in personal beings but in abstract ideas, principles and forces. There prevails here a striving to reduce various phenomena to one principle, a striving which already within the theological stage leads over from fetishism to polytheism and from polytheism to monotheism. The metaphysical stage continues this striving by assuming as many forces as there are peculiar groups of phenomena, thus a chemical force, a vital force, etc. At last the striving is then to reduce all these various forces to one fundamental force, one fundamental being, Nature, an equivalent of the concluding unity that monotheism made possible at the theological stage. Common to these first two stages is the tendency to seek absolute solutions. Metaphysics, just as much as theology, seeks to explain the inmost nature of things, the origin and destiny of all things, and the manner in which the phenomena are produced. The difference is only that the abstract is put in place of the concrete, argumentation in place of fantasy. As great a preponderance as fantasy had over observation at the first stage, so great a preponderance has argumentation here.

Comte conceives the metaphysical stage essentially as a stage of transition and as a process of dissolution. Argumentation bores its way into the theological circle

of representations, points out the contradictions there, puts constant ideas or forces in place of the incalculable wills, but thereby weakens the living impression and the authoritative influence of the powers that are assumed to rule in nature and in human life. Something new it cannot build up; real equivalents it cannot give. In practical respects the dissolution shows itself in the rule of doubt and egoism. The individual is torn out of the living connection with society, and the understanding is cultivated at the expense of feeling; Comte even speaks in his later years of the long revolt of the understanding against the heart. In politics it is the age of the peoples, as the first stage was that of the kings; the jurists are the leading men; society is conceived as having arisen by contract, and the state is built on the principle of the sovereignty of the people. *[331]*

At the *positive* stage both fantasy and argumentation are subordinated to observation. Every proposition that is put forward concerns a fact, either a special or a universal fact. Agreement with facts is the only criterion. Not, however, to be understood as though it were the single, isolated facts that are stopped at. Positivism is as far from empiricism as from mysticism: just as little as it departs from the facts in order to lose itself in supernatural beings or in abstract principles, just as little does it scatter itself in incoherent observations. But instead of seeking absolute *causes and* tracing the production of things out of them, it seeks the *laws* of phenomena, that is to say, the constant relations that subsist between the observed phenomena. Whether it is thoughts and feelings or impact and gravity that are to be understood, the relations under which they appear are the only thing our knowledge can ascertain. Science builds on the invariability of the laws of nature, which first dawned on thought with the founding of mathematical astronomy among the Greeks, but in modern times has been found again in one domain after another. Bacon, Galilei and Descartes stated this principle and thereby stand as the founders of the positive philosophy. Even if the laws that hold in the particular have not yet been found everywhere, an irresistible analogy now leads us to apply that great philosophical principle of the invariability of the laws of nature to all phenomena and events.

Whereas the first two stages showed us a striving to reduce everything in existence to an absolute principle, whether this was thought of in theological or in metaphysical form — as God or as Nature —, it follows from the nature of the positive philosophy that such an absolute and objective conclusion is not possible. The strict demand for confirmation by experience makes it impossible to reduce everything to one principle; experience always shows us only a limited connection, and there will always be many phenomena and events which it will not be possible to bring into connection with the

[332] others. There are groups of phenomena (at least as many as there are different sciences that cannot be reduced). The many laws cannot be reduced to a single one. Only a subjective, not an objective unity can our knowledge reach. The subjective unity consists in the same method being applied everywhere, which brings with it uniformity and convergence of the various theories. (Quant à *la source intérieure* des théories humaines . . . nos connaissances tendent vers une entière systématisation. Discours p. 24). Subjectively regarded, then, we get only a single science. And it can become common to all human subjects. The positive method brings about unity not only within the consciousness of the single subject, but also between the various subjects. Thereby the positive philosophy becomes the intellectual foundation of society among men. Even now it is only on subjects that have been drawn into the domain of positive science that agreement prevails among men, while otherwise opinions go in the most various directions. Catholicism, as long as it ruled without intellectual competition from the side of scientific thought, offered a spiritual community which stands as a model of what the positive philosophy will one day bring. The positive philosophy finds in the subjective unity that lies in the concept of humanity the only possible equivalent of what the concept of God was for the theological philosophy and the concept of nature for the metaphysical philosophy.

The connection of theory and practice is far closer at this stage than at the first two. For knowledge of the laws of phenomena makes it possible for us to intervene in the future development of phenomena, and the wish to be able to do this is a contributing motive in the transition from the first two stages to the third. See in order to be able to foresee (voir pour prévoir!) is the motto of positive science. To this third stage therefore corresponds industry, in the sense of man's working-up of nature. Yet the consideration must not here be restricted to external nature. If there are also laws for man's states and actions, individual and social development, no less than the purely physical, can come to
[333] be determined by knowledge.

Comte points out (see especially "Discours sur l'esprit positif" p. 41—44) that all the various meanings in which the word *"positive"* can be used fit the philosophy he calls the positive. Positive can mean the same as *real,* and the positive philosophy will, after all, first and foremost build on facts. It also means *useful* as opposed to idle, and the positive philosophy, after all, aims at the improvement of our individual and social existence, seeks to be more than the mere satisfaction of curiosity. By the positive we also often mean *the certain and indubitable, and* it is, after all, the task of the positive philosophy to lead us beyond the constant doubt and debate of the earlier philosophies. The positive

can further mean *the precisely determined, and* the positive philosophy will, after all, put the laws, the constant and determined relations, in place of the indeterminate and shifting representations of earlier stages. Finally, "positive" is also used as the *opposite of negative, and* that fits the third stage, whose task it is to organize, whereas the second stage essentially signified a process of dissolution. The positive philosophy cannot, indeed, use the explanations with which the theological philosophy rests content; but it wages no direct battle with them. It even concedes that it is as impossible to bring a proof for the rejection of the beings in which one believes in the theological philosophy as to bring a proof for their acceptance. No one has ever brought a proof that Apollo or Minerva did not exist; the belief in them fell away when it no longer fitted together with the whole spiritual condition. Not only does the positive philosophy not polemize against the theological; it also seeks to do it full justice, by investigating historically the conditions of its rise and evaluating its influence on human development. Such an understanding was not possible at the earlier stages, where one absolute theory stood over against another, and where each single system had to regard it as its task to annihilate the others.

Only one peculiarity of the positive stage does not lie in the word "positive" itself. It is that it everywhere puts *the relative* in place of the absolute. But this relative character *[334]* follows for Comte necessarily from the other marks of the positive philosophy. When it puts the law in place of the cause, it stops at a mere relationship, at a relation. It asks: how? not: why? The inner production and the first grounds it does not seek. The single laws, moreover, cannot be reduced to a single law, and in any case there remains the relativity that arises from our always regarding the world from man's point of view. All knowledge presupposes an opposition between the individual and the surrounding world. Kant therefore rightly distinguished between the subjective and the objective, and deserves an eternal admiration because by this distinction he put an end to the absolute philosophy, although he was not sufficiently positive in his thinking to be able to prevent his successors from returning to it[81]). Although the world presupposes man if it is to be known, it can nevertheless exist without him. And although man depends on the world, he is nevertheless not a result of it. (L'homme dépend du monde, mais il n'en résulte pas. Catéchisme positiviste. *2.* éd. p. 146). In vain has materialism sought to abolish the independence and self-activity (spontanéité) of organic life by exaggerating the significance of the influence of the inorganic surrounding world. The duality remains.

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The law of the three stages Comte derives from the history of the sciences, thus by

the way of experience. But after he has set it up, he seeks to show that it can be derived from what we know of the nature of the human mind; the induction is confirmed by a deduction. For the human mind cannot do without modes of view and concepts that can bring connection among the single facts. Before the bond between the phenomena can be found by investigating the phenomena themselves, as at the positive stage, it must be found in the mythological representations that arise involuntarily, or in the metaphysical ideas that are formed by giving these representations a more abstract and constant form. At the same time the human mind has an inclination to conceive everything after its
[335] own likeness, to carry its own inner feeling over into things. That is an easy way of gaining an explanation. But if men did not trust that the explanation could easily be found, they would not engage in any investigation. —

It would be interesting to make a comparison between Comte's law of the three stages and similar thoughts in Kant, Fichte, Hegel and Saint-Simon, indeed already in Rousseau and Lessing. Beside characteristic divergences there would turn out to be agreement at least on the one point, that the spiritual life of man, after the period when authority ruled in theory and in practice, has had a period when criticism, reflection and doubt performed a dissolving work, after which the point now is to reach a standpoint with a positive and common foundation for faith and conduct of life. It is a great historical experience that has been formulated by the thinkers named in the law of the three stages. As far as Comte is concerned, it is really only the first and the third stage that stand out with a definite characterization. One often gets the impression that Comte really assigns to the metaphysical stage all such endeavors in thought and life as he harbors an antipathy toward. He sympathizes with the old Catholic system, in which he sees one of the most marvelous productions of the human spirit, and he sympathizes with that system of the future which, on the ground of empirical science, was to lead to a new spiritual community. The intervening critical and revolutionary period he looked on with aversion, although he had to concede that it had performed an indispensable work. The various marks he adduces of the metaphysical stage do not necessarily hang together. In particular it will not be possible to find a connection between the tendency to explain the phenomena by assuming specific forces or faculties and the overvaluation of the intelligence or the predominance of egoism. That tendency appears at a certain point within the history of each single science and cannot be used to characterize the common human development; in any case it can very well be united with the recognition of the central significance of the life of feeling and of the reality of the sympathetic feelings. An overvaluation of the intelligence is found perhaps quite as much among cultivators

of the special sciences as among speculative philosophers. At the conclusion of his main *[336]*
work Comte did, after all, feel vividly the opposition in which he stood to scientific specialism, and he expressed it so strongly that it had drastic personal consequences for him. But for this opposition between the positive philosophy and positive special research he finds no place in his doctrine of the three stages. And yet it is a very essential characteristic of modern times that the division of labor in the scientific domain has advanced so far that an intellectual unity, a common world-view, becomes harder and harder to reach. Here there surely lies a greater problem of the history of culture than in any of the defects Comte finds in the metaphysical stage. It was Comte's enthusiasm and sanguineness that kept him from seeing this clearly. The problem of the relation between positivity and generality — of the possibility of reaching a general world-view on "positive" ground — did not stand sharply enough before him. That is connected, as will appear later, with his shortcomings in respect of epistemology. The positive stage itself is therefore unclearly drawn in several respects. Thus it remains unclear how narrowly the positive philosophy binds us to the facts before us. How far may we depart from these by the way of hypothesis? What justification has a hypothesis that is a logical consequence of given experiences but cannot be verified? What justification has a faith that cannot indeed be derived from experience, but yet does not conflict with experience? Such questions, extraordinarily important for the relation between science and view of life, Comte passes by.

Absolute boundaries between the various stages Comte does not really assume, although he has not sufficiently emphasized the unity of human thought through all the stages. It was not his opinion that the theological stage was wholly without positivity. The influence of experience has always asserted itself, e.g. in the transition from fetishism to polytheism and from polytheism to monotheism; but as long as the circle of experience was small, its influence could not become one of principle. It is nevertheless the forward-driving power of experience that expresses itself in the metaphysical principles. These *[337]*
represent in reality groups of experience which are demanded to be withdrawn from mythological arbitrariness. Metaphysics systematized, as Comte says in one place (Discours sur l'esprit positif, p. 36), the opposition of nascent science to the old theology. As long as the positive spirit was still only little developed, it could not formulate its own tendencies, and it then used metaphysics as its organ. Thus knowledge moves forward through continuous transitions. A complete breakthrough and a general dominion of *esprit positif* Comte by no means expected at the beginning of his career. He writes to Valat on 30 March 1825: "Although I hope for some use from my endeavors, I do not hide

from myself that in my lifetime they will not be able to produce any remarkable result, even if they should be able to give an impulse to all minds that are capable of taking part with effect in the great work, something which only the future will be able to show." Later, when under the influence of his cerebral hygiene and of the mystical absorption in his ideas he had lost all understanding of what was going on around him in the scientific and the political world, he cherished a greater and ever increasing sanguineness in this respect.

Whether Comte laid most weight on the negative character of the second stage or on its significance as a necessary intermediate link, it was his conviction that the decisive battle in the spiritual world would come to be fought between Catholicism and positivism. Repeatedly in his letters to Stuart Mill he expresses the wish that a direct debate between these two tendencies — with the exclusion of Protestantism, deism and other inconsistent intermediate forms — might come about. It was to him a significant sign of the times that the Catholics now demanded freedom of teaching, while the "metaphysical" school (Cousin, Guizot, Thiers, Villemain) demanded that the state monopoly be maintained. Both schools had thereby denied their principles, the former the principle of authority, the latter the principle of freedom. But if discussion and teaching became free, he hoped
[338] that the metaphysical school's fear would be fulfilled, while Catholicism would then stand over against a new adversary it had not known before. He only regretted that the representatives of Catholicism had become inferior since de Maistre's time. That the oppositions sharpened, and that Catholic orthodoxy became ever stricter and more systematic, was for him fully understandable, after the revolutionary movement had betrayed the tendency to build spiritual and social development on an entirely different foundation from that of authority. In the face of these radical prospects de Maistre had inculcated ultramontanism as the only consistent religious system, and it had succeeded in shattering Gallicanism, whose fate other intermediate standpoints too would come to share. Comte therefore rejoices that his friend Valat, in a breakthrough of religious faith, took the step fully back to pure Catholicism, without stopping at any intermediate standpoint, although in the beautiful letter he wrote to Valat on that occasion (25 August 1843) he made no secret of his doubt that Catholicism could now bestow the peace and harmony which in its classical time it could give to those who had the highest intellectual and moral development of that day. In particular he refers in this respect to the misgivings about the dogma of the eternal damnation of those who practice the highest morality even if they reject every article of faith, a dogma which is nevertheless indispensable to the whole Catholic organism.

c. The Classification of the Sciences

By philosophy Comte understands the whole system of human concepts. This systematizing can, as we have seen, take place in three different ways. Hitherto the theological and the metaphysical method have ruled; Comte himself believes he is the first to carry out the systematization by the positive method, that is to say, by the same method as that by which the concepts or laws are won within the single sciences. It is the task of the positive philosophy to gather and order the laws that have been extracted from the facts. By thus connecting what occurs scattered in the various sciences, it seeks to *[339]* remedy the drawbacks the division of labor brings with it in the domain of knowledge. Comte regards it as hopeless to attempt to reduce all special laws to a single one, as the theological and the metaphysical philosophy had each in its own way attempted. The positive philosophy would certainly become more perfect if that were possible; but experience speaks against it, since it presents diversities that cannot be reduced. While the unity of the positive philosophy does not consist in all laws being reduced to a single one, it comes out clearly, on the other hand, in the common method, which everywhere proves applicable, and which, in spite of the irreducible diversities, points to a uniformity in the various classes of phenomena. To each of these classes there corresponds a peculiar science, and the classification of the sciences therefore becomes the most important task of the positive philosophy.

Comte's classification of the sciences orders them in *the sequence in which each of them historically entered the positive stage.* Here comes first mathematics, then astronomy, physics, chemistry, biology and sociology. But this sequence shows us at the same time a progressive transition from *simplicity* in the phenomena considered to *compositeness* (complication). The simpler a science's content is, the more quickly will it be able to run through the various stages; the more complex the phenomena that are to be investigated, the longer it will be before the childhood diseases are overcome. Biology and sociology therefore come last in the series. The simpler the relations a science concerns, the more *universal validity* will it at the same time have; for the simpler relations recur in the more composite ones. Mathematics holds for all phenomena, and the most concrete sciences, biology and sociology, have the *narrowest extension.* (This relation between généralité décroissante and complication croissante Comte could have found demonstrated in logic's doctrine of the inverse relation between the content and the extension of concepts). Finally, the methods of the various sciences also present a corresponding sequence. The simpler and more universal a science's foundation is, the greater the preponderance the *deductive* method gains over the *inductive.* Thus mathematics is the most deductive *[340]*

science, sociology the least deductive. In sociology the demonstration of the historical course of development plays the greatest part; after this has been established by the inductive way, the task then becomes to deduce it from human nature, as Comte himself has attempted by his setting up of the law of the three stages. The inductive foundation of mathematics is so simple that one often overlooks it and regards mathematics as a pure science of reason. But a pure science of reason there is not at all. Even mathematics is a natural science; its concepts, just as much as those of all other sciences, stem from experience. But the facts mathematics concerns are so simple that one can consider them as easily in fantasy as in reality, and that one can abstract from the physical and chemical properties of bodies. Number, extension and motion are properties we can represent to ourselves just as well in an indeterminate medium thought of as including all the bodies of the universe as in the real bodies themselves. This explains the apparent independence of mathematics from experience. On the foundation of fantasies of space and number the mathematical deductions can unfold unconcerned about experience. If mathematics had no foundation in experience at all, it would be altogether impossible to understand how the mathematical deductions could be applied to the study of real nature. — Between the as good as purely deductive method of mathematics and the as good as purely inductive method of sociology lie the other chief methods, each corresponding to its chief science. Through astronomy's use of deductively set up hypotheses that are confirmed by observation, we come to the experimental method of physics, to the rational method of division of chemistry, and to the comparative method of biology; this last forms the transition to the historical method of sociology.

The six groups of concepts or laws that Comte thus sets up in the "Cours de philosophie positive" he regards as irreducible. In the transition from one sphere to the other a leap takes place; a new principle comes into force which cannot be derived from the preceding one. It is in Comte's eyes materialism to seek to derive the higher (i.e. more
[341] complex, more limited and more inductive) science from the lower. Just as little can any of the single groups be made the all-embracing one. And the same discontinuity that thus for Comte prevails between the six groups of fundamental concepts prevails also within each single one of them. The various branches of physics will thus — he asserts — always remain separate from one another. It is with the organic species as with the physical forces. The concept of species would, says Comte, entirely lose its scientific significance if one were to admit transformations of one species into another under the influence of the external conditions of life. Between the plant and the animal world in particular he maintains "a real and deep discontinuity, which cannot be effaced by any

transition whatsoever."

This point in Comte is an instructive example of how tendencies of thought can turn over into their opposites. The difference of the positive philosophy from the metaphysical was to consist, after all, in laws being put in place of ideas or forces, and relative explanations being put in place of absolute ones. But Comte in reality himself sets up a Platonic world of ideas through his irreducible groups of concepts. He takes the discontinuity too dogmatically, instead of taking it as a mere fact, a fact which is perhaps due only to the imperfection of science. By his emphasis on the discontinuity in existence he sets his positivism in the sharpest opposition to German Romanticism, which sought to dissolve all differences in ideal continuity. It is, as Comte also concedes, the constant task of science to reduce the diversities, the breaks in the continuity of the phenomena, to the least possible. Every discontinuity can therefore signify only a provisional limit. And the later development of science, precisely in the period that begins with the conclusion of Comte's main work (c. 1840), has indeed led to demonstrating or conjecturing continuity where Comte had considered it impossible. Thus the law of the conservation of energy, in particular the doctrine of heat as a form of motion, and the demonstration of the identity of light and electricity have given an insight into the unity of nature for which Comte had no room in his system. In the organic domain the hypothesis of evolution *[342]* has led to the well-founded assumption that species signify only stages in, or offshoots of, connected processes; in particular it has become harder and harder to maintain sharp boundaries between the animal and the plant world. Comte already pronounced against the hypothesis of evolution as it existed in his time (in Lamarck's form); and his disciples (Charles Robin, Littré) have since been eager opponents of it. This much can be said without anticipating anything: that the relation between continuity and discontinuity has proved to be more intricate than Comte had supposed in his philosophy.

The connection between the various domains rests, according to Comte, on this, that the preceding domain, as the simpler and more universal, lies at the basis of the following, more complex and special one (even if the latter cannot be derived from the former). Mathematics is the most abstract and universal science, and within mathematics, again, the theory of number is simpler and more universal than geometry and the theory of motion; it therefore becomes the rational basis of the whole system of our positive knowledge. Comte has here overlooked that there are fundamental concepts of our knowledge that are still more simple and universal than the concept of number, namely the concepts of identity and difference, as these are conceived in logic: not only as identity and difference between magnitudes, but also between qualities. In his series

of branches of knowledge Comte has thus not got the first member in. Moreover, a difficulty comes up for him when he teaches that geometry and the theory of motion — which in abstraction and universality come right after the theory of number — hold for all phenomena, the former in so far as these are conceived in equilibrium (from the static point of view), the latter in so far as they are conceived in motion (from the dynamic point of view). Extension and motion are forms only of material phenomena, not of mental phenomena, and since Comte himself rejects materialism because it introduces the points of view of the lower science into the higher and seeks to *derive* man from the world, not merely to point out his dependence on the world, the science of mental phenomena does not fit into Comte's sequence of the sciences, in which each preceding
[343] one is to hold for the following ones, although these are more special. Logic and the theory of number hold directly for the science of mental phenomena as well as for the science of material phenomena; but this cannot be said of geometry and the theory of motion, as long as one does not, with consistent materialism, regard the phenomena of consciousness without further ado as extended in space. The classification of the sciences is not so simple a matter as Comte presupposed. His system presents here a difficulty similar to the one we have already pointed out in Hobbes (in the first part of this work, p. 264 f.). Comte himself would not concede this. He does not regard psychology as a special science and therefore gives it no independent place in the series either. He denies the possibility of self-observation. One observes — he says[82]) — all phenomena with one's mind (esprit); but with what, then, is one to observe one's mind? One cannot, after all, divide one's mind into two parts, of which the one acts while the other investigates how it goes about it! Yet Comte concedes that the feelings can be observed more easily than thinking, because, as he believes, they have a different organ from it. Our intellectual activities must be studied in their products and results or in the organs to which they are tied. Instead of setting up psychology as an independent science, Comte distributes the psychological investigations between biology and sociology. Biology alone would not let the fact that the individual's mental life is determined by the influence of history and of society come into its own. It is thus by a purely objective way that Comte wants the life of consciousness studied. It was meritorious that the objective method was emphasized so strongly in opposition to the purely subjective and spiritualist psychology of Cousin and Jouffroy. But Comte has not seen that the subjective method in reality always lies at the basis of the objective one, and he has on the whole not let the peculiarity of the phenomena of consciousness come into its own. Whereas elsewhere he emphasizes discontinuity more strongly than is defensible, here he emphasizes it too little. There

would, as already indicated, also have arisen a difficulty for his classificatory system if the qualitative peculiarity of the phenomena of consciousness were to be stressed; *[344]* they would break the sequence, since, even if they are in and for themselves simple and elementary, they are in fact tied to the most complicated material phenomena. Should they then be placed first or last? — Comte sets himself in definite opposition not only to spiritualism but also to the psychology of Condillac, which he reproaches for conceiving the single individual as purely isolated, and also for taking account only of the external influences, not of the inner conditions. The single individual is a scholastic abstraction; only the species is a reality. (Cours[2] VI, p. 590). In biological respects Comte found the most important work done in Gall's physiology of the brain, if one disregarded his unfortunate doctrine of localization. Gall's list of mental faculties would indeed, Comte concedes, have to be considerably reduced; but no fewer than ten could be assumed; between ten and fifteen faculties was what he was most inclined to assume (Lettres à Stuart Mill. p. 51—55). Gall, Comte believes, first brought the study of the intellectual and moral functions into the positive stage and maintained the significance of the inner dispositions. This admiration of Comte's for Gall conflicts with his own conception of the relation between the metaphysical and the positive stage; for Gall's doctrine of faculties presents all the marks of "metaphysics." — If one asks how Comte unites the assumption of 10—15 peculiar faculties with the unity of the life of consciousness, the answer is that this unity is, according to Comte, not original but a result, resting on the harmony between man's various tendencies and forces (Cours[2] III, p. 545). — The problem of the relation between discontinuity and continuity extends into the life of consciousness and appears here in a way that demands its own peculiar investigation, an investigation, however, that Comte does not come to take up, because he carves psychology up into a biological and a sociological part, without the central psychological questions getting room to come forward. On this point Comte is supplemented by the new English school, which lays a peculiar weight precisely on psychology. *[345]*

d. Sociology and Ethics

More than half of Comte's chief work — the last three and largest volumes — treats of the science of society (sociology — as he calls it with a word of his own coining, which in spite of all philological misgivings has now made its way through). It comprises an essential part of psychology, the whole of political economy and ethics, together with the philosophy of history. Just as Comte protests against treating the psychology of the single individual apart from the spiritual development of the whole species, so he also

protests against political economy and ethics being isolated from the general theory of society, and the later development of these sciences has proved him right in this protest[83]). Neither psychology, political economy nor ethics can be treated without regard to human development through history. In all the different scientific domains Comte points out the relation between statics and dynamics. The world is regarded statically in geometry, dynamically in the doctrine of motion. Within physics and chemistry forces are regarded partly in equilibrium, partly in activity. In the organic domain statics is represented by anatomy, which investigates organization, dynamics by physiology, which investigates functions. To sociology there belong partly a social statics, which studies the constant conditions of existence of society, partly a social dynamics, which studies the laws of the steady development of society: the fundamental idea of the former is order, of the latter progress. Statics and dynamics belong closely together, for order and progress mutually condition each other — something which neither the reactionary nor the revolutionary school has seen.

a) Social statics. Society forms a whole, within which all elements stand in the most intimate interaction, so that one of them cannot be altered without more or fewer of the others undergoing a corresponding alteration. Thus, e.g., the political and social constitution is closely connected with the whole of civilization. There is a close connection between ideas, manners and institutions, and no external authority — whether of revolu-
[346] tionary or reactionary kind — is able to accomplish anything by forcing upon society institutions which do not correspond to the prevailing ideas and manners. Naturally the institutions in turn react upon ideas and manners; but this reaction presupposes a longer undisturbed subsistence and is most strongly prominent in the childhood of the human race. The ideas and the manners, too, stand in mutual interaction. It is the task of political institutions to regulate the common life which has developed involuntarily during the progress of humanity in intellectual, moral and physical respects. The elements which play the greatest part in this development will in the end become the ruling ones in society. Authority rests on free co-operation, not the reverse.

Within social statics *ethics* finds, on an essential side, its place. The ethical laws express the solidarity in all human life. This solidarity is discovered involuntarily, as men follow their need of fellowship. It is self-contradictory to explain the origin of social life out of calculation of the utility man can have from living together with others. Utility can only show itself after a sufficiently long common life has taken place, and can therefore not be the original motive for the common life's taking place. Comte here sets himself in opposition to the explanation frequent in the eighteenth century, which stopped at the

shrewd calculation of isolated individuals. In his view an instinctive impulse to social life, independent of personal calculation, asserted itself originally; feeling here, as on the whole, precedes knowledge. In this view of his Comte is, by his own account, a disciple partly of Hume and Adam Smith[84]), who led him beyond the current theory of egoism, partly of Gall, who assumed definite organs of the brain for the social instincts. Comte finds the first germ of sociability at the stage in the animal world where the sexes are separate and where care for offspring appears. Yet even in man the egoistic inclinations at first have the upper hand over the social inclinations, which latter Comte, in order to designate them in their marked opposition to egoism, designates as altruism (from alter, one's neighbor). Nor is personal interest to be rooted out. Altruism would become a vague and unfruitful love if it did not acknowledge the need of personal satisfaction *[347]* both in others and in the individual himself. What matters is that the egoistic instinct be subordinated, and this happens through the successive development of intelligence and sympathy. Through sympathy intelligence is freed from working in the service of egoism alone, and through intelligence the eye for social solidarity is sharpened. The social state favors the sympathetic feelings. As the individual extends his feeling to embrace the whole species, he obtains through this social expansion satisfaction for his need to make himself eternal, seeing the continued life of the species as a continuation of his own life.

As isolated and single the individual is only an abstraction. The social unit is the family, where the first germs appear of the dispositions which characterize the social organism. Here the individual learns to go beyond himself, to live in others, at the same time as it obeys its most energetic instincts. It is the most intimate society. It is a unity, not a combination (union, not association). Co-operation no doubt also plays a part in the family, but not the chief part, as in the larger societies, which rest on common labor and on the supplementing made necessary by the division of labor. By performing a definite labor each individual and each family exercises a social function. The authority which is to direct this function must proceed from society itself and found its power on the confidence it inspires and the free adherence it finds. Every society will inevitably come to have a government. While in the individual life it is the personal instincts, and in the family sympathy, that rule, in the larger society it will essentially be the intellectual faculties that act to order it. A certain intellectual community is necessary for a society; the co-operation of interests and immediate sympathy are not sufficient. Positive science has a very great significance for ethics, for it will be a matter of determining as exactly as possible the real, direct or indirect influence which every action, inclination or feeling has on human existence, both on the single individuals and on society as a whole.

[348] It was, according to Comte, the great merit of Catholicism that it emancipated morality from politics, to which in the polytheistic period it had been wholly subordinated. This emancipation found its expression in the doctrine of the two powers and of the independence of the spiritual power from the temporal. But Catholicism showed itself unable to satisfy the growing intellectual needs of men, and metaphysical criticism began its dissolving work. Moreover Catholicism furthered egoism by letting the individual be occupied with his own salvation, whereby it no doubt gained a mighty motive, but hindered the free and pure unfolding of the sympathetic and high-minded feelings. Only the positive intellectual tendency directly favors the unfolding of these feelings by impressing upon us that our whole development takes place *within* society, so that it is the individual and not society that is an abstraction. In social life the egoistic inclinations must in many ways be checked; only the social drives can unfold freely, and the heightened activity which the unlimited horizon thus opened brings with it contains a source of happiness, of inner satisfaction, which is independent of external rewards. The concept of duty springs from the view of the whole (esprit d'ensemble) which positive philosophy asserts, and which shows the single individual as a member of the whole species, so that the rules for its action must follow from a universal order of things, not merely from purely individual interests. The highest idea in the ethical domain is the idea of humanity as that whose development is uninterruptedly determined by the co-operation of all individual and social organs. Comte polemicizes against the sharp separation between private and public functions. This distinction is purely empirical and comes forward only in times of transition, when a new civilization is developing, and when the new elements have difficulty in finding their place. It did not exist among the Greeks and Romans, nor in the theocracy of the Middle Ages; it first came forward at the end of the Middle Ages, especially after the industrial upswing which followed the abolition of serfdom. The masses of the proletariat have since that time not truly been incorporated into the social system. And only with such an incorporation comes the
[349] right personal self-respect, which consists in feeling oneself a fellow worker within a great whole. One day such a feeling will ennoble even the humblest occupation, namely when through *positive* education the consciousness has become alive of the significance of every single endeavor for society as a whole.

It was Comte's conviction that what was needed for the time being was political quiet, the avoidance of great political upheavals, so that ideas, feelings and manners could undergo the involuntary change which the positive stage will bring with it. The most important social difficulties were in his eyes not political, but chiefly moral, and

could be overcome only through the influence of ideas and manners. In him the concept of duty decidedly preponderates over the concept of right. His philosophy contains no elements for a philosophy of law and political theory proper. For he holds the belief that institutions will easily develop once ideas and manners are changed. Here, already in his first and proper chief work, there asserts itself a utopian tendency, arisen as a reaction against the preceding period's strong emphasis on individual rights as well as against the many constitutional struggles with which the age was occupied. He thought that a comparatively short-lived "progressive dictatorship" would suffice to bring about the quiet ripening of ideas and feelings which was a necessary condition for the Occident to arrive at its lasting social institutions, and for the great Revolution to obtain its positive counterpart.

β) Social dynamics. One important law which belongs to social dynamics we already know: the law of the three stages. In the intellectual domain, as Comte emphasizes, progress can most easily be demonstrated. But to the different stages of intellectual development there correspond, as we indicated in the exposition of the three stages, stages of development of a social and political kind. To the theological stage corresponds the *military.* Its significance consists in this, that regularity and discipline, necessary conditions of political organization, are implanted and developed. Forces are gathered about pressing common purposes. Just as the first spiritual authority necessarily had to be theological, so the first governments necessarily had to be military. Only external *[350]* force could gather and hold together. And only through war does society grow at the earliest stages of development. War in turn brings slavery: in order that the warriors might have free disposal of their force, the economic labor had to be done by slaves. To the transitional period which in intellectual respects is designated as the metaphysical there corresponds in the social and political domain a period in which the *jurists* ("the legists") rule. A defensive military organization takes the place of the earlier offensive one. The warlike spirit gradually gives way to the productive spirit. The middle classes push forward and demand political rights. It is the task of the ruling jurists to weigh the different claims against one another. It is a vague and restless transitional stage, the social phase in which we still find ourselves. To the positive stage corresponds the *industrial* phase, in which the producing forces determine the arrangement of institutions and the distribution of power. Social questions now take the place of political ones. The proletarians gradually discover that political revolutions cannot solve the great social problems whose sting they feel most of all. They will naturally sympathize with the endeavor of positive philosophy to set duties above rights, so that the attention of all

may be directed to solving the task which is the properly social one: to procure for all access to spiritual development and access to work. There will therefore be a natural understanding between proletarians and positive philosophers.

This demonstration of humanity's solidary course of development becomes of great significance for ethics, which thus belongs not only to social statics but also to social dynamics. Not only is the social feeling strengthened by the demonstration of the solidary development, but an essential part of the content of ethics can be determined only on a dynamic basis. For the task of ethics is to contribute to the development of the specifically human qualities — in contrast to the animal and vegetative ones. And here ethics has its presupposition not only in what social dynamics teaches about the different
[351] stages, but already in the results of comparative biology, which show how the animal functions, the higher one comes up in the series of animals, play an ever greater part in relation to the vegetative ones. In man the specifically human qualities, intelligence and sociability, in turn develop in relation to the animal ones, which Comte — following Gall's interpretation of the brain — finds expressed in the stronger development of the frontal region of the brain in relation to the part of the brain which lies nearest the spinal cord. Within the human species Comte assumes an ever further-reaching development of intelligence and sympathy (altruism), the condition for the single individual's being able to make himself one with the species. Although Comte is an opponent of Lamarck's doctrine of evolution, he nevertheless acknowledges the great influence which uniform and steady exercise has on faculties and qualities. The nobler tendencies in our nature will through social life be ever further developed; the baser instincts will gradually be weakened, partly by self-control, partly by lack of exercise, or they will gradually be compelled to serve the social order itself. — A more definite form Comte is unable to give his ideas on this point, partly because he lacks the doctrine of the displacement of motives founded by Spinoza and Hartley, which English psychology later carried further, partly also because he could not make use of the close connection which evolutionism assumes between the development of the individual and of the whole species.

In concluding his exposition of human ethics, Comte declares that when one wishes to compare the morality of positive philosophy with religious morality, one must not forget that while the former is scarcely thought out and cannot work through regular institutions, the latter is fully developed through centuries of spiritual labor and has for long ages found support in a great social apparatus. In his later years — after the remarkable nervous crisis in the middle of the forties — he became more sanguine and no longer held fast to the conviction that we stand only at the beginning of the development.

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e. Theory of Knowledge

Although Comte has not made the theory of knowledge the object of special investigation and exposition, the whole of positive philosophy nevertheless rests clearly enough on definite epistemological presuppositions, and in some places (especially in the last volume of "Cours" and then in "Discours sur l'esprit positif") he expresses himself on it at greater length. It will be of great interest to attempt a determination of Comte's epistemological standpoint.

Positive philosophy is, according to Comte's express and repeated declarations, not empiricism. Already in 1825 he shows in an essay *(Considérations philosophiques sur les sciences et les savants),* that an absolute empiricism is impossible. Science does not consist in a mere accumulation of facts. Its essential task is the connection of facts; it consists of laws, not of mere facts. No isolated fact is incorporated in science; it becomes so only when the isolation is removed by its being connected — at least by means of a hypothesis — with another fact. Comte even makes the interesting remark (Cours[2] IV, p. 300) that no isolated and purely empirical observation can be certain. It was indeed also, as we have seen, through the necessity of a bond between phenomena that Comte made it intelligible that our knowledge must pass through the theological and the metaphysical stage before it can reach the positive. At the positive stage the laws of phenomena are sought, that is to say a factual connection between them. The connection can be brought about in two ways. Either it is a question of simultaneously given phenomena, and then the explanation is gained by demonstrating similarity in the relations and laws which hold for different groups of phenomena. Or else it is a question of successively appearing phenomena, and then the task is to demonstrate a continuous connection between them. In the first case we have a static, in the second case a dynamic explanation. Whether we explain *par similitude* or *par filiation,* we connect phenomena in such a way that we can foresee their occurrence. And in both cases we satisfy our mind's need of unity and find again the constant in the midst of the variations. (Discours sur l'esprit positif, p. 20—21). *[353]*

Comte's theory of knowledge does not go beyond these developments, interesting in themselves, on this point. He does not pursue the connecting activity of the mind more closely, although he regards it as fundamental: *tout se réduit toujours à lier!* A closer investigation would here have led him further into subjective psychology than he wished and considered possible. Nor does he enter into the question what validity can be ascribed to those laws of similarity and of succession. He expresses himself somewhat

waveringly on this point. Now he says that science has nothing to do with the first fundamental propositions; they spring up involuntarily in the human mind and cannot be discussed. (Discours sur l'esprit positif. p. 46). Here he approaches Reid's theory of knowledge built on common sense. Now, on the other hand, he says that the principle of the invariability of natural laws, on which positive science rests, cannot be grounded a priori, but will always in the end prove to rest on simple observation and induction (Discours. p. 17). This last consideration would evidently, if carried through consistently, drive him over into pure empiricism and into the difficulties in which it finds itself with regard to the causal principle, difficulties which first came clearly to light through Stuart Mill's significant resumption of Hume's problem. — When Comte expects that the principle of the invariability of natural laws will gradually make its way in all domains, it is, as already remarked, because he thinks that an irresistible analogy will compel us to carry it through. It will, as he declares in one of his older essays, in the long run be impossible for the human mind to think positively in some domains, metaphysically or theologically in others. It will strive after unity in method and theory. Here, then, the power of habit is at work, to which Hume already pointed. But habit can explain only the principle's psychological power over the mind, not its real validity. The sting of the problem of knowledge proper Comte has not felt. He sought to systematize positive
[354] knowledge; but he did not set himself the task of discussing the ultimate foundation of this knowledge.

From another side, however, Comte approaches the problem of knowledge, namely in his emphasis on the relativity of knowledge. He finds it, as we have seen, to be an essential characteristic of positive philosophy that it everywhere puts the relative in place of the absolute which was sought in the earlier forms of philosophy. The relativity of knowledge he grounds in two ways. In the first place, positive science can teach us only relations of similarity and of succession between things, not the absolute causes of things, nor the very inmost nature of the things between which those relations obtain. The relations themselves, too, stand as mere facts which cannot be fathomed further. In the second place, all our knowledge is determined by the relation between our organism and its environment — Comte even speaks of "the great elementary dualism between the understanding and the milieu" — and in such a way that the environment does affect and determine the understanding and gives it its material, but the working up takes place, as in the process of nutrition, according to the laws of our own organization and in its forms. There is therefore a relation partly to the subject, partly to the object, which asserts itself in all knowledge. In this biological epistemology[85]) Comte feels himself a

successor of Aristotle, Leibniz and Kant (Cours[2] VI, p. 620 ff. Catéchisme positiviste, p. 150 f.). It leads him to see that our knowledge can reach only an approximation to reality; but on account of his practical standpoint he does not come to discuss the question how far our knowledge can rightly be said to be a reflection of reality. It is enough for him that the knowledge we have can be used practically for orientation. On the other hand he lays great weight on this, that the very subject to which knowledge stands in relation undergoes change and development. All knowledge is determined by the stage of development which the individual and the species have reached. All our knowledge therefore has a historical character. Sociology leads, as already the doctrine of the three stages has shown, no less than biology to an emphasis on the relativity of knowledge. As long as biology and sociology were not yet developed in positive form, the relativity *[355]* could be overlooked, as happened in the whole period when mathematics stood as the chief science. Now the scepter has passed to sociology. It is sociology to which it belongs to state the concluding scientific points of view.

That Comte thus came to stress the biological and sociological point of view for our knowledge, or in other words came to see knowledge as determined by man's nature, needs and degree of development, brought with it an important change in his philosophical standpoint. From the first he had laid the chief weight on "the necessary and rational subordination of the concept man to the concept world" (Cours[2] III, p. 188); now he lays more and more weight on the subjective side of knowledge. Knowledge comes to stand for him essentially as the satisfaction of a subjective need, and he regards every means of reaching such satisfaction as justified. The need of knowledge becomes for him an artistic need. He asserts the justification of following the simplest hypotheses and disregards the necessity of a confirmation by experience. In what relation the satisfaction of the need of unity and simplicity in the conception of the world stands to positive reality he does not discuss, occupied as he more and more became with the mystical world to which he had gained access after the nervous crisis and the "méditation exceptionelle" it brought with it. His plan now became to set a subjective system in place of the objective system which had been set forth in "Cours". Hitherto he had explained man by the world; now he would explain the world from man.

f. Comte as Mystic

Comte's great work as a thinker was done with his exposition of positive philosophy. But later this stood for him only as an introduction to the highest mysteries. Of the psychological causes of this change in his views we have already spoken. We shall now

merely mention briefly what the ideas were which occupied him in his last years. If
[356] they have no significance as scientific thoughts, they nevertheless have significance as symptoms.

In his positive philosophy Comte had gone from the world to man; he had advanced from the abstract and universal relations to the more complicated — and man is the being that lives under the most complicated relations. Now he will go from man to the world, will with full consciousness take up the subjective standpoint which men in the childhood of the species took as their basis without being conscious of it. The world is now regarded as the supporting foundation of human life. From the most complicated — and therefore the highest — standpoint in existence the lower stages are now to be looked back upon. Within human nature it was in the exposition of positive philosophy necessarily intelligence that had come to stand in the first line. It is through intelligence that we know the laws of existence, and it is its influence on feeling that leads to new manners and institutions. Now feeling is set in the first line, and it becomes the task to enlighten intelligence by means of the heart. In place of analysis and the specialties synthesis, the unified conception, is now set. But only a subjective unity in our knowledge is possible. The forces of existence cannot, as positive philosophy has established, be reduced to absolute unity. But a unity in conception can be gained by regarding the whole of existence (so far as we know it) in relation to humanity. But Comte now feels the necessity of making a change in his earlier classification of the sciences, namely that of separating ethics from sociology and setting it up as the seventh fundamental science. For when humanity is to be the point of unity about which all thoughts gather and from which they go out, then the task is to immerse oneself in human nature in its whole fullness. But in sociology it is properly only thought and action that become of significance; the feelings play a part there only as motives, and since the different motives can on the whole balance one another, there will from a purely sociological point of view be no reason to lay great weight on the feelings. Ethics now becomes precisely the most complicated science of all by bringing forward an element
[357] which in sociology still stood as subordinate. The addition of ethics as an independent member in the system of the sciences thus takes place in accordance with the law of advance from the more abstract to the more concrete[86]). All sciences are now to be regarded as parts of ethics and to be cultivated with it in view, so that each single one is treated as preparation for the following, more complex one, until we land in ethics, the concluding science. Care must be taken that analysis does not again gain the upper hand, and all studies should be dropped which do not further our knowledge of the order of

nature and our ability to adapt ourselves to it, or which cannot serve as guidance for our activity. New chief works should be composed which can reduce the fundamental sciences to their necessary extent. — Characteristically Comte now (in *Politique positive* and in *Catéchisme positif)* calls the chief doctrines of positive philosophy "the positive dogma". His thought seeks rest, and he becomes impatient with every investigation that leads away from immediate absorption in the religion of humanity. Feeling stands for him high above knowledge and action, which have value only through their results, which moreover are dependent on external circumstances, while feeling causes direct and inner satisfaction. And from this follows in turn the precedence of art over science. The source of art is in feeling. It gently leads the abstract considerations of the theorists back to reality, while it instills in the practical men a noble interest in great ideas. Its physiological origin lies in the involuntary movements which are bound up with the feelings and express our inner states, while they also react upon these. It forms ideal types which by steady contemplation perfect our thoughts and instincts. In the priests of the religion of humanity philosophy and poetry are to melt together.

The religion of humanity, of which Comte regarded himself as the founder, is a worship of humanity as *le grand être,* in which we all have a share, the sum of all deceased, now living and future beings who in a free way have worked for human progress and happiness in small or great circles. It is an ideal concept; for only those will live in the continued memory of the species who have devoted themselves to it *[358]* and worked for it. In drawing up a positivist calendar, in which every month and every day is named after men who have had significance for the development of humanity, Comte himself has given directions for a public cult, which consists in periodically recurring, enthusiastic and grateful remembrance of the benefactors of the species. In the private cult persons who have stood nearer to the individual will represent for him the ideal of the species. Those who have put themselves in the way of progress — as, e.g., the Emperor Julian and Napoleon, "ces deux principaux rétrogradeurs que nous offre l'ensemble de l'histoire" — will be remembered in the positivist cult only in order to receive "the deserved periodical flogging". The single individual's dignity rests on its absorption in "the great being" through the smaller and larger circles (family, fatherland, etc.) within it. Toward the preservation and perfecting of this being all thoughts and actions are to be directed. Altruism (vivre pour autrui) contains at once the highest duty and the highest happiness.

The direction of the cult as well as of education is, according to Comte, to be entrusted to a priesthood consisting of encyclopedically trained philosophers, who are also to be

poets and physicians. Through their authority, resting on free acknowledgment, they will gain influence on public opinion. Comte seeks on the whole to carry over into the religion of positivism analogies to "everything great and heartfelt which the Catholic system of the Middle Ages has realized or at least sketched". Sociocracy, the social order he has in mind, is far more akin to theocracy than to the intervening individualist period with its "endless agitation". Indeed, Comte even seeks further back. Precisely because positivism has completely worked itself free of the old prejudices, it will be able without misgiving to resume the fetishistic view of nature and ascribe life and soul to everything in nature. That will give language force and favor the artistic imagination, and it will quicken the feeling for everything that can serve the subsistence and development of "the great being". Every animal species is regarded as a human being that has been arrested
[359] in its growth (un grand être plus ou moins avorté). It is the time of images that now comes again after the long critical period. The new religion then regards cosmic space as "the great medium" in which the earth, "the great fetish", has formed itself. The great fetish has in turn let its mighty elementary forces rest and has sacrificed itself out of longing that "the great being", in which the highest perfection appears within a narrow frame, should develop. (The highest beings are, as Comte already taught in "Cours", also the most dependent). The great medium, the great fetish and the great being make up the positivist trinity. —

Comte also gives hints concerning the constitution of the future sociocracy. The idea of right is to disappear completely. No one possesses any other right than that of doing his duty. Individuals are regarded not as separate beings but as so many organs of the great being. The directing power in external affairs belongs to the industrial chiefs (bankers, manufacturers and landowners). These *patricians* are by their wealth freed from covetousness and will be able to ennoble labor by taking it on by free choice, motivated by the highest personal feelings. Their task is to direct industrial labor in such a way that all can come to share in family life, which is the foundation of human happiness. It is in the interest of the proletarians that capital be concentrated in the hands of a few patricians; thereby a common and intelligent direction becomes possible. The time of the middle class is past. The patricians correspond to the nutritive organs in the organism. Over against them stand the cerebral organs, the organs of reason *(the philosophers),* of heartfelt feeling *(the women)* and of energy *(the proletarians).* While the patricians chiefly represent order, the proletarians represent progress. It is the business of the philosophers (or priests) to find out what the welfare of "the great being" demands; the business of the women is to awaken the right feeling to drive the work. Philosophers

and women will sympathize with the proletariat and find support in it against possible encroachments on the part of the patricians, and the proletarians have in public opinion and in refusal of co-operation a means of preventing abuse on the part of the spiritual or temporal authorities. — *[360]*

Comte's utopia has the interest which such descriptions can have when they come from eminent minds with an eye for the tendencies of the age. One traces his historical presuppositions, just as in Plato's Republic a multitude of features from Greek conditions have been found again. His sketch of a religion of the future has significance partly as the testimony of an earnest and heartfelt man that all is not done with criticism and negation of the views of life of the past, partly in that it is the idea of humanity and of love that he makes the center. What he would not admit was that, provided only this idea can come to dominion, the religious representations would be able to develop with individual freedom and in great multiplicity among those who feel a need of them. Hardly many will share the taste Comte displays in his arrangement of a cult of the future, apart from some brilliant thoughts, e.g. about a new calendar with the names of historical personalities in place of the unknown names of saints. Many will think that the real religious problem only begins where Comte's religion ends, namely with the question how the development of the world is related to the development of the human species and its ideals. Comte's new trinity gives no answer to this. — Just as little as Comte will give individuality freer play in religious respects, just as little will he do so in political respects, so that he even wholly excludes the concept of a legal order and of a representative constitution. Public opinion and the strike will hardly be sufficient correctives when there are no fixed forms of public and private life to which those in power are bound. — Most characteristic is the suppression of free inquiry in the face of the closed "positive dogma". It can be excused only as an effect of the cerebral hygiene which the tired thinker, after his energetic labor, had had to impose on himself. But it would conflict with the spirit of his own most excellent work if his proposals were followed. *[361]*

B. John Stuart Mill and the Renewal of English Philosophy in the Nineteenth Century

In England the clash between the thought of the eighteenth and of the nineteenth century was not so violent as in France and in Germany. The English people's wonderful ability to carry out radical upheavals without breaking the continuity of its development displays

itself also in the philosophical domain. There does indeed appear (as already remarked in the first part), after the time of Hume and Adam Smith, an ebb in English philosophy. In the domain of epistemology Hume had drawn such radical consequences from the preceding English philosophy of experience that there was hardly anything to be done for the time being from that standpoint. The practical, political and religious interests, moreover, came into the first line. The men who held fast to the philosophical interest at the transition to the new century, Jeremy Bentham and James Mill, are accordingly guided essentially by practical interest. The ethics and psychology of the older English school were taken into the service of the ideas of reform. And here it is remarkable to see in how different a way essentially the same circle of thoughts is made fruitful in England and in France. There is not the heaven-storming quality in Bentham and James Mill that there is in the French philosophers of the Enlightenment; but there is a concentration of thought on definite purposes and a sense for practical application of general principles, which forms a salutary contrast to the pointed phrases and declamations of the revolutionary Frenchmen. The English do not like to fire off their artillery into the air; they care about hitting their target, even if the thunder of the cannon sounds less resounding. Through their practical sense and their steadfast upholding of the principles of the older English school those two men mark significant extensions of the best in the eighteenth century into the new time. They brought up a young generation which could receive the thoughts of the new time without forfeiting the yield of the preceding time's intellectual
[362] labor. Stuart Mill's personality as a thinker obtains a central position in the history of English philosophy by the fullness and understanding with which he is able to take up into himself the thoughts of the old and of the new time and to work them over in such a way that with him a whole renewal of English philosophy sets in. He resumes Hume's problems on a broader basis, at once with greater consistency and with greater many-sidedness. With conscientious energy he holds fast to the points of view of the older school as long as it is possible for him, without concealing the difficulties and without lacking an eye for the significance of the new experiences. He has, as regards both theoretical and practical problems, carried out a work of survey and of reckoning of the greatest significance for the spiritual and social development of the nineteenth century, and in the forties and fifties he was undoubtedly the greatest philosophical thinker of the age. Then came the doctrine of evolution with its new points of view.

Critical philosophy and Romantic speculation, however, also made their way into the English life of the spirit and gave it mighty impulses. A deepening and an enlargement became possible for which the older English school, left to itself, could hardly have

prepared the way. Against empiricism the tendency which Schiller had called "the philosophy of the ideal" asserted that there were spiritual values which the philosophy of experience had not been able to explain. In England this tendency was represented by Coleridge and especially by Carlyle. In the latter Goethe's humanism and Fichte's doctrine of personality are transformed into a peculiar historical view of life, which has intervened powerfully in the development of many individuals and given forceful guidance for preserving, in the struggle with the problems of the age and amid the century's sharp oppositions, the inmost kernel of the personal life. There was one question, however, which Carlyle always pushed aside, in so far as he did not answer it by cutting the knot. It was this: how this assertion of the value of the personal life is related to the attempts to find a thoroughgoing conformity to law both in the spiritual domain and in the rest of existence. Here lies the epistemological problem which in the spirit of critical philosophy is taken up by inquirers such as Whewell and Hamilton, and *[363]* which Stuart Mill then treats in the direction of absolute empiricism.

1. Philosophy in England before 1840

a. The Philosophy of Reform

Jeremy Bentham (born in London on February 15, 1748, died June 8, 1832) belongs rather to the history of jurisprudence and of philanthropy than to that of philosophy. But he intervened powerfully in the development of philosophical ethics by the energy with which he declared and applied the principle that every action and every institution is to be judged by the tendency it has to further happiness and keep unhappiness away. He would found all ethics and all jurisprudence on the one fundamental proposition that well-being is to be preferred to ill-being. Only prejudices, especially religious prejudices, and the lust for domination could, he thought, hinder the open acknowledgment of this principle, and we therefore also see men follow it everywhere where they use their reason without external and internal hindrances. The principle of the greatest possible happiness for the greatest possible number, or, as it is also called, the principle of utility (by utility is understood the tendency to produce happiness), is according to Bentham a self-evident principle: it grounds all practical valuation, but itself needs no grounding. So he sets it up already in his first writing, which appeared in the year of Hume's death. (*A Fragment on Government.* 1776. Chap. 1. § 48. Here *The principle of utility* is said to be "the principle which contains the only reason [reason] that does not in turn depend on any higher reason, but is itself the one, fully sufficient reason [the sole and all-sufficient reason] for every kind of action".)

This principle, however, Bentham did not discover himself. It had been set up in a similar formulation by Hutcheson and by several after him. Bentham himself declares (in *Deontology),* that he has it from Priestley. But it was, he says, in an interesting statement
[364] about his course of development (Fragm. on Gov. Chap. 1. § 36 Note), the reading of the third part of Hume's "Treatise," that made the scales fall from his eyes. He had been brought up in an orthodox Tory family, in strictly conservative views. He had regarded Charles I as a martyr and rebellion as an impiety. His legal studies had later led him to the natural-law doctrine of the original contract, from which it was deduced that when the prince broke his obligations, the subjects were no longer obliged to obey him. This doctrine he now broke with, influenced by Hume, partly because no conclusion of any such contract could be demonstrated in history, partly because, even if one assumed an original contract, there still remained the question to be answered *why* one should keep contracts or promises at all. The only possible answer to this question is: the keeping of contracts is to the advantage of society, and therefore each single individual ought to keep his promises; if he does not keep them, he ought to be punished, for the suffering the punishment brings him is amply outweighed by the good that the keeping of promises brings society (Fragm. Chap. 1. § 42). Thereby the theory of natural law is replaced by the theory of utility, the original contract by the principle of utility. And this transition has, as Bentham himself emphasizes, the great significance that we now pass from the world of fictions over into the world of facts. For whether an action or an institution is useful or not must be demonstrable in experience. Discussion will therefore now consist essentially in the ascertaining of facts. The right of free criticism of actions and institutions is therefore of extraordinary significance. There could — Bentham thinks — be no danger in such freedom of expression. For every institution that is really useful will have those as its defenders who draw utility from it, and so there will be no lack of answers to charges. — In his later writings Bentham preferred to speak of *the principle of happiness* rather than of the principle of utility.

Through his principle of utility Bentham came to make front on two sides. He came forward as a severe critic of the traditional institutions and especially of the chaotic English legislation. His proper field is the censure of the existing legal and social order
[365] (censorial jurisprudence, see Fragm. on Gov. Preface). He brought forward and chastised existing abuses with such effect that it has been said of him that he worked against his own fame: the evils he attacked disappeared to such a degree that later one could not sufficiently appreciate the significance of the labor that made them disappear! — In the Italian philosopher of law Beccaria he saw in this domain his most important

predecessor. Beccaria had already set up the principle of the greatest happiness for the greatest number as the purpose of laws and applied it in his criticism of criminal law. Bentham gave it a far more comprehensive application. While he thereby came into sharp conflict with the conservatives, he opposed on the other hand the French revolutionaries' appeal to natural law and their setting up of general rights of man. Rights the individual has, in Bentham's view, only in so far as the welfare of the whole society brings this with it. And he thought that the proclamation of the rights of man led to favoring egoism, which was strong enough already, while what mattered was precisely to get the individual to submit to the sacrifices the good of the majority demanded. When nevertheless in revolutionary declamations people spoke of natural and inalienable rights, that was for Bentham only "nonsense upon stilts".

In his chief philosophical work *Principles of Morals and Legislation,* which appeared in 1789, Bentham goes more into detail with regard to the application of the principle of utility. He investigates which feelings of happiness are to be preferred to others, and in this demands regard for the certainty of the feeling of pleasure, for its intensity, duration, propinquity, purity and capacity to propagate itself. He investigates further by what rewards and punishments men can be led to perform actions that bring happiness, and in close connection with this, what different motives determine men's actions in general, and what moral value they have. Everywhere he shows himself strong in distinction, definition and classification. In his domain he is something of a scholastic. Of particular interest is his investigation of motives. For it is by no means the case that he overlooks their significance. On the contrary, the principle of utility itself gives him a standard *[366]* for the valuation of the inner sources of actions. Those motives are to be called good which prove to work in the direction of harmony between the individual's own interests and those of others, bad those which lead to separating the interests. The motive that will most surely of all lead in the direction of what the principle of utility demands is benevolence (good-will, benevolence): "the demands of utility are neither more nor less than what the most comprehensive and enlightened benevolence requires". Then comes the need of others' esteem, the wish to win their love, religion, and finally the individual impulse of self-preservation and the wish for enjoyment, advantage and power. (Princ. of Mor. and Legisl. Chap. X. § 29—42).

Bentham's chief interest was the reform of legislation in a humane direction: codification of the laws (the very word "codification" he coined), so that everyone can know and understand them, and so as to keep abuses and costs away, improvement of the prison system, and the development of the constitution into a democracy through the

introduction of universal suffrage. In his struggle for the reforms he uses the principle of utility as his starting point. And since he constantly proceeds from it, he takes it as a dogmatic principle, as settled, and feels no need to investigate on what its acknowledgment is grounded. He does not see that a similar question can be put to him to the one he himself put to the adherents of natural law. As he asked them why one ought to keep one's promise, so one can ask him why one is to work for the happiness of the many. Logically self-evident it is not. If one investigates what possibilities of an answer to this question are to be found in Bentham, one gets no clear result out of it. As we have seen, he declares the demands of the principle of utility to be the same as the demands of comprehensive and enlightened benevolence. So far one might believe that the acknowledgment of the principle of utility as a standard springs, according to Bentham too, from sympathetic feelings. In Bentham personally it probably was so. He was a man easily moved to pity and sympathy, just as he was also, for his own part, very susceptible to pain. But he lacked the ability to enter into the moods and states of others, and moreover the many prejudices and hindrances springing from
[367] egoism which he met on his way did not give him any great notion of the power of disinterested sympathy in the world. While his own sympathy led him not only to work his whole life long, theoretically and practically, to further human happiness, a labor which brought him himself neither advantage nor external honor, and while it moreover inspired in him a great hope that the demands of the principle of utility would make their way, he nevertheless built, both in his activity and in his hopes, rather on there proving to be among men's egoistic interests a harmony which made it possible for the general happiness to be reached, provided only each individual energetically and shrewdly worked for his own happiness. It is the harmony of well-understood interests on which he properly builds. In a writing which was composed on the basis of his notes by one of his disciples and appeared after his death *(Deontology)* this side of his view comes out especially strongly. He here takes a standpoint which recalls that of Helvetius; he also always rated this French author very highly. Although disciples of Bentham have protested against the views developed in the said writing being without further ado ascribed to Bentham himself, it is nevertheless not improbable that it contains a part of Bentham's views of whose relation to other parts he himself had not made himself clear. All virtue is here traced back to individual prudence, which sees that one must serve others in order to receive services from them in return, and the hope for the future is built on public opinion gaining greater and greater significance, and on its becoming more and more insight into the harmony of well-understood interests that determines

the judgments of public opinion. The notion of this great harmony inspired Bentham, just as Adam Smith was inspired by the harmony of economic interests. Bentham has properly extended what in Smith is political economy to make up the whole of ethics. He builds on a conception of the species as consisting of isolated individuals, each striving eagerly to procure for himself as many goods as possible at as little cost as possible. Even so great an admirer of Bentham as Stuart Mill has nevertheless declared that he knew only "the business side of human life" (the business part of human affairs)[87]). Only it *[368]* must be held fast that Bentham himself did not stand in any "businesslike" relation to his views and endeavors[88]).

Early on Bentham's ideas gained influence on men who were active in public life, and when he learned what resistance his ideas of reform met with on the part of state and church, he himself became more politically interested and more anticlerical than he had previously been. But no school proper or closed party formed itself around him. He lived very retired and worked properly only through his writings. Toward the end of his life his tendency got its own organ (Westminster Review), which entered into sharp opposition to the journals of the old parties (Quarterly Review and Edinburgh Review). Bentham's friend James Mill and the latter's son Stuart Mill were the most active contributors. The adherents of Bentham's principle of utility were commonly called Utilitarians, a word which Stuart Mill (Levned. Danish trans. p. 81) believes he brought into currency, although Bentham himself had already used it. Like the word positivism, the word utilitarianism, too, designates rather different views; for an ethical view is not completely characterized by the standard it applies in valuation. The ethical problem contains more questions than this. A utilitarian (in a wider sense) need not, in particular, conceive the psychological foundation of ethics in the same way as Bentham.

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Among Bentham's collaborators James Mill stands in the first rank. His philosophical significance consists especially in this, that he seeks to supply the psychological foundation which Bentham's ethics lacked. In this respect his education at the University of Edinburg, where he heard lectures by Dugald Stewart, a disciple of Reid, had given him a preparation. Later he is influenced by Thomas Brown, whose philosophical standpoint forms a transition between the older Scottish school's appeal to induction, instinct or common sense and pure association psychology. But on James Mill's definitive standpoint the study of Hartley had the greatest positive influence. He was born in the south of Scotland on April 6, 1773, and grew up in humble circumstances. His father was a *[369]* village shoemaker. His mother was of a more well-to-do family, and it was owing to her

self-respect and ambition that the son's good intellectual abilities received sufficient development. Later James Mill received support from a landowner, Sir John Stuart, in whose family he became tutor. In Edinburgh he studied theology, but it seems that he early gave up the thought of becoming a clergyman, even if his theological views only much later (and still later than his son relates in his Autobiography) developed in a radical direction. In his thirtieth year he went to London, where he made his living by literary work. These became straitened years for him, when he married and had a numerous family which he had to support with his pen. In this hard struggle for existence he nevertheless did not give up the struggle for humane purposes, any more than he abated his gradually developed radical views. The chief work from this period of his life is *History of British India,* in which he gives a sharp criticism, but one built on thorough knowledge, of the East India Company's government in India. It is characteristic of English conditions that Mill was nonetheless appointed in the Company's service when he applied for a vacant post; one held to the knowledge and was not afraid of the criticism. Soon he rose to a very high position and gained great influence on the Indian administration. Of great significance for him was his acquaintance with Bentham, which became a firm friendship that lasted in spite of passing ill feelings and clashes between the two sharply marked and self-assured men. Mill worked for the application of the principle of utility in a whole series of philanthropic and political domains. His endeavors in these directions are still interesting and instructive to follow; Alexander Bain gives in his *work James Mill. A Biography* (London 1882) a detailed survey of his labors from year to year. Mill worked not only through his writings, but quite as much through his conversations with a circle of younger men who gathered around him, members of the radical political party and contributors to the Westminster Review. He is the spiritual father of the first parliamentary reform. With a zeal and a sharpness which at times became too strong
[370] even for Bentham, who otherwise did not mince matters, he attacked the aristocracy and the hierarchy. From his study he directed the great class struggle which brought English politics onto new paths. He showed himself a more practical politician than Bentham by not proclaiming universal suffrage without further ado, but restricting the program to the emancipation of the middle class. It was his conviction that the suffrage could be extended to larger circles only as enlightenment and instruction spread. From a politics which rested on general popular enlightenment and suffrage and was guided by the principle of utility, he expected an unlimited progress for the human race. For him, as for Bentham, the decisive thing was that men should be enlightened about their interests and be allowed to follow their reason. He proceeded, like Bentham, predominantly

deductively. The principle of utility was for him too an eternal truth from which it was only a matter of drawing the consequences. Neither he nor Bentham had a sense for the manifoldly nuanced concrete circumstances under which the general ethical and legal-philosophical principles were to be applied. To give laws was for Bentham far from so difficult a matter as Montesquieu had thought. Bentham did indeed write a special essay "on the influence of time and place on legislation"; but he nevertheless thought that — especially under the influence of prejudices and of zeal to defend traditional abuses — the significance of historical circumstances was exaggerated. What in James Mill marks an advance in relation to Bentham is the endeavor to give a more definite foundation for the general principle of valuation from which the ethical and political propositions were to be deduced. To this he was led by the psychological studies which he steadily continued during his official activity and during his philanthropic and political endeavors, and which found their conclusion in his famous work *Analysis of the Human Mind* (1829).

This work is of great significance in the history of psychology as the most thoroughly carried-through attempt to trace all mental phenomena back to association between representations. It resumes the attempt which Hartley had in his time made in the same direction. There is far greater clarity and fullness in James Mill's exposition than *[371]* in Hartley's. He not only explains all phenomena in consciousness as arisen through association, but all association he in turn traces — in a somewhat artificial way — back to association between representations which have frequently occurred together (what has later been called association by contiguity). It is the principle of simplicity that is at work here. James Mill sets up as a rule that as few original facts as possible are to be acknowledged. Just as Bentham would build the whole of ethics on the one proposition that pleasure is to be preferred to pain, so James Mill would build the whole of psychology on the one proposition that what has once been experienced can be recalled again when experiences which occurred together with it in time or space are repeated. — If this proposition could bear the whole of psychology, psychological theory would undeniably gain a quite extraordinary simplicity and perspicuity. And at the same time the prospect would be opened of so consistent a carrying through of the philosophy of experience as the older English school had not yet known. Not only all single representations each by itself, but also all combinations of representations would then be wholly determined by what had appeared to consciousness from without. An unlimited prospect of influencing the human life of the spirit and determining its direction opened here; for through legislation, education and in general the ordering of external circumstances one determines the combinations of representations which come

to rule in men. Association psychology thus gives access not only to understanding men's representations as they have once and for all combined and formed themselves, but also to deciding which representations and combinations of representations are to rule in the future. It could be the foundation both for criticism and for new formation, a mighty weapon against prejudices by demonstrating their origin, and a means of progress by putting new, justified combinations of representations in the place of the prejudices. — Beside Hartley's influence that of Helvetius shows itself here. The book "De l'esprit" was in Mill's circle a favorite work, just like "Observations on Man".

When now association psychology, in the form in which it here appears, is to serve
[372] as the foundation for the acknowledgment of the principle of utility, James Mill — following Hartley — lays strong weight on this, that association can lead not only to one representation calling up another representation or awakening a feeling of pleasure or pain, but also to several representations and feelings entering into so close a combination that they cannot be separated, and that the new whole they together make up has qualities which the parts each by itself do not have. The new whole formed by association can, as James Mill expresses himself in the clearest exposition he has given of this (Appendix B to the polemical writing *Fragment on Mackintosh,* an essay of classic significance in psychology), itself become a supporting principle in human nature (a substantive principle of human nature). One can therefore not infer from the nature and value of a feeling to its origin. It has given occasion to many disputes and misunderstandings that these two things, value and origin, have been confused. It has been believed that a mental faculty must either be absolutely original, or else that the factors to which it owes its coming into being must be homogeneous with it. Mental processes have been compared with mechanical combinations, and it has been overlooked that they must often be conceived by analogy with chemical processes. This mistake has led to either (as in "the selfish system") tracing, for the sake of clarity and simplicity, all feelings back to egoism, or assuming several different original kinds of feelings (some selfish, others disinterested). By means of association psychology, on the other hand, the view can be grounded that the disinterested feelings have developed out of elementary feelings just as much as the egoistic ones. As Butler already demonstrated, egoism proper cannot be original, since it presupposes conscious calculation; but just as little is disinterested benevolence original: it develops through pleasure and pain being very early transferred from the nearest and more elementary causes to more remote ones, which stand in connection with them as conditions and means of their working. Secondary feelings form themselves, which are attached to something that was at first a means but gradually

comes to stand as a purpose. Thus motives which at first have value only as conditions of action can acquire independent value; and the unconditional value we ascribe to *[373]* the moral feeling or conscience thereby obtains its explanation. The welfare of others, which at first was only a means to the individual's welfare, can secondarily become a purpose for him. James Mill expected, by virtue of this possibility, great things of the political education a people undergoes by living under a good constitution. He says of this in an essay ("Education", written 1818): "When the political machine is such that the great objects of desire are seen to be the natural rewards of great and noble conduct, of considerable services to humanity and of the high-minded and benevolent cast of mind from which great endeavors in the service of humanity proceed, then one will also naturally find zeal to acquire the qualities which lead to admirable actions spread abroad in humanity: such qualities as great intelligence, perfect self-control and all-conquering goodness!" — And secondary feelings and qualities can be just as firm and immediate as the primary ones. Nor will — Mill asserts — the analytic investigation of the origin of such secondary feelings and qualities through association weaken the value they can have for us: "Gratitude remains gratitude, resentment remains resentment and generosity generosity in the consciousness of those who feel them, — after the analysis as before it. He who can trace such feelings back to their elements does not cease to have them, any more than he who never thought of investigating them . . . . They are constituent parts of human nature. How we are affected by having them is a matter of experience which everyone knows from himself. Their working is what it is, whether they are simple or compound. Does a compound motive cease to be a motive as soon as it is discovered that it is compound?" (Fragm. on Mackintosh. p. 51 f.). "The disinterested nature of a feeling is not impaired by the consideration that it has come into being through one of the great psychical processes by which we with great probability explain the coming into being of the other habitual states of human consciousness (habitual states)" (ib. p. 421). Mill here answers an objection which both then and later was often made against the psychological explanation of the origin of the moral feeling — not only against the *[374]* explanation the older association psychology gave, but also against the one the doctrine of evolution has given.

But in the very statement of Mill's just quoted it is implied that the deductive application of association psychology has its definite difficulties and limits. First the concrete states must be studied in their *historical* coming into being before their elements can be discovered. By the reference to immediate experience it is admitted that the matter is not so simple as it might seem according to the principles of association psychology.

The new qualities of the product cannot, after all, be deduced from the factors! And the historical element which thus asserts itself in the individual's life of consciousness as a — at least provisional — hindrance to analysis appears in far higher degree in the social and political domain as a barrier to the deductive application of the principle of utility. The Utilitarians of the older school satisfied only in a very imperfect way the principle of experience which it was their task and their pride to assert.

It was now precisely the irreducible, that which resists analysis, on which the philosophy of Romanticism laid weight. With great passion the most important representative of the Romantic school in England in particular asserts that in life which does not seem intelligible by mechanical explanation, the original and unique which all personal and historical development presents.

b. The Romantic Philosophy of Personality

Bentham and James Mill are two mighty rocks that jut into the new century, and against which the waves of Romanticism break. It was of extraordinarily great importance for the life of the spirit in England that the new currents met such superior representatives of the old time, so firm of character. The interaction between the old and the new thereby became as fruitful as it seldom is. The clash between the two tendencies was sharp and cut deep into many a personal development, but it gave birth to forms and endeavors in
[375] the view of life and in thought that have epoch-making importance for the life of the spirit in Europe as a whole.

Through Samuel Taylor Coleridge (1772—1834) the new German literature and philosophy of the age of Romanticism gained influence in England. In his youth he had (as he relates in his *Biographia litteraria*) been an adherent of Hume and Hartley. But he was repelled by the eighteenth century's attempt to lead all mental phenomena back by analysis to elementary functions, and by the endeavor to demonstrate mechanical laws for the life of the mind: the unity and activity of the mind were lost! The study of German philosophy led him decidedly away from the English school. As a good Romantic he reveled in absolute ideas, in which the differences and oppositions of the finite world faded and vanished. He set the whole, intuited by presentiment, against the piecemeal work of empiricism, synthesis against analysis. In his own opinion he anticipated some of Schelling's ideas; so thoroughly had he lived himself into the speculative line of thought that he could continue it on his own in the same direction as its original founders. Occasionally, in editing his writings, he confused pieces he had translated from Schelling with his own compositions, which gave occasion to charges of plagiarism. This poetic,

enthusiastic spirit, who was quite as much poet and preacher as thinker, could not properly distinguish between what he had himself conceived and what he had found in others. What he took from German philosophy was above all Kant's distinction between reason (in the narrower sense), as the faculty of ideas of the unconditioned, and understanding, as the faculty of categories that make limited knowledge possible. This distinction he used above all to find a reconciliation between thought and religion. All the objections the preceding age had held to be decisive were now — not exactly rejected, but — set down to the account of the "understanding" and pushed aside by an appeal to the deeper conception of "reason." Behind this appeal there lay hidden the awakening sense for what in history and in life escaped, or had hitherto escaped, scientific explanation. It was a protest against the sufficiency of the science developed hitherto, a *[376]* protest that was given and taken as a new solution, a new foundation. Coleridge did not get beyond presentiment and annunciation. He worked mostly through conversations (or rather through his monologues before listeners). His great work on the agreement between philosophy and Christianity, to which he referred his listeners at the difficult points, he never got written. He placed himself in opposition both to the philosophy of the eighteenth century (Hume and Voltaire) and to the ecclesiastical theology that had stiffened into outward forms and formulas. Like the German systematizers of Romanticism, he wished to bring about a higher unity of reflection and dogmatic faith. In his fundamental thoughts he recalls the so-called right wing of Hegel's school. He found an explanation of the Trinity, e.g., by applying the schema thesis, antithesis and synthesis, which from Fichte's time on played so great a role in German speculation. (Under the evident influence of Schelling's later doctrine, however, he is in danger of deducing a quaternity, since he posits God as "the absolute will or identity" = prothesis, as the foundation for the interplay of thesis and antithesis that is harmonized in synthesis! See *Table Talk*, 8 July 1827.) But coherent thinking was not Coleridge's affair. It is surely apt when he has been called a religious epicurean. He reveled in religious thoughts and feelings and in ideas that could furnish symbols for them. But he was a weak character. He could master neither his conduct of life nor his line of thought. He fled, as Carlyle (who in *Life of Sterling*, chap. VIII, gives an interesting characterization of him) has said, from life into the world of theosophic dreams; what he gave was "transcendental moonshine," which showed many a young doubting and seeking nature the old things in a new light, but did not contain any really new life. A man who was himself for a time strongly influenced by Coleridge, the friend of Carlyle and of Stuart Mill, John Sterling, has said of him: "His misfortune was that he appeared at a time when there was a man's

work to be done — and he did not do it. He had not sufficient strength of character; doctrines he had ceased to believe in he professed, in order to avoid the unrest of strife."
[377] What this charge, which Sterling repeats in a letter to Carlyle, is aimed at is not clear. But when it is speculation and fantasy that become the chief organs of religious thought, it easily happens that more doctrines are acknowledged than one can really turn into personal property.

Meanwhile, Coleridge had awakened a new sense and new ways of seeing in many of the younger generation. No less a man than Stuart Mill, who was otherwise not favorably disposed toward German speculative philosophy, has placed him beside Bentham and regarded these two men as the two great fertilizing minds (seminal minds) of their time. Mill's own great and honest attempt to learn from both of them we shall later have occasion to bring forward. Most of those whom Coleridge influenced for the time being would not learn from Bentham as well, but were led through the "moonshine" to orthodoxy, and soon to a more pronounced orthodoxy than the old. Perhaps Carlyle is right when he says that had there been no Coleridge, there would have been no Anglo-Catholic movement (Puseyism). Several of the men of this movement at any rate began with poetic and speculative ideas before the positively ecclesiastical line of thought gathered speed[89]). That the oppositions in the domain of the view of life sharpen in the course of the nineteenth century — at any rate as regards the outward formulas — is something to which the life of the spirit of every country bears witness.

Thomas Carlyle (1795—1881) found nourishment for his spirit in a wholly different side of German philosophy. Not a higher speculative knowledge, but a more vigorous assertion of the value of personality, a new faith, was what he gained through the study of German poetry and thought. Through the German poets he was led to the German thinkers; through Schiller and Goethe he was led to Kant and Fichte. He saw clearly that the Scottish school had not overcome Hume's doubt by its appeal to instinct *(Signs of the Times* 1829), whereas Kant and his successors went to work more thoroughly and attacked Hume, and the mechanical philosophy in general, at the center, not in the outworks. Like Coleridge, Carlyle lays particular weight on the distinction between
[378] understanding and reason; it was for him from the first the most characteristic thing in Kant's philosophy *(State of German literature.* 1827). But Goethe was and remained the great prophet of his youth. However different his heavy Scottish nature, in which the mists did not wholly vanish even on clear days, was from Goethe's bright humanism, Goethe was nevertheless his model, and his confession of faith was the same as Faust's. From Kant he took into his definitive philosophy above all the distinction between

the thing in itself and phenomena. All that nature shows us is only phenomena. The "philosophy of clothes," which he sets forth in his profound and humorous work *Sartor Resartus* (1833), rests on the thought that just as little as the clothes make the man, so little has one fathomed existence because one knows the phenomena. Existence is and remains an unfathomable wonder, which must fill us with reverence. The world is the garment of the Deity. Natural science merely demonstrates the outward mechanism, but the kernel of existence is not thereby reached. Nature is a great symbol, a revelation of ideas that cannot be grasped by means of any scientific method. The world is no dead machine, as science would make it. Even the purely outward connection — the clothes — is inexhaustible and beyond survey from our little corner of existence. And however far our experience and our thought might be able to reach, beyond the forms of space and time we do not get, and they are after all only forms of knowledge; they too belong to the garments of the Deity, which are forever woven anew. The wonder of existence hides itself from us because from dull habit we take the forms for realities, and because our first fundamental propositions are often only assumptions we are accustomed to and therefore do not call into doubt. But what is philosophy other than a constant struggle against habit? It is "transcendental" precisely because it oversteps (transcends) the sphere of blind habit. It is then the task of philosophy to awaken anew the feeling for the wonder of existence when the mind is dulled by mechanical representations. "The man who cannot wonder, and who does not constantly wonder and worship — were he President of innumerable scientific societies and did he carry in his single head the whole Mécanique céleste and Hegel's philosophy and the sum of the work of all observatories and laboratories — is but a pair of spectacles behind which there is no eye. Let those *[379]* who have eyes look through him, then he may be of some use." (Sartor Resartus. I, 10).

It is easy to see that Carlyle applies the distinction between the thing in itself and phenomena not in the spirit of Kant but in the spirit of Romanticism. For Kant it was the task of science to find the lawful connection of phenomena, and the concept of the thing in itself marked the limit of this endeavor. But Romanticism despises and cancels the scientific endeavor to link phenomenon to phenomenon according to definite laws. One never comes to an end that way, after all, and one does not reach the kernel of things! Carlyle himself, like his hero in Sartor Resartus (Book II. Chap. 3 and 7), had a time in his youth when the world was to him a dead machine. (Froude: *Thomas Carlyle.* I. p. 101)[90]). The study of Hume, Gibbon and d'Alembert had robbed him of the religious conception of his childhood. For a time he was wholly absorbed in mathematical studies, and his first literary work was a translation of Legendre's Geometry. It was then that he was led

through Mme. de Staël to German literature, and Goethe's great world opened itself to him. "Four years ago," he writes to Goethe in 1824, "I read Faust on the mountains of my Scottish homeland, and then it was my dream that I might reveal to you, as to a father, all the pains and wild wanderings that my heart, whose innermost secrets you seem to know so completely, has gone through." Like Faust he grew impatient with science. What is the use of being able to dig a few feet deeper than one has reached hitherto, when there is still an infinitely long way left? Or, as he later said in a conversation with Charles Darwin, how ridiculous is it not to care whether a glacier moves a little faster or a little slower, or moves at all! As a good Romantic he believed in Goethe's optics and would not grant the physicists the right to criticize it. His hints of a view of nature recall Schelling's, especially the eleventh lecture on "Methode des akademischen Studiums," where it is stated, among other things, that "the ideas *symbolize themselves* in things,"
[380] and that "empiricism apprehends reality quite independently of its meaning, since it is the nature of the symbol to have a life of its own in itself": natural science thus takes as absolute reality what is only symbol.

But it is not outward nature that interests Carlyle. Man is for him the real revelation, the true Shekinah, the highest symbol. No analysis can exhaust his essence. Locke and his successors sought to analyze and mechanize the mind through and through. And in particular men sought to make the whole conduct of life mechanical, to lead it back to the mechanics of the feelings of pleasure and pain, to make the principle of utility the rule, and to exclude every original, spontaneous, independent action. Bentham and utilitarianism are objects of Carlyle's sharp criticism and bitter scorn. He ascribes to the principle of utility only a negative, dissolving significance. It may be serviceable for criticizing the old order of things, but not for producing a new one. It puts a dead mechanism of interests in place of the living, personal morality that springs from the heart's yearning to fulfill its inner ideals. As eagerly as Carlyle polemicizes against what he calls the "philosophy of cause and effect," just as eagerly does he polemicize against an ethics that builds on "profit and loss" (virtue by profit and loss), and he sees in Bentham's theory a solution of the problem: "Given a world full of knaves, to show how virtue becomes the result of their united endeavors." *(On Heroes and Heroworship.* Lect. VI). What Carlyle holds against the analytical psychology and the doctrine of utility is not least that they lay too great weight on conscious reflection. Everything great comes into being and ripens in silence. Only a man who can keep silent becomes a great man; and a great deed that is done with full consciousness of its greatness is in reality a small deed. Rightly do the poets sing hymns to the night. Full, clear consciousness makes

everything small and mechanical. The highest truth can exist for man only in the form of the symbol: the symbol speaks and is silent, reveals and conceals at once.

Toward theological dogma Carlyle is no more favorably disposed than toward natural science and empirical philosophy. Only in the form of the symbol can man have the truth. But symbols grow old and wear out like clothes, and new ones must be formed in place *[381]* of the old. Philosophy, which is a struggle with habit, has in particular to struggle with the habitual and mechanized religious representations. Carlyle describes his standpoint as *natural* supernaturalism. It is divine forces that work in us and outside us. But they work in an inward, natural way. As the living garment of the Deity is constantly woven anew in outward nature, so the inner sense constantly fashions new forms for the life of the spirit. Each individual has to find his symbol and his religion, just as he has his deed to cast into the ever-flowing stream of time. "By *religion*," says Carlyle *(On Heroes and Hero-Worship.* Lect. I), "I do not understand the church-creed a man professes, the articles of faith he will sign . . ., but that which a man practically believes (often *without* declaring it to himself, and still less to others) — that which a man practically lays to heart and holds for certain concerning his inner relation to this mysterious universe." To speculate further on the essence of the Deity is useless. We have each of us enough to do in performing the deed that lies in our power. Carlyle's religious conception is nevertheless clearly enough the same as Goethe's, and as Fichte's in his later writings ("Grundzüge des gegenwärtigen Zeitalters" and "Wesen des Gelehrten"), to which he even refers explicitly. Fichte was the one among the German philosophers with whom Carlyle most sympathized. The thought of the Deity pervades everything, the spiritual and the physical world; every spiritual being is a spark from it. Whether his concept of God is theistic or pantheistic, he will not decide. John Sterling, in a letter that contains a criticism of Sartor Resartus, had objected among other things that the God spoken of in the book is no personal God; Carlyle declares it an abstract question that he cannot or will not discuss. (Life of Sterling. Part II, Chap. 2). — In his later years he declared that it was best for the time being to keep silent about religion. Its time will come again; but the dishonesty must first be done away with (Froude: Thomas Carlyle. II. 2. p. 394).

Carlyle's protest both against empirical and critical science and against theology was bound to make him lonely. He felt himself also a stranger in his time. It was a time *[382]* dissolved by skepticism and analysis, a decrepit (decadent) time, the age of mechanism. Spirit was lacking and faith was lacking. Faith is the expression of the truth of life, mysterious and indescribable like all the activities of life. But next after the end of the Roman world there has been no time so skeptical, untrue and decadent as the eighteenth

century, whose after-effects reach into the nineteenth century. It is a time in which no ideal can grow or flower. (On Heroes. Lect. V. — History of the French Revol. Vol. I, Chap. 2). The law of the three stages appears plainly here in Carlyle, who very likely has it from Fichte and the Saint-Simonians. But more strongly than any of the other writers in whom this law appears, he speaks out about the wretchedness of the time. Both in speech and in writing he complains of having come into being in so miserable a time. His mighty fantasy, his intense and easily stirred mood, and his deep melancholy, which was constantly reinforced by bodily infirmity (dyspepsia), made him dwell in particular on the dark sides of the time, which he exaggerated in comparison with those of earlier times. He found it hard to imagine that this time could be a time of transition to a better, positive time, as according to the law of the three stages it should be. He himself really assumes no gaps in development: it is a misunderstanding, he says, to believe that the phoenix burns itself up completely and then lies as a heap of ashes until a new bird comes into being in a miraculous way; no, destruction and creation go hand in hand, and the new garment is woven in the very midst of the process of the old one's dissolution (Sartor Resartus Chap. 7). But Carlyle was not himself able to find the new threads. He is convinced that none of the spiritual works of the past is lost: "Whatever mechanical science and spiritual torpor may say, man and his universe are eternally divine, and what the past has possessed of nobleness and of divine revelation can and shall not be lost." (Life of Sterling. Part I. Chap. 8). This conviction of the subsistence of values he will not, like Coleridge and Hegel, ground by a symbolic interpretation of dogmas and customs. But he wants the same reverence, the same wonder, the same
[383] energy in renunciation and in work, to which the faith of the past partly bore witness and which it partly conditioned, preserved for the conduct of life of the future. By this endeavor his standpoint is understood, his opposition at once to scientific criticism and to ecclesiastical faith. He seeks the foundation for the preservation of values much farther within personality than analysis and tradition can reach. The living garment of the divine force that never dies must be woven from within. But will these inner forces now also be capable of being awakened after the time of dissolution in which we are placed? Carlyle is thrown back and forth between faith in the subsistence of values, or of the inner forces, and the dark picture the age presented. The picture became, as it seems, even darker and darker to him. In his youth he was indignant at Bentham's philanthropy, in his old age at Gladstone's zeal for England's political and material development. In a conversation in March 1867 (see Caroline Fox's diaries) he declares that the whole thing is going downhill and means nothing to him. Like "a mourning Jeremiah" he strikes

up the old lament he had begun to chant forty years earlier, and still can find no bright spots. The more vigorous faith of his younger days weakened in his later years.

The cause of Carlyle's dark view of the time cannot be sought merely in his own melancholy and easily wounded heart. It lies also in the very unrest and doubt of the time, in the ever sharper opposition between the contending powers in the domain of the spiritual life, the ever advancing settlement between the old and the new, a settlement in which debit and credit are so extraordinarily difficult to apportion. We have seen a whole series of thinkers, right from Rousseau, Lessing and Kant, stand before this problem. Few could by their whole nature feel the sting as sharply as the Scottish peasant's son who, starting from the strict Presbyterian faith, had lived through the spiritual powers of the time and tested them in his personal life. The problem will follow the race on its way until "the third kingdom" comes — if it comes. That Carlyle saw nothing but darkness around him came from the strict demands he made on the solution of the problem. It was his indwelling ideality, his own inner light, that made the darkness around him so deep. He says somewhere that to every young and fiery mind that enters this world with *[384]* a great goal before its eyes and with a clear view of life, the world will appear as bad: for what else should its strength and heroism be used for? Were the world good, he would be of no use! It is thus human strength and ideality that make the world appear as evil: evil is the shadow of ourselves, a shadow that does not stay outside us but stretches into our own mind. Man's misfortune comes from his greatness, from the infinite that stirs in him and to which he cannot give vent in the forms of finitude. (Sartor Resartus. II, 9. — Life of Sterling. I, 5).

But the cause lay also in Carlyle's being an incorrigible Romantic. He had all Romanticism's lack of ability to discover the ideality in the constant, faithful work that adds link to link and through the small is led to the great. He has only scorn for science and for all coherent endeavors in the practical domain. And yet precisely he, who has spoken such beautiful, and often such loud, words on the significance of silence and of quiet preparation, should naturally have been able to come to understand that proud buildings can rise from stones each of which by itself is insignificant. That is the faith that lies behind the detailed investigations of science. But inquiring natures Carlyle did not understand. For men like Stuart Mill, Faraday, Charles Darwin and George Grote he has only superficial mockery to spare. Froude (II, 2. p. 259) says of him: "He fought against Darwin's doctrine of the transformation of species; but I could see that he feared it might be true!" His idealism was shy, because he lacked understanding of the significance of intellectual and practical work. His faith in the personal life was not so

firmly established that he could trust that it would surely subsist, however much analysis and criticism were applied to it. The kernel of life can surely bear a purgatory. Carlyle is right that we cannot live on criticism, but he did not see the ideal forces that could lie behind the critical work. On this point the Romantic became a philistine.

When Carlyle speaks out on what is to bear the spiritual and social life in the future,
[385] he points to what in his conviction has also borne it in the past: reverence for heroes, for the great men. The infinite force that works and symbolizes itself in everything appears above all in man, and most especially in the great man. As in Fichte, faith in the rise and significance of the chosen spirits stands in close connection with faith in the working of the Deity in all finite beings. By heroes or great men Carlyle understands the leaders and models of men, the founders of everything that the multitude of men strive to accomplish and attain. All the works we see realized around us in the human world have had their first germ in the mind of the great man. There the invisible forces work that, after the time of quiet conception and pregnancy, at last set the new into the visible world. The history of great men is therefore the soul of world history. The hero can appear as prophet, poet, statesman, but in all his shapes he represents the great, collected force of life, as opposed to all externality, partiality and limitation. Before them the deep need of reverence, the divine spark in all, breaks forth. (On this point too we are reminded in Carlyle of Fichte, especially of the latter's "Reden an die deutsche Nation"). It is the hero who finds the hidden thoughts of existence and of the times and proclaims them to others in word and deed, and thereby leads the race forward. More than ever, such heroes are needed in our time, an enlightened aristocracy to lead men. The great social gulf that the whole of history proclaims grows broader and broader in our day. Already in "Sartor Resartus," before his social writings proper, Carlyle points to it (in the chapter "Helotage," III, 4), lamenting the bodily and spiritual hunger of the many. Popular education and emigration on a great scale are the means he points to for relieving the distress. He misses new prophets and new Hengists and Horsas to set these means to work. But to go more closely into the social question was not Carlyle's bent; for that he despised philanthropists, political economists and politicians far too much. In its place he turned to history, in order to bring forward great figures and models, in
[386] case they might have an awakening and concentrating effect on the slack and divided race. It is through his historical writings above all that Carlyle has worked on wider circles. It is through these writings that his thoughts work on more and more people in our day. For many he has himself been one of the heroes he wrote about, one of those who have the ability to feel, see and express the needs of the time and the thoughts of

the time. Of his significance as a historian we shall not speak here; it belongs before another forum. Only we find Romanticism again in him in the excessive weight he lays on what is original, spontaneous, inexplicable in the great figures. He makes no real attempt to show the dependence of the great on the race — although greatness does after all show itself also in the great capacity to run into debt, because the spiritual sense and consumption of the great are greater than those of others. Only a mystical explanation of "the hero" does Carlyle give, by regarding him as an incarnation of the infinite force that works in everything. Here too the Romantic shyness of analysis and of the "philosophy of cause and effect" shows itself.

Carlyle does not, however, go as far as a later conception that finds "the goal of history" in the great men. He regards them as the great causes, and thus as the great servants. He proclaims the worship of heroes — but he sees clearly that to "be worshipped by" or to rule over a stupid mass is no heroic deed. "Hero-worship," he says *(Past and Present.* I, 6), "yes, my friends, but consisting first and foremost in our having a heroic mind ourselves. A whole world of heroes — not a world of bunglers, where no hero-king can rule — that is what we strive for."

c. Critical Philosophy

In the same year in which James Mill's "Analysis," which marked a vigorous resumption of the work of English philosophy of experience, appeared (1829), the "Edinburgh Review" brought out an essay by William Hamilton, *Philosophy of the Unconditioned,* by which the critical philosophy was transplanted onto English ground. Hamilton was born in Glasgow in 1788 and had studied there and at Oxford. When his epoch-making essay appeared, he was professor of history at the University of Edinburgh, after having lived *[387]* for a number of years as an advocate. His starting point as a philosopher lies in the Scottish school founded by Reid, which marked a reaction against Hume's skeptical and negative results. Reid and his successors sought to correct these results by more exact psychological observation, but were led in their zeal to set up a whole series of original faculties and instincts. All too often they dismissed the results of analysis and criticism with an appeal to "common sense" — an appeal that called forth Kant's apt remark that one proves one's common sense by using it, not by invoking it, and that Hume's common sense might be just as sound as Reid's. Kant appears here (in the introduction to the "Prolegomena") as advocate for Hume, to whom he owed so much. But he sought in his own way to refute Hume, and thereby came to have the same opponent as Reid and his disciples. William Hamilton now seeks to combine the doctrines of Reid and

Kant. In the essay mentioned and in a series of essays following upon it, which were likewise published in the "Edinburgh Review," and among which *Philosophy of Perception* (1830) is especially to be noted, he developed his peculiar views, which were of no small interest in epistemological and religious-philosophical respects. These essays he later published collected under the title *Discussions on Philosophy.* In 1836 he became professor of philosophy at Edinburgh, and there, until his death (1856), he gave well-attended lectures on psychology and logic, which became of great importance for philosophical development in Scotland. The lectures were later published under the titles *Lectures on Metaphysics* (2 vols.) and *Lectures on Logic* (2 vols.). Hamilton was an acute thinker, an eager seeker after truth, akin to Lessing in spite of his Scottish gravity; he was also in possession of comprehensive learning in the history of philosophy.

In his essay on the philosophy of the unconditioned, which is a criticism of Schelling and Cousin, and in part also of Kant, Hamilton seeks to show that only the conditioned and limited can be an object of knowledge, and that the attempts to set up a philosophy
[388] of the unconditioned cannot be carried through. But he gives not only criticism; he gives the outline of a whole epistemology, which on particular points he worked out further in his later writings. His main proposition is: *to think is to set conditions* (to think is to condition). By this he means that everything we are to be able to apprehend and comprehend we determine by its relation to something else that conditions and limits it. An absolute whole we cannot apprehend; every whole is for us always part of a greater whole. An absolute part we can just as little apprehend; every part can in turn be thought of as divided and is thus a whole. An infinite whole we cannot apprehend either; for an infinite time would be required to go through all its parts. Thus neither the absolutely limited nor the absolutely unlimited can be an object of knowledge. Our knowledge concerns the conditionally limited. The philosophy of the unconditioned, which credits itself with the ability to construct a scientific concept of the unconditioned as that which is to be at once absolute totality and infinity, sins against the nature of our knowledge. In a later appendix to the essay (Discussions p. 577 ff: *Conditions of the Thinkable*) it is developed how all knowledge consists in a judgment that connects two terms: therewith the relativity of all thought is given. All knowledge rests, moreover, on a relation between subject and object; beyond this opposition we cannot get, unless with Schelling we are willing to sink into the absolute unity in a mystical way, whereby all knowledge is excluded. Another relation that plays an essential role in our knowledge is the relation between thing and property: the thing and the property are known only in relation to each other. In the closer determination of things we determine them in time

(protensively) or in space (extensively) or with respect to the degree of their properties (intensively); but all these determinations are relative: the one determination of time, space or degree presupposes the other. The concept of cause too Hamilton brings under what one might call the epistemological law of relation *(law of relativity).* He takes it as a limitation of our knowledge that we understand things only by finding their causes. We are not able to think an unconditioned beginning. When a phenomenon begins, this *[389]* can for us be only apparent. Just as little can we think an unconditioned cessation. The disappearance of a phenomenon too can be only apparent for us. When we seek the conditions of the phenomenon, this means that we connect it with other phenomena, regard it as a term in a relation — because of our impotence to think an unconditioned! Every existence is for our knowledge relative, is bound to a preceding existence. As soon as something new is added to what exists that is more than a transformation of the earlier into a new form, and as soon as something is taken from what exists that does not come again under a new form, we stand before the incomprehensible. Cause and effect cover each other and can be found again in each other. The concept of cause is thus not, as Reid and Kant each in his own way assumed, an independent, isolated concept. It is a particular form of our consciousness's general way of working, a particular example of the fundamental law that our knowledge by its nature can grasp only the conditionally limited.

By this epistemological law of relation Hamilton has cast light on an essential side of our knowledge. In place of Reid's appeal to the instincts of common sense and in place of Kant's scholastic table of categories he has put an analysis of the fundamental forms of the knowing consciousness and has demonstrated a common characteristic mark for them. Psychology is for him the fundamental philosophical science. Consciousness is for the philosopher what the Bible is for the theologian, and the original testimonies of consciousness must be taken as true, since they lie at the foundation of all our knowledge. On this point Hamilton sometimes speaks just like Reid. "The root of our nature cannot be a lie" — therefore the assumption of an opposition between consciousness and its object, which immediate consciousness according to Hamilton entertains, must be true. This is Hamilton's simple solution of the problem of the existence of a world independent of consciousness. The problem is thereby hardly allowed to come forward. "Natural realism" has, for Hamilton, right on its side.

On the particular psychological and logical questions that Hamilton has treated with great acuteness and learning we cannot enter here[91]). We must, on the other hand, dwell *[390]* on the consequences Hamilton draws from the limitation of knowledge. The value of

philosophy cannot for him, naturally, depend on its bringing us absolute truth. Its value depends altogether not on the finished results, but on the constant release of mental energy into activity and striving. By speculation our mental forces are exercised. Lessing is right when he places the eternal search above the possession of truth. A waking error is better than a sleeping truth. Philosophy does not lose its value because it ends with an acknowledgment of our ignorance grounded on insight into the conditions of our knowledge (docta ignorantia, as Hamilton calls it with Cusanus). Before it reaches this its conclusion, it has developed and exercised the noblest mental forces.

This ignorance concerning the absolute essence of existence rests on the fact that our thought ends in a dilemma whose horns are equally inconceivable. Were we to know the unconditioned, it would have to stand before us either as an unconditionally limited, an absolute totality, or else as an unconditionally unlimited, an infinite in time, space and degree; but neither of the two can be grasped by us, because both conflict with the law of relativity. Hamilton does not think, however, that we must or can remain stuck in this dilemma. For by virtue of the logical principle of the excluded middle between two mutually contradictory possibilities, one of the two assumptions *must* be true. (The logical principles thus have for Hamilton not — like the law of relation — validity only for our knowledge, but are laws of things, not merely laws of thought. See on this Discussions p. 577 f.). There is then here not merely room for a faith, but necessity for a faith. We have learned that our knowledge is no measure of existence, and we have learned that a choice must be made. The idea of an unconditioned is not, as Kant thought, a positive idea that has its natural root in the nature of our knowledge; it signifies on the contrary the negation of all knowledge. While Coleridge held to Kant's doctrine of ideas, Hamilton parts from Kant precisely on this point. He finds the possibility and the necessity of a faith given in this, that, as regards the idea of the unconditioned, we
[391] stand before two possibilities that are both inconceivable, and between which we must nevertheless choose with logical necessity.

The choice is determined, according to Hamilton, by practical, moral motives. We need an unconditioned being that can uphold and secure the subsistence of our mind. And in the representation of this being we lean on the analogy with our own being. Hamilton does not agree with Kant in his criticism of spiritualistic psychology. Consciousness, he says (Lectures on Metaphysics. I. p. 158), is indeed the condition of all inner phenomena; but it is itself only a phenomenon; there must lie something behind it whose property it is, and this must be something different from what lies behind material phenomena. The same relation as there is between our mind and our body we assume — by an analogy —

between the unconditioned spiritual being we believe in and the world of relativities that knowledge presents to us. Through this analogy Hamilton makes the transition from philosophy to theology. The end of philosophy is the beginning of theology. But he says at the same time that no difficulty arises in theology that does not already appear in philosophy. And his own doctrine confirms it: for that unconditioned being which is to be the object of faith is set up by an inference from analogy and is thus to *stand in relation to* the world as the soul to the body; but then it becomes, after all, conditioned, relative! Hamilton has not shown the possibility of escaping, by passing over to faith, the epistemological difficulties he himself has so energetically demonstrated. He agrees with Fichte that a God who could be comprehended would be a conditioned being; but a God who is placed in relation to a world would, even if he could not be comprehended, be a conditioned being, and if he is to be maintained as an unconditioned being, his concept would contain a self-contradiction. — As regards the content of faith, there is confirmed here in Hamilton what we have been able to establish in a whole series of thinkers from Leibniz's time on: that every attempt at a speculative or religious world-view must, consciously or unconsciously, lean on an inference from analogy. — Hamilton's whole standpoint, which is characterized by his psychological starting point, his critical analysis of the conditions and limits of knowledge, at once with an adoption and a correction *[392]* of Kant's line of thought, and finally his philosophy of religion grounded on analogy, recalls such German inquirers as Fries and Beneke.

Two years after Hamilton's death his disciple Henry Mansel gave a series of lectures at Oxford, which are published under the title *Limits of Religious Thought* (Danish translation by N. Teisen: "Grænserne for den religiøse Tænkning." Odense 1888). Here he draws from Hamilton's doctrine the consequence that a scientifically grounded theology is impossible, because our knowledge cannot reach the unconditioned. But at the same time he maintains that no objections can be raised from scientific thought against revelation; only if we had an absolute knowledge, a philosophy of the infinite, could revelation be refuted. And in agreement with Joseph Butler's "Analogy" — a work that has altogether played a great role in religious discussion in England in our century — he maintains that the difficulties and contradictions that can be demonstrated in the theological assumptions occur within every attempt at a conclusive world-view, even if it moves on the foundation of nature and reason. From this he concludes that the difficulties arise — not from revelation, but from the limitation of reason. We must then believe what we cannot comprehend. It is our duty to assume the personality of God, although it is for us self-contradictory that an unconditioned being should be a personality, since

personality presupposes opposition and limitation. We must believe in the dogma of the atonement and in the dogma of eternal punishment, even if these dogmas conflict with our notions of love and justice. What is love and justice in our eyes is perhaps not so in God's! We see only the part, not the whole. If our horizon were widened, we would apprehend everything differently! Just as Hamilton had already taught that it is owing to an imperfection of our nature that we apprehend things as causes and effects, so Mansel teaches that human ethics hangs together with limited human nature, for which reason no inference can be drawn from the ethics of man to the ethics of the
[393] Deity. For man it is, e.g., a duty to forgive, because he needs a bridle on his selfishness; but this grounding of the necessity of forgiveness cannot hold for God!! — Both human thought and human conscience are thus here cut off from all influence on the valuation of religious representations. Without rudder and without compass, according to Mansel, man is to sail the ocean of religion. Whether the theological voyage will go better in this way is perhaps a question — until such time as the superhuman thought and the superhuman conscience that are to escape the logical and ethical difficulties have been demonstrated. Mansel's philosophy of religion was at first received with great approval as a good weapon against rationalism and speculation. But people seem to have come back somewhat from the first enthusiasm[92]), having discovered that it was a weapon that easily wounded the one who used it. It is also clear, after all, that however much one may wish to inculcate the necessity and the possibility of leaping from thought and conscience into faith, man cannot, even by the boldest salto mortale, once and for all leap over his own shadow. —

Besides Hamilton and his disciples there must still be named an inquirer who in a fruitful way brought Kant's epistemology to bear in England. William Whewell (1795—1866), who worked at the University of Cambridge first as a natural scientist (mineralogist), later as a philosopher, published in 1837 his work *History of the Inductive Sciences,* which some years later was followed by *Philosophy of the Inductive Sciences, founded upon their History* (1840). In these works he sought by the historical road to establish the correctness of Kant's fundamental thoughts, which, to be sure, he did not always apprehend with sharpness and clarity. He stands on the ground of the English school in maintaining that all knowledge develops on the foundation of experience. Both the human sciences and natural science are inductive sciences. But induction does not mean merely the collecting and juxtaposing of facts; it means the grouping of the facts under a guiding thought, their reduction under a general law. And the history of the inductive sciences now proves that this grouping and reduction become possible only

when there are already at work in the inquirer ideas and points of view that can make *[394]* possible the discovery of the law of the facts' connection. The facts must be connected by a mental act whose possibility must lie in the nature of the mind. Every inquirer therefore starts from presuppositions that cannot be derived from the particular facts each taken by itself. By a closer investigation of the way in which inquirers such as Kepler and Newton came to their discoveries, this conception of the method of induction is further illuminated. Kepler had the concept of an ellipse beforehand, just as Newton had the concept of attraction, and only because these concepts were present in advance did their discoveries become possible. It is not Whewell's opinion that such concepts lie fully finished in the mind, but that no discovery can be made without the cooperation of a mental activity that follows its own laws. Even if the special ideas (as, e.g., of ellipse and of attraction) have themselves developed under the influence of experience, we nevertheless come back at last to fundamental concepts that are the expression of nothing other than the laws of the cognitive faculty itself, which assert themselves in all experience, from the simplest sense perception to the most comprehensive induction. By attending to the way in which our knowledge works, we find a series of forms or fundamental concepts — the concepts of time and space as the foundation of mathematics, the concept of cause as the foundation of the mechanical sciences, the concept of purpose as the foundation of the organic sciences, and the concept of duty as the foundation of ethics — concepts that according to Whewell cannot be reduced to simpler forms or derived from observation. He contents himself with setting them up in a series without subjecting them to a closer examination, and his conception thereby acquires a likeness to Reid's, whereas Hamilton, as we have seen, made an interesting attempt to demonstrate the various fundamental concepts as various expressions of the epistemological law of relation. Nor does he apprehend the relation between philosophy and theology with the critical sharpness that distinguishes Hamilton. His significance lies in the preparatory work he has contributed toward a theory of induction. Here he has in a significant way illuminated the method of induction as a *method of discovery,* whereas — as his great *[395]* opponent Stuart Mill showed — the method of induction as a *method of proof* did not come into its own with him. For one who occupied himself particularly with the history of the inductive sciences, the point of view so energetically stressed by Whewell was bound, quite naturally, to come especially into the foreground.

2. John Stuart Mill

a. Biography and Characterization

John Stuart Mill, James Mill's son, was born on 20 May 1806 in London. He is an example of early intellectual development, but he also developed under a high pressure of intellectual influence that a less vigorous and original nature could hardly have endured, and that left lasting consequences in him too, both in his intellectual tendency and for his bodily health. His father taught him himself. At three years old he began Greek and soon after arithmetic; at the same time he naturally learned English reading and grammar. Latin he began when he was eight. History he learned on his own from detailed works, and then gave his father an account, on walks, of what he had read. After a great part of Greek and Roman literature had been gone through, he began logic when he was twelve years old, and here too what he had read on his own was gone through and discussed thoroughly on walks. Then followed reading in political economy and a careful study of Demosthenes and Plato, with particular regard to argument and method. Very early the boy had to teach younger brothers and sisters, and thereby got practice in mastering and applying what he had learned. There was no cramming. The father endeavored that understanding should keep pace step by step with instruction, indeed preferably go ahead of it: "What could be found out by thinking," says Stuart Mill in his autobiography *(Levned.* Danish trans. by Vilh. Arntzen, p. 34), "my father never told me until I had
[396] exhausted my strength in finding it out myself." — Mill recommends the way in which he himself was taught for imitation. He thinks that what he achieved here any boy or girl of fairly good abilities and a healthy nature could achieve. Even if what was demanded often lay beyond the child's ability, Mill nevertheless holds it pedagogically right to make such demands: "A pupil from whom nothing is ever demanded that he cannot do never does what he can do!" He would never admit that this upbringing, which, as he says, had given him a quarter of a century's start over his contemporaries[93]), had exercised any questionable influence on his development and on his health. And yet there is hardly any doubt that a mental crisis he later went through, as well as a nervous debility from which he suffered all his life from his thirtieth year on, were in great part consequences of the one-sided and forced development of his childhood years. That at least was the view of his friends (see A. BAIN: *John Stuart Mill.* A Criticism. London 1882). Mill would never admit that work could be harmful. Although an eager philosopher of experience, he was nevertheless so spiritualistic that he denied all essential influence of physiological dispositions and states. Therefore he not only believed that mental development could be hastened without harm to health, but he also did not acknowledge

any original difference between individuals. It was always a foundation of the hope he had for the future of humanity that all qualities of character were effects of upbringing and of outward, social conditions, so that by pedagogical and social reforms one would be able to alter human characters to indeterminable degrees and in the same direction. In this respect he constantly paid homage to the doctrine of Helvetius, of which his father and Bentham were such eager adherents. — Apart from the physical consequences of the forced upbringing, it suffered also from the defect that it was only the understanding, not feeling and fantasy, that was developed. The whole spontaneous unfolding of moods and images was checked under this strictly rationalistic upbringing. Whereas in his old age, in his autobiography, he recommended the way in which he had been brought up, he had at an earlier stage himself felt its drawbacks. In a conversation of 1840, which the *[397]* Quaker Caroline Fox records in her diaries, he says expressly that he will not recommend the early intensive strain in the upbringing of all children, since it easily comes at the expense of the liveliness of childhood and favors reflection at the cost of activity. "I have never been a boy," he said, "have never played cricket; it is better after all to let nature go its own ways."

Another point on which Mill's autobiography must also be corrected by what has come to light by other roads about his development is what he relates about his religious notions in childhood. He relates in the autobiography that he received no religious upbringing at all, since his father had long since given up all religious faith. Supported by definite recollections on the part of members of the family, Bain (in his biography of James Mill) has shown that Stuart Mill remembered wrongly as regards the definitive change in his father's religious standpoint. In Stuart Mill's childhood his father still went to church, and the son with him, and the rest of the family continued to do so later too. Definite utterances of Stuart Mill as a boy are preserved that show that he read the Bible with enthusiasm. But the father began very early to discuss religious questions with the boy. He taught him not to break off a line of thought or let go of a problem out of a feeling of religious devotion before he had found out whether it could be solved or not. He impressed upon him what was for him the decisive difficulty not only in the faith of the church but also in faith in a good and almighty Creator: the impossibility of reconciling the experience of the physical and moral evil in the world with such a faith. It was Butler's "Analogy" that had led James Mill to the purely negative standpoint he took in religious respects in his later years. Not for the first and only time has a work of defense here had the opposite effect from that intended — here, however, by its depth and consistency. The line of thought into which Stuart Mill was thus early led

determined his later position on the religious problem, even when, after a mental crisis, he took a more positive standpoint, and even in his posthumous essays in the philosophy
[398] of religion the after-effects of the father's conversations with the little boy are to be felt.

A year of his early youth (1820—21) Stuart Mill spent in the south of France with a brother of Bentham's, who had an estate there. He ascribes to this stay a great influence on his development. Above all there was awakened in him a love of France and an interest in French literature and politics to which he remained faithful all his life. The French historical writers later, at an important point in his development, gained great influence on him, and he found in the French a capacity for enthusiasm and for self-renewal that he often missed in his own countrymen. After his return home he pursued legal studies and studied Bentham's writings. In Bentham he found an idea that could bring coherence into all notions and endeavors. He felt himself a new man when he had made a penetrating acquaintance with Bentham's "principle of utility." He says of this in the autobiography (Levned. p. 68 f.): "This principle gave unity to my notions of things. I now had a conviction, a faith, a doctrine, a philosophy, indeed a religion in the best sense of the word, the spreading and inculcating of which was to be the chief task of my life. I pictured to myself a great number of changes in the condition of men that were to be carried through by this doctrine . . . . . The prospect of progress that Bentham opened for me was sufficiently wide and brilliant to make my life bright and to give my endeavors a definite form."

With zeal Mill joined the band of young men who strove to work for the philosophical and political ideas of Bentham and James Mill. Hartley's psychology, Malthus's political economy and Bentham's moral philosophy were the foundation on which they worked. The "Westminster Review" was their organ, in which Stuart Mill's first larger essays were printed, and what they fought for first and foremost was freedom of expression and the extension of the suffrage. They hoped that if only full access were given to enlightenment and knowledge, all social problems would be solved. By prudent self-control the workers would be able to prevent the strong increase of population, which is
[399] a chief factor in pressing wages down. Class rule would be broken through the extension of the suffrage. Human notions would be altered, and through them in turn human character, by virtue of Hartley's doctrine of inseparable associations. Despite all the zeal with which these ideals and prospects for the future were held fast, Mill nevertheless states that the expression "thinking-machine," which has often wrongly been applied to Bentham's adherents, really fitted him in this period. As he expected the regeneration of the human race not through disinterested love of justice but from "the enlightening

influence of the cultivated understanding upon the selfish feelings," so in himself it was essentially the understanding that had the floor. "My zeal," he says in his autobiography, "was in this period of my life not much else than zeal for speculative opinions. It did not have its root in any genuine benevolence or sympathy with the human race, though these qualities had their due place in my ethical conception. Nor did it stand in connection with any ideal elevation. And yet my fantasy was very receptive to this feeling; but there was at that time a stoppage in its natural nourishment, poetic culture, while there was a superabundance of what stood in conflict with it, logic and analysis. Add to this that my father's teaching had a tendency to undervalue feeling." (Levned p. 112). It was the rationalism of the eighteenth century that here, through Stuart Mill's upbringing, continued its effect far into the new century, nourished and goaded by the constant struggle against prejudices and sentimental declamations.

There came upon Mill, when he was twenty, a mental crisis that on essential points gave him another view of life. The foundation of his whole mental life suddenly collapsed for him. He fell into a state of dejection in which he asked himself whether he would now really feel happy if everything he expected of intellectual and political progress were realized — and he had to answer No. Life now became heavy and desolate to him. He performed his work purely mechanically, but it seemed to him as if the source of mental life in him had dried up. Constant analysis had made everything cold for him. *[400]* The psychology by which he had been brought up had indeed seen that the feelings are influenced and altered by the development of representations, but had not seen that deep and strong feelings can be produced and preserved in this way only when they get sufficient nourishment through spontaneous experiences and through constantly fresh and immediate influence. Now he felt like one who is stranded at the beginning of his voyage, with a well-equipped ship and a rudder — but without a sail! — It was a reaction against the violent strain of childhood and early youth and against the one-sided development of the understanding. And it was at the same time a reaction against the line of thought of the eighteenth century, a reaction that was bound quite naturally to come forth in a young man who had been brought up in the thoughts of the old time and stood in the midst of those of the new, even if he had not yet let them come in over his threshold. — The mental crisis was resolved for Mill through new experiences, new ideas and a new human relationship. — He was deeply moved by a touching passage in a biography and so learned that the source of feeling was not exhausted in him: "I was no longer hopeless; I felt that I was neither a stick nor a stone." Thereby he gained new courage. He immersed himself in the poets, of whom Wordsworth above all came to have

great significance for him. He learned what indispensable spiritual nourishment poetry can give; hitherto he had been inclined, with Bentham, to fear that it would impart false notions to men. He learned to give the unconscious and the spontaneous their due in life, and found out how important it is not to stare at one's own happiness as the final purpose, but to set oneself a higher purpose: then happiness will be found in the endeavor to realize it. "Only those are happy who have their gaze directed on something other than their own happiness . . . . Ask yourself whether you are happy — and you will cease to be so!" — To this were added new studies, especially of the French historians (Guizot, Michelet, Tocqueville), who taught him that institutions and laws develop according to natural laws; they correspond to the stage of development of society and of views, and their perfection must be regarded in relation to this stage, and only to a certain degree
[401] can the order that progress follows be altered by arbitrary intervention. In particular the writings of Comte and the Saint-Simonians made a strong impression on him. He came to know the difference between critical and organic periods, and Carlyle's works acted on him as a mighty poetry of personality. He saw that one could not, as he had hitherto believed with his father and Bentham, treat human affairs purely deductively, or ascribe to outward conditions a wholly determining influence. He came — which was after all also the most consistent course for an adherent of the philosophy of experience — to give experience its full due. In his religious views too this crisis and its resolution intervened, although he does not speak of this point more closely in the autobiography, where he says only that he now saw the significance of the development of the inner man. The foundation of the religious views he developed in his posthumous essays was laid in these years, as will later be shown more closely. By Bentham's friends he was regarded as an apostate, while in other circles he was received as one who had broken with sectarian prejudices in order to grasp the truth. In a remark recorded by Caroline Fox, Mill's friend John Sterling says of him that he has sacrificed the position of undisputed leader of a mighty party for the higher honor of being a private in the army of truth, and Sterling hopes that Mill's books will work to give the feeling of reverence the place in life that so many have denied it. — But what a stirring of feeling and self-forgetfulness had begun, and what a widened historical view and religious sense had developed, gained its proper fullness in his life only through his acquaintance with the woman who later became his wife, and to whom he himself ascribes a wholly overwhelming influence on his mental development, and whom he regards as a supremely superior mind. By his overflowing eulogies of her he set his friends, and — especially by his autobiography — his readers, a riddle, since he seems to ascribe to her everything he has accomplished. The secret is no

doubt that Eros, here as so often, has been the great teacher, who has given an eye for ideals and a sense for intimate, personal qualities, and has set feeling in a motion that does not stop at the object that has awakened it. This explains why those nearest to Mill *[402]* and his friends could see nothing extraordinary in the woman who for him was a genius of the highest rank. One of his brothers, who knew her well, used to say that she was an able and remarkable woman, "but nothing like what John took her for." Where Mill gives a closer account of what he owes her, he exempts his purely scientific works (his Logic and the theoretical parts of the Political Economy) from her influence, whereas he characterizes her as the one who had the freest and boldest eye for the ideal and for what the future might bring, while at the same time she possessed a great sense for concrete, factual conditions and a lively ability to make them vivid, just as she also had a sure apprehension of the relative significance of different points of view[94]). One understands that Mill here found something he had missed all his life, and to which his upbringing had made him less receptive, until it came to meet him in personal reality. He found his Beatrice, as Comte found his.

Stuart Mill entered practical activity early, being employed already at seventeen in the service of the East India Company, where he gradually rose to fill the high post his father had held at the last. For a time he directed the education department, later the Indian government's correspondence with the Indian princes and with foreign states, and at last, as Chief Examiner of the Indian Correspondance, had the superintendence of the whole administration. His activity in these positions has won great recognition from competent quarters, and on the dissolution of the India Company he was offered a seat in the new "Indian Council," which, however, he declined for reasons of health. In his leisure he continued his studies. In his younger days he was an eager member of a debating club in which Bentham's disciples discussed philosophical, literary and political questions with young men from other circles, partly Tory lawyers, partly Coleridgians. As an author he worked for a long time only in reviews and newspapers. His most important essays he collected in a series of volumes under the name *Dissertations and* *[403]* *Discussions.* His definitive views on philosophical questions were founded around 1830. Of influences from a later time must especially be named those he received from Comte, with whom he stood for a long time in lively correspondence, and from Carlyle, whom he admired as a prophet but could not follow as a philosopher. — In those essays one finds the astonishing balance and many-sidedness that distinguish Mill's line of thought. After the strict school of clear thinking he had gone through in his childhood had been supplemented by richer experiences, he was able to show appreciation and to cast light

in many directions.

The reaction against the line of thought of the eighteenth century, and in particular against the ways of seeing that derived from Bentham and James Mill, acted more strongly on Mill's social, literary and religious conception than on his general philosophical (epistemological and moral-philosophical) standpoint. In Caroline Fox's diaries we meet him in the circle of eminent men who frequented the house of the gifted Quaker family in Falmouth[95]). Mill had gone out to Cornwall to nurse a brother who was ill with consumption. The utterances and conversations of his that are reported bear an intimate, personal stamp, and bear witness to a deep seriousness about life and its tasks. He speaks much in a Carlylean spirit, and in that circle one hardly had the feeling of being in the company of one of the most radical men of the century. Later, when the book "On Liberty" appeared, the eager listener felt deeply indignant (as, for that matter, she had also felt at Carlyle's "Life of Sterling"). For Mill there was no contradiction between the philosophy of personality into which he had been led and the strictly rational consideration of knowledge and life that he applied in his inquiry. His significance rests in an essential respect precisely on his having freed the philosophy of personality from the Romantic character, shy of knowledge, that it had in Carlyle. For him personal inwardness and enthusiasm were not incompatible with an untiring search for grounds and causal explanation. He learned — after his crisis — that one can have a great view of life without therefore fearing or despising thought. That Carlyle could never get into his
[404] head. "Poor fellow," he said of Mill, " he has had to work his way out of Benthamism, and the emotions and sufferings he has gone through have helped him to thoughts that never came into Bentham's head. But he is all the same far too eager to prove everything. If John Mill got to heaven, he would hardly be content until he had found out how everything was arranged. For my part I should not care much about the machinery of the place, but should quite give up the question whether there was to be an artisan class or a working class among the angels." *(Memories of old Friends from the Journals of Caroline Fox.* 6 June 1842). In humorous form Carlyle has here aptly indicated the difference between himself and Stuart Mill. Yet it should perhaps be added that if heaven is not reserved especially for Romantics, insight into and work with the mechanism of life would be necessary to prepare the way to heaven. And besides, heaven must be assumed to furnish satisfaction for the intellectual need as well as for every other personal need. What Carlyle reproaches Mill with, he could just as well reproach Socrates with, who imagined, after all, that in the other world he would spend the time as here in testing and inquiring. — A comparison between the way in which Stuart Mill speaks of Carlyle

in his autobiography and the way in which Carlyle in conversation and letters (see e.g. Froude II, 1. p. 74; II, 2. p. 419) spoke of Mill, even just after the latter's death, establishes which of the two men had the better ability and the more honest will to understand and to appreciate. Carlyle, who himself always had the valve open and loudly preached the gospel of silence, could not reconcile himself to the subdued quality in Mill's personality. And when he had had to give up the hope of getting an unconditional adherent in Mill, he had almost nothing but scorn and mockery to spare for him. —

Stuart Mill attached an extraordinarily great practical significance to the purely theoretical debate carried on in modern philosophy between empiricism and speculation. All false assumptions and tendencies in the moral, religious and social domain would — he thinks — be invincible if one did not refute the claim that truths can be won by immediate intuition, by the road of pure thought, independently of experience and observation. *[405]* This claim in reality makes an assumption its own proof! Never has a better means been devised for securing the preservation of all ingrown prejudices! In opposition to this Mill sets the derivation of all knowledge from experience and the explanation of all intellectual and moral qualities by means of the laws of the association of representations. Here there are tasks enough to solve, whereas the intuitive philosophy, which marks the reaction of the nineteenth century against the eighteenth, favors human indolence while at the same time being a bulwark for every kind of conservative prejudice. It was this line of thought (see Levned. p. 226 f. 277 f) that lay at the foundation of the working out of his two chief philosophical works *System of Logic* (1843) and *Examination of Sir William Hamilton's Philosophy* (1865). — In his Logic he has for the first time given a theory of induction, systematized the methods of experience, just as Aristotle systematized the methods in which deductive reasoning moves. Aristotle had as his predecessors the Greek philosophers and sophists, and as foundation the eager discussions that were held in Athens, especially in the Socratic schools. Stuart Mill built on the history of modern natural science through three centuries and on an analysis of the forms in which its work has moved. His work marks the most thoroughgoing presentation of empiricism in epistemology that has been given, just as his father in his "Analysis" had carried empiricism through in the psychological domain. There appears in the foundation of Mill's Logic a clear and fateful influence of his father's psychological theory, a theory that he later (partly in the notes to a new edition of the "Analysis," partly in the work on Hamilton's philosophy) gives up, without becoming aware of what effect this giving up really has on epistemology. His great love of truth, which shows itself in his untiring readiness to go into objections and let them have their full effect, has also opened his

eyes to truths for which at the beginning of his career he did not yet have an eye. He began writing his Logic in 1830, and still in the last edition of the work on Hamilton's
[406] philosophy, at the end of his life, he makes modifications in his conception that arose from his uninterrupted work with the problems. Whereas he starts from James Mill's conception of consciousness as a series or heap of independent elements, which are brought into connection with one another in a purely external and really inexplicable way according to the laws of association, he ends by acknowledging the unity and connection in consciousness as a fundamental psychological fact. "The uniting principle," which Hume wished to push aside, Mill has now acknowledged as a cornerstone. This marks a break with the older English school whose range Mill was not aware of.

Besides the purely philosophical writings named (to which must still be added an interesting book on Auguste Comte), there is a whole series of significant writings of ethical, social and political content. To these belong his *Principles of Political Economy* (1848), in which work he conceives political economy as a part of the whole theory of society and carries through a distinction between the laws of production and the laws of distribution, a distinction that makes it possible for him to acknowledge what is justified in the socialist systems. (It was, however, only in the second edition of the book, 1849, that this acknowledgment properly came forward.) Whereas he had hitherto, in the footsteps of his father and Bentham, fought against class rule and for personal freedom, it was now clear to him that behind political emancipation there stood a far more serious question, the social one. The Saint-Simonians had already opened his eyes to it. That he did not lose sight of the significance of personal freedom is shown by the book *Om Friheden* (1859). In the essay *Utilitarianismen* (translated by G. Brandes under the title: Moral, grundet paa Nytte- eller Lykkeprincipet) he defends the ethical doctrine of utility against its opponents, and in the work *Om repræsentativ Regering* (1860) he gives his political theory.

Mill had a seat in Parliament 1865—1868 and won recognition for his clarity and expert knowledge, although he spoke above all on the most unpopular questions. Gladstone has in a letter (of 19 Dec. 1888, cited in GOMPERZ: *John Stuart Mill.* Ein Nachruf. Wien 1889. p. 46) expressed himself on Mill's parliamentary activity in the following way: "Mr.
[407] Mill's eminent gifts of mind were well known to us all before he entered Parliament. What his conduct there revealed to us, to me at least, was the uncommon nobility of his character. I used at that time in conversation to call him the saint of rationalism . . . . On the other hand he was indeed a philosopher, but not a man of fixed ideas. He united the sound sense and practical tact of the statesman with the lofty independence of a solitary

thinker. I need not say that I greeted his appearance with joy and deeply regretted his disappearance, and that in the interest of the whole House of Commons. He was a boon to us all. In every party, within every political tendency, such men are, I must confess with pain, rare." — The reason Mill was not re-elected was partly that his religious views were used against him, partly that his radical political ideas went too far for the class of voters who had elected him the first time.

During a stay in Avignon Stuart Mill died on 5 May 1873. In him passed away one of the greatest, most honest and noblest minds of the nineteenth century. The course of his life, as he himself has depicted it, is a source of instruction for everyone who strives, just as his writings cast light on some of the most important subjects of human thought[96]).

b. Inductive Logic

Mill's chief strength as a thinker lies in his untiring discussion, his unwearied turning and twisting of problems in order to reach their ultimate presuppositions. In the later editions of the *System of Logic,* in which account is taken of the objections raised against his theories, his exposition takes on a dialogical character. Just as the author of a dialogue, when it truly deserves its name, endeavors to let the different standpoints come forward in as clear and characteristic a form as possible, so it is also Mill's endeavor to do full justice to the views and objections of his opponents; he does not really regard them as opponents, but as fellow workers. That his understanding of other standpoints *[408]* had its limit goes without saying; it was the same limit that his personality and his historical starting point as a thinker set to his inquiry altogether. This limit stands out to a critical examination all the more clearly as the psychological foundation of his whole epistemology shifts under him, without his seeming to have become aware of it. Whoever would assess Mill must first and foremost see him as the great seeker. His merit in philosophy does not stand or fall with the answer to the question whether he succeeded in carrying through empiricism in so absolute a form as even Hume had not yet conceived it.

It is the logic of experience that Mill sets over against the logic of pure thinking, partly in opposition to it, partly as an extension of it. Pure thinking cannot extend our knowledge, but only lead us to preserve agreement among our representations. New truths, however, are obtained only by observation and experience, and the question then arises by what paths, on the basis of observation, we can prove the new truths. It is on the proof, not on discovery itself, that Mill lays the chief weight. His guiding interest is to let all assumptions pass through a purgatory before they enter the heaven of truth. How

assumptions arise in the first place interests him less; it is the justification that matters. The significance and practical worth of philosophy rest for Mill, as for Carlyle, on its being a struggle with habit. The assumptions which have formed in the individual by involuntary association of representations, or which he has received by tradition, must be sifted and tested, so that one can exclude prejudices that hinder progress. For his great hope of the advance of the human race the consistent tracing back of all assumptions to experience stood as an essential support.

Apart from such cases in which a general proposition is laid down by an authority, as with theological and juridical propositions, every general proposition, Mill maintains, is only a summing up of a series of particular observations. Deductive logic, which begins
[409] with general propositions, therefore presupposes an inductive logic, which shows how we arrive at these general propositions from the particular observations. Pure thinking always presupposes experience. If, e.g., I reason thus: "All men are mortal — the Duke of Wellington [who was still living when Mill wrote his Logic] is a man — therefore the Duke of Wellington is mortal", — then it is clear that, in order to have the right to set up the first proposition, about the mortality of *all* men, I must really be certain that the Duke of Wellington too will die. In reality, says Mill, I do not infer from the death of *all* men to Wellington's death, but I infer from a great series of experiences of the death of particular men to the death of this particular man, Wellington. If I really knew that *all* men were mortal, I should not need to infer at all, since Wellington's mortality would be included in the general mortality. My inference is then really this: John is mortal, Thomas is mortal, etc., etc. — so Wellington is presumably mortal too. At the bottom of every syllogism whose major premise is not established by authority there lies, then, in reality an *inference from particular to particular.* This is the form of inference from which both induction and deduction spring, — first induction, and then deduction. The beginning of the whole process of knowledge consists in two phenomena (man and death, e.g.) occurring together for me. The next time the first of them occurs, the expectation of the second will then occur. If this expectation is confirmed, I sum up all these experiences in a *general* proposition, that is to say, in a proposition which gives me *all my experiences* in brief compendium. More it cannot give me, if I am not to make an unjustified generalization. All inference is therefore from particular (or particulars) to particular (from particulars to particulars). This inference occurs already in children. When the child sees the flame on which it has burned itself before, it draws its hand back, not because it has set up any general proposition, but because the sight of the fire immediately calls up the representation of pain. Animals too can infer in this way, for

not only the burned child, but the burned dog too shuns the fire. (Logic. Book II. Chap. 3 § 3). *[410]*

Mill of course sees clearly that this immediate transition from an observation to a representation or an expectation is merely an association, whose justification has first to be tested. It is an association of the kind to which James Mill sought to reduce all association, association "by contiguity": we have often seen A and B together, and now expect B because A occurs again. But can *proof* now be given of the validity of such an expectation? For what is at issue in logic is not what we acknowledge as valid, but what we *ought* to acknowledge as valid (Evidence is not that which the mind does or must yield to, but that which it ought to yield to. Logic. Book III. Chap. 21. § 1). By what right, then, do we infer from the occurrence of one phenomenon (A) to the occurrence of another, different from it (B)?

On the basis of the history of the empirical sciences, a basis in whose use Mill acknowledges having had predecessors in Comte, Whewell and John Herschel (On the Study of Natural Science. London 1830), he now sets up four chief methods by whose help one can distinguish between valid and invalid connections of representations. The very detailed exposition of these methods (whose indefinite main features Herschel had given) is one of the most important parts of Mill's work. Here we can keep only to one main point, the weight Mill lays on the negative cases, the experiences in which a phenomenon does not occur, although the circumstances otherwise, save for a few particular ones, are the same as in the experiences in which it does occur. In such cases the *method of difference,* the chief inductive method, finds application. (It is suggested by Bacon as one of the most important "instantiæ solitariæ"). In the particular circumstances by which an experience that presents the phenomenon differs from an experience that does not present the phenomenon, we must then see something that stands in causal connection with it (either as its cause or its effect or in such a way that both are effects of the same cause). Only then, therefore, can we rightly expect B after A, when it has been shown that with the absence of A follows the absence of B. And we then need only two *[411]* cases, one positive and one negative, to establish a law concerning the relation between A and B.

Only in very simple cases, however, is the inference so simple. In complicated conditions, where many different elements act together, we must first resolve the phenomenon into its particular elements, then by simple inductions investigate how each of these acts, then again by deduction find out what the result of their combined action will be, and finally by observation show that what we have inferred or calculated also really

agrees with experience. The whole process of knowledge thus consists of three members, induction, deduction and verification. — So far is Mill from failing to recognize the significance of the deductive method that he even sees a proof of a science's perfection in the role deduction plays within it. What he insists on is only that all deduction rests in the end on inductions and must be confirmed by the agreement of its results with experience. He rejects pure thinking only when it would begin without a foundation in experience and end without confirmation by experience.

The method of difference itself, which is to bear the whole edifice of knowledge, is, as Mill clearly sees, demonstrative only if we proceed from the assumption that there is such a connection in nature that what has once happened will happen again when exactly the same conditions occur. The uniformity of nature, or the law of causality, is thus the presupposition of the validity of all our inferences concerning real phenomena. In discussing how the causal principle is proved, Mill takes up the great problem which Hume and Kant had discussed. He answers it in Hume's spirit, but in such a way that he would illuminate it from the logical side, whereas Hume declared it logically insoluble and went straight over to giving a psychological explanation of how the assumption of a causal connection could have arisen.

Mill denies that the causal principle is founded on immediate belief, intuition or instinct. In the first place, belief or instinct are not proofs. A strong and firm association
[412] of representations can produce a conviction that cannot be shaken by any counter-proof; but its strength is not itself a proof. On the other hand, Mill holds, every assumption of cause can be suspended and "instinct" pushed aside. Anyone who is accustomed to abstraction and analysis, and whose imagination can move freely, will not find it impossible to conceive an absolute chaos in which phenomena follow one another in ever-changing ways, without any definite law. (Logic III, 21, 1. 4). And often we are compelled to stop at mere "collocations", a sum of factually given conditions which cannot in turn be shown to be caused. Nor, indeed, have men at all times believed in the law of causality; they have assumed "chance" and ascribed reality to "free will". Nor, in order to understand the validity of empirical science, need we assume that the law of causality holds for all phenomena, if only we assume it valid for the domains in which our investigation moves; the motion of the planets may, e.g., be subject to definite laws, even if wind and weather are not. And we lack all right to extend the validity of the law of causality further than to the part of the universe known to us. Experiences can be grounded only by experiences; the very foundation of empirical science must be established by way of experience. Experience must tell us how far we can rely on

experience; we have no higher authority to appeal to. We must make experience its own measure. *(We make experience its own test.* Logic. III, 4, 2).

The causal principle itself, then, is to be grounded on experience and proved by induction. Our conviction that a phenomenon does not occur unless a certain definite other phenomenon goes before it is grounded on our having experienced such a connection between phenomena countless times. The situation with the conviction of the validity of the law of causality for the particular phenomenon is thus the same as with the conviction that the Duke of Wellington will die. Here too it is an inference from particular to particular. But Mill holds that the law of causality rests on so many experiences that it may rightly be said to be the most comprehensive generalization we possess. By giving our more limited pieces of knowledge a support in it, we raise their certainty to a higher level. *[413]*

Yet it was precisely the justification of connecting two representations (e.g. A and B) in the future, because they had previously occurred together, that inductive logic was to establish; the "inference" from particular to particular is, after all, only an association. But we have in reality come no nearer to seeing that justification by being told that it is grounded on the causal principle, when this principle itself in turn proves to be grounded on inference from particular to particular — that is, on association, however countless the repetitions that strengthen this association. (In the causal principle A is "the occurrence of a phenomenon", B "the occurrence of a certain definite other phenomenon"). Mill moves in a circle[97]), or rather: he does not really get off the spot. What he has really attained by his reference to the causal principle as the ultimate basis of the particular inductions is only to establish that the most indissoluble associations will be able to hold up the looser, more unstable ones; a new habit can be strengthened by an old habit of kindred nature. But that is no proof.

Mill has surely let himself be carried too far by his zeal to root out all a priori principles. He has — like so many of his opponents — confused the assumption that a principle springs from the very nature of consciousness with the proof of its real validity. He is right that, however much a principle has its ground in the nature of our consciousness, its real validity is not thereby proved. But it might be that there were hypotheses which we were driven by the nature of our consciousness to set up, and which it was the task of empirical science to test in detail. The task of scientific knowledge will then be partly the making precise, partly the verifying of involuntarily posited presuppositions. Mill harbored suspicion every time it was maintained that involuntary assumptions could become of significance for knowledge; he feared that one would at once smuggle in a

dogma. A march on the spot was the nemesis his suspiciousness brought down on his own epistemology.

An inference from particular to particular does not in reality exist, if by inference
[414] one understands a process that leads *with necessity* from the one assumption to the other, and if one will not abolish the difference between inference and mere association of representations. If the transition from the one particular to the other is to be justified, this can only be done by the likeness between the first particular and an earlier one: I infer that, since B_1 corresponded to A_1, B_2 will correspond to A_2. It is an analogy that here guides the transition, and on a test of the analogy's validity the decision as to the justification of the transition depends. Mill too comes to admit the significance of the relation of likeness for all inferring (Logic. II, 3, 3 and III, 3, 1); but he does not see that this abolishes the possibility of an inferring from particular to particular such as he has described it, and that the principle of identity becomes the ultimate presupposition of all inference, whether it appears in inductive or deductive form. A blind coupling together of representations gives no logical connection: only where the relation of identity can be shown is the blindness removed[98]).

But the logical fundamental propositions themselves Mill regards, in consequence of his pure empiricism, as grounded on experience. I experience early that light and darkness, motion and rest, past and future are different and incompatible predicates. I experience that I cannot at the same time believe something and not believe it. Belief and non-belief are incompatible mental states. From this I then derive by generalization the proposition that what is self-contradictory cannot be true. But to this proposition there can belong no necessity that goes beyond what experience teaches. Even if we stop at certain assumptions which we cannot reject without feeling self-contradiction, we may not therefore infer that they are in themselves, in reality, impossible. There may be associations of representations that are so firm that we cannot dissolve them. But the history of science shows us examples enough that what at one time was regarded as unthinkable was nevertheless true. When we see that inconceivable things in the course of time become conceivable, the inconceivability of the opposite of an assumption cannot be any proof of the correctness of this assumption. What is possible and real we learn only by way of experience, not through subjective criteria. — Of the two
[415] questions, whether there are self-contradictions that are more than factually indissoluble associations of representations, and whether we have the right to assume that the self-contradictory cannot exist, it is evidently the latter that has the greater interest for Mill. He even expresses himself in his later writings (Examination of Sir William Hamilton's

Philosophy. 2. ed. p. 67) doubtfully as to whether the incompatibility of two mutually contradictory assumptions is grounded on the original nature of our consciousness or springs from experience. To change the logical principles, he remarks (ib. chap. 21), experiences would be required which the conditions of our life do not make possible. But in any case he holds fast that a subjective necessity cannot prove an objective reality. What Mill overlooks here is the stimulating, forward-driving force of self-contradiction. It is the self-contradiction in some of the inferences we draw from experience that sets us problems and spurs us to work to procure the self-contradiction away — by extending experience. If we did not constantly use self-contradiction as a criterion, our thinking would soon fall asleep.

The mathematical fundamental propositions are for Mill, like the logical fundamental propositions and the causal principle, mere generalizations from experience. Mathematics is indeed a science of reason which by way of thinking leads to necessary results. But it rests on fundamental propositions that can only have been taken from experience. That two magnitudes which are equal to one and the same third are equal to each other is a proposition of experience and nothing more. Likewise it is a proposition of experience that two straight lines that have intersected will always diverge. The geometrical definitions contain elements from experience, although these elements of experience are represented as more perfect and more exact than experience presents them. The definition of the circle thus implies that all the radii are equal, which they cannot be shown to be in any real circle. All real circles nevertheless contain greater or lesser approximations to the ideal circle about which geometry proves its propositions, and the greater the approximation, the greater the validity the application of these propositions acquires in reality. We make a leap from the approximately equal to the absolutely equal, because we can thereby derive consequences; and in applying these *[416]* consequences we then take the degree of approximation into account. Geometry thus rests on hypotheses, or, if one will, on fictions. If geometry is to have real validity, the presupposition is that the nature of space corresponds to the observations we have used to form our ideal hypotheses or fictions. But whether space is so constituted everywhere in the world we cannot know a priori. — Had Mill gone more closely into the idealization which leads to the setting up of the hypotheses of geometry, his whole epistemology would perhaps have taken on another character, since he would have seen that one may very well acknowledge an a priori element in our knowledge without therefore assuming that assumptions about the nature of existence can be founded on it without further ado.

Mill's attempt to carry through empiricism and to make earnest of the doctrine of

the blank tablet cannot, then, be said to have succeeded. But by venturing the attempt he has, through his acute and ramified investigations, thrown light on the nature of our knowledge. The attempt itself may be regarded as a counterpart to Hegel's dialectical method: whereas Hegel would develop truth by the self-unfolding of pure thought, Mill would develop truth by piecing together independent representations taken up from without. —

A peculiarity of Mill's epistemology is the dependence in which it stands on his psychological presuppositions. He builds his logic on his father's psychology. Association by contiguity as the fundamental form of all connection of representations, and indissoluble association as the explanation of what stands as inconceivable: these are the means he operates with. *Psychology* is for him the fundamental science which bears all other sciences. The laws of association are the fundamental laws of all our knowledge.

Mill has not himself given any exposition of psychology, but has in his Logic treated its method and in other writings ("Examination", 1865, and in the notes to James Mill's
[417] "Analysis", 1869) treated particular psychological questions in an interesting way. The exposition of psychology with which in his later years he most nearly sympathized was the one his friend and disciple Alexander Bain gave in his two chief works *The Senses and the Intellect* (1856) and *The Emotions and the Will* (1859). But in these works the strict association psychology, as James Mill taught it, is abandoned, and the relation of likeness is — under the influence of William Hamilton — acknowledged as the condition of all association of representations, association by contiguity included. Mill associates himself with this in his notes to the "Analysis". The relation of likeness should thus, to be consistent, now have gained a greater influence in epistemology than Mill had granted it in his Logic, where, as we have seen, he grounded all inference on association by contiguity. He did indeed already in the Logic (I, 3, 11; 5, 6) acknowledge the relations of likeness and of difference as distinct, irreducible relations; but he did not investigate more closely the role these relations play in all connection of representations and all thinking.

Even in his later writings Mill would hold fast to Hume's and James Mill's conception of consciousness as a series of states, only he adds: — real or *possible* states. But he finds a difficulty at which his predecessors had not stopped. In memory and expectation the presupposition is contained that *I myself and no other* have had at another time, or will have, such a state of consciousness as I now have the representation of. There must then be more in consciousness than the particular members that make up the series. How can a series know that it is a series, — that it has past members and will have new

members? The presence of *a bond* is here made evident between the particular members of consciousness, the particular sensations and representations, a bond that is just as real as the particular members themselves and is not a mere product of thought. If we are to have a name for this original connecting element in our consciousness, we must call it our ego or our self. (Examination. Chap. 12. The most characteristic and decisive statements were first added in the later editions). In the notes to the "Analysis" (II. p. 175) Mill expresses himself thus: "There is a certain bond between all the members of the series which makes me say that these were sensations of a person who was the *[418]* same person throughout . . . . and this bond constitutes for me my self". Further than this — Mill adds — we can at least not get by way of psychological analysis. With this, association psychology is definitively abandoned. Mill has, as has been aptly said, opened a trapdoor in the middle of his own philosophy. For Hume "the uniting principle" consistently stood as a riddle, because he proceeded from the particular sensations and representations as the only reality. When Mill now acknowledges the uniting bond as just as real as the particular elements, he corrects the whole concept of consciousness from which Hume, and after him James Mill, proceeded. The laws of association can now only be special forms of "the uniting principle", and the English school has thus by its untiring analysis corrected its own principle[99]). It brings contradictions into Mill that he shows the relation of likeness and the bond to be realities, although he builds his logic on purely external and accidental association. But it bears witness to the honesty and acuteness of his search that he thus arrives at results which lead beyond his own original presuppositions. —

A peculiar application that Mill makes of the laws of association must still be mentioned. He seeks to show how belief in an external world can be explained from them. — What I know of the world, I know through my sensations. But does there then exist in reality a world in the sense of something different from my sensations? — What is really given are the sensations I have at this moment. But besides, in memory and expectation I have representations of possible sensations, representations which by repetition and association can come to form fixed, coherent groups. And these groups continue to stand before us in the same way in our representation, whether they are given as sensations or not. We thereby get the representation of something that stands fast, whether we perceive it or not. What is the so-called "external" object other than the regular possibility that certain sensations recur in the same way as I have had them before? — In the assumption that there exists something independent of my consciousness I am *[419]* confirmed when I experience that other sentient beings have a similar regular series of

sensations to my own. — What I mean by "matter" is then nothing but a permanent possibility of sensations. It is the fixed and common associations of representations that lead us to set up the concept of matter. And since we find that the concept of cause holds within the series of our sensations and representations, we involuntarily apply it also to the whole series, the whole sum of our sensations and representations, and regard matter as their cause. — The practical belief, the strong inclination to assume an external world, is for Mill no proof. By virtue of the principle of fixed association he holds that such an assumption is not necessary. One need not assume anything existing other than consciousness — only that besides the consciousness really given one must also assume the *possibility* of new states of consciousness. It is clear that in this "possibility" Kant's "thing in itself" lies hidden. When Mill says (Examination. 2. ed. p. 189): "The not-self might be merely a form in which consciousness represents to itself the possible modifications of the self", then it must be asked whether the self can itself turn these possibilities into realities, so that its changes of state could be understood out of itself. If the answer is yes, the self will stand as creating its inner world out of nothing; if the answer is no, one must either deny the causal principle or assume a reality besides the self. Mill does not go more closely into this side of the matter and in general brings out too little the connection in which the problem of the reality of an external world stands with the problem of causality.

c. The Principles of Ethics

Nowhere does Stuart Mill's peculiar position within the history of thought show itself more clearly than in his ethics. Brought up in the spirit of the eighteenth century, he seeks with honest will to make the points of view of the new age his own; but since the presuppositions are taken from the old school, while the goal he would reach and the
[420] points of view he acknowledges are due to the influence of new motives, there appears at decisive points in his line of thought a duality which he himself does not always notice. In the ethical domain this duality consists in his holding fast to the principle of utility, which he had learned to acknowledge in Bentham's school, but wishing to supplement it by bringing it into agreement with the philosophy of personality, whose ethics is of a more subjective kind. He saw clearly that this task is the most important one for the future of ethics, although he did not succeed in giving a theoretical solution of it. The spiritual crisis he went through at a certain point of his development was conditioned precisely by this problem. That in practice, in the art of life, in his great and significant activity, he united in a rare degree regard for the inner development of the personal life

with regard for the outer effects of particular actions, his biography bears witness. But the theory of this art he was not able to give.

The reason why he did not succeed in this must doubtless be sought chiefly in his great piety toward Bentham and his father. He had at one time, in the subjectively strongly agitated period after that crisis, spoken out strongly against Benthamite utilitarianism, which he reproached for taking a purely external business standpoint. Now he believed he had got beyond the reaction against the view into which he had been initiated in his early youth, and he now came forward as its defender — without being clear how far he himself had altered the theory whose defender he would be. He did not distance himself sufficiently from the older utilitarianism, and at particular points his theory is also determined more by it than, to be consistent, it should be. That is why his treatise "Utilitarianism" has become one of his least clear writings.

A clear and natural measure in ethical valuation we obtain — of this Mill is constantly convinced — only by keeping to the effects of actions. If we ask *why* an action is good, the decision will in the end always depend on whether it produces feeling of pleasure at some point or other and in some extent or other. Human nature is once and for all so constituted that it desires nothing which is not either the whole of happiness, a part of *[421]* happiness or a means to happiness. Every action is judged and must be judged according to the degree in which it leads toward what is thus always the ultimate object of human desire. In this consists the application of the principle of utility.

To the question *whose* happiness it is that is to furnish the measure of valuation, Stuart Mill answers: "Not the agent's own greatest happiness, but the greatest total sum of happiness". And this assertion he grounds not, as Bentham in the "Deontology", by pointing to the harmony of well-understood egoistic interests, but by giving a psychological explanation of the origin of the moral feeling. It is the moral feeling that makes it possible for us to strive to produce happiness, even if it is not our own happiness that we produce.

Stuart Mill does not take the moral feeling to be innate, but to be a highly composite product. Its most important elements are sympathy, fear, religious feeling of various kinds, experiences of the effects of actions, self-esteem, desire for the esteem of others. In its extraordinarily composite character must be sought the cause of the mystical character with which the notion of moral obligation appears. There is a great quantity of feelings that must be rooted out before one can do anything that conflicts with what one regards as right. The association between those various elements is besides so strong that the feeling appears with an indissoluble unity. And since the laws of association

are laws of nature, the moral feeling is a natural feeling, notwithstanding that it has a history of its genesis. It is, after all, natural for men to speak, to draw inferences of reason, to till the soil and to build cities, and yet these skills are not innate, but acquired. If any single element in the moral feeling were innate, it would have to be sympathy. But the greatest weight must be laid on the fact that social life accustoms all individuals to having a common interest, to working with united forces, to constantly taking account of one another. There forms a kind of instinct that leads individuals to solidarity in feeling and in action. And the more social life develops by the barriers between the
[422] different classes being broken down, the more solidarity grows. The feeling of unity can become a religion when it is constantly favored by education and by the ordering of institutions and is enjoined by public opinion. There takes place through social life an education of egoism, so that even if regard for others was at first only a means, it later becomes itself an end. Whereas Bentham — at any rate in practice — proceeded from self-interest as the universal motive, Stuart Mill was convinced of the reality of disinterested feelings. Thus he remarks, in his interesting essay on Plato, apropos of the doctrine of the dialogue Gorgias that it is better to suffer than to do wrong: "The advance which 'Gorgias' marks is one of the greatest that has ever been made in moral culture, namely the development of a disinterested choice of duty for its own sake, such that this choice signifies a higher state than that in which selfish inclinations are sacrificed to a more remote selfish interest". (Diss. and Disc. III, p. 340).

Stuart Mill thus embraces an individual doctrine of development, like Spinoza, Hartley and James Mill. The principle of indissoluble association serves him — like his predecessors — to ground and explain the inner foundation of ethics, while the principle of utility serves to ground the judgment of particular actions. In clarity and simplicity, however, his exposition cannot measure itself with Spinoza's and James Mill's. There is a certain wavering in him as to what is original and what is acquired, and he also entangles himself in a difficulty his predecessors did not know, by assuming differences of quality between feelings. One feeling of pleasure is to be able not only to be stronger, more lasting, more unmixed with pain, more fruitful in happiness-bringing effects, more common to many than another feeling of pleasure; it is also to be able to be of a qualitatively higher kind, which shows itself in that whoever has experienced it seeks it, even if it can be obtained only by going through great suffering, and prefers it to the greatest possible quantity of another feeling of pleasure. The question is whether the "difference of quality" Mill here posits does not find its explanation precisely in certain feelings of pleasure being
[423] sanctioned and favored by the moral feeling whose origin Stuart Mill himself has sought

to explain. In any case the question whether there are qualitative differences between feelings is far too complicated a question to be solved by the simple proof Mill adduces. —

While Stuart Mill, by laying so great a weight on the subjective foundation and by assuming a difference of quality between feelings, moves far away from Bentham's standpoint, he is still far too much a Benthamite in distinguishing between the value of the action and the value of the acting personality and in holding that ethics has to do only with the former. Precisely from a utilitarian standpoint this distinction must be untenable. For since every motive contains possibilities of far more actions than the particular one being judged at the moment, valuation must extend to it as well. Even if the particular action does not give rise to effects that conflict with the principle of welfare, actions may nevertheless spring from the same motive which must be condemned, and the motive itself becomes in such cases ethically questionable. It is consistent that valuation goes back to the source. In the book "Om Friheden" the distinction mentioned plays, as will appear, a large role too.

d. Social Ethics

Of the whole series of ethical and economic questions which Mill has treated in his various writings and essays, only a few will be brought forward here to illustrate his ethical standpoint.

α. *Individual and Society.* (Om Friheden. 1859). — Mill fought for emancipation and self-development. But he experienced that political freedom does not bring with it real, spiritual freedom and independence. In place of the physical compulsion that is gradually displaced, public opinion, the moral police, easily steps in. And this tyranny is more dangerous than the political one, since it leaves fewer ways of escape open, penetrates into daily life and makes the soul itself a slave. One follows the current, even when it concerns only which pleasures one is to enjoy, instead of following one's own drive *[424]* and one's own judgment. In England especially, says Mill, the yoke of public opinion is heavy, while the yoke of the law is lighter than in other countries. A great danger threatens here: the rule of the masses, the rule of collective mediocrity. The mass must certainly always be led by individuals — but it prefers to take its opinions from people who do not stand too high above it. And yet the impulse to everything that is noble and wise always comes from individual outstanding people. The few original people are the salt of the earth! — Mill will, however, have no cult of great men. All they can demand is full freedom to utter the new ideas. Power to compel others would first and

foremost corrupt the great themselves. The means of avoiding that danger, which lies in the tyranny of public opinion and in the lowering of the level by the rule of the masses, is *freedom,* the only constant, never-failing source of progress, because it creates as many independent centers of improvement as there are personalities.

As a general fundamental proposition for the interference of others in the affairs of the individual, Mill lays down that an individual's freedom of action may be restricted only when an interference is necessary to prevent his conduct from bringing suffering on others. Care for the individual's own good is not sufficient ground for such interference. The only part of his conduct for which the individual owes society an account is the part of it that concerns others. — By interference Mill understands partly physical compulsion, partly the spiritual compulsion that lies in the judgments of others. In the latter respect he distinguishes between "moral disapproval or condemnation" and "expression of displeasure or withdrawal of esteem". (Om Friheden. Danish trans. p. 142). This, however, is a very fine distinction, since to withdraw my esteem from an individual is to inflict a pain on him — if, that is, he sets store by my esteem; and if he does not, my "disapproval" will surely not pain him either. The distinction is just as difficult as that between actions which have effects only for ourselves and actions which have effects for others as well. Whoever does not develop his personality, or who shows
[425] a lack of prudence and personal dignity, can thereby deprive others of a power to which they rightly had a claim. Precisely when one maintains, as Mill does, that what matters is to get as many independent starting points as possible, it is impossible to mark off any domain that could not acquire significance beyond the limits of individuality. This distinction between what concerns the individual himself and what concerns others is as little tenable as the previously mentioned distinction between the value of the agent and the value of the action. And in both places Mill would precisely by virtue of the principle of utility have been able to arrive at a more correct conception. —

The importance of as great a freedom as possible Mill insists on as regards both opinions and actions. — If an opinion is true, what harm is there then in its being freely discussed? Should it contain only a part of the truth, then only free discussion can decide how great a part of the truth it makes up. And only by being kept alive through discussion does it gain influence on characters and actions. Even if, finally, an opinion could be useful without being true, its utility must nevertheless be discussed just as well as its truth; in our days — Mill holds — the utility of opinions plays quite as large a role as their truth. — As for actions, they cannot of course have as great a freedom as opinions. But Mill maintains that the value of the different ways of living must be tested

in experience, and that this can only be done by giving differences of character as much scope as is possible without causing others suffering. It is a condition of the individual's own happiness as well as of individual and social progress that its own character, and not custom and usage, determine its manner of acting. Strong drives and desires are a good; they are the stuff of which heroes are made, and men do not act badly because their desires are strong, but because their conscience is weak. —

Even if the boundary between individual and society is not so easy to draw as Mill thinks, he has nevertheless by his investigation (which, moreover, takes up Kant's fundamental thought in his "Rechtslehre") shown that the burden of proof always lies on whoever would have freedom restricted, just as he has impressed the responsibility *[426]* that is attached to every moral judgment.

β. The Woman Question. (Kvindernes Underkuelse. 1869). — The question of woman's right is for Mill only a single side of the great question of emancipation. Here too what essentially matters is the struggle against the abuse of power, the removal of the hindrances to the free unfolding of personality, the abolition of the right of the stronger and of all external authority where this is not necessary as protecting and educating. Besides, in his investigation of this question he finds application for his favorite theory of indissoluble association, to explain the opinions that hinder development. The notions people have of the female nature rest, he maintains, only on habit and tradition, not on real experience. What we call the "female nature" is only an artificial product. Only by opening the barriers that have hitherto been set to the development and use of female capacities can one come to know the real female nature. It is the same line of thought as in the book on liberty: compulsion must go, so that experience can be gained. It is a practical application of the method of difference. But Mill believes he knows the result beforehand; he maintains that the mental capacities of women and men are equal — an assumption which is of course not necessary to ground equality of rights.

γ. The Question of Representation. (Om den repræsentative Regering. 1861). — In discussing this question too Mill pursues his main thought: how freedom can be secured not only against the despotism of individuals and estates, but against the despotism of the majority. He discusses the dangers and advantages of democracy. The constitutions between which the struggle now stands are in reality only two: democracy and bureaucracy. The only possible solution is that democracy take bureaucracy into its service and merely reserve to itself superintendence and control. Parliament should represent only the decision of will in the state; for drafting laws it is not suited; for that commissions of experts should be appointed. To secure the right of the minority in

[427] decisions Mill recommends, among other things, proportional representation. Clear as his eye is, as the book on liberty also shows, for the defects of democracy, he nevertheless sees clearly how shortsighted are those who sigh for a despotism in order to get their plans of reform carried through. They forget that the development of the people itself is the most important condition of continued progress, and they do not see that a good despotism in a fairly civilized country is even more harmful than a bad one, because it is far more enervating to the spirit and strength of the people.

δ. The Social Question. (Principles of Political Economy. 1848). — In the course of Mill's development it became more and more clear to him that the social question takes precedence over the political. His democratic conviction he never gave up; but partly, as we have already seen, he came to have an open eye for the defects of democracy, partly there formed for him an ideal of the future which lay far beyond the democratic program. Without great social changes individual and political freedom would in any case not really be able to benefit all.

Already in a letter from the year 1842 (to Robert Barcley Fox, printed in Caroline Fox's Diaries. II. p. 272) Mill speaks of the significance of the social question: "I believe that ever since the changes that took place in the constitution through Catholic Emancipation and the Reform Act, a great part of the ruling class, especially younger men, have gradually had their eyes opened, and the progress of Chartism is producing, I believe, the feeling that those who govern must strive, both from duty and from prudence, to a far higher degree than they have long been accustomed to, to look after the temporal and spiritual interests of the poor". But he must add: "As to the means, on the other hand, of curing or even merely alleviating great social evils, people are just as far from harbor as they were before". To a definite solution Mill did not come later either. He acknowledges openly that here there are two fundamental thoughts, both of which he must hold fast, although in their consequences — at any rate apparently — they conflict with each other. They are the ideas of individualism and of socialism, of which Mill (Levned. p. 259) declares that they are diametrically opposed to each other. How it is then possible to unite them,
[428] he declares he does not know; that he leaves to the future. It was his conviction that we do not yet know either of the two systems in its best possible form. We know neither yet what free self-activity in its highest degree and form can accomplish, nor what possibilities there are for a thoroughgoing social ordering of the external conditions of life. Nowhere perhaps do Mill's great qualities show themselves so clearly as here. Together with a great hope for the future, indeed perhaps precisely supporting himself on it, he takes up a critically discussing position toward the different possibilities that

experience presents. What is merely handed down and customary does not impress him; he calmly investigates the conditions of its subsistence and looks to see whether these conditions might not be replaced by others.

In his investigation of the socialist systems he took up a far more unprejudiced standpoint than the economists of his time were wont to. That followed from the spirit of his philosophy. He could not, like the usual opponents of socialism, see in the right of inheritance and the right of private property dogmas that did not need to be grounded. The way in which the yield of production is distributed depends, in his conception, to so high a degree on the actual social order and on the habit and will of men that it was a great piece of sleight of hand when the economists were wont to regard the mode of distribution now prevailing as grounded on eternal natural necessity. As regards private property in particular, it was not introduced for any of the reasons for which it is now upheld. Originally society, for the sake of peace, maintained the individual in the possession of what he had once and for all seized. Whether this arrangement is to be retained for the future or replaced by an arrangement such as socialist systems conceived, that is now an open question, and a question which is the object of discussion through all classes in the civilized countries. Mill finds precisely what is peculiar to our time in this, that all principles are discussed, and that even those who suffer most under the existing institutions take part in the discussion of the value and justification of these principles. It is the first time in history that this has been the case. What will the result be? Mill answers (Principles of Polit. Economy. II, 1, 3): "If I may venture a conjecture, *[429]* the decision will probably depend chiefly on one consideration, namely which of the two systems is consistent with the greatest degree of human freedom and self-activity. Once the means of sustaining life are secured, the need of freedom is the strongest of all the personal wants of men, and it grows in strength, instead of diminishing, the more the intellectual and moral faculties are developed . . . An education which taught men, or social institutions which commanded them, to give up control over their own actions in order to obtain comfort and abundance, be it in ever so high a degree, or to renounce freedom for the sake of equality, would rob them of one of the most exalted characteristics of human nature". Now Mill does think that this objection to socialism is often exaggerated, and he maintains that the compulsion this system would entail would deserve to be called freedom in comparison with the compulsion under which the great working class now lives. But he sees no reason to give up the system of private property, if only the laws will do as much to prevent its drawbacks as they now do to aggravate them. The present order favors the development of selfishness to a degree

which is not in itself necessary, even if the system of private property is retained. The socialists wrongly blame competition for all social evils. They forget that where there is no competition, there is monopoly and with it taxation of the industrious in favor of the idle. Competition among the workers themselves certainly presses wages down; but all other competition is to the good of the workers by making the necessaries of life cheaper. Not competition, but the subjection of labor to capital is the cause of the drawbacks. (IV, 7, 7).

What according to Mill essentially matters is, by vigorous intervention on the part of the state, to raise the whole level of the working class, its demands on life and on itself. What matters is to reach by peaceful means such an advance as the French working class gained by the French Revolution. By improved education, by the parceling out of land
[430] and by an emigration carried through on a large scale the working class will be able to attain such a social position and to have acquired such ideal and material needs that, in order not to lower the level and diminish wages, it will show the necessary self-control and avoid improvident increase of population, even if this effect only occurs in a later generation that has grown up under the more favorable conditions. (II, 11, 2). But Mill also expected much of the free associations (trade unions and especially productive associations) and followed their development in France and England with great interest. Their chief significance he saw in their favoring the virtues of independence: justice and self-control. In such smaller circles experiments in a socialist direction could moreover be made which might become of significance for the continued discussion of the social question.

e. The Religious Problem

In the writings Stuart Mill himself published he touches only in passing on religious questions. Most fully in the work on Hamilton's philosophy. He here disputes two conceptions. The one (which Hamilton too, from his standpoint, had disputed) was the one Schelling and Hegel had asserted, that by way of pure thinking a scientific concept of God could be set up, a concept of an absolute and infinite being, the author and end of all things. Mill maintains that in the religious question too we must proceed from experience, and that it must be the contemplation of nature that leads us to the assumption of a God. He seems also to hold that it is possible to ground such an assumption, and in accordance with this he maintains (in his work on Comte and positivism) that even if one assumes with Comte the law of the three stages, there will still remain on the positive stage *open questions,* and that one may very well hold fast to theological assumptions, provided only

these are framed so that they do not come into conflict with what experience teaches. The other conception Mill disputes is the one Hamilton and especially Mansel embraced, that even if the scientific investigation of the concept of God as the absolute and infinite, *[431]* almighty and all-good being should lead to self-contradictions and to consequences that conflict with what the human conscience must maintain, we must nevertheless believe in such a being, since neither our logic nor our ethics can find application to the Deity. Against this doctrine Mill expresses himself with great indignation. "If", he says (Examination. 2. ed. p. 103), "instead of the glad tidings that there exists a being in whom all the excellences which the highest human thought can conceive are found in a degree inconceivable to us, I am told that the world is governed by a being whose attributes are infinite, without our knowing what they are, any more than we can understand what principles it follows in its government, except that 'the highest human morality which we are able to conceive' does not sanction them, — convince me of it, and I will bear my fate as well as I can. But when I am told that I am to believe this and at the same time call this being by names which express the highest human morality, then I say in plain words that I will not. Whatever power such a being may have over me, there is one thing it shall not be able to do: it shall not compel me to worship it. I will call no being good who is not what I mean by it when I apply this word to my fellow creatures; and if such a being can sentence me to hell, well then, to hell I will go".

From this forceful statement, which aroused great attention and offense and was used against Mill by political opponents, one will see what for Mill was the kernel of the religious problem, namely the moral impossibility of acknowledging a higher measure of moral valuation than the human one. Already Bentham had (in Deontology. Part. I. Chap. 8) spoken out sharply against calling an attribute in God love which in man would be the opposite of love; that is to call a dagger thrust a kiss! And according to what Stuart Mill relates in his Levned (p. 42), it was the impossibility of reconciling the evil in the world with belief in an almighty and all-good creator that determined James Mill's rejection of all religious assumptions and made him feel a certain sympathy with the Manichaean assumption of a good and an evil principle that struggle with each *[432]* other for dominion in the world. For his own part Stuart Mill did not express himself definitely in any of the writings that appeared in his lifetime. It was certainly not from fear of men that he refrained from uttering his views. What he did utter was enough to brand him as a man who departed to a high degree from the usual religious notions. His utterances were indeed used as a means of agitation against him, and a church organ declared, when he was dead: "His death is no loss to anyone, for he was a gross

unbeliever, however nicely he behaved, and a very dangerous person. The sooner the 'lights of thought' who hold the same opinions as he go to the same place where he has gone, the better it will be both for church and state". What opinions Mill held, even the church organ, which was hardly in doubt about Mill's fate after death, would have had trouble saying. The explanation must be sought in his not himself finding his ideas on the religious problem fully developed; the question was for him in reality open, and he would not speak publicly about it before he had satisfied his own demands for clarity and thoroughness. Even his friends knew no more of his religious standpoint than has been set out above, until his *Essays on Religion* appeared some time after his death. Of the three essays this work contains, two (on the utility of religion and on theism) were not finished; only the third (on nature) did he himself consider worthy of publication. — A contributing reason for his reticence may perhaps also have been that he would not mix several problems together, but have them discussed each by itself. He felt no need to fire off his artillery in all directions at once. The secret of the great influence which, in spite of his radical standpoint, he gained in his time has partly lain in his taking one point at a time. As he himself said to a friend: "I spare no prejudice, but I attack one at a time".

The fundamental thought of the essay *on nature* is that one cannot set up nature, as it
[433] is without human intervention, as a model for man, any more than one can infer from it an all-wise, all-good and almighty author. Nature can impress us by its might and power; but its conduct bears the stamp of terrorism and injustice; it gives to those who have, and is merciless to those who have not. The only possibility of uniting belief in a God with experience of the real world consists in the assumption that the Deity is indeed good, but not almighty. Omnipotence must be sacrificed to goodness. In the essay *on theism* this is developed more closely. The author of the world order has been obliged to conform to conditions that were independent of his will. The matter and the force in the world are not created; their properties and laws are independent of the will of the orderer of the world. It is then no wonder that there are so many imperfections! Everything that checks the tendency, traceable in nature, to produce something purposive and good must be put down to the account of the material hindrances with which the Deity has to struggle. The mood that will be peculiar to the religion of the future must be the inspiring feeling of being the Deity's fellow worker, a feeling that cannot without self-contradiction be united with belief in an almighty God. What the Deity's will is can only be discovered by attending to everything in nature that tends toward general welfare and a higher development of life. In this struggle between the good and the evil forces in the world

even the slightest contribution can be of significance. To do something, however little it may be, to further the victory of the good is the most living and strengthening thought that can animate a man. This thought will be a leading thought in the religion of the future! — Already in a letter of 1841 (to Robert Barclay Fox, see Caroline Fox's Diaries. II, p. 260) Mill mentions this thought, referring to the Saint-Simonians' doctrine of continuing the work of creation.

One point in Mill's polemic against Hamilton first becomes properly intelligible when one knows that he conceived the Deity as a limited being. For he cannot understand all the difficulties Hamilton finds theoretically in the concept of God, namely not how *[434]* the law of relativity can be a hindrance to holding fast this concept in thought. God is thought in relation to the world — what contradiction is there in that? asks Mill. He surely presupposes here his own concept of a limited God; for the self-contradiction is of course present only so long as the Deity is to be thought as an absolute and infinite being — and yet in relation to something that is not itself, so that it is thus conditioned and limited, i.e. becomes relative! — It was not logical but ethical reasons that led Mill to his standpoint in the philosophy of religion, which recalls the one Voltaire and Rousseau took.

When Mill, from the tendency to purposiveness and heightened life which can be traced in the world alongside wholly opposite tendencies, infers the assumption of a Deity, this inference is shaken if a purely scientific explanation of purposiveness and the development of life were possible. Mill rejects the dogma of creation because every phenomenon must be explained by another phenomenon; but must not for the same reason the inference too which he believes he can draw from purposiveness and the development of life be rejected, since here too definite natural causes must be sought in each particular case? It is these causes that the hypothesis of evolution seeks to show, and Mill does indeed declare that if the hypothesis of evolution is adopted, it will not indeed make impossible the assumption of a creative intervention, but will nevertheless greatly weaken the proof of such an assumption. "Let us leave this remarkable hypothesis to the fate which the progress of science reserves for it".

Even if religion could not be proved, it will — in Mill's conception — not fall away so long as it is of benefit to man. Mill investigates in a separate essay *the utility of religion.* Religion has this in common with poetry, that it springs from a need of grander and more beautiful images than those prosaic life brings with it. Human life will, in spite of all progress, continue to be so wretched and narrow that man will always need greater breadth and height for his destiny. Imagination must be freed from the limitation that

[435] positive experience brings with it. Even if religion differs from poetry by the weight it lays on the real significance of ideals, the essential thing in all religion is nevertheless that feeling and longing are directed in a forceful and earnest way toward an ideal object. This condition is fulfilled by Comte's religion of humanity better than by any of the religions that embrace belief in God as the author of the world. The feeling of unity with humanity, of deep sympathy with its welfare and progress, is a disinterested feeling which does not contain anything that can become questionable in a logical or ethical respect. Only one form of supernatural belief is there that is neither illogical nor immoral, namely the one described above, which regards nature as the product of a struggle between a good and wise being on the one side and on the other either matter (as Plato assumed) or an evil principle (as the Manichaeans assumed). Whoever embraces this belief can be sure that the evil in the world is not a work of the being he worships. Even if this belief could not be proved, even if it is rather a hope than a belief, true wisdom of life nevertheless demands that it be held fast with all one's might. It brings with it a bright and cheerful way of taking life, and it does not exclude reason and criticism. By one-sidedly and needlessly turning one's gaze toward what is dark, narrow and unhappy in life one only hampers one's capacity for action. The hope that beyond the limit set to our experience there are good forces and an immortal life has its essential significance in giving the scale of the feelings a greater extent. Mill ascribed significance to belief in immortality especially for the sympathetic feelings; he even at times placed the whole significance of religion at this point, as when he said to a friend who had lost one of those nearest to him: "In my view the only lasting value of religion consists in alleviating the feeling of complete separation which is so dreadful in a great sorrow". But in his work on the philosophy of religion he dwells also especially on the significance Christianity has had and has in setting up an exalted model for the race; it is far more by its image of Christ than by its idea of God that Christianity has worked. And
[436] the influence of this model will — Mill maintains — not fall away because one conceives it in a purely historical, human manner. —

Mill's religion has this in common with Kant's, that it is rather a hope than a belief. He is quite clear that the decisive question is this: "Is it irrational to let oneself be determined by the imagination to a hope for whose realization one may perhaps never be able to give any probable ground? Must one combat this hope as a departure from the principle of reason which bids us regulate our feelings as well as our opinions according to strict proofs?" Mill holds that on this point there will long be dispute among thinkers, and that each of them will decide it according to his temperament. But he holds as

well that the question itself has not received the serious discussion its great significance demands.

Mill's own solution leaves one side of the problem standing. If the good principle struggles with material chaos, there must nevertheless be a world order that embraces them both and makes the struggle possible. Struggle is interaction, and interaction presupposes an order of things that makes it possible for the different acting forces to meet. Thus the old problem rises again behind Mill's solution. Mill's critics in the philosophy of religion in England[100]), who for the most part operate with Hegelian weapons, have especially brought forward this point, while they have not felt the ethical sting of the problem as strongly as Mill. — Psychologically regarded, Mill has passed over a factor that plays a large role in religion, namely the need of absolute absorption and rest in a being that is not itself placed in the struggle of life. This need is in the religions curiously at odds with the need of having in the Deity a living and struggling model. The religious problem is sharpened when both spiritual needs demand satisfaction at once.

But Mill's greatness does not lie in his results. His pure empiricism he succeeded in grounding and carrying through as little as Plato succeeded in carrying through his pure idealism. This comparison lies all the closer as Mill was all his life a great admirer of Plato. One can apply to him what he himself says of Plato (Levned p. 24): "I have always *[437]* felt that the name Platonist belongs with far greater right to those who have been trained in and have endeavored to apply Plato's mode of investigation than to those who have only adopted certain of his dogmatic propositions". Mill's significance in the history of philosophy consists precisely in his mode of investigation, in the way in which he has set experience and critical thinking to work in a whole series of theoretical and practical domains. He has awakened a philosophical spirit which is of greater significance than any single one of his results. — And with this we leave the thinker at once all of a piece and ever striving, logically clear and strongly moved by feeling, who in his development at once reflects the problems of his time and gives weighty help in their treatment.

C. The Philosophy of Evolution

The general fundamental thought on which the positivist philosophy rests is that our views are to be built on observations. In explaining the phenomena that experience presents, we are then to rely on causes that can themselves be demonstrated in experience. Positivism is really only a carrying-through of the demand for "true causes" that Kepler

and Newton made. As Auguste Comte has shown, the difference between the various main forms of philosophy depends on which causes one assumes. But even if one adheres to this principle, there may still be room for significant oppositions. Comte and Stuart Mill stopped at the conception of the causal relation as a relation between two different phenomena which for us simply were, as a matter of fact, connected. They laid no weight on that side of the causal relation by which, the more it is deepened and seen through, it shows us a continuous connection between the phenomena. Therefore they did not come to see the close connection there is between the concept of cause and the concept
[438] of evolution. In the social domain this latter concept does, to be sure, play an essential role with them; but it does not become determinative for their whole view of nature. They laid too great weight on the differences that experience presents, and did not see that all knowledge aims at reducing them as much as possible. They stopped at the problem of how new forms arise. Within the individual they assumed a development, and in the species likewise, through the influence of traditions and institutions. But that the concept of evolution could become a leading thought for the whole conception of the world they did not see, and the limits of their thinking were in great part determined by this.

The change in the view of nature that Charles Darwin's discoveries and hypotheses brought about may with good reason be compared with the changes caused by Copernicus and Bruno, Galilei and Newton. Copernicanism enlarged the world and made it infinite; the earth and human life no longer stood as that about which everything turned. Newton showed that a fixed lawfulness holds in the whole great world, since the distant globes follow the same laws that hold here on earth. Darwin's view of nature signifies a similar widening of the gaze as regards organic life. In the domain of living beings no great connection, no general law of origin and development, had yet been demonstrated. It had been shown by Caspar Friedrich Wolff and Karl Ernst Von Baer that the single organism develops through a series of stages from a germ that bears no likeness to the fully developed individual. It has been shown by Spinoza, Hartley and James Mill that a psychological development can take place in the single individual, through which, by virtue of the laws of association, mental forms take shape that resemble the original foundation as little as the forms of the grown organism resemble those of the germ. And the historical school, founded by Montesquieu and flourishing after the Revolution especially in France and Germany, had asserted the concept of development with regard to the origin of social and political forms. In the philosophy of Romanticism the concept of evolution likewise played a great role. The inward connection of the individual

thing with the whole was held fast, and existence was conceived as a series of steps *[439]*
on which the content of the world appeared in ever higher forms. But the use that was made there of the concept of evolution was a purely ideal one. Development was not conceived as a process by which one phenomenal form arose out of another; the continuity was thought to lie in the ideal kernel of existence, not between the single phenomena. And no one asked after the efficient causes that could lead over from one step to the next. A great example of how new forms and states can take shape by virtue of laws that constantly hold was, on the other hand, given in KANT's and LAPLACE's hypothesis of the development of the solar system out of a primeval nebula through the influence of definite physical and chemical relations, and in LYELL's theory of the origin of the present state of the earth's surface through the constant action of the same physical and chemical causes that are at work to this day. Darwin's doctrine holds that the organic species arise because constant variations of the organic forms take place, but that only those variations can survive which under the given conditions of life (physical conditions and relations to other living beings) offer certain purposive properties. It is this that he figuratively designates by the expression "the struggle for existence." When the conditions of life then continue to be the same through long periods, the purposive properties will be fixed and heightened. Darwin has widened the view of nature and let us discover an immeasurable course of development out of which the species now living have come forth. And on the other hand he has sharpened the eye for small things, in that he has taught us to see, in the often inconspicuous causes that are at work around us, the forces by whose quiet but uninterrupted working the living species have arrived at the forms under which they now appear to us.

What was new in Darwin's doctrine was not the general thought that organic species and forms have arisen through natural causes. This thought had come forward in many shapes before him. His significance lies in his having supported it by demonstrating definite causes that work in the direction of such a new formation. The concept of evolution itself is, as already indicated, so closely bound up with the concept of cause and the concept of continuity that it could come forward as a leading idea before Darwin gave *[440]*
it the mighty confirmation of experience through his investigations. Thus HERBERT SPENCER had, several years before Darwin's epoch-making work appeared, already adopted the hypothesis of evolution, and in particular he had (in the first edition of his Psychology) shown what significance this hypothesis had through the possibility that the properties and faculties that cannot find their explanation in the single individual's experience could be explained by the experience of the whole species. It was an extension

of the philosophy of experience, by which it became possible for it to let points of view that hitherto only the speculative and critical philosophy had asserted come into their own more fully than had happened before. Later Spencer has, in a detailed, systematic exposition, sought to show the concept of evolution to be a principal concept to which our experience and our thinking lead us back from all sides.

1. Charles Darwin

a. Biography and Course of Development

The justification for drawing this great naturalist into the history of philosophy lies — in a similar way as with Copernicus, Kepler, Galilei and Newton — first and foremost in this, that his method and his results have significance far beyond his special field and mark a turning point for scientific research and for the view of nature as a whole. Quite shortly after the idea of his hypothesis had arisen in him (1837), twenty years before his famous work appeared, he wrote in his notebook: "My theory will lead to a whole philosophy." But it was not only through its consequences, not only by raising the question of how the new theory bore on the whole world-view, that Darwin acquired significance for philosophy. He has also himself treated psychological and ethical questions and expressed himself on the limits of knowledge. And as we know him through his autobiography
[441] and his letters, he stands before us as a peculiar Socratic figure in the research of our time, venerable by his energy, his love of truth and his feeling of humanity.

Charles Darwin was born at Shrewsbury on 12 Feb. 1809. After having tried medical studies in Edinburgh and then theological studies in Cambridge, without having got anything out of it, his early awakened interest in natural history led him to go along on a voyage round the world that the ship "Beagle" made from 1831—1836. The observations he made on this voyage form the first foundation of his theory. He had compared the present fauna of South America with the extinct species and wondered at the likeness in structure alongside the differences, especially in size. He had compared the animal species in the northern part of South America with those in the southern part, and here too found likeness and difference combined in a peculiar way. But above all he had wondered to find on the Galopagos Islands, some 150 miles from South America, an animal and plant world that quite recalled South America's, although it consisted of species that were found nowhere else in the world: the species occurred only on these islands, but they belonged to genera that were found on the nearest mainland. It was as if one and the same fundamental form had been transformed partly to live on the mainland, partly on the islands. And besides, each one of the islands had its peculiar

species, which were not found on the other islands. How is this kinship on the one side, this difference on the other, to be explained? With this problem Darwin returned home from his great voyage, and he spent his whole life answering it. He lived in the country near London under the happiest circumstances, uninterruptedly occupied with collecting facts that could throw light on his problem. The question to be solved was why precisely such forms and properties are preserved and developed as are of benefit to animals and plants under the conditions in which they live. What had struck Darwin was precisely the variation in forms and properties in accordance with the conditions of life. He started from the assumption that there must be a natural cause of this — but which? The reading (in the year 1838) of MALTHUS's work on the population question set his thoughts going. Malthus showed that living beings have a tendency to multiply *[442]* to a higher degree than the quantity of the means of nourishment can increase. But then, Darwin concluded, living beings must surely be one another's opponents, must fight or compete with one another for access to what they need in order to live. Life is and must be a "struggle for existence," and the individual or the group of individuals that for some reason or other has acquired a faculty or an organ that the others lack, and that suits the circumstances, will more easily hold its own in the struggle than the others. The multiplication of the species will take place chiefly through them, and forms that lack this faculty or this organ will gradually die out. And when a group of living beings has to live in a limited space, this will be all the more easily done, the more different forms there are within the group, since their wants will thereby more easily be able to be satisfied. When all have to live on one and the same thing, not so many can live as when there is difference in food. It is therefore an advantage in the struggle for existence that a capacity for variation is present in a species.

Already at the beginning of the forties Darwin worked out a treatise in which he set forth his theory, but he laid it aside in order to undertake further investigations. In the middle of the fifties he began the final redaction of his work, and this appeared in 1859 under the name of *Origin of Species.* (Danish translation by J. P. Jakobsen. 1872). To it was joined a long series of other writings, of which the most important in philosophical and psychological respects are *Variation of Animals and Plants under Domestication* (1868), *Descent of Man* (1871, Danish translation by J. P. Jacobsen 1874) and *Expression of the Emotions in Men and in Animals* (1872). — Darwin's long, faithful work in the service of science ended only with his death on 19 April 1882[101]). Three years before, he had written in his autobiography: "I believe that I have acted rightly in devoting my life to science. I feel no remorse for having committed any sin, but I have again and again

[443] regretted that I have not done more direct good to my fellow men." —

b. Theory and Method

The result of Darwin's research was a victory for the principle of natural causes and for the assertion that nature makes no leaps. It had indeed earlier been assumed that the varieties are due to definite natural causes; but the species themselves were regarded as supernaturally produced or at any rate as inexplicable. Continuity was now demonstrated in a domain where hitherto one had either assumed supernatural intervention and interruption or had stopped at original, inexplicable differences, or at the most had appealed to an inner impulse of development that was to lead from one step to the next. By the explanation Darwin gave of the origin of species, through the struggle for existence, he moreover threw a light on the conditions of life, not only physical but also mental life, which is quite as important as the solution of the special task he had set himself. It was an old English thought that he, without knowing it, took up again. For his "struggle for existence" recalls Hobbes's "war of all against all." The English have always had an eye for the conditions of life and for the necessity of putting force into the effort to master them. By this is conditioned the peculiar mixture of empiricism and idealism that is often regarded as a contradiction. Their sense for practical circumstances as they simply are does not, however, dull their conviction of the value of the inner forces, even if they do not express it in the mystical or speculative way into which the thinkers of the Continent often lapse.

The explanation Darwin gives of the origin of species is entirely in the spirit of positivism, and it was therefore inconsistent when Comte's disciples took up a dismissive attitude toward the new hypothesis. The cause to which Darwin appeals is given, *positive*. He declares in a letter that to say that the species were created thus or thus is no scientific explanation, but only a pious way of saying that it is as it is. And according to other statements, when at the close of the "Origin of Species" he spoke of the first forms of life
[444] as having been created, he meant only that we know nothing of how life arises; later he expressly regretted that out of regard for common opinion he had used the word "created."

The expression *the struggle for existence* is figurative. It is meant to designate the dependence of one being on another and on the conditions of life, and in such a way that not only the single individual is thought of, but also its offspring. A plant that grows on the edge of a desert must struggle for life, that is to say, its subsistence depends on how little moisture it can make do with. An even greater role than the relation to physical

conditions is played in the organic world by the relation to other living beings. Mistletoes on the same branch "struggle" with one another for room and nourishment; but they also struggle with other fruit-bearing plants, since their multiplication depends on whether the birds swallow their seeds rather than those of the others, so that sowing can take place to a higher degree. While the expression the struggle for existence designates the organism's adaptation to the living or lifeless environment, Darwin designates by the expression *natural selection* the other side of the relation, namely the way in which the conditions of life favor certain properties — "the principle according to which every variation, however slight, that is useful to the individual is preserved." There is struggle everywhere in nature. It is always a matter of: to be or not. Every being subsists by virtue of a victory it has won at some time or other of its life. And the struggle for existence is in turn a consequence of the strong multiplication of organic beings. The whole economy of nature becomes unintelligible to us if we for a single moment forget that each single species seeks to increase extraordinarily strongly, and that there is always some obstacle or other that checks or stops its spread, even if we cannot always catch sight of it. It is because of this collision between spread and obstacle that a selection becomes necessary. The individuals that offer purposive variations survive, the others die out; thus the rise of differences is favored, and there take shape at last the species that are now in the eyes of so many essentially different. We have here a key to understanding how forms of different kinds can be related to one another, since with increasing divergence of character they develop out of a common fundamental form. What C. F. Wolff had shown *[445]* for the different organs of the same individual, that in spite of their unlikeness they develop out of certain simple rudiments, Darwin showed for the different species within the organic world as a whole.

Darwin's method is an interesting model for inductive research. His course of development as a researcher presents with unusual clarity the three main stages that a truth built on experience runs through. The first foundation of his theory was, as we saw, the observations made on the voyage. They called forth the idea, the provisional hypothesis, and — after the reading of Malthus — the explanation by deduction from the strong impulse of multiplication in its relation to the conditions of life. The third link in the series of labors was the verification, the confirmation by experience. It consisted essentially in four groups of facts. The English animal breeders had for many years produced new races by "artificial selection." They noticed small variations, suited to their purposes, in the different individuals, and then, by seeing to it that only individuals which offered such variations propagated the species, succeeded in breeding new forms.

What happens here consciously and methodically happens through "natural selection" unconsciously and in a more irregular way. The second fact that confirms the theory is the kinship of extinct species with species now living. They can as a rule be ranged as links in groups now existing and be conceived as lower forms of development of these. The third fact lies in the geographical distribution of the species, which is best understood by assuming that different forms and properties in related beings have arisen through the spread of the stock over tracts of different character. That the lowest beings are the most widely distributed is explained by their seeds and eggs being small and well fitted to be carried far and wide by water and current. Finally the agreement there can be in the embryonic state between animal forms that, when fully grown, differ greatly from one another, testifies to a distant kinship, and rudimentary organs (which are found especially in the embryonic state) likewise become intelligible only on the assumption of such a kinship.

[446] There was a remarkable kinship between Darwin's character and the direction and method of his research. What distinguished him as a researcher was his open and childlike mind, for which nothing was mean and insignificant. Everywhere he divined a connection. Nature was to him not a dead collection of objects to which the naturalists were to give name and number, but a living reality, where the subsistence and growth of the one depends on that of the other. The insect and the flower, the bird and the plant, the mold and the worm, the life of animals, their structure, their ornaments, their love and their fights, all this stood for him in mutual connection and could only in an unnatural way be separated. The word natural history he has made a truth. With the childlike openness toward nature there was joined in him the manly understanding that fixed, definite laws hold everywhere, and that even the highest in nature is not withdrawn from such laws, but precisely shows its highness by developing by virtue of these laws. He used to say that one could not be a good observer if one could not speculate. It was not possible for him to see anything new in nature without at once forming a hypothesis about how it could have come to be. But on the other hand he had a peculiar capacity for seizing an objection and holding it fast. He followed what he called the golden rule, of noting down every fact and every thought he came across that conflicted with the results he had reached. Very few objections were therefore made against his views that he had not himself noticed and sought to answer. He was in this respect more critical than several of his adherents; these sometimes thought that he laid too much weight on the objections that could be made against his doctrine. He was fully aware that his hypothesis really cannot be proved directly. Of one of his critics he

writes in a letter: "He is one of the very few who see that the change of species cannot be directly proved, and that my doctrine will sink or swim according as it is able to group and explain the phenomena. It is remarkable how few judge it in this way, which is the one right way." The proof of his doctrine he found in the "thread of thought" (the intelligible thread) with which it connects a whole series of facts. His doctrine was for *[447]* him not a dogma but a means of work, a light over nature that could produce more light. The significance of scientific hypotheses is, after all, not least that of giving occasion for continued research: we try whether it is possible to connect *all* our experiences in the way we have succeeded in connecting *some* of them; we learn what we are to ask nature about. In this respect Darwin's doctrine of the origin of species by natural selection has been extraordinarily fruitful. It teaches us everywhere in natural science to trace the significance that a property, a faculty or a form has in the struggle for existence. It starts from the assumption that nothing can subsist or develop that does not have a definite significance in the whole connection of life.

c. The Limits of the Theory

Darwin acknowledges a limit to his research in the question of the first origin of the individual differences between which both artificial and natural selection take place. A selection presupposes something that can be selected, differences and variations. Darwin takes it as a fact that such variations arise, and the more frequently, the more favorable the conditions of life are, and the more developed a species already is. He admits that "we are very ignorant on all points concerning the causes of variability," and the enigma of the origin of the original variations became for him for a time all the greater, as he was inclined to ascribe to external circumstances a very slight direct influence. In a letter to Huxley shortly after the "Origin of Species" had appeared, he writes (25 Nov. 1859): "You have very acutely found a point that has greatly perplexed me: if — as I believe — external conditions produce little *direct* effect, what the devil determines each particular variation?" (See already the letter to Hooker of 23 Nov. 1856). Later he came to ascribe to the conditions of life a greater direct influence on the variations (see his letters to Carus 4 May 1869, to Wagner 13 Oct. 1876, to Neumayr 9 March 1877). In " Variation of Animals and Plants under Domestication" (Chap. 22—26) Darwin *[448]* discusses the question more closely and emphasizes not only the direct action of the conditions of life but also the effect of the use or non-use of organs and faculties. (See also the "Descent of Man" ch. 4). — There is all the more reason in studying Darwin to be attentive to this point, as his critics have often confused the original origin of the

variations with the natural selection of properties that have come to be through such varying. "Natural selection," says Darwin against this (Variation of Animals and Plants. London 1868. II p. 272), "depends on the best endowed individuals remaining alive under different and complex circumstances, but has nothing whatever to do with the original cause of any change whatsoever in structure." Darwin's hypothesis concerns chiefly the effects of selection among the variations, not the origin of the variations. Naturally he assumed that there were causes of this too; but it was not the main thing for him to find them. Every hypothesis must build on a certain foundation that it cannot draw into the proof along with it. There is then no contradiction in Darwin's taking the variations essentially merely as factually given, and it is a strange objection that has sometimes been directed against him that he builds on "chance." Darwin does indeed use a few times the expression "chance-variations," thus in the letter to Hooker mentioned above, but by it he means variations whose cause is not known, and he himself declares the expression inexact. Variability depends more on the nature of the organism than on the character of the surroundings (More Letters from Ch. Darwin I. p. 377); this explains its mysterious character. Darwin declares (in a letter of 1881: More Letters I. p. 391) that the discovery of the causes of variability will be one of the most important tasks of the research of the future. — Just as Darwin regarded the first origin of the variations as enigmatic in several respects, so he also declared the origin of life as a whole to be an unsolved enigma.

On the other hand Darwin found no reason to assume that in the origin of the human
[449] species from lower forms quite peculiar forces were at work. As soon as the conviction had dawned on him that the species had arisen by natural development, it became clear to him that the human species could form no exception to this. But in the "Origin of Species" he contented himself with hinting that the new doctrine would also be able to throw light on man and his history; there was no room here for special investigation of a single species of living beings. When Darwin later took up this question in detail (in the work "On the Descent of Man"), it was, as he declares in a letter, partly because he was accused of not daring to come forward with his views on this point. He stood here even more alone than on the general question of the origin of species. Even men like Lyell and Wallace, who otherwise adhered to him, became doubtful here. Darwin himself felt no conflict between feeling and the impulse of research. Man's real worth and highness was for him not diminished by his having developed from lower forms. Against the theological and romantic conception that regards man as a fallen angel he set the realistic conception of man as an animal that has developed into a mental being.

Neither in mental nor in bodily respects did he assume more than differences of degree between man and animal. There is, he asserts, a far greater leap between the mental faculties of one of the lowest vertebrates (the lamprey or the lancelet) and those of one of the highest apes than there is between the apes' and men's mental faculties. And he shows how difficult it is to draw the boundary between mere instinct and reason proper. The very fact that animals can learn by harm shows that one dare not deny them reason. And the case is similar with memory, the sense of beauty and sympathetic instincts.

It has sometimes been concluded from Darwin's doctrine that there must really take place an uninterrupted progress in perfection in all living beings, and one has then found a great objection to the doctrine in the fact that the lowest organic forms continually go on existing. Here too Darwin himself has already answered the objection. Natural selection does not necessarily bring progress with it. What use would an intestinal worm *[450]* or an earthworm have of more perfect organs than it has? An organ that does not help in the struggle for existence takes up room and force to no use. Where circumstances for some reason or other exclude strong rivalry, it is perfectly intelligible that a form of life remains as it is through immeasurably long periods of time. (Origin of Species ch. IV. p. 154—155. Cf. ch. X. p. 422 f). Natural selection can only make every organic being as perfect as it needs to be in order to rival or "struggle for existence" with other organic beings, and the remarkable thing, according to Darwin, is that experience shows us so few cases where such perfection is not present. But perfection must always be measured by the relation to the conditions of life. (The natural selection of each species implies improvement in that species *in relation to its conditions of life.* Letter to Lyell. 25 Oct. 1859). In this it is then also implied that natural selection can sometimes bring about a regression to simpler and more elementary forms of life, namely when the conditions of life for some reason or other are simplified, so that various organs become superfluous, if they do not become harmful. There is no inherent or necessary tendency in organic beings to rise upward on the ladder of organization. (Variation etc. I. p. 8).

d. Ethical-Religious Consequences

Apart from the objections that point to supposed self-contradiction or disagreement with experience, objections of an ethical-religious kind have also been raised against Darwin's theory, and that not only from the theological side; even so radical a thinker as Eugen Dühring has here taken offense on ethical grounds and expressed himself in the most violent way against the theory. The very idea of the struggle for existence seems to many incompatible with an ethical conception of life: how are love of mankind and

conscience to be reconciled with this idea?

[451] As regards the ethical objection, it is so far from being the case that Darwin has overlooked it, that he has with full consciousness discussed the ethical problem in connection with his theory. As a moral philosopher he stands at a standpoint similar to the one that Shaftesbury and Hutcheson founded, and that in the nineteenth century Comte and Spencer carried further, only that Darwin's natural-historical point of view and his doctrine of natural selection give a broader basis for the whole standpoint.

Darwin finds in the fact that man is the only being that can with certainty be designated a moral being the greatest difference between him and the animal. But from this it does not follow that the moral feeling should not have developed in a natural way, and there is nothing in it either that conflicts with the struggle for existence or with natural selection. One must only hold fast that the properties and faculties that natural selection fosters are not only such as are for the good of the single individual, but also such as are for the good of the whole group or species. In all such animals for which it is most fortunate to live in close association, those individuals that found the greatest pleasure in the company of others would most easily escape dangers. And since the continued subsistence of the species depends on the preservation of the very often helpless offspring, it will be easy to understand how the love of parents for their offspring can have been developed by natural selection. Experience shows that animals sometimes expose themselves to danger to save other animals. A tribe or group of animals or men in which mutual compassion and the need to help one another prevail will be especially fortunately placed in the struggle for existence, more fortunately than other groups where each cares only for himself, and where the forces are not united toward a common end. Thus in the single individual, by virtue of the law of natural selection, properties can be preserved and developed that are beneficial to the subsistence of society, not especially to that of the single individual. Not only the selfish but also the unselfish feelings have their natural history. However great the distance may be between what stirs in animals when they show love and self-sacrifice for one another, and the highest
[452] human morality, there are between the two countless degrees, and one has at no point the right to assume that natural development is interrupted. Within the human world itself there are, after all, very great distances in moral respects. There are states and forms of human life that stand even far below what animal life can show us. Darwin declares that he would rather be descended from the ape that risked its life to save its keeper than from the savage who delights in torturing his enemies, kills his children without remorse, treats his wives as slaves and is himself a slave of the most dreadful

superstition[102]).

The moral feeling presupposes, according to Darwin, besides the desire to live together with others (sociability) and the capacity for compassion with others (sympathy), also the capacity to remember and compare. When these presuppositions are present, past actions will be able to be drawn forth and judged by what the strongest (in the moment of remembrance, if not at all moments the strongest) feeling demands. When the faculty of speech is developed, mutual praise and blame will act on the single individuals. A public opinion will take shape. Further, constant habit and practice in working for common interests will root and strengthen the social motives and instincts. Perhaps an inheritance of dispositions in this direction will also take place. — A confirmation by experience of his moral-philosophical theory Darwin finds in the investigation of the properties that at different times and among different peoples have been acknowledged as virtues, and of the different extent of the circles of individuals whose weal and woe the moral feeling has at different times taken into account.

The very theory of natural selection thus shows us that not every adaptation, every selection, every form of the struggle for existence is approved. From lower forms of this struggle we are led over to higher forms, first and foremost to a stage where a valuation of the forms of the struggle can take place[103]). —

It was found not only that Darwinism was an immoral doctrine, but also that it was a materialistic and godless doctrine.

Darwin has nowhere expressed himself on the relation between the mental and the material. He keeps to the fact that mental life in men and animals is, as a matter of fact, *[453]* bound up with the activity of material organs and so far can be studied as natural history like other organic phenomena. He therefore investigates the development of mental life through the struggle for life from lower to higher steps, and the definite laws of this development. In this there is nothing materialistic. Just as he declares the origin of life an unsolved problem, so he would surely also have declared the origin of mental life unexplained; perhaps the two problems coincided for him. Darwin has made interesting special psychological investigations, namely, besides those on the moral feeling, on the sexual feelings and instincts, on instincts in general, on the expression of the emotions, on the development of the child in its first year. One finds all through his writings, often where one does not expect it (e.g. in the little book on the influence of earthworms on the formation of mold), interesting psychological traits. But to discuss the principles of psychology itself was not his business.

If by materialism one merely means the reduction of phenomena to definite natural

laws that exclude supernatural intervention, then Darwin is certainly a materialist. The organic forms have, according to him, not stood as fully finished ideals before the Creator's imagination, to be thereupon set into material reality, but they are results of long processes of development from the most inconspicuous beginnings, under the constant action of the conditions of life. Darwin has widened the domain of natural connection; he has brought it about that the habit of thinking positively and doing without theological causes has increased among naturalists and thereafter also in wider circles. In principle nothing new has in a certain sense been added by this; for the principle of natural causes had long since been laid down — but it is not frequent in the history of science that this principle finds so brilliant an application as here. When in recent times several English theologians have accepted Darwin's doctrine, they help themselves by placing the creative act at the beginning of life, when the primeval form or forms arose, and natural selection then becomes for them the way, fixed by the Deity,
[454] in which the single species arise. To this expedient Darwin himself, however, could not resort. In no less a degree than Stuart Mill he found it impossible to reconcile the suffering and the disharmonies in the world with the providential government of an almighty being. He was no pessimist; he had the conviction, which to be sure he found very difficult to prove, that happiness in the world decidedly outweighed unhappiness. But natural selection, which brings suffering and unhappiness with it and breeds hatred and cruelty no less than love and gentleness, could not in his eyes be regarded as a means in the hand of a providence. Should the ichneumon wasp have been created on purpose to devour living caterpillars, the cat to play with the mouse? Already in his writings (especially "Variation." II. p. 432) he expressed himself in this direction, still more strongly in his letters and in his autobiography. It was the fact of evil that he again and again came back to when he was invited to express himself in religious respects. When he came home from his great voyage he still believed in revelation, and when he published the "Origin of Species" he was still a theist. But little by little, without any painful break, his views changed, and he at last declared himself (in a piece of the autobiography written in the year 1876) an *agnostic*, that is to say, one who is conscious that our knowledge cannot solve the problem. The expression had first (1859) been used by Darwin's friend and disciple Huxley (who on this point was influenced by William Hamilton). As little as he could reconcile himself to the thought that the world as it is was due to conscious intention (design), so little could he reconcile himself to the thought that it should be a result of chance (chance or brute force). "The safest conclusion," he wrote to a young man who wanted to know his opinion, "seems to me to be that the

whole question lies beyond the reach of the human understanding. But man can do his duty."

Agnosticism really means that what stands for us as a dilemma need not be one in itself, since there could be more possibilities besides those that are for us the only ones: design—chance. Darwin ends with the same result as Kant in the "Kritik der Urtheilskraft," who declared that the distinction between mechanism and teleology perhaps belonged *[455]* to the oppositions that we must set up by virtue of our knowledge, but that we had no right to take as valid for existence itself.

2. Herbert Spencer

a. Biography and Characterization

The year before Darwin's famous work appeared, Herbert Spencer drew up an outline of a systematic exposition that was to show the significance of the concept of evolution in the various domains. His earlier works had led him to conceive this concept as a principal concept in our knowledge, and he now wished to devote his life to carrying out this conception in detail. As he was without means, it was his wish to be enabled with the government's help to carry out his plan. He was warmly recommended by men who knew his earlier works, men such as the philosophers Stuart Mill and Fraser (Hamilton's successor in Edinburgh), the historian Grote, the physiologist Huxley, the botanist Hooker and the physicist Tyndall. The work was reckoned first at seven, later at ten volumes; in the actual execution it came to comprise nine volumes. Although Spencer obtained no government support, he has nevertheless with great perseverance, and in spite of his often failing strength, carried his plan through: the first part of the work *(First Principles)* began to appear in the autumn of 1860, and the last part (the conclusion of the *Principles of Ethics)* appeared in the spring of 1893. The intervening parts treat biology, psychology and sociology. It will be of interest to see how the thought of this comprehensive work arose and matured[104]).

Herbert Spencer was born at Derby on 27 April 1820. His father was an excellent schoolmaster, who laid great weight on the view that sound mental development comes only through self-education, and therefore sought to get his pupils to think and observe on their own. Spencer showed an early interest in natural science and history. It was his delight to follow the development of insects; in his father's library he made himself *[456]* acquainted with books of various content, and together with his brothers he eagerly discussed scientific, political and religious questions. His parents were Methodists; but the father gradually became dissatisfied with the clerical organization of this communion and

joined the Quakers, without, however, adopting their special doctrines; what appealed to him was their "unpriestly system." The son Herbert then went on Sundays with his father to the morning service of the Quakers and with his mother to the evening service of the Methodists. The result was that in the end he could not bear to hear Bible texts. Later Spencer came to live in the house of an uncle who was a clergyman (of the Broad Church tendency), an excellent man who took an eager part in the agitation for the repeal of the Corn Laws and had merits in the regulation of poor relief. Spencer's critical position toward all far-reaching attempts to order social relations by the intervention of the power of the state, and his great faith in free development, were founded and nourished by the religious and political views he met with in his early youth. It was intended that he should be a schoolmaster, like his father; but for a number of years he worked as a civil engineer. Mathematical studies and mechanical inventions together with political agitation filled a great part of his time. The first works mentioned from his hand are articles on the proper sphere of governmental authority and a treatise on the nature of sympathy, in which he set forth a theory similar to the one Adam Smith had in his time given. Already earlier he had, through his natural-historical studies, become an adherent of the theory of the natural development of species. He has himself declared that the study of Lyell's geology, which in the older editions combated Lamarck's theory, was precisely what led him (in the year 1839) to see the correctness of the theory of natural development. This also had an influence on his religious notions, which changed without his being able to trace the decisive change back to any definite point of time.

In his first considerable work, *Social Statics* (1850), he conceived social development
[457] by analogy with organic development, a conception that also plays a great role in his later writings. The perfect unfolding of life is here for him a divine idea that is to be realized, and of which nature offers intimations and approximations. Spencer has himself declared that in this work he was influenced by Coleridge and through him by Schelling. This theological mode of consideration he later abandoned for a purely factual demonstration of development under its various forms. The greatest influence on his theory of development he himself ascribes, already for this period, to "the truth that Harvey's embryological investigations first dimly indicated, that was later grasped with greater clarity by [the anatomist] Wolff, and that received its definite form from von Baer, the truth that all organic development is a change from a state of homogeneity to a state of heterogeneity." This truth, which had been found valid for the development of the single organism, stood for Spencer as valid for development in all domains, and it dawned on him at the same time that in all domains development takes place. In a little

essay on the development hypothesis from the year 1852 he makes a comparison between the theory of development and the theory of creation, and assumes — relying on the variations of domestic animals and cultivated plants, on the difficulty of distinguishing between species and variety, and on the likeness of different forms in the embryonic state — that the species have acquired their present forms by development under the influence of external circumstances. Decisive for his whole point of view was the working-out of his *Principles of Psychology* (first edition 1855). What is significant about this work is that — although it takes its stand on the ground of the philosophy of experience — it insists on the impossibility of explaining the individual consciousness out of the single individual's experiences alone. All earlier empiricism had trusted that it was only a matter of finding the experiences the single individual has, in order to understand its conscious life. It was a conviction that had not only theoretical but also practical significance. For if man's conscious life is determined merely by the individual's own experience, there opens up a prospect of giving it, through purposive arrangement of education and social relations, the stamp one wishes. Stuart Mill and Comte built to a great extent their hopes for the *[458]* future development of the species on this conviction, although both these thinkers' deep understanding of historical development stood in a remarkable conflict with it. Spencer now arrived at the conviction that development in the mental as well as in the material domain proceeds very slowly, through countless intermediate links and stages, of which none can be skipped. It is the constant influence of the conditions of life that carries life forward to new forms, among others also to the forms with which it now appears. We cannot therefore understand the conscious life of the single individual from the experiences to which it has itself been subject, but must go back to the experiences of the species. And it is not here only the traditions that are at work; in the innermost structure of our mind, in the way in which representations are combined, and in the direction in which feelings and drives unfold, hereditary tendencies are at work that become intelligible only by assuming an after-effect of the experiences of preceding generations. Heredity had hitherto stood as a curiosity to which no significance was ascribed outside single striking examples; now it came to stand as a constantly co-operating factor even in the very highest forms of life. It was this widening of the horizon in the psychological domain that led Spencer to search out the general laws of development, and at the same time to investigate whether these laws might not be derivable from the fundamental laws of our knowledge itself. With this the idea of his philosophy of evolution was given. The first outline of it appeared in an essay *Progress, its Law and Cause* (1857). (Danish translation in "Mindre Afhandlinger af Herbert Spencer." Copenhagen 1877). Here he

carried through the thought that all development is a transition from homogeneity to heterogeneity, and sought to derive it from the law of the conservation of energy. In the first edition of "First Principles" he still sets up only this one form of the law of evolution; later he supplemented it with two others: evolution as a transition from dispersion to coherence and as a transition from an indefinite to a definite state. And thereafter the publication of the great work went forward.

There were several preliminary works besides those mentioned. In a series of essays
[459] that were published in various journals, Spencer had already, before the working-out of his great work, treated various subjects of a philosophical, natural-scientific, sociological, aesthetic and ethical kind. His systematic works are only further, broad (often far too broad) elaborations of what is laid down in these short, clear and often also in form masterly essays, which are published collected in three volumes of *Essays*, and which are perhaps the part of Spencer's production that will last longest.

During the working-out of the first edition of his Psychology, Spencer through overexertion brought on himself a nervous debility that he has never since got over. He came to suffer from chronic sleeplessness. At last it threatened that he would have to give up all work. But in later years (from 1890) he has in his advanced age not only been able to complete his work, but also with life and vigor to take part in the newest discussions (called forth by Weissmann's hypothesis) in the domain of the biological doctrine of evolution. The last thing he has published is *Facts and Comments* (1902), a collection of short essays. He declares that it will be his last book.

The most decisive question in evaluating Spencer's theory becomes whether, by the extension he gives to empiricism and positivism with the help of the concept of evolution, he has succeeded in reaching the reconciliation of hitherto mutually conflicting views at which he aims. But even if he should not have succeeded in this, — even if old problems, after having passed through the purgatory of the philosophy of evolution, rose up in a new form — his widening of the psychological horizon and his demonstration of the concept of evolution as the principal concept of special research do not lose their significance. It must, in evaluating his work, be held fast that it was in this widening and this demonstration that his real new thought consisted, not in the setting-up of any definite epistemological or psychological theory. Moreover it cannot be otherwise than that during the working-out of a work that extends through some thirty-odd years unevennesses and inconsistencies in the treatment must arise, especially when his views
[460] on single points change before the completion of the work.

Spencer's works do not give us much access to knowledge of his inner life. The

impression one gets is that he is not so subjective a nature as Mill and Comte, in spite of all their positivism, were. Yet one finds between his long inductions and deductions, between his formulas of evolution and his statements of fact, utterances of warm feeling and quiet enthusiasm. He does not have, like Mill, the need to influence men directly. The doctrine of evolution has opened his eyes to the hard struggle in which everything living is placed, and to the difficulties with which especially man's spiritual and social life has to contend. It has taught him that ideals that to the preceding generation still seemed able to find their realization in the nearest future presuppose a long series of intermediate links that must all be gone through, and therefore come to lie as an immeasurably distant goal for the wandering of the species. Therefore resignation comes to play a great role in the view of life that arises on the basis of his doctrine. Not as if it were his opinion that the present should altogether fade or be sacrificed for the distant goal. But in such a way that the present is not understood merely by being seen in its connection with the past out of which it has developed, but only when one at the same time holds fast that it is a transition to a more developed state. Spencer surely belongs himself among those of whom he speaks in the words with which his great work closes (Princ. of Eth. II, p. 433): "The highest ambition of the good man will be to have a share — even if it is an extremely small and hidden share — in the development of what is human ('the making of Man'). Experience shows us examples of interest arising from the pursuit of altogether unselfish ends; and in the course of time there will be more and more of those whose unselfish end is the further development of humanity. When from the heights of thought they behold the distant form of the life of the species, which they themselves will never come to enjoy, but which will first fall to the lot of a distant posterity, they will feel a quiet joy in helping toward its coming."

The year after Spencer had conceived the idea of his philosophy of evolution, Darwin's book on the origin of species appeared, through which the doctrine of evolution acquired a wholly new and firmer basis. Darwin names Spencer in the preface among *[461]* his predecessors and later calls him (in a letter to Ray Lankester 15 March 1870) "the greatest living philosopher in England." Darwin's whole theory fitted excellently into Spencer's system, only that Spencer — with all acknowledgment of the significance of natural selection — laid more weight on direct development through the use of the faculties under the influence of circumstances than Darwin did in his earlier years. — Of no small interest is it that Stuart Mill, who at first was doubtful about Spencer's evolutionist psychology, later declared himself convinced that not only in the individual but in the race, through inherited dispositions, a mental development takes place. This

changed conviction he expressed the year before his death in a letter to the physiologist Carpenter (printed in the latter's Mental Physiology. p. 486).

b. Religion and Science

Spencer begins his systematic exposition *(System of Synthetic Philosophy)* with a section on the Unknowable[105]). It contains a doctrine of the limits of knowledge and at the same time examines the relation between religion and science. It is to show what validity can be ascribed to the scientific world-view that the remaining parts of the system seek to build up. There is here something inexpedient in the manner of exposition of which Spencer has not been aware. Neither epistemology nor the philosophy of religion can attain full clarity and grounding when they are not connected with psychology — but psychology comes only far along in the system!

The line of thought in the first section is as follows. In recent times religion and science stand as opposites. This comes from religion's wanting to solve problems that only science can solve, and from science's wanting to intrude upon domains where only religion belongs. Every religion has hitherto sought to give a theoretical explanation of existence, presenting itself as an unveiling of the riddle of the world. Different as the
[462] religions are, they nevertheless agree in this, that they would be revelations of something that would otherwise be unknown. The difference between the lower and the higher religions consists in this, that the lower — fetishism, and polytheism as well — think that it is very easy to form a notion of what is at work in existence. It is above all the belief in the spirits of ancestors that forms the foundation of primitive theology. One has, no doubt, a presentiment that the world is at bottom a riddle; but this presentiment is very weak, and the riddle is held to be easily solved. In the higher religions ever more weight is laid on the mystery, and it comes at last to this, that every image, every thought becomes inadequate to express the innermost of all things, and it is declared a blasphemy to think the Deity in the way in which alone we *can* think it. The religious development will be complete when it is recognized that the mystery is greater than the religions have hitherto assumed. The religions assumed a relative mystery, a mystery that could be unveiled. But the mystery is absolute when no image, no thought any longer suffices. The contradictions to which the religious notions constantly lead arise from the wish to assume an absolute cause. Whether one will assume that the world has produced itself, or that an external cause has produced it, one ends in the contradiction that something should be able to be the cause of itself; in the first case the world would have to be the cause of itself, in the second case the being that has produced the world. The more

special contradictions in the religious notions — between omnipotence on the one side, goodness and justice on the other, between justice and grace etc. — are, as Mansel has already shown, only special consequences of that fundamental contradiction.

But it now fares, Spencer continues, with the underlying scientific concepts just as it fares with the religious attempts to express the innermost nature of existence. Concepts such as time and space, motion and force, consciousness and personality are clear and serviceable so long as we keep within the limited and relative world of experience, but lead to contradiction when we would use them as expressions of the nature of an absolute being. Our scientific knowledge moves — as regards both the outer and the inner world — in the midst of a multiplicity of constant changes, without our being able *[463]* to see either beginning or end, either the first cause or the final end. It follows, moreover, as Hamilton and Mansel had already shown, from the very nature of our knowledge that we comprehend only the finite and limited. All knowledge presupposes a distinguishing; but the Absolute cannot be distinguished from anything else, since there can be nothing besides it. Spencer adds that all knowledge also presupposes an apprehension of likeness, a referring of what is to be known to something kindred; but the Absolute cannot be kindred with anything else, since there can be nothing besides it.

Yet Spencer does not grant Hamilton and Mansel that the Absolute is a purely negative concept. Knowledge must assume that there is other and more than it can comprehend. Its activity consists in distinguishing, apprehension of likeness, determining and limiting; but there must be something that is determined and limited, something that is formed, and that has subsistence independently of the determinate form it receives for our consciousness. What thus forms the constant foundation, for us indeterminate, of the whole content of our knowledge, we represent by analogy with what we sense as our own *force* in muscular exertion. There is a force to which it is owing that anything at all can become an object of our consciousness. It is this force which not only produces the individual changes that we distinguish and apprehend, but which also underlies that in us which persists through all changes. A concept of this force we cannot form; we can only approach it by successively thinking away the limits under which the unknowable force appears in each individual case in our experience. It is, then, the primitive sensation of force that in the end gives our knowledge its content, both in the shifting changes and in what remains constant while these take place (something constant under all modes).

Religion and science will in the end be able to meet in the conviction that the innermost essence of existence is unknown and incomprehensible to us, whereas we can have a scientific knowledge of the way in which it manifests itself in the world of *[464]*

experience. Our consciousness of existence is, seen from the one side, religion, seen from the other side, science. The strife will cease when the limit of knowledge is fixed in the right way; then religion will not draw anything that can be comprehended under *its* authority, and science will not draw anything that cannot be comprehended under *its* authority. There can be no strife between our intellectual and our moral faculties: it therefore cannot be our duty to believe something that contains a self-contradiction. It cannot, as Mansel assumed, be our duty to think God as personal and yet infinite, when personality and infinity exclude each other. Our duty can only be to submit to the limits of our knowledge and to acknowledge the mystery that really is there. When the concept of personality cannot serve as an expression of the Absolute, it is not because it is too high a concept, but precisely because it is too narrow a concept. A divine psychology we cannot give, and the strife between religion and science continues so long as the adherents of religion claim for themselves an intimate acquaintance with what is an eternal mystery, and therefore set up concepts that can be attacked by criticism.

Spencer is fully aware that his attempt to mediate between religion and science will meet with resistance. Men need living and concrete notions, need to represent what underlies all things as a being akin to themselves, and only when this need can find expression in their faith can their faith gain influence on their actions. So long as there has not developed an *organic morality,* which lets man act ethically from inner need, so long is it of importance that the influence of religious opinions should not be lost. Every faith stands in a definite relation to the stage of development on which man stands, and a hasty rejection of religious faith will therefore easily let the imperfections of our nature come forward more than they did while the old faith ruled. In any case so profound a spiritual revolution cannot take place without pain. He who is impatient because he finds that the process of transition lasts too long must remember that in every religion,
[465] even the lowest, there is "a soul of truth", — that even the lowest religions give their adherents something they could not get in any other way, — and that all forms of faith are links in a great process of development that is still far from its end. If he considers this, he will more readily put up with the sophistical reasons by which traditional opinions are defended, with the unworthy flattery with which men worship their deity, with the arrogance of ignorance, which far surpasses the arrogance of science, and with the condemnation of actions that proceed from unselfish compassion and pure love of the good. —

One misgiving about this doctrine of the relation between religion and science, which Spencer does not touch upon, is that the Absolute and the Relative are distinguished

in an all too external way[106]). Through this relation of opposition the Absolute itself becomes relative. That the opposition between the Absolute and the Relative cannot consistently be so external on Spencer's own conception is seen from his statement that a completed concept of the Absolute could be reached only by the removal of *all* limits and relations: such a removal is, after all, an infinite process, and we cannot, therefore, in reality set the Absolute and the Relative over against each other as two *finished* concepts. When, moreover, Spencer states in some places that the Absolute, which for him is one with the Unknowable, works through all phenomena, the inner as well as the outer (thus First Principles § 34, 46 and 93), then its essence must manifest itself in the forms and laws under which the phenomena appear; it cannot in reality be wholly unknowable, and Spencer is wrong to declare the chief concept of experience, the concept of evolution, wholly invalid as regards "the Absolute". It would be a self-contradiction if the general laws and forms that experience presents should have no significance whatever for what lies behind or beyond experience. No proof can be given that evolution should concern only the shell of existence and not its kernel at all. A dualism has here remained in Spencer. *[466]*

c. Philosophy as Unified Knowledge

The problem of the relation between religion and science arises not only every time a special science draws a new phenomenon within its domain, but also when an attempt is made to unite the various truths that the special sciences have found into universal truths. It is, according to Spencer, the task of philosophy to carry further the work of demonstrating unity in the multiplicity of phenomena, which the special sciences each carry on in its own domain, by seeking laws that are common to all our various domains of experience. The point is to find a highest truth from which the fundamental propositions of mechanics, physical and psychological principles and social laws can be derived. We should then have founded a philosophy as completely unified knowledge (completely unified knowledge).

Philosophy begins with a provisional assumption of the validity of the fundamental propositions on which all thinking rests. This provisional assumption is justified later by the consequences drawn from it agreeing with experience. The validity of an assumption can only be established by its agreement with all other assumptions. Truth can for us consist only in complete agreement between our representations (representations of things) and our observations (presentations of things). When our expectations and our observations do not agree, the assumption from which we started has been invalid.

The complete establishment of an assumption's validity by its agreement with all other assumptions leads us again to that working together of all knowledge into unity in which philosophy consists. But this also implies that every demonstration of the validity of individual assumptions presupposes the validity of the activity of thought by which we find things like or different. This activity of thought underlies all knowledge, both observation and inference, whatever it concerns. Every proof presupposes this *primordial act,* and its validity therefore cannot be disproved; a disproof is, after all, also a proof.

[467] We think always in relations. Next after the relation of likeness and difference, the relation of succession and that of coexistence are of importance. The relation of coexistence is derived from the relation of succession: we represent as simultaneous such different phenomena as can be represented just as well in the one order as in the other. The fundamental experience that underlies every apprehension of likeness or difference, succession or coexistence, we already know; it is the sensation of force: of something that offers resistance or produces change, and that we represent to ourselves by analogy with our own sensation of effort. Matter and motion are only manifestations of force, and time and space are forms of manifestations of force. Both in the domain of inner and in that of outer experience the concept of force is the ultimate concept to which we come back. We rightly distinguish between the self and the not-self; but both are only different ways in which the concept of force manifests itself. The matter and content of our thinking are all the different forms of force.

The concept of force is thus of a symbolic kind. It is the ultimate symbol (the ultimate symbol). It points back to our subjective experience, by analogy with which we apprehend everything else. Spencer here recalls the metaphysical idealism that uses this analogy to solve the problem of existence. He concedes that if we had the choice between reducing the mental elements to the material or the material to the mental, we should have to choose the latter. It is absurd to reduce the known to the unknown; an intelligible hypothesis we get only when we reduce the unknown to the known, and thus regard the objective, material elements as being in their essence of the same nature as the subjective elements of our consciousness. In his "Principles of Psychology" § 63 Spencer says: "Let me say here, once for all, that if we were obliged to choose between the two possibilities, of reducing mental phenomena to physical phenomena and of reducing physical phenomena to mental phenomena, this latter possibility would stand as the more reasonable (the most acceptable) of the two." But such a reduction Spencer considers impossible, since we cannot avoid making the mental elements clear
[468] to ourselves by means of forms and relations that are taken from external nature. A clear

intuition of the nature of consciousness we get only by using images from the material world. Moreover, we cannot derive and explain the physical phenomena from the mental phenomena. We must therefore stop at the difference between psychical and material phenomena and content ourselves with showing that the same laws of experience hold for both kinds of manifestation of force.

Now it is, according to Spencer, a fundamental presupposition of all science that force neither arises nor passes away in existence. All thinking consists in setting something in relation to something else. If, now, force were to perish, or if force arose out of nothing — which in outer experience would show itself in matter's being annihilated or arising, or in motion's being interrupted or beginning, — then we should have a relation between something and nothing or between nothing and something; but a relation one of whose terms has vanished from consciousness or has not arisen in it is a self-contradiction. The force that must thus always be assumed to persist is not any relative, phenomenal form of force, but the absolute force that stirs in all things, and so we here come back again to the thought that an Absolute must underlie all the phenomena of experience, the fundamental thought common to religion and science. It is the business of empirical science to demonstrate the special transitions that take place between the various forms of force. But the principle of the conservation of force itself cannot be proved by way of experiment, since every experiment, on closer reflection, turns out to presuppose its validity. All weighing and measuring presupposes, after all, that the unit we take as a basis does not change while the weighing or measuring goes on. If the force with which the weight tends toward the earth varies while the chemical experiment that is to determine the atomic weight is being performed, then the conclusion drawn from the experiment is invalid. The conservation of force is therefore a principle or a postulate on which all investigation of the real world rests.

For Spencer, as for Hamilton and for Kant, the principle of the conservation of force coincides with the causal principle. We shall also find that several of the natural scientists who set up the principle of the conservation of force on experimental grounds assumed *[469]* that it was really a proposition of reason, so that what mattered was to show how it manifests itself in experience. So Robert Mayer, Joule and Colding. There is, however, surely a distinction to be made here. For there could, after all, be lawfulness in nature even if force arose and force passed away, provided only that this arising and passing away were bound to definite conditions. One could imagine that when the condition A was present, B always arose, although B in quantitative respect might be put = A + x, and that when the condition C was present, D always arose, although D was = C—y. In that

case the causal principle would hold without the principle of the conservation of force holding. This shows that this latter principle is a more special principle than the causal principle, when the latter merely expresses a rule for the occurrence of phenomena. All regularity in nature would not be gone because the law of the conservation of force did not hold. It is another matter that we wish, as far as possible, to establish that just as in the logical inference there is no more than was already in the premises, so in the real effect, too, there is no more than was already in the cause; for thereby the principle of causality and the principle of grounding will be brought into the greatest possible agreement with each other. Even without this complete analogy with logical grounding, however, the causal relation could have validity. Therefore Spencer's deduction does not suffice, even if he is right that every attempt to prove the conservation of force experimentally in a certain sense presupposes it.

From the principle of the conservation of force Spencer derives that motion goes in the direction of greatest attraction or least resistance, — that all motion is rhythmical, — and that all phenomena are subject to evolution and dissolution. By this road he passes from the consideration of philosophy as unified knowledge to his consideration of philosophy as the doctrine of evolution. But before we follow him to this consideration, we must first ask what consequences Spencer draws from the law of the conservation of
[470] force with regard to the relation between the mental and the material phenomena. It has not been Spencer's intention to discuss this problem in particular, and one finds in his writings, as they now stand, a certain wavering on this point, which is explained by Spencer's having changed his conception in the interval between the various editions of his books, without this change having been carefully carried through at every point in the new editions. Hence arise the contradictory utterances that are found in him[107]). At first (in the first edition of "Principles of Psychology" and "First Principles") he conceived the relation between the mental and the material after the likeness of the relation between the various forces of nature, and assumed a transition from the material to the mental just as from motion to heat. The arising of sensations is thus explained by the law of the conservation of force. Insofar as one should wish to find (as some critics did find) materialism in this conception, Spencer reminded them that matter and motion are for him in the end only symbolic expressions for the unknowable force that stirs in all things. But later he became aware that the arising of the phenomena of consciousness could not be derived from the principle of the conservation of physical energy. In particular, the arising of sensation out of motion, as an equivalent for the latter, is incompatible with the doctrine of the continuity of motion. In later editions, therefore, Spencer conceived

the relation between the mental and the material according to the identity hypothesis, in such a way that they stand as two phenomenal forms of the unknowable force, mutually irreducible. With regard to the doctrine of evolution, which is the main thing for Spencer, it comes to the same which hypothesis one holds to. For whether one assumes a relation of interaction or a relation of identity between the mental and the bodily, evolution in both domains can present the same fundamental features, and that is what is decisive for the philosophy of evolution. According to the identity hypothesis, to be sure, mental evolution cannot be explained by deduction from the conservation of physical force; but the material process corresponding (obverse) to mental life can nevertheless be explained in this way[108]). When Spencer in his doctrine of evolution explains all evolution by the laws of matter and motion, and yet uses examples from psychology and sociology no *[471]* less than from astronomy and physiology to illustrate the forms of evolution, this is justified by the fact that the mental phenomena, whichever hypothesis one starts from, are assumed to be bound in a lawful way to material phenomena.

d. Philosophy as the Doctrine of Evolution

The unified knowledge that philosophy seeks is won, according to Spencer, not only by a demonstration that there is a common presupposition on which all inquiry rests, but also by the demonstration of a law that is common to all phenomena that experience presents. Spencer systematizes positivism partly by reducing all positive knowledge (knowledge of facts) to a common presupposition, partly by setting up a common law or form for everything positive, all phenomena. Every phenomenon has its history: it comes into being for us, and it vanishes again for us. Each single science describes the history of *its* phenomena; the point now is to examine whether these various historical processes do not present common features, so that *a general law of evolution* can be set up. It then turns out that all evolution more or less clearly presents three different characteristic features, which together give us the complete concept of evolution. — As already remarked, Spencer had at first set up only one characteristic feature of evolution, namely the transition from homogeneity to heterogeneity. Later he saw that there were quite simple forms of evolution in which this feature was wholly subordinate, and that, moreover, in order to be able to distinguish sharply between processes of evolution and of dissolution, it was necessary to set up yet a third characteristic feature.

1) *Evolution as concentration* (or integration). When a phenomenon comes into being, there takes place a gathering, combination and concentration of elements that previously occurred in a dispersed state. When a cloud forms in the sky or a heap of sand on the

beach, an evolution of the simplest kind has taken place, the process consisting almost exclusively in a gathering and heaping up. Such a process of concentration took place,
[472] according to the hypothesis of Kant and Laplace, at the first coming into being of our solar system out of the primeval nebula, whose parts were previously in a more dispersed state. All organic growth takes place through the taking up into the organic tissue of elements that were hitherto found dispersed in the surrounding world. A psychological example is had in every generalization, every formation of general concepts and laws: we thereby concentrate in a single thought a whole multitude of different observations and representations. Social evolution consists first and foremost in a joining together of individuals or groups of individuals who formerly lived in a more dispersed state.

2) *Evolution as differentiation.* Only in the very simplest cases can evolution be described as a mere process of concentration. There will not only take place a separation of the whole mass from its surroundings; but within the mass thus set apart special concentrations will in turn take place, so that the evolution becomes compound. And in the course of evolution these special concentrations will come forward more and more, so that on comparing the earlier and later stages we find a transition from homogeneity to heterogeneity. In the evolution of the solar system there occurs a separation between different globes, each with its peculiarity. Organic evolution advances from the homogeneous germ to the organism furnished with different kinds of tissue and with differently built and differently acting organs. The whole of organic life has, according to the hypothesis of Lamarck and Darwin, been more homogeneous at earlier stages, the differences of species now existing being due to evolution from common ancestral forms. The senses develop, when we compare earlier and later stages, from a less distinct and less exact faculty of apprehension to ever greater distinctness and exactness, so that more differences can be apprehended. Mental life is measured not only by its concentration but also by its richness. In the evolution of society different estates and classes are formed through the division of labor.

3) *Evolution as determination.* In a process of dissolution, too, differences arise in a hitherto homogeneous mass. To distinguish evolution from dissolution it must then
[473] be added that in evolution there occurs a transition from a more indeterminate and disordered state to a more determinate and ordered state. Evolution advances from a chaos whose parts are dispersed and homogeneous to a gathered whole whose parts are heterogeneous and at the same time stand in a determinate connection with one another. Thus the solar system, the organism, consciousness and society are more or less ordered wholes. This third point of view really arises from the combination of the first

two: an ordered whole is one in which differentiation of the parts and concentration of the whole go hand in hand with each other.

Everywhere in existence — in the great and in the small, in the mental and in the material world — processes of evolution of the kind now described take place. The philosophy of evolution formulates, on the basis of a comparative consideration of these processes, the fundamental features of the general history of every phenomenon. But what has thus been found inductively must now be grounded deductively by being derived from the law of the conservation of force.

In the first place, a concentration of homogeneous parts will arise when they are acted upon in a homogeneous way by the same force. This happens when heaps of withered leaves or heaps of sand or clouds form under the influence of the wind. Natural selection works in this way by requiring quite definite variations of the living beings that are to be able to subsist under definite conditions: thereby these are set apart as new species from those not fit to live. In the second place, when a homogeneous whole has formed, differences will arise when the homogeneous parts are acted upon by heterogeneous forces. And when difference in the state of the parts has thus been produced, homogeneous forces will act differently upon them. An organic species will vary under different physical conditions, and even if the varieties are subjected to a homogeneous influence, this will nevertheless not have the same effect on them all. — This follows from the conservation of force, which would be abolished if the same causes produced heterogeneous effects on homogeneous objects, or heterogeneous causes produced homogeneous effects on homogeneous objects, or the same causes homogeneous effects on heterogeneous objects. *[474]*

In order to understand Spencer rightly one must remember that his doctrine of evolution does not apply to the world as a whole, of which, owing to the relativity of knowledge, we can form no notion, but that it applies to the individual wholes that arise within the circle of our experience. It has sometimes been objected against him that his "evolution" becomes incomprehensible if it is to begin as an absolute homogeneity; whence — it is asked — are the differences then to come? Spencer himself declares expressly (First Principles. § 116, 149, 165) that he speaks only of limited phenomena, and that he assumes only a relative homogeneity and heterogeneity, just as the concepts of concentration and differentiation, simplicity and complexity, must also be taken in a relative sense. "Since we know," says Spencer, "no absolute dispersion or absolute concentration, change can never be anything but a transition from a more dispersed state to a less dispersed state, from looser coherence to firmer coherence. Likewise,

since no concrete phenomena present absolute simplicity, — since nothing is absolutely uniform, — since we nowhere find complete homogeneity, the transformation in reality always takes place in the direction of greater compositeness, or increased multiplicity, or a higher degree of heterogeneity. This qualification the reader must constantly keep hold of"[108 b]). *If* one could imagine the whole universe in a state of complete equilibrium, where wholly homogeneous centers of force were distributed with absolute homogeneity over the whole of space, then everything would remain in equilibrium for all eternity. But such a thought is excluded already by the fact that no limit can be set to space. Of all homogeneous masses that we know it holds that they will necessarily become differentiated in a lower or higher degree. —

Evolution must (naturally, when it is not interrupted from without) necessarily lead to a state of equilibrium in which both concentration and differentiation have reached their culmination. As regards human evolution this state means the highest perfection and bliss, consisting in the greatest possible harmony between men and nature, and
[475] among men themselves. But since the external influences constantly continue to exert their effect, this state of equilibrium will not be able to maintain itself in the course of time. Evolution will be followed by dissolution when energy is lacking to preserve, under the continued influences, the harmony between concentration and differentiation. And through the various stages of dissolution one ends up in a new chaos. — Just as processes of evolution go on uninterruptedly within the circle of our experience, so processes of dissolution of greater and smaller wholes also go on uninterruptedly. Even if our solar system too — and all solar systems —, as some natural scientists think, should have the germ of dissolution within it, there will always be a possibility of new formations of systems, since there will always be external forces to set the processes of evolution going. All motion is rhythmical, and evolution and dissolution will therefore alternate endlessly.

While Hegel in his idealist doctrine of development guaranteed an eternal progress, the "higher unities" (which correspond to Spencer's harmony between concentration and differentiation) always becoming the foundation for new evolutions, Spencer, as a good positivist and by virtue of the doctrine of the relativity of knowledge, cannot say anything about whether evolution or dissolution is the stronger process in existence. By virtue of his dualistic conception of the relation between the Absolute and the Relative he will not even ascribe any absolute significance to evolution (nor, therefore, to the whole rhythm of evolution and dissolution). Here, as already shown, a contradiction comes out in him, since he acknowledges a connection between the phenomena and the unknowable order of things that underlies them (the connection between the phenomenal order and the

ontological order). Only this he holds firmly, that just as little as we can set limits to time and space, just as little can we set limits to the force that works in the world of phenomena. But his proper task he has accomplished by his characterization of the history that every phenomenon runs through. *[476]*

e. The Concept of Evolution in the Domain of Biology and Psychology

It is in the biological and the sociological domain that Spencer moves with the greatest expert knowledge. A biological idea gave, as we have seen, the impulse to his whole system, and this biological idea (of evolution as differentiation) he in turn connected directly with a sociological idea (of the division of labor). Social life is for him a form of life that obeys the general laws of life, and the same holds for mental life and the ethical life. Biology is thus the dominant concrete science in his system. It is the evolution of life at all stages and under all forms that his inquiry will follow.

Life consists in a constant adaptation of inner relations to outer. In a living body an impression from without produces not only a direct effect but also an indirect effect, a state being called forth whereby the living being becomes able to meet a following change in the surrounding world. An inner activity is set going which comes, or can come, to be of use on a later occasion. The organic tissue has two apparently opposite properties which in combination make adaptation possible. It has a plasticity that lets the impressions from without send a tremor through the whole organic mass, or at least through large parts of it; and it has a polarity, a definite manner of arrangement of the ultimate organic particles, which Spencer calls the physiological units and regards as far simpler than the cells but more compound than the chemical molecules, a polarity which brings it about that the effect of the impressions on the organic mass is determined predominantly by the mass's own constitution. The formation of the organic mass in the first place is an obscure question. Spencer rejects the assumption of the arising of organic *forms* out of inorganic matter. He is, on the other hand, inclined to assume that organic mass of a wholly *structureless* kind was formed at a certain definite point in the cooling process of the earth's surface. Since this original organic mass was without definite structure, this assumption is thus not one with the assumption of a first organism. There has been organic life without organization. (See especially the Appendix to the first volume of *[477]* "Principles of Biology".) The organic forms have arisen successively under the action of outer conditions on inner. By virtue of the general laws of evolution, constant external influences produce differences in the mass, first and foremost a difference between the surface and the interior; and through the different ways in which the different parts

react to the impressions, they gradually take on a different nature. Spencer even sets up the proposition that function precedes structure, and that it is the constant functioning in a definite way that produces the definite structure that the individual organs possess. This direct influence of the surroundings on the organic tissue is really presupposed by the doctrine of natural selection; this selection takes place precisely among the incipient structures called forth by the direct influence, among the various variations that the organic mass undergoes under the external influence. Even where the variations arise from other, unknown causes, it is nevertheless only when favorable modes of action are bound up with these so-called "spontaneous variations" that natural selection can preserve them. (Princ. of Biol. § 61). The change that can take place in the structure through a certain constant functioning will be able to be inherited when it comes to produce an alteration in the organic polarity, that is, in the manner peculiar to this particular organism in which the physiological units are arranged.

Darwin's doctrine of natural selection (which Spencer prefers to call the doctrine of the subsistence of the most purposive forms of life: the survival of the fittest) Spencer acknowledges in the fullest measure and with great admiration. But just as little as Darwin himself does he think that selection can explain everything. Without the assumption of the inheritance of acquired characters the origin of species is for him incomprehensible, and he has therefore in recent years carried on a polemic against "the natural scientists who would be more Darwinian than Darwin himself", especially against the Weissmannian hypothesis, which denies all influence of practiced functions on the cells that contain the germ of the offspring, and must therefore derive all evolution from natural
[478] selection. Especially in the case of the higher organic beings, which have a developed nervous and muscular system, he thinks that the influence that the use of the powers has on the organization plays a considerable part. He ascribes to the question not only biological but also ethical and social significance. "If", he says in his essay *Factors of Organic Evolution* (1886), "a nation is altered as a whole by the inheritance of the effects that the nature of its members undergoes through the forms of daily activity that its institutions and conditions of life determine, then these institutions and these conditions of life must mold the citizens far more quickly and completely than they could if the only cause of adaptation is the more frequent survival of individuals that have varied in a favorable direction". The ethical and social responsibility thus becomes greater when one assumes the inheritance of acquired characters than when everything rests on selection. It is this that gives the problem so great an interest in Spencer's eyes. —

Consciousness too is an activity that arises in the adaptation of living beings to

their surroundings. When the impressions grow in number, the adaptation to them can only become exact if they are ordered in a series, as happens in consciousness. Psychology is to that extent a part of biology. But Spencer here distinguishes between objective and subjective psychology. It is really only objective psychology, the one that studies the life of consciousness through the material functions to which it is bound, that belongs in the series of the concrete sciences. Subjective psychology, which builds on immediate self-observation and wins its results by analysis of it, stands in the same relation to all other sciences as the subject in general stands to the object; it is the science of subjective existence, while all other sciences together concern objective existence. Therefore subjective psychology is a science wholly unique of its kind. (Princ. of Psychol. § 56. — Classif. of the Sciences. 3. ed. p. 26 b).

Like Stuart Mill, then, Spencer maintains the independence of psychology in opposition to Comte's overlooking of it. And he differs at the same time from the older English school in maintaining that one cannot regard consciousness as a mere sum of sensations *[479]* and representations, since there is always something that connects these and maintains the unity of the circle of representations against attempts to divide it. There must — Spencer teaches — be something substantial in consciousness as opposed to all shifting forms, but this substantial something we cannot know, since it does not come forth in any single state, although every state is a particular form of it.

For all the peculiarity with which the life of consciousness appears in subjective experience, it nevertheless presents the same fundamental features as life and as evolution in general. Its evolution consists in progressive concentration, differentiation and determination. All the transitions take place gradually, and it is therefore wrongly that faculties differing in essence have been distinguished in the soul. From reflex movement through instinct and memory to reason, from the simplest distinguishing and recognizing to the highest scientific act of thought, there is a continuous series of stages. The qualitative and quantitative richness of consciousness corresponds to the richness of the living being's relations to the surrounding world. There is always a correspondence between the life of consciousness and the external relations the individual has to do with. There is a continuous series of stages from the intestinal worm in an organic tissue to a Newton or Shakespeare whose thought embraces the world.

With regard to the question of the origin of knowledge Spencer makes a stand against Leibniz and Kant on the one side, Locke and Mill on the other side. He combats empiricism, which does not see that the matter of experience is always taken up and worked over in a manner determined by the original nature of the individual, and which

moreover comes to lack a criterion of truth. An original organization is needed to explain the influence that impressions have on the different individuals; and there is needed, every time one would ground an assertion, a criterion consisting in this, that the opposite would contain a self-contradiction. In the innate nature of the individual and in the logical principle on which we rely in every inference we thus have something a priori, something that cannot be derived from experience. To that extent Spencer grants Leibniz
[480] and Kant that they are right as against Locke and Mill. But only so long as it is a question of the experience of the single individual. For the single individual there exists an a priori, but not for the species. For the conditions and forms of knowledge and feeling that are found originally in the single individual, and therefore cannot be derived from its experience, are due to inheritance from earlier generations. The forms of thought are one with the accumulated and inherited changes of structure that are latently present in every newborn individual and are gradually unfolded by its experiences. And their first origin, then, is empirical after all: the fixed and universal relations between things must in the course of evolution form fixed and universal connections within the organism; by the incessant repetition of absolute external uniformities there are bred in the species necessary forms of knowledge, indissoluble connections of thought, which express the net yield of the experience of perhaps millions of generations right down to the present time. The single individual cannot take apart what has thus become an ingrown connection in the organization of the species, and therefore it brings with it in its innate nature the foundation on which the necessary truths are built. (See Princ. of Psychol. § 208, 216 cf. First Princ. § 53: Absolute uniformities of experience generate absolute uniformities of thought). Although the inductive school has, in Spencer's opinion, gone too far when it would derive everything by way of induction, whereby induction itself comes to hang in the air, he would nevertheless, if he must choose, rather count himself of Locke's school than of Kant's. For *in the end*, according to him too, all knowledge and all forms of thought nevertheless stem from experience. Max Müller has held that when Spencer conceded that there is something in consciousness that is not a product of one's own experience, he is a pure Kantian (a thorough Kantian). But Spencer has answered this: "The evolutionist point of view is wholly empirical. It distinguishes itself from the older view of the empiricists by the extension it gives this view . . . . But Kant's view is evidently absolutely unempirical". (Essays. III. p. 274. 277).

Spencer is, however, struck here by his own criticism of empiricism when he assumes
[481] that the species at any stage whatever of its evolution has been able to receive influences from without without there being, for receiving these influences and determining their

results, any previously given organization. There is here an unclearness in his psychology that goes back to his biology. For his proposition that function determines structure seems unable to be understood otherwise than as saying that the organic tissue at the lowest stage behaves absolutely passively toward the influences that produce activity in it. Such a relation of absolute passivity is, however, excluded by his own definition of life as adaptation — and does not in reality take place even where it is inorganic beings that are acted upon, since surely the stone is affected otherwise by heat than the wax. — Moreover, Spencer has overlooked that the *validity* of our knowledge is not absolutely guaranteed by the first presuppositions' being derived from the experiences of countless generations. This establishes at most that these presuppositions have hitherto proved practically serviceable in the struggle for existence. A proof of their absolute truth cannot be built on it[109]). —

Even more than knowledge, feeling, according to Spencer, presupposes a foundation that is original for every individual and received from the species. The constant interaction with the surrounding world, the constant struggle with the conditions of life, engenders a tendency to moods for which one does not always succeed in finding clear expression in representations. So it is with the sympathetic feelings, e.g. the little child's pleasure at a smiling face, its pain at a sorrowful or threatening face. The feeling of justice, the immediate indignation at encroachment, can arise without there being any distinct concept of the limits of the individuals' freedom of action. Spencer objects to the earlier utilitarians that they let the conscious insight into the effects of utility precede the appearance of the actions and motives that are valued. In order to experience that sympathy is beneficial, one must first act sympathetically without deliberation. It is thus feeling that makes it possible to discover utility, not the discovery of utility that calls forth the feeling in the first place. It is from the constant education to which man is subjected through the interaction of the conditions of life that Spencer expects continued changes and progress in human emotional life, not from the mere communication of *[482]* finished knowledge. He opposes the proposition from which Comte, especially in his earlier writings, started, that ideas rule the world. It is feeling and character that drive the work, and which ideas become dominant will depend on the character of the people, which in turn rests on a foundation inherited from earlier generations.

f. The Concept of Evolution in Sociology and Ethics

For Spencer's sociology the relation between individual and species that his biology and psychology have found becomes of decisive significance. When every individual has an

original foundation for its character that goes back to the earlier history of the species, the evolution of social life will not be able to be changed suddenly by the influences that one may be able to exert on the single individuals through reforms in education and constitution. Spencer has not so immediate a hope for the future as that which animated Bentham, Mill and Comte. Evolution advances slowly, because it is the very natural foundation in the individuals, not merely their thoughts and knowledge, that must be changed. During his stay in America Spencer, in a conversation, strongly censured the abuses in the public life of the United States, which he derived from the fact that the country had got its constitution, good in itself, by a happy accident, but also had only *got* it; it had not grown forth spontaneously, and therefore it was no wonder that its consequences turned out other than had been imagined. When an American friend asked whether instruction and the spread of political knowledge would not remedy this, Spencer answered: "No. It is essentially a question of character, and only in a subordinate sense a question of knowledge. It is a common illusion that instruction can be a universal remedy for political evils". Changes of character take place, according to Spencer, only through a race's standing for generations in direct interaction with the real conditions of life. Only through practical adaptation, through the development and exercise of the powers that are used in the struggle of life, do firm characters and healthy feelings
[483] arise. In his work on education Spencer therefore impresses the necessity of the child's gaining experiences on its own account, coming itself to know natural relations and to feel the effects of its own actions. Screens are to be used as little as possible, since they involve double work: first adaptation to the screen, and then to what lies behind the screen, the real relations. The child had best burn itself; otherwise it does not rightly learn to shun the fire. And what holds of the character of the individual holds of that of the race, and of the whole species. In his largely and broadly planned work *Principles of Sociology* Spencer seeks to illustrate this by means of an extensive material, drawn especially from the races standing lowest. Together with a circle of collaborators he has published, under the title *Descriptive Sociology*, a series of tabular surveys (supported by references to sources) of the conditions of cultural history among various peoples. In a lively written book, *The Study of Sociology*, he has discussed the difficulties with which a purely scientific sociology has to contend.

It is a favorite proposition of Spencer's that societies and constitutions grow, but are not made. He demonstrates in detail the analogies between a society and an organism. The single individuals correspond to the cells, or rather perhaps to the physiological units. Both within society and within the organism the common life is to a certain

degree independent of the fate of the single units each taken by itself. But there is a great and significant difference between society and the organism: while in the latter consciousness (where it is present) is bound to the central organs, in relation to which the other organs and units have only subordinate significance, in society it is precisely the single units that have consciousness, while the central organization as such has no consciousness of its own. In the organism the parts exist for the sake of the whole; in society the whole exists for the sake of the parts. On this difference between society and organism, according to Spencer, no less weight must be laid than on the analogies between them. (Cf. Princ. of Sociol. § 222. cf. Princ. of Ethics. II. § 102, 137). For on it rests the fundamental proposition, whose correctness the history of social life establishes, that centralization always has significance only as a necessary means, while the decisive *[484]* thing always remains what goes on in the single individuals, where human life proper is led. In social life as in individual education the point is to reduce screens and tutelage to the least possible, since they always complicate and retard evolution by necessitating a double adaptation, first to the artificial, then to the natural relations. Experience shows that wherever demands that have their root in nature are enjoined by external authorities, obedience to the authorities comes to take the first place at the expense of the practical relation to natural reality. And experience shows as well that artificially set up and appointed authorities do not work as ably and purposively as the free, individual forces themselves. The social machinery uses too much force to reach its end — and easily reaches other ends than those intended. The single individuals experience far more urgently than the authorities of the state a direct connection between labor and yield. Although Spencer theoretically holds fast (what he had already taught in "Social Statics") that the land is the property of the whole people, he declares himself, from aversion to "officialism", against the state's taking over the cultivation of the land, as the "Nationalists" propose. Quite apart from the financial difficulties of the state's taking over the land, difficulties that in Spencer's view would in and of themselves make the proposal impossible to carry out, the notorious imperfection of state administration in comparison with private administration is in his eyes sufficient to reject it. (See Appendix B to Princ. of Ethics. Part. IV).

Experience shows us societies at the most different stages of evolution; the degree of evolution is measured here as everywhere by concentration, differentiation and determination. But there is an opposition that in sociological — and no less in ethical — respect stands in the first line. It is that between militarism and industrialism. Two types of society appear here, of which the first prevails especially at lower stages and

gradually, but very slowly and with many fluctuations, gives way to the second. The military society arises from the necessity of defending the social group against external
[485] enemies, often also from the drive to acquire livelihood and power at the expense of other groups. Characteristic of this type is the absolute subordination of the individuals to the whole. They are means, not ends. Obedience becomes the highest duty. Peaceful labor at procuring the means of subsistence is left to women and slaves. In industrial society, on the other hand, labor stands in the first line. The main thing here is the free, personal intercourse between the individuals, their working together in the service of common interests. Instead of the mixture of savagery and submission that the military type fosters, the industrial type sets the individuals freely over against one another and lets them learn in daily intercourse to attain their ends in such a way that they acknowledge the equal right of others to attain theirs. It is a discipline that gains influence on characters, manners and constitutional conditions. While the regulating apparatus under the military type was all in all, it now becomes more and more restricted to the office of maintaining peace and right among the members of society. For carrying out the functions that the state formerly attended to as a kind of providence, voluntary associations are formed, insofar as the spontaneous working together of individuals is not sufficient. While formerly it was society that stamped its impress on the single individuals, it is now more the single individuals who order society according to their needs. — The struggle between militarism and industrialism is still undecided. After long years of peace militarism got a new upswing on the Continent, in which for the second time the Bonaparte family, the greatest of all modern curses (that greatest of all modern curses, the Bonaparte family), intervened in the course of things, and now it flourishes in all countries more or less and introduces its spirit and type again in other domains too than the purely military. Compulsory regulation now takes the place of free self-development in a series of domains. In the ideal of social democracy, of an army of workers, each with his appointed occupation and his appointed wage, the same type expresses itself, and it is then no wonder that socialism, like militarism, thrives best in one and the same country, Germany. (Princ. of Ethics. II § 26, 72). And the prospects of
[486] improvement are for the present not bright: " So long as the European nations divide among themselves the parts of the earth that are inhabited by lower peoples, with cynical indifference to the rights of these peoples, it is foolish to expect that within each single one of those nations a tender regard for the right of individuals should be shown on the part of the government". (ib. § 119). — Yet Spencer descries the possibility of a third type, which stands as high above industrialism as the latter stands above militarism. The

industrial type is easily carried into the one-sidedness of everything's being directed to labor for the sake of gain. The third and higher type of human life will be one in which free devotion to activities that are immediately satisfying, and do not merely serve as a means of existence, comes to occupy a far greater place than now. Within our present social order it is especially efforts and institutions for the furtherance of intellectual and aesthetic ends that contain indications in the direction of the third type[110]).

The closer working out of what this highest type of human life contains within it belongs to ethics, not to sociology. For this type of the future is, according to Spencer, the goal toward which all ethical striving moves. So long as this goal has not been reached, no strict ethics is possible: the realization of absolute ethics presupposes a perfect human life in a perfect society. According to Spencer, scientific ethics is a system of deductions from the highest laws of biology and of the general doctrine of evolution. But he concedes that such deductions cannot hold without further ado. At the imperfect stages the conditions are so intricate that only approximations to the absolutely right and compromises between the absolutely right and what the subsistence of life under the actual conditions requires become possible. But relative ethics must constantly be checked and regulated by keeping the ideal principles of absolute ethics in view. Ethics must construct a picture of the conditions on which the perfect life rests, without at once directing itself by whether these conditions are already actually present now. If we compare ethical development at various stages, we find that it bears the general characteristic marks of all evolution. Ethical action presents greater concentration and coherence than unethical *[487]* action: compare e.g. self-control with dissoluteness, truthfulness with mendacity! It likewise presents greater richness in differences, greater differentiation: for he who merely looks after his own egoistic interest gets a narrower range of vision and range of action than he who works for others as well; all of a man's faculties and possibilities do not come to development when he will work only for selfish ends, and the highest degree of development can therefore be present only where the individual's activity furthers the welfare of others. Finally, the more perfect ethical conduct also bears the stamp of greater determinateness than the less perfect: it takes definite considerations into account and sets definite limits to the individual's own impulses and to those of others, which in themselves might spread into the indeterminate; conscientiousness, justice and moderation are examples of this.

In the perfect type of life the development of the individual will be limited only by the equal right of others to development; the individual will, moreover, spontaneously, from his own inner need, refrain from encroachment on the healthy development of

others, and for his part help it forward as much as lies in his power; finally, it will not be necessary, for the attainment of more remote ends, to undertake labor that does not bring immediate satisfaction.

The construction of these conditions for the perfect form of life rests on the principle of welfare. Spencer does indeed criticize the utilitarianism of Bentham and Mill; but only because it is too empirical, too inclined to keep to the nearest effects of actions without catching sight of the more remote ones, which can be established only by way of deduction. Here, as in his theory of knowledge, he seeks to show that the empirical and the a priori conception can be united by means of the philosophy of evolution. The significance of a priori ("intuitive") ethics he finds partly in its maintaining the significance of deduction and setting up ideal principles that are not directly built on experience, partly in its finding a deeper psychological foundation for ethical feeling than the individual's own experience can afford. In his construction of the doctrine of right he comes to the same result as Kant: that the fundamental right is freedom for the
[488] individual, insofar as he does not come to commit encroachment on the equal freedom of others; this is, after all, as stated, the first characteristic mark of the perfect type of life. This agreement of Spencer's with Kant, which surprises him himself (see Appendix A to Princ. of Eth. Part. IV), cannot surprise us, since we have seen Kant's ethics and doctrine of right proceed from an evolutionist line of thought. The great significance of a priori ethics consists in not letting itself be entangled in regard for the nearest effects of actions; but it is mistaken in thinking that the ethical principles are not in the end determined by regard for whether the actions cause happiness or unhappiness. And even if there is an a priori foundation for ethical feeling, independent of individual experience of happiness, this very foundation is in turn to be explained as the result of the doing and suffering of earlier generations. In his closer conception of ethical feeling Spencer parts from Kant by assuming that the feeling of duty belongs only to a certain stage of evolution. The feeling of duty consists in an inner control that one feeling exerts over another; but with advanced evolution such control will be unnecessary; there will have been formed an "organic morality" which makes the performance of the actions required by the ethical principles as spontaneous and as immediately satisfying as the mother's care for the child and the artist's absorption in his work already are now. Man will then be completely adapted to the social state — and the latter to him.

Until perfect evolution is reached, we must make do with compromises. As a strict science ethics is not possible before the highest stage of life. Spencer thus really assumes that no ethics can be given before it is — superfluous! Paradoxical as this may seem,

there might nevertheless lie in it a correct recognition of the difficulties of principle with which every attempt at a scientific ethics has to contend under the intricate conditions under which we have to lead our life. Perhaps one must even go still further than Spencer, in his great confidence in the victorious course of evolution, has thought necessary; in particular, he takes too little account of individual differences and their influence on ethical determinations[111]). — We shall bring forward some examples to illustrate what *[489]* Spencer means by the opposition between absolute and relative ethics.

Sociology has shown us that the present stage of development of the human race is characterized by a conflict between militarism and industrialism. In this period of conflict the freedom of the individual will in many ways be more limited than absolute ethics permits. Slavery is an institution that belongs to militarism. With the development of industrialism personal freedom comes into force in ever wider circles. But through the worker's dependence on the employer something constantly maintains itself of the relation of obedience in which the slave stood to the master, although the relation now rests on contract, and so on consent and obligation on the part of both parties. Whether this can ever be wholly changed we cannot know; the task of relative ethics, on the other hand, is to impress the necessity of as great an approximation to the ideal relation of equality as the conditions in any way permit. — While slavery existed, the poor enjoyed the protection of the masters, which was often paternal. With the abolition of unfreedom has followed the abolition of protection, and with it in turn the sufferings that the struggle for existence brings. By state compulsion, then, men have sought to remedy this through the arrangement of poor relief, since it was clear that the principle of the subsistence of the best adapted could not be allowed to prevail in all its severity. But the intervention of the state has here caused greater evils than were to be remedied: it has protected the weak and incapable, made it possible for them to bring offspring into the world, and maintained them at the expense of the able and industrious! Men have wanted to soothe their feeling of the suffering that is in the human world by applying a system that on closer consideration makes evil worse. The present system of state aid is a kind of social opium-eating. The sufferings that evolution once and for all brings with it cannot be escaped. And they are not escaped by letting feelings that have their proper domain in more personal relations and within the family gain influence on the arrangement of political and social relations. Family ethics and state ethics must not be confused; the former concerns the rearing of helpless offspring, the latter the ordering of the mutual relations of adults. When adult age is reached, no state aid is to hinder *[490]* the conditions of life from taking effect. It is another matter with free charity, which

uses its own means, while the state *extorts* its means. But it too will only be able to hinder superfluous suffering and soften the necessary; if it goes further, it weakens the vital power of the race. — Spencer must himself concede, however, that just as there are many transitional forms between the child and the adult, so there is room for many compromises between family ethics and state ethics. These systems do not form so thoroughgoing an opposition in principle to each other as one would think from many of Spencer's utterances. Spencer's correct fundamental thought is that compassion with suffering must not exclude the discipline through interaction with the real conditions of life that is necessary for the healthy development of the individual and of the whole race.

In its crudest form the struggle for existence favors egoism. But altruism, which has its root in sympathy, slowly works its way forward, first in the form of care for the helpless, later also in other and higher forms. There is no absolute opposition between them: evolution will lead to such a change of human nature that the individual finds his highest happiness in devotion, without, however, hampering by his intervention the independent development of the other, and that, on the other side, no one is so egoistic as to wish to be the object of an absolute self-sacrifice by another individual. What now characterizes the exceptionally highly placed may be expected one day to characterize all: for what the best human nature is capable of lies within the compass of human nature as a whole, and evolution is not ended so long as there is still a possibility of a fuller and richer life through the unfolding of faculties that at once bring the individual himself immediate satisfaction and do others good. Free activity and unfolding of life which is more than a means to a future end is, as we have seen, what is characteristic of the higher type of life to which Spencer looks forward. Already now there are, as we have also seen, actions and efforts that bear this stamp, so that "the third kingdom", to
[491] which Spencer with so many other thinkers in the nineteenth century looks forward, does not lie only in the future.

Appendix

In English literature there remain to be mentioned some works of a more special nature that are of interest for philosophical thought in general. These are the logical works of Boole and Jevons and the ethical work of Sidgwick.

After Stuart Mill had treated inductive logic in such a way that the limits of the probative force of the purely inductive method came clearly to light, George Boole in his work *An Investigation of the Laws of Thought* (London 1854) gave a new exposition

of deductive logic, in which he assigned to deduction the significance of deriving all the combinations of certain concepts that are logically possible when a definite connection between these concepts is given. The task of deduction is thus to give exhaustive information about the logical value of a given judgment. Boole's method is acute, but very artificial in its form. His fundamental thought was carried out in a far simpler form by Stanley Jevons in a series of works that began with *Pure Logic or The Logic of Quality apart from Quantity* (1864) and ended with *Principles of Science* (1874). Jevons takes the content of concepts, not their extension, as his basis, and thereby brings out more sharply the difference of pure logic from pure mathematics. He also carries through (what Hamilton and several before him had already begun by quantifying the predicate) the principle of identity far more sharply, already in the formulation of judgments, than had hitherto been done in logic — if some long unprinted treatises of Leibniz are excepted. His logical work goes far more into particulars in its exposition of the methods of research than Mill's logic. Jevons's logic has a considerable epistemological interest through the clear way in which it establishes that a deductive inference underlies every induction, the proof of the correctness of an induction always consisting in this, that deduction from the proposition tentatively set up leads back to the given experiences, *[492]* neither to more nor to less than them. Since deduction in turn presupposes the validity of the logical principles, the incorrectness of pure empiricism is thereby established. In the last years of his life (1877—79) Jevons published in "Contemporary Review" a very sharp criticism of Stuart Mill's philosophy, in which he sets pure empiricism in the sharpest opposition to evolutionism and attaches himself decidedly to Spencer's philosophy.

Henry Sidgwick by his work *Methods of Ethics* (1877) gave ethical discussion new life and new clarity. In particular he drew attention to the fact that behind the name utilitarianism there were hidden two different ethical systems, of which the one took egoism, the other altruism as its basis, while they both saw the ethical standard in the principle of welfare. He undertakes a careful investigation of how far the two systems can each reach by itself. He also seeks to show that the current moral propositions (the morality of Common Sense) all lead back to utilitarianism, and that the gaps or contradictions found in them can be made to vanish when one carries through this unconscious utilitarianism[112]).

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Tenth Book

Philosophy in Germany 1850–1880.

Of the two great tendencies in nineteenth-century philosophy it was positivism that *[493]* had best maintained its connection both with the thought of the eighteenth century and with empirical science. The philosophy of Romanticism, by contrast, was a definite and conscious attempt at reaction against both these tendencies; it wanted even to do over again what had been founded with the rise of natural science in the seventeenth century. In Germany, the hearth of Romanticism and of its philosophy, that tendency was the dominant one at the middle of the century. The adherents of the critical undercurrent were the only ones who held fast to continuity with the other sciences — leaving aside Schopenhauer and Feuerbach, who at that time still stood as solitary thinkers for whom the hour of recognition had not come.

As little as positivism was to be explained as a reaction against Romanticism, so little is the philosophical movement which takes place in Germany after the middle of the century to be explained as a continuation of Franco-English positivism. It has its presuppositions in Germany itself. It receives its problems above all from the renewed rise of natural science toward the middle of the century. Not only do natural-scientific studies and results now become the object of more general interest than in the earlier part of the century, when poetry, religion and speculative philosophy had laid claim to men's minds; natural science itself now returned, with clearer consciousness, to the great *[494]* principles its founders had in their day set up. The demand for an understanding of outer nature as a series of demonstrable causes and effects — in other words, for a mechanical explanation of nature — was now asserted again, as in the seventeenth century. And at the same time a new and comprehensive law of nature was discovered and demonstrated: the law that in physical nature no energy arises or perishes, but that what happens in every apparent arising or perishing is a conversion of energy into another form, and in such a way that the different forms of energy stand in a determinate quantitative relation to one another. This law — which, together with Darwin's doctrine of the origin of species

through the struggle for existence, is the most significant result of natural-scientific investigation in the nineteenth century — could not but set philosophical thinking in motion, just as in their day the Copernican theory and Galilei's founding of mechanics had led philosophy into new paths. For German philosophy in particular it had to become a principal question: how far can the thoughts developed by the philosophy of Romanticism be held fast, if one accepts the new natural-scientific way of regarding things? This question was answered in three different ways. Modern materialism rejected those thoughts as pure illusions and proclaimed the doctrine that only the material exists, as a necessary consequence of natural science. Lotze and Fechner, by contrast, attempted to exhibit the fundamental thought of speculative philosophy of religion as the last and concluding presupposition of the picture of existence that can be built up by the help of the natural-scientific method. Finally Albert Lange and Eugen Dühring approached critical philosophy and positivism, by bringing out the significance of the problem of knowledge and by maintaining practical idealism's independence of empirical science, while at the same time insisting upon experience's right to determine the actual content of the world-picture.

1. Robert Mayer and the Principle of the Conservation of Energy

[495] Not only in philosophy, but in the other sciences too, knowledge moves forward in rhythmic fashion, through actions and reactions. At the close of the eighteenth century natural science had a great and fruitful period, in which some of the most important truths in the domains of chemistry and physiology were discovered. Lavoisier introduced the quantitative method into chemistry, and by its help the truth of the old thought was established that no matter arises or perishes, but that the same mass of matter subsists through all changes, under different forms. Chemistry was founded as an exact science. Priestley, Ingenhouss, Senebier and Saussure found the principal laws of the metabolism of plants and animals, and thereby founded the great doctrine of the circulation of matter in nature, whereby the organic and the inorganic world appear in their close mutual connection. It was discovered that it is under the influence of sunlight that organic matter is formed in the green plant cells by the taking up and dissolution of the carbonic acid of the air. The carbon stored in the plant cells then serves animals for nutriment; by the animal functions a combustion takes place, and the carbonic acid thereby formed passes out into the air in respiration, whence the same circulation can

begin anew. A great cosmic connection had here been exhibited.[5] The picture of the world sketched by Copernicus, Bruno, Kepler and Galilei, which Newton had enriched by exhibiting universal attraction as a binding and ordering tie, hereby received a new and significant enrichment.

But the age was too much taken up with revolution and world war, with Romanticism and speculation, and soon with orthodoxy and mysticism as well, for these great ideas to gain the influence upon general opinion that they deserved. Even within natural science their victory was hindered by other tendencies. It was within the science of organic life that resistance was chiefly offered to the new points of view, because men were still governed by the reaction against the Cartesian doctrine of the organism as a machine. The *[496]* expressions peculiar to the organism were derived from a peculiar vital force, supposed to be wholly different from the other forces of nature. This mode of explanation, so-called vitalism, could not but lead to a rejection of the attempts to draw the organic phenomena into the universal circulation of matter.[6] Besides, the prevailing interest among botanists and zoologists was to describe the forms and determine their systematic place, not to find the developmental process through which the forms come to be, and the causes that determine that process. Romantic philosophy of nature favored this aesthetic and formal conception — and had itself proceeded from it. When "philosophy of nature" is often given the whole blame for the fact that the points of view of the close of the eighteenth century did not sooner prevail in botany and zoology, it must not be forgotten that it was quite as much an effect of the condition of natural science as a cause of it.

From about 1840 a change came about in this respect. Life began to be conceived as something other than a play of forms or a revelation of ideas. Dumas and Liebig insisted upon the significance of the chemical processes for plant and animal life. And from the medical side too a strict mechanical explanation began to be demanded of the processes that go on in the organism. In some of his earliest writings Hermann Lotze worked in this direction, especially in *Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften* (1842) and in the treatise *Leben, Lebenskraft* (1843, in R. Wagner's Handwörterbuch der Physiologie). The task was set of making physiology a purely mechanical science. Most epoch-making, however, was the fundamental proposition set up by the physician and physicist Robert Mayer (b. 1814, d. 1878) in his work *Die*

[5] On "the period of the great discoveries" at the close of the eighteenth century see *Rasmus Pedersen*: *Planternes Næringsstoffer.* Historisk Indledning. København 1883, p. 44 ff. *[596]*

[6] On vitalism in its various forms see *Louis Peisse*: *La Médicine et les Médicins.* Paris 1857. I, p. 226–297. — It must be firmly held that by vitalism is here meant a tendency which finds the *explanation* of vital phenomena in the assumption of a "vital force" — not a tendency which maintains that there is much *[597]* in the vital phenomena that has hitherto not been explained by physical and chemical causes. See my treatise *Om Vitalisme* (in "Mindre Arbejder").

organische Bewegung in ihrem Zusammenhange mit dem Stoffwechsel (1845): that in the process of life there takes place only a transformation of force as well as of matter, never a creation of them. Mayer here points back to the great law he had discovered a few years earlier, when he set the law of the conservation of force beside the law of the conservation of matter deriving from Lavoisier.

[497] Robert Mayer's life as an investigator is borne by a single thought, which he came upon early; but it was the tragedy of his life that he was unable to carry it through in detail by experimental means. As a child he had tried to construct a *perpetuum mobile*, and it made a strong impression upon him that he did not succeed. From that time he brooded constantly upon the relation between cause and effect — first and foremost within physiology, which as a medical man lay nearest to him, but thereafter within chemistry and physics as well. On a voyage to the East Indies as ship's doctor he arrived at the new idea of the imperishability of force in nature, under the influence partly of investigations into the origin of organic heat, partly of the fact that heat arises in the wave-motion of the sea. His point of departure is thus definitely observation. But that observation led him to ask whether a law similar to the one in chemistry might not hold in physics too, so that not only matter but force also was imperishable, and every change a mere conversion. And he then came back to his earlier broodings upon cause and effect. It was for him a self-evident proposition that there could not be more in the effect than in the cause: *causa æqvat effectum*! It follows from "the law of the logical ground"! Mayer thus does not distinguish, any more than the dogmatic philosophers do, between ground and cause; thereby he comes, like William Hamilton and Herbert Spencer, to regard it as a matter of course that the cause cannot become nothing when the effect arises, but that the effect must furnish an equivalent for what seemed to vanish with the cause. Hitherto men had rested content with the fact that a motion ceased when it met a sufficiently strong resistance, and that heat arose from frictional resistance. But — Mayer asked — does the motion then become nothing, and does heat then arise out of nothing? If that is so, then "the red thread of science" is cut through, just as it would be if in chemistry we were to assume that oxygen and hydrogen became nothing when they combined, and that water then came to be out of nothing. Just as we assume that oxygen and hydrogen are transformed into water, so we must also assume that the motion does not become
[498] nothing, but is transformed into heat.[7] And by the expression *transformed* Mayer meant

[7] Cf. *Mayer's* letter to Griesinger of 20 July 1844: "*Wenn ich sage, Wärme lässt sich in Bewegung verwandeln, und umgekehrt, so will dies nichts meinen, als dass zwischen Wärme und Bewegung finden hin und her dieselben qvantitativen Beziehungen statt.*" (*Kleine Schriften und Briefe* von *Robert Mayer.* Herausg. von *Weyrauch.* Stuttgart 1893, p. 225.) — In his paper *Robert Mayers Entdeckung und Beweis des Energieprincipes* (in the Festschrift for Sigwart. Tübingen 1900) A. *Riehl* illuminates Mayer's doctrine from

to designate only that a constant quantitative relation subsists between the cause, which disappears, and the effect, which comes in its place. If the train of thought put forward is correct, experience must establish such a relation of equivalence between the different forms of force that succeed one another. On the basis of experiments already available, Mayer sought already in his first paper (*Bemerkungen über die Kräfte der unbelebten Natur*, printed in Liebig's "Annalen der Chemie und Pharmacie", 1842) to determine the quantitative relation between heat and motion. It was this idea of an invariable relation of magnitude that was the main thing for Mayer, and that in the following period proved so fruitful for natural science. It was a significant extension of the abstract law of the conservation of force, already maintained by Huyghens and Leibniz, so as to hold for the relation of the different forces of nature to one another. Mayer derived from his train of thought that there really exists only a single force, which appears under different forms, and these stand to one another in a determinate quantitative relation.

Although Mayer's attempt to give a philosophical derivation of the law of the conservation of energy does not hold, his train of thought has nevertheless an epistemological interest. For it was in itself fully legitimate to raise the question whether between the event we call the cause and the event we call the effect there might not obtain a relation similar to that between logical ground and logical consequent. And Mayer's train of thought leads naturally to raising that question. Experience must then decide what the answer is to be. — It is interesting that the Danish physicist Colding, who in the year after Mayer's first paper was printed set up the proposition of the conservation of energy by a road of his own and confirmed it by experiment, proceeded from the view that it was really a proposition of reason; and likewise that Helmholtz, in his paper *Ueber die Erhaltung der Kraft* (1847), proceeded from an epistemological postulate. The Englishman Joule, who like Colding and Helmholtz had come by a road of his own to the same result as Mayer — a remarkable example of how several may simultaneously be on the track of one and the same discovery — proceeds predominantly experimentally, *[499]* but nevertheless indicates the a priori improbability that force should be annihilated without a corresponding effect.[8]

an epistemological point of view.

[8]"If the resistance to electrolysis which is over and above that due to chemical change were not accounted for elsewhere, *it would prove the annihilation of a part of the power of the circuit, without any corresponding effect.* We shall see that this is not the case, but that in the evolution of heat, where the excess of resistance takes place, an exact equivalent is restored" (*Joule*: *On the Heat evolved during the Electrolysis of Water.* 1843. Scientific Papers. I. London 1884, p. 115). "*We might reason, a priori, that absolute destruction of living force cannot possibly take place,* because it is manifestly absurd to suppose that the powers with which God has endowed matter can be destroyed ... but we are not left with this argument alone, decisive as it must be to every unprejudiced mind" (*Joule*: *Matter, Living Force and Heat.* 1847. Scientif. Pap. I, p. 269). — Joule's standpoint is therefore not in principle different from Mayer's and

The new proposition had difficulty in winning recognition, but gradually made its way in the course of eighteen years (1842–1860), especially as it proved fruitful in leading to new investigations and discoveries. Its great importance consists precisely in the determinate questions it leads the investigator of nature to ask, every time a peculiar manifestation of force appears or ceases. Just as it is the importance of the general causal proposition that at every change which occurs it leads us to look about for a preceding change of which the new one might be the effect, so it is the importance of the proposition of the conservation of force — or, as one now prefers to say, following the proposal of English investigators, of energy — that one is at once brought to set oneself the task of finding out in what relation the manifestations of force that succeed one another stand to one another.

Philosophically regarded, the question had above all to become how the spiritual phenomena stood to the new law. It is characteristic in this respect that almost all its discoverers proceeded from decidedly spiritualistic and teleological views. Mayer repeatedly declared himself a definite opponent of materialism, and expressed as his conviction that the scientific truths stand to the Christian religion as brooks and rivers to the ocean. At the congress of natural scientists in Innsbruck in 1869 he made a declaration to this effect, to the offense of Carl Vogt and his friends. Colding's standpoint is seen from the following utterance: "My first thought that the forces of nature must be imperishable I drew from the view that the forces in nature must be akin to the spiritual in nature, to the eternal reason as well as to the human mind. It was thus the religious conception of life that led me to the thought of the imperishability of the forces of nature." Colding and Joule conceive that God at the creation laid a certain sum of force into nature, and that this sum can neither be increased nor diminished as a whole, but only
[500] distributed in different ways — as Descartes in his day assumed of motion. And both assumed that with the conservation of energy the conservation of what is valuable in the world was secured, since they no more than Leibniz distinguished between energy itself and its employment in favor of life and development.[9]

Colding's. For the latter compare an utterance in his *Undersøgelse om de almindelige Naturkræfter og deres gensidige Afhængighed.* (Videnskabernes Selskabs Skrifter. Femte Række. Naturv. mathem. Afd. II, p. 129): "The thought that an activity could disappear in the corporeal without appearing again as operative cause seems to me contrary to reason." — It is of interest in this connection to recall that *Lavoisier* too proceeded in his day from a fundamental proposition he regarded as a matter of course, namely that the weight of a compound body is equal to the weight of the substances of which it consists. Self-evident as that presupposition was, it nevertheless had to be proved to those who took heat for a substance; for if that assumption were correct, the release of heat which accompanies a chemical combination would have to entail a diminution of weight. Compare *Ernst von Meyer*: *Geschichte der Chemie.* Leipzig 1889, p. 137.

[9] *Mayer*: *Kleinere Schriften und Briefe*, p. 339 f., 460. *Die Mechanik der Wärme.* 3. Ausg., p. 356 f. — *Colding*: *Naturvidenskabelige Betragtninger over Slægtskabet mellem det aandelige Livs Virksomheder og de*

But a closer investigation and grounding of these presuppositions the natural scientists named do not undertake; they proceeded, like the philosophers of the seventeenth century, from given religious representations. A closer investigation of the relation of the law exhibited — and of the knowledge of nature so far won generally — to the phenomena of spiritual life had therefore naturally to be called forth. And it is then no wonder that the ways parted, and that different tendencies stepped forth against one another. All that was given was this: that when one kind of physical energy ceases, another kind of physical energy enters, in a certain determinate quantity, in its place. The question was what significance this knowledge must consistently come to have for the conception of the world.

2. Materialism

The strongest reaction against the philosophy of Romanticism — apart from crass orthodoxy — is marked by the materialist literature which flourished in Germany at the middle of the century. If there the Idea had been everything, now matter became everything. Enthusiasm for natural science, and for the great points of view it had succeeded in carrying through in the doctrine of the conservation of matter and of force, could naturally lead to regarding that doctrine itself as a sufficient philosophy, and to explaining every side of existence by its help. The simplest way of solving the question what significance the law of the conservation of matter and force has for our world-view would undeniably be to say that this law contains our whole world-view. The philosophy which modern materialism sets up will be nothing other than the simple consequence of natural science. Thereby a firm basis seemed at the same time to have been won for all our representations and for our conduct of life. In place of a mystical or spiritualistic *[501]* foundation we now get a wholly tangible foundation to build upon in theory and in practice. And it is a doctrine easy to popularize. It is childlike, intuitable, easily accessible, and its presentation gives good occasion to bring forward a mass of natural-scientific facts of general interest. Several of the spokesmen of German materialism were able and independent investigators of nature, such as Carl Vogt and Jacob Moleschott. Others, like Louis Büchner, worked especially by clear, tasteful and enthusiastic presentation. It is in its popularizing tendency that this literature has its greatest merit. A

almindelige Naturkræfter. (Oversigt over det kgl. danske Videnskabernes Selskabs Forhandlinger. 1856), p. 155, 166. — *Joule*: *Scientific Papers.* I, p. 269, 273. — As regards Mayer, it appears from his address at Innsbruck (*Die Mech. d. Wärme*, p. 357) that he did not assume any break in the series of material *[598]* phenomena, but assumed a parallelism between the spiritual processes and the brain processes.

wealth of knowledge has been spread abroad by it in wide circles. Büchner's *Kraft und Stoff* is one of the most read popular-scientific books of our century; from 1855 to 1899 nineteen German editions appeared, and it has besides been translated into several other languages. Even if there is something dogmatic about materialism, it has nevertheless done good service by setting its dogmatism against ecclesiastical dogmatism, and thereby awakening reflection upon problems which had been pushed aside at the dissolution of Romantic philosophy. The whole materialist movement in Germany was moreover borne by an idealistic interest in humanity and progress, and Büchner was fully justified in protesting against the confusion of materialism as method and theory with materialism in the sense of a practical direction of life. Materialism can very well acknowledge the value of the highest and noblest ideas and feelings — even if it holds that they, like all spiritual phenomena, are only products of, or forms of, material processes.

When materialism claims to be the simple consequence of the results of natural science, it is of interest to observe — not only that the discoverers of the law of the conservation of energy proceeded throughout from spiritualistic presuppositions, but that the most important collisions in the materialist controversy took place between parties who all stood upon the ground of natural science. This does not necessarily show that materialism is wrong; but it shows how difficult it is to draw the true consequences, and how many different motives come to influence the world-view of individuals.

[502] Moleschott's famous book *Der Kreislauf des Lebens* (1852) is directed against Liebig's theologizing utterances in his "Chemische Briefe", in which he had particularly attacked an earlier proposition of Moleschott's: "Ohne Phosphor kein Gedanke!" At the same time a controversy broke out between the physiologist Rudolph Wagner in Göttingen and the zoologist Carl Vogt in Geneva. This controversy reached its height when Wagner, at a congress of natural scientists in Göttingen in 1854, declared himself in favor of the assumption of an ethereal soul-substance which moves the brain's fibres as the musician moves the pianoforte, and which is propagated from parents to children by division [!] — an assumption he supported upon the Bible's teaching, but which he admitted he could maintain only by a sharp distinction between faith and knowledge. For when one goes beyond the domain of exact knowledge one has, according to Wagner, only an unprovable faith to hold to — and he adds: "In Sachen des Glaubens liebe ich den schlichten, einfachen Köhlerglauben am meisten." Against this Vogt came forward with a robust polemic: *Köhlerglaube und Wissenschaft* (1855). In a later work, *Vorlesungen über den Menschen* (1863), Vogt gives a more scientific presentation of his standpoint. He lays weight above all upon the brain's being the organ of consciousness, and upon

consciousness standing to the brain as every function stands to the corresponding organ. Although he admits that it cannot be explained *how* consciousness arises in the brain's cells, he holds fast that it is indissolubly bound to them; and he does not give up the teasing proposition he had already set up earlier, that thought stands in the same relation to the brain as bile to the liver or urine to the kidneys — notwithstanding that a function stands in a different relation to its organ than a product does to the place where it is produced.

Moleschott relies, as the title of his work shows, especially upon the doctrine of the conservation of matter. It is for him the great thought which the Encyclopedists of the eighteenth century had already brought forward, which subsequent natural science has confirmed, and upon which the faith of the future must be built. He feels reverence for the great circulation in nature. The miner fetches phosphate of lime up out of the earth in the sweat of his brow — and perhaps thereby the matter destined for the best head and the highest thoughts passes through his hands: the farmer manures his soil with the *[503]* phosphate of lime; this becomes a constituent of the wheat that nourishes men's bodies and brains! With matter, life circulates through all parts of the world; with life, thought — and from thought there springs in turn the will to make life better and happier. If we can but supply the organism and the brain with the best substances, thought and will too will attain their highest development. The investigator of nature is our age's Prometheus, and chemistry is the highest science. The social question finds its solution if one can but [sic] find the right distribution of the substances to which the life of thought and of will is bound. — Only the first edition of Moleschott's book has been taken into account here; in later editions it was altered and expanded. Moleschott, who was born in Holland in 1822, was working as a lecturer in Heidelberg when the book appeared; but when his freedom of teaching was interfered with there, he moved to Zürich; later he worked as professor of physiology in Turin and in Rome, where he died in 1893. After his death his autobiography appeared (*Für meine Freunde.* Lebenserinnerungen von Jacob Moleschott. Giessen 1895). It gives a fine picture of the idealistically minded investigator of nature, striving after an all-round culture. For a right understanding of his standpoint the following utterance is of service, which comes after the train of thought of "Kreislauf des Lebens" has been developed: "One-sidedly 'materialistic' this is only for those who can imagine a matter without force or a force without material bearer. I was clearly conscious that one could turn the whole way of representing it round: since all matter is a bearer of force, is endowed with force or penetrated by spirit (*geistesdurchdrungen*), one may just as well call it an idealistic view" (p. 221). Moleschott's standpoint is rather to be

called monism than materialism: "It is a question of a true, indivisible two-in-oneness, and the dispute is not between materialism and spiritualism, but between two-in-oneness (*Zweieinigkeit*) and divided duality (*zwiespältige Anschauung*)" (p. 222).

Louis Büchner (1824–1899) does indeed declare that he will not explain what the relation is between spirit and matter, force and substance; he will only maintain that
[504] there is a necessary and indissoluble relation between them. But even this is more than can be scientifically proved; and Büchner moreover harbors no doubt that spirit is only a property of matter, force only a property of substance. The impulse to his famous book "Kraft und Stoff" he had received from reading Moleschott's "Kreislauf des Lebens", and it is the eternity of matter that is the ultimate foundation in him too. He holds that the conservation of force is really a self-evident consequence of the conservation of matter, and could therefore already have been drawn from Lavoisier's chemistry. It was only in the fifth edition of his book, however, that he inserted a special chapter on "the immortality of force", in which the circulation of force is set as a necessary correlate beside the circulation of matter, so that "the two together form, from eternity and to eternity, the sum of phenomena which we call the world". Yet he constantly holds fast to the assumption that "in matter dwell all the forces of nature and of spirit; matter is the primal ground of all being". And as regards the spiritual, he regards it — notwithstanding that he has disclaimed all insight into its relation to matter — as a mere product. "As the steam engine produces motion, so the complicated organic composition of force-endowed substances in the animal body produces a total sum of certain effects which, when they are bound together into a unity, are called by us spirit, soul, thought" (Kraft und Stoff. 7. Aufl., p. 130). In a later utterance Büchner goes — without appearing to notice it — a step further. Thus it runs, in an utterance directed polemically against the author of the present work: "The opposition between physical and psychical energy is tenable only so long as one conceives the concepts of body and spirit, or more generally of matter and force, in a dualistic way, whereas for the materialistic-monistic conception they coincide into one and perhaps present only *different sides or forms of expression of one and the same primal ground of all things*. It is precisely the law of the conservation of force that in the problem of the soul leads necessarily to materialistic consequences, as I, supported by physiological-psychological facts and considerations, believe I have established to evidence in several of my writings, and recently again in the fifth of my
[505] letters on *Das künftige Leben und die moderne Wissenschaft* (Leipzig 1889). I believe I have there shown that psychical activity *is not, and cannot be, anything other than the radiation, taking place between the cells of the gray cerebral cortex, of a movement initiated*

by outer impressions" (from an article in the weekly *Menschthum.* Gotha 1889, No. 46).

It is clear that Büchner in this utterance confuses two different theories: the one, which conceives spirit and matter as forms of expression of one and the same substance, and the other, for which matter is "the primal ground" and psychical activities are therefore consistently *nothing other than* motion (or "radiation of motion"). In one and the same breath he professes both. The tireless champion of the influence of natural-scientific results upon our world-view has not seen this inconsistency, which perhaps in his eyes did not much matter, since his real aim was not to inculcate any one determinate philosophical theory. In his great zeal he has also believed that he had "established to evidence" more than can be established.

With sharper criticism Heinrich Czolbe (1819–1873), a physician like Büchner, sought to determine the consequences of the teaching of natural science. In his first works he is a consistent materialist. In *Neue Darstellung des Sensualismus* (1855) and *Die Entstehung des Selbstbewusstseins* (1856), as well as in the paper *Die Elemente der Psychologie vom Standpunkte des Materialismus* (Zeitschrift für Philosophie. XXVI), he develops the view that materialism conflicts with natural science so long as the latter assumes the doctrine of the specific energy of the sensory nerves, and thus assumes a difference between the sensations and the outer processes corresponding to them. In opposition to that doctrine Czolbe sets up the assumption that it is essentially one and the same motion which propagates itself from the surrounding world, through sense-organs and nerves, to the brain. At no point does any qualitative change take place; the difference between the different senses is to be explained by the different intensity with which the motion proceeds in the different organs. The unity of consciousness is to be explained by the movements in the brain acquiring a direction that runs back into itself; *[506]* by making such circular movements possible, the brain is the organ of consciousness. Thus Czolbe would conceive both sensation and self-consciousness, plainly and simply, as movements in space. But — clear and consistent thinker that he is — he becomes aware that if he identifies sensation and self-consciousness with motion without more ado, then one must also be able to reverse the proposition and say that wherever motion of a certain intensity and a certain direction is present, there consciousness is present too, so that the consequence is a universal ensoulment of nature. Thus the consistent carrying through of materialism led beyond it.[10]

[10]The consequence Czolbe draws from materialism was already drawn by *Henry More*, who (*De anima.* Roterodami 1677, p. 64) sets up the following fundamental proposition: "If sensation can be ascribed to matter, then the motion of one material part, or its reaction upon another material part, or at any rate the continuation of that motion, is one and the same as sensation and apprehension. This proposition ... is confirmed by the voice of my chief opponent, Hobbes." He then refers to Hobbes's doctrine of sensation as

Later — in the work *Die Grenzen und der Ursprung der menschlichen Erkenntniss* (1865), and in the exceedingly interesting paper *Die Mathematik als Ideal für alle andere Erkenntniss und das Verhältniss der empirischen Wissenschaften zur Philosophie* (Zeitschrift für exakte Philosophie, 1866) — Czolbe saw that it was impossible to explain existence from a single principle, whether one finds that principle with Büchner in matter, or with the speculative philosophers in spirit, or with the theologians in God. We reach an explanation only by proceeding from several elements, calling that an element which we are unable to analyze. Such elements, not reducible to one another, are the material atoms, the organic forces, and the mental elements (which last make up in sum the world-soul). Between these three kinds of element a harmonious co-operation takes place, by which a purposive connection in nature is realized. Not in its origin, but in its direction and tendency, does existence establish itself as a unity. — Thus it appeared that the problem of existence was more intricate than materialism supposed. In his later investigations, which are only partly published, Czolbe approached Spinoza's fundamental thoughts, of which he wished to give an empirical execution. — On the different stages of development of this energetic thinker's philosophizing, reference must here be made to Vaihinger's paper *Die drei Phasen des Czolbeschen Naturalismus* (Philosophische Monatshefte XII).

[507] What lies at the ground in Czolbe from first to last is the demand for clarity and intuitability in all fundamental concepts. He wanted everything mystical and supersensible away, and sought so far as possible to think all ideas with geometrical clarity. It was more with respect to intuitability than with respect to stringency of proof that he proclaimed mathematics the ideal of science. In his youth he was enthusiastic for Hölderlin's poetry, and had set himself the task of maintaining Greek clarity and joy in life against all mysticism and dualism. Intuitability in the domain of thought was for him closely akin to joy in the natural world. Just as it may require labor to carry through the principle of intuitability in thinking, so it may cost resignation to carry through joy in actual life in practice; but in the theoretical domain as in the practical, Czolbe would combat what he calls "der Blödsinn der Transcendenz". He admitted already in his materialist period that the innermost motive of his philosophy was the subjective need for intuitability and for holding fast to the actual; materialism was to him a postulate — which was later replaced by other postulates. —

Ernst Haeckel (b. 1834, from 1865 professor of zoology in Jena) is often reckoned to materialism, but has himself determined his standpoint as a monism which leads beyond the opposition between spiritualism and materialism, and which takes as its

a motion. Hobbes, however, denied that consequence, saying expressly that the proposition that sensation is motion cannot be reversed. (*De corpore* XXV, 5).

ground the great pantheistic thought of the unity of nature. For monism, he says, there exists neither spirit nor matter in the usual sense, but only one thing which is both at once. It was in his *Generelle Morphologie* (1862–1866) that he first uttered his general philosophical views. The mental is for him — as for Czolbe — an original element in existence, though it exists in exceedingly different degrees, from the atom-soul and the cell-soul to the souls of the highest organisms. This theory of ensoulment becomes questionable in Haeckel through his often letting the soul's action serve as an explanation of organic movements, instead of seeking a purely natural-scientific explanation. On the other hand he sometimes declares mental phenomena to be mere effects of the most complicated and most unstable carbon compounds, the albumen molecules in nerve *[508]* tissue. He believes himself to be beyond spiritualism and materialism, and yet one finds in him both propositions that sound spiritualistic and propositions that sound materialistic. He has not arrived at his principles through a sharp thinking-through of the laws of the conservation of matter and of force. It is Darwin's doctrine that set his thinking in motion, and he has sought a highest principle that could be united with the great unity in nature to which the new doctrine pointed. Monism leads, according to Haeckel (Gen. Morphol. II, p. 445–451), to the most exalted idea of God. In the great, all-embracing law of causality the Godhead reveals itself as embracing the whole of nature and operating in every natural phenomenon. The usual conception of God, by contrast, is belief in two gods — amphitheism, not monotheism — because it sets the natural causes beside the Godhead.

Haeckel was one of the first German investigators of nature to attach himself to Darwin, and he fought with great ardor for his theory, to which he ascribed a certainty, and gave an extension, that could not always be approved by the theory's critical and level-headed founder. He was untiring in constructing genealogical trees for the species now living, and he had no scruple about assuming a constant arising of organic matter. He has no clear eye for the limits of hypotheses and for the necessity of verification; only thereby is it explained that he can without more ado set Darwin's hypothesis alongside Newton's. He accordingly also reproached Darwin with laying too much weight upon the objections to the theory. The great investigator shook his head at his young adherent's zeal: "Your boldness sometimes makes me tremble", he wrote to him (19 November 1868). — In Haeckel's most recent "philosophical" work (*Die Welträthsel.* Gemeinverständliche Studien über monistische Philosophie. Bonn 1899) the qualities mentioned appear in a still more glaring form than in his earlier utterances. —

The doctrine of the conservation of matter and of force, and the doctrine of the

development of species through natural selection, were — as already remarked — the natural-scientific results which in the period after the middle of the century had above all
[509] to set reflection in motion. Having now considered the attempts made by investigators of nature to found a philosophy upon these results, we shall go on to examine the way in which professional philosophers took up their position with respect to them.

3. Idealist Constructions on a Realist Basis

a. Rudolph Hermann Lotze

Idealist philosophy has in Lotze its most important representative in the last half of the century. In his personality and in his development he stands as one who has taken up into himself, in an interesting way, both the ideal motives upon which the philosophy of Romanticism built and that strict carrying through of the mechanical view of nature which toward the middle of the century was working its way forward in science. Lotze is a master of conceptual development — of discussing a thought so that all its nuances come out, of taking up a problem again and again and treating it from different points of view. His ideal was in the last instance the same as the one that floated before the philosophy of Romanticism: to derive all development and all connection in the world from an eternal Idea, which should contain the ultimate ground of all that happens, as well as of the value of what happens. But he saw clearly that such a derivation exceeds what human thought can accomplish, and that when the thinkers of Romanticism nevertheless believed they had given one, it was because in them the poetic-religious tendency involuntarily merged with the philosophical. Lotze therefore separates the poetic-religious element from the speculative. His own fine poetic sense let him understand that their Romantic confusion was not a happy one. And this poetic sense is one with a sense for individual nuances and relations, which speculative philosophy all too often dissipated by its abstractions. Through this sense Lotze's speculative interest hangs together with his
[510] realist tendency. What matters for him is first and foremost to apprehend the phenomena in their concrete nature and in their determinate, law-governed connection; it is then the task of philosophical thinking to find out the presuppositions upon which that real connection rests. The poetic, the natural-scientific and the philosophical element in Lotze thus hang closely together, and he was equipped as few have been to treat the task he had set himself, and which the intellectual conditions of the age suggested: to attempt a reconstruction of an idealist philosophy upon a realist basis. It was his conviction that the philosophy of Romanticism had overlooked the real conditions and the mechanical

connection of nature, without which even the most significant ideas are helpless idols; just as it was his conviction that materialism made *that* into the be-all and end-all which was only a — quite certainly necessary — form and frame for the valuable content of existence. His philosophy is essentially an analysis of the very concept of natural mechanism, designed to show that this concept leads with necessity to the assumption of an ideal principle of existence, and that it at any rate does not exclude the assumption that such a principle is the eternal source of all that is good and valuable in the world.

In Lotze's development the different motives and interests that animated him appear clearly. He was born in Bautzen in Saxony on 21 May 1817, in the same region from which Lessing and Fichte come. At the university of Leipzig he studied philosophy, medicine and physics. He was here initiated into the two circles of thought which it was later to be his task to unite, after having worked through them separately. It was the aesthetician and philosopher of religion Chr. Hermann Weisse who was his teacher in philosophy, and Lotze later said that he owed to Weisse not merely many particular influences, but that he was in particular introduced by him into a circle of thoughts which he has since felt no need to give up. Weisse was the most eminent representative of philosophical theism. Through him Lotze's religious-philosophical conception stands in historical connection with Schelling's — and through that with old Jacob Böhme's. Lotze emphasizes, in a retrospect upon his development (*Streitschriften.* 1857, p. 5 ff.), that *[511]* he appropriated the ideas of speculative philosophy not as a closed doctrinal edifice but as a peculiar form of intellectual culture. The presupposition that the ultimate ground of things could be thought only as spiritual was early laid down in him and never given up. Medicine and physics he studied under E. H. Weber, Volkmann and G. Th. Fechner. Here he came to know the natural-scientific method and conception at first hand. In one and the same year he took the doctorate in philosophy and in medicine, and now came forward as lecturer in both subjects — also after a professorship in philosophy had been conferred upon him in 1842. After working some years in Leipzig he became professor in Göttingen, as Herbart's successor, and here fall his most important works. The year before his death he received a call to Berlin, but there soon succumbed to an illness from which he had long suffered (1881). His life was quiet, devoted to study, thought and academic teaching. With a rare union of many-sidedness and thoroughness he knew how to orient himself in highly different fields — something to which not only his medical and philosophical works but also a whole series of smaller papers and reviews (published after his death in four volumes under the title *Kleine Schriften*) bear witness. From his more rigorous scientific labors he rested in occupation with art and literature: thus he

occasionally translated "Antigone" into Latin verse.

As a medical writer Lotze set himself, as already touched upon, the task of maintaining physiology's character as a mechanical natural science. He would not have the peculiarity of the organic phenomena explained by appeal to a mysterious vital force, but by exhibiting the determinate, law-governed manner in which the general forces of nature operate within the organism. As everywhere in nature, so here too we must find the explanation in the interaction of real elements. Not by being withdrawn from the mechanical connection of nature, but by the peculiar manner in which connected series of effects arise, does organic life distinguish itself from the inorganic world. In speculative philosophy of nature the idea of the organism had been made the type for the
[512] whole conception of the world; that this idea was corrected had therefore to be of general philosophical significance, and one understands that not only Lotze's medical but also his philosophical interest was satisfied by his labors in this direction (*Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften.* 1842. — *Allgemeine Physiologie des körperlichen Lebens.* 1851). These works even gave some materialists occasion to see a fellow-thinker in Lotze, notwithstanding that he stated definitely that mechanism was for him only a part of his conception of nature, not the whole. It is characteristic of Lotze that he thus divides the problems. His general philosophical views he had developed in his *Metaphysik* (1841). But he by no means felt himself finished when he began his teaching activity, and especially as regards the concluding ideas he constantly referred to future works. In *Medicinische Psychologie oder Physiologie der Seele* (1852) he expressed himself in detail upon the relation between the spiritual and the material, and at the same time gave psychological investigations of lasting interest. A long-cherished plan he carried out in the following period (1856–1864) with the publication of his *Mikrokosmus* (3 vols.). This work, conceived as a counterpart to Humboldt's "Kosmos" and Herder's "Ideen", gives a psychology in close connection with physiology and cultural history, and ends with a development of Lotze's cosmological and religious-philosophical ideas. For all his interests, for all the different roads his thought and feeling had traveled, he had here found expression. The work is laid out as a popular presentation and has won a considerable circulation. After publishing a spirited *Geschichte der Aesthetik in Deutschland* (1868), Lotze proceeded to give a concluding systematic presentation of his philosophy. But only two parts of it did he succeed in finishing (*Drei Bücher der Logik.* 1874. — *Drei Bücher der Metaphysik.* 1879). The third volume, which was to comprise aesthetics, ethics and philosophy of religion, was not worked out. His philosophy is thus unfinished; it was not granted him to set the crown upon the work and show how

the many threads of thought he had spun could be woven together into unity. — Short presentations of Lotze's teaching in the different philosophical fields are to be had in the *[513]* dictated notes (*Grundzüge*) from his lectures, published after his death. Among these must especially be singled out *Grundzüge der Logik und Encyclopädie der Philosophie*, which (in its second part) gives a short and clear survey of his whole system. Here we give a characterization of the three most important points in his philosophy.

α. The Mechanical View of Nature

Lotze's thinking has two points of departure: a living feeling for the value of spiritual life, for the fact that the highest thing for us is bound up with spiritual development and its ideals — but at the same time a firm conviction that a system of mechanical causes and laws is necessary to the realization even of the highest ideals. He can, as he says somewhere, believe in ideas that operate, but not in ideas that bewitch. The two points of departure do not, however, stand wholly isolated in him. He seeks, by proceeding from the one, to show the possibility of acknowledging the other. The most characteristic and the most scientific feature of his philosophy is his attempt to establish, through an analysis of the very concept of mechanism, that its ultimate presupposition is the concept of a principle which — as soon as we conceive it in a way intelligible to us — shows itself as the bearer of, and the source of, the highest ideas. Lotze means to take the bull by the horns: by drawing out realism's consequences he will found idealism. Only investigations conducted in realism's spirit can, in his view, bring us nearer to the goal idealism sets itself: the knowledge of the world as the expression of a valuable idea. Romanticism's philosophy had believed it could derive the forms of reality from the highest Idea. That has shown itself to be an insoluble task. We can, by contrast, attempt by way of inference to go back from what is given to its presuppositions. Not a *deduction* but a *reduction* is possible. Thinking must always apply its forms to a given material. Both general culture and the several sciences here operate with a mass of concepts whose origin, meaning and validity they do not investigate more closely. Such concepts are cause and effect, matter and force, end and means, freedom and necessity, *[514]* matter and spirit. It is philosophy's task to bring unity and connection into the circle of representations by making these concepts — presupposed in practical life and in the several sciences — the object of a peculiar investigation, and by determining the limits of their validity (*Grundzüge der Logik und Encyclopädie der Philosophie*, § 88).

Of these concepts the most important is the one we presuppose in every investigation of reality — every time we make an experiment, every time we seek an explanation: the

concept of a pervasive, all-embracing causal connection. This concept is not grounded upon experience, but is presupposed in all experience. But since the understanding of nature so far won rests upon its reality, it can itself be described as the expression of a fact; and that fact — that the single element in our experience is bound by a law-governed connection to other elements, the fact of mechanical connection — is what philosophy has to think through and to draw all the consequences of. Deduce it, philosophy cannot; but it may perhaps discover what lies in it. A *multiplicity of real elements in mutual interaction* is the foundation upon which the mechanical view of nature builds; and Lotze was, as we have seen, so convinced of that foundation's inevitability that he himself worked with zeal to gain it recognition within physiology, where hitherto vitalism had reigned with its appeal to a single forming and directing "vital force".

But because mechanical connection is an inevitable feature of our conception of the world, it need not be the only, all-determining feature. On the contrary: even if its validity is unlimited, it may perfectly well have a subordinate significance in the world as a whole. (*Die unbeschränkte Gültigkeit des Mechanismus, aber zugleich seine durchaus untergeordnete Bedeutung im Ganzen der Welt! Drei Bücher der Metaphysik*, p. 462.) And a closer analysis of the concept of mechanism leads us to see that this is precisely how the matter stands.

[515] The mechanical view of nature stops — when it is made into a concluding world-view — at a multiplicity of elements (atoms) in mutual interaction. It proclaims a pluralism. But how do the elements stand to the connection in which they stand? Can they have independent subsistence apart from that connection, so that it is a relation indifferent to their essence — or are they not determined right through by the combination in which they stand with the world as a whole? Interaction and connection cannot, after all, float freely above or between the elements, but presuppose an inner unity of them. For if I assume that the two elements *A* and *B* are wholly independent, interaction between them becomes unintelligible. An effect cannot be transferred as a fully finished state from *A* to *B*. What goes on in *A* can acquire significance for what goes on in *B* only if *A* and *B* are not in reality independent, absolutely separated beings, but if their states are states in one and the same, more comprehensive being. A multiplicity of independent beings would make mechanical interaction unintelligible; intelligibility comes only with the assumption of an infinite, all-embracing being, of which the single elements are moments, or points of action. The concept of a force's or an influence's "passing over" from one independent element to another is untenable. Not a *transient* cause (*causa transiens*), only an *immanent* cause (*causa immanens*) is intelligible. The states of one

and the same being can stand to one another as ground and consequent; the states of two mutually independent beings cannot. — Among the numerous presentations Lotze has given of this train of thought, *Logik und Encyclopädie* §§ 99–100 and *Drei Bücher der Metaphysik* §§ 50–81 must be singled out.

By analysis of the concepts of the causal relation and of interaction — the fundamental concepts of the mechanical view of nature — Lotze has thus arrived at the idea of a primal substance, an all-embracing principle. His thinking here moves along the same paths as those which in their day led Spinoza to his concept of substance, and which Kant walked especially in his youthful writings, without however abandoning them in his later works. He is reminiscent of Leibniz too at several points, without there seeming here, any more than in Spinoza's case, to have been any influence through study. When he heard that *[516]* he was being placed alongside Spinoza and Leibniz, he said (see Falckenberg: *Hermann Lotze*. Stuttgart 1901, p. 85) that he did not feel himself to stand in any inner relation to them, though he admitted that there was occasion enough to institute a comparison.

The concept which Romanticism's philosophy would lay at the ground, and from which it would derive everything, becomes for Lotze the last postulate — or, as he also expresses it, the last fact for our thinking. A closer, intuitable development of the universal principle presupposed in the simplest relation of interaction is impossible. We stand here at a *limiting concept* which we can neither do without nor develop. (*Drei Bücher der Metaphysik*, §§ 73 and 246.) It is nevertheless this concept that makes it possible for Lotze to hold fast to what was for him the essential thing in the idealist train of thought into which his teacher Weisse had introduced him. By it he found a connection between the opposed currents of the world of thought. Neither materialism's absolute atoms, nor Leibniz's monads, nor Herbart's reals could for him mark the conclusion of thinking: pluralism must necessarily give place to *monism.* As regards the atoms in particular, Lotze maintains that natural science's interest leads only to assuming elements which are in fact indissoluble for our experience, but that the assumption of a multiplicity of extended elements — even if these are thought of as vanishingly small — cannot be any conclusion for thought. Either one must give up the atom's unity or its extension: within an extended atom every effect will take time and pass from part to part — and these parts will then be more fundamental unities than the whole "atom". In the concluding concept of the atom we must abstract from all extension and think of the atoms as centers of force which — according to the analysis of the concept of mechanism given above — are each of them points of departure for the primal substance's operation. (*Drei Bücher der Metaphysik*, §§ 190–191 and 245; cf. the interesting review of Fechner's

"Atomenlehre" in *Kleine Schriften* III, p. 215–238.)

[517] Lotze's assertion of the mechanical view of nature, and the consequences he draws from it, make up the most significant part of his philosophy — notwithstanding that it is as a rule not brought forward when Lotze is praised. It is the spiritualistic and theologizing tendency in him that is most frequently emphasized, by admirers and opponents alike. And yet he has earned his greatest merit as a thinker by his analysis of the fundamental concept of the scientific view of nature. It is a defect in him that he does not undertake an epistemological investigation of the limiting concept he arrives at, but lets it stand in the form in which Romanticism's philosophy and the earlier dogmatism had conceived it. Lotze had in general no understanding of the importance of epistemology.

β. Metaphysical Idealism

In Lotze's train of thought, so far as we have presented it hitherto, there is still a certain indeterminacy. It might still be asked whether we then have no way at all of forming a closer representation of the elements and of the primal substance. In now bringing forward Lotze's answer to this question, it must first of all be emphasized that he is entirely clear that any answer to it rests upon analogies, and is determined by motives other than the purely theoretical. The law of analogy, and the need to find our own spiritual being again in existence — which have shown themselves to lie at the ground in all metaphysical idealists, and especially in Leibniz, Berkeley, Schelling, Schopenhauer and Beneke — appear with full consciousness in Lotze.

Lotze is an atomist — but he does not conceive the atoms as material, because extension, as well as all other sensible qualities, must be explained by the interaction of atoms, which therefore cannot themselves have that property. Like life and all qualities of experience, the sensible fact of extension is due to an interplay of force-points, which are in turn to be conceived as initial points of the infinite primal being's inner operation. Lotze does not regard it as impossible that wholly new beginnings may occur in the *[518]* course of the world: this does not conflict with universal lawfulness, for the law always expresses only the order in which the states arise, not an outer fate. The number of the elements need not be constant, but may be thought to rise or fall according to the demands of the highest world-purpose. (*Drei Bücher d. M.* §§ 33 and 235.) Nor need the single elements be wholly homogeneous with one another. A certain agreement or commensurability must indeed be present if one world-order is to embrace them all; but a complete likeness is not necessary. A law of nature may very well bind together qualitatively different elements for which we are unable to find any common denominator.

It is in the end not a *logical* but an *aesthetic* necessity that makes existence intelligible to us (*Drei Bücher d. M.* § 59): not a formal consistency in the primal being's operation, but the richness and fullness with which that operation proceeds, is decisive for the constitution of the single elements. At this point Lotze plainly enough passes over from purely theoretical motives to motives of feeling — without either the one or the other enabling him to determine more definitely how great *differences of quality* are compatible with the *commensurability* which the fact of interaction presupposes.[11]

If we would now form a representation at all of the inner nature of the elements, we must conceive them on the analogy of our own spiritual being. The mechanical view of nature really instructs us only about the mutual relation between the elements, not about their inner nature. It concerns the outer circumstances, moves the whole time around the outside of things — is a *cognitio circa rem.* It is perhaps even inclined — like the popular conception of the world — to hold that it is wholly indifferent to things whether we apprehend them or not. In opposition to this Lotze maintains (already in his youthful work *Metaphysik*, 1841, p. 313) that subjectivity is just as much a part of reality as the outer objects are: *Nicht nur was zwischen den Wesen, sondern auch was in ihnen sich zuträgt, ist ein wahres und wirkliches Geschehen.* In his later writings — first in *Medicinische Psychologie* — he goes a step further, maintaining that our own *[519]* subjective being is the only case in which we know a thing's interior and have a *cognitio rei*, not merely a *cognitio circa rem*: the only way of forming a representation of the interior of other things is then to conceive them on the analogy of ourselves — that is, as feeling beings (if not as representing beings, since feeling is a more original expression of consciousness than representation). The use of this analogy is the only means of thinking things as real beings subsisting for themselves, which are more than mere images for us. We thereby merely make use of the general method that leads the unknown back to the known. (*Drei Bücher d. M.* §§ 96–98.) And the "aesthetic necessity" demands this too, since it cannot be reconciled with an existence in which a great part — perhaps the greatest — should be merely a dark foundation for a mental life extended to only a few regions (see already *Allg. Physiologie* § 129). The elements of existence must therefore be assumed to be ensouled in manifold degrees.

As it stands with the elements of existence, so it stands also with the all-embracing world-ground that makes their interaction intelligible. If we follow the law of analogy and attempt to lead the unknown back to the known, then according to Lotze we must conceive the world-ground as a spiritual being. When inner states are to be bound into

[11] On the indeterminacy in Lotze at this point see *Max Wentscher*: *Lotzes Gottesbegriff und dessen metaphysische Begründung.* Halle 1893, p. 19–29.

unity and connection, our own inner, spiritual life is the only example we know in which the possibility of such a preservation of unity through the different states is realized — at least approximately realized. Following his teacher C. H. Weisse, Lotze conceives the world-ground as an absolute personality, and he defends this transference of the concept of personality to the absolute being on the ground that only an absolute being can be a personality, because only it has inner independence and originality, while the concept of personality finds only imperfect application to finite beings, which are dependent upon outer conditions. Lotze admits, to be sure, that to a personal life there belong resistance to be overcome, and a capacity to suffer and to receive as well as to act. But
[520] when one then asks how an absolute being, subject to no limits, can suffer, Lotze answers (*Grundzüge der Religionsphilosophie* § 34, cf. § 52) that what sets the Godhead's feeling in motion must be "inner products of its own creative imagination"! — It is surely a great question, however, whether such a self-created resistance can mean anything serious — especially as it can be annihilated at whatever moment one pleases. The personal life we know has in any case precisely not self-created and not so easily removable limits to struggle with, and the analogy upon which Lotze relies therefore seems to fail at just the decisive point. And to this it must be added that Lotze — on the most probable interpretation of his obscure and shifting utterances[12] — differs from Weisse in that he does not hold the form of time applicable to the absolute being. A personal being that does not develop in time — a timeless life, and a timeless suffering and acting: these are concepts that make far too great demands upon our capacity for analogy! — It need only further be remarked that Lotze — as regards both the elements and the world-ground — presupposes that the opposition between spirit and matter is contradictory, so that no

[12] *Reinhold Geijer*, in his paper *Hermann Lotzes tankar om tid och timlighet i kritisk belysning* (Lund 1886), drew attention to the shifting standpoints Lotze seems to have taken on the question of time's absolute validity. In my paper *Lotze og den svenske Filosofi* (Letterstedtske Tidsskrift 1888) I associated myself with Geijer's interpretation, though I had to conceive the problem itself otherwise than my Swedish colleague. Yet the matter seems to stand otherwise than Geijer, and I with him, had reason to assume. According to what *Richard Falckenberg* has shown in his paper *Die Entwickelung der Lotzeschen Zeitlehre* (Zeitschrift für Philosophie und philosophische Kritik, vol. 105), the *Grundzüge der Metaphysik* published after Lotze's death date from as early as 1865, and with that an essential support falls away for the assumption that Lotze first denied the reality of time toward the close of his life. When in earlier writings he speaks without qualification of the time-relation as absolutely valid, this is according to Falckenberg a legitimate use of popular representations where the context did not require a more stringent usage. Lotze's definitive conception was the one presented in *Drei Bücher der Metaphysik* (1879) and in *Grundzüge der Religionsfilosofi* (from 1878–79), according to which succession holds for finite beings, while the Godhead is raised above every difference of time. *This* view departs from his earliest presentation (in *Metaphysik*, 1841) in asserting this distinction, which admittedly does not make the whole question clearer — though the *psychological* distinction between succession and abstract time-form may well be defensible. — As little as my Swedish colleague am I able to connect any meaning with the expression "ein zeitloses Geschehen und Wirken" (Grundzüge der Metaphysik § 58). My German colleague, by contrast, finds a significant thought in it.

further possibilities could be thought than the two: only on that presupposition is there compelling necessity to conceive the elements and the world-ground as spiritual beings, when one cannot or will not conceive them as material. But there is no logical, only a factual necessity for choosing between spirit and matter in determining the concluding concepts. We in fact know only these two forms of existence; but there would be nothing *unthinkable* in one or more other forms existing. Therefore metaphysical idealism cannot be proved.[13]

Lotze is now also fully conscious that the conclusion of his whole world-view rests upon motives of feeling, not upon strict thinking. The necessity of going back from the multiplicity of the elements to the unity of the world-ground he asserts, indeed, on purely theoretical grounds: "*Gar kein Weltlauf, weder ein harmonischer noch ein unharmonischer, ist mir ohne die Voraussetzung jener Einheit begreiflich, die alles gegenseitige Wirken* [521] *des Einzelnen erst möglich macht; die Störungen der Dinge durch einander bezeugen die beständige Gegenwart dieses Einen eben so eindringlich, wie das Zusammenstimmen der Kräfte zum Zweck*" (*Drei Bücher d. M.* § 233). But when he leads everything back to a single principle, he knows very well that we cannot construct the world-connection and the elements out of it: we always lack the second premiss! (*Drei Bücher d. M.* § 93.) Only in practical conviction, then, can the thought be held fast that all beings and all events have their ultimate ground in the highest purpose maintained by the world-ground. What that purpose is we do not know — any more than we know how the conflict of forces in nature and the reality of evil in the moral world are compatible with the validity of the world-plan. (*Mikrokosmus*, Buch IX, Kap. 5. — *Drei Bücher d. M.* § 233.) But in an ethical idea Lotze found, at the beginning as at the close of his career, the final conclusion for his thinking: metaphysics goes back for him to ethics; in what *ought* to be lies the ultimate explanation of what *is*. —

If by pantheism one understands the doctrine which maintains an inner relation between God and the world — whether the Godhead is conceived as a personal being or not — then Lotze's religious-philosophical standpoint is to be described as ethical pantheism. He himself felt closely akin to Fichte; the kinship with Spinoza he would not properly acknowledge. In the use of theological words and turns of phrase he went — especially in *Mikrokosmus* and in the *Grundzüge* of philosophy of religion and of moral philosophy published after his death — further than was fitting and warranted. By his

[13] Compare my *Psykologi*[4], p. 80; *Psykologiske Undersøgelser*, p. 91 f.; *Filosofiske Problemer* (Universitetsprogram 1902), p. 62–65. — It is interesting that the founder of metaphysical idealism, Leibniz, in his [599] earlier period, like Spinoza, laid great weight upon the duality spirit–matter being purely factual, not logically necessary. *Philos. Schr.*, ed. Gerhardt I, p. 237, 242, 268. Compare *Ludw. Stein*: *Leibniz und Spinoza*, p. 93 f.

usage (for it is nothing more) it came to look as though he stood nearer to the popular religious circle of ideas than in fact he did. He does, however, occasionally emphasize (especially *Mikrokosmus*, Buch VIII, Kap. 4) the difference between free religiosity and the positive religions.

γ. Spiritualistic Psychology

[522] The elements whose interaction produces the phenomena that experience presents need not — as we have seen — according to Lotze be absolutely homogeneous. There need be only such homogeneity as the interaction itself presupposes. By this very indeterminate rule Lotze makes it possible for himself to hold fast to the popular assumption of an interaction of soul and body. The elements which the concept of mechanism presupposes are for him partly psychical, partly physical. He disputes the assumption of a continuous series of physical phenomena. At certain points the physical motion is broken off, to be "absorbed", that is, to pass over into psychical states. It is indeed doubtful whether Lotze, when he first put forward this claim (in the year 1851; *Allg. Physiol.* § 424), had any living conception of the significance of the newly discovered law of the conservation of energy — though neither did he later find in it any reason to depart from the standpoint he had once seized. There were for him compelling grounds which made the assumption of a peculiar soul-substance necessary. Only two possibilities presented themselves to him: either to derive the psychical phenomena from a soul as a principle peculiar to them, or to explain them by the co-operation of the physical forces. (See *Selbstanzeige der medicinischen Psychologie*, Kleine Schriften III, p. 4.) Since the latter is impossible — inasmuch as the interaction of physical forces cannot explain the unity that stamps even the simplest expression of mental life — only the first way out remains for Lotze. A great part of Lotze's psychological investigations is therefore devoted to finding out the manner in which the psychical elements stand in interaction with the physical ones. His psychology here recalls Descartes's. There are, on his view, certain regions in which the physical elements act upon the soul; thereafter the mental processes may for a time run their course exclusively according to their own laws, and they may then in turn release a mechanical force for the production of new physical changes. (Kleine Schriften III, p. 7.)

[523] Notwithstanding that the present condition of nerve physiology and of psychology would seem to make a separation of these different stages hopeless, Lotze nevertheless proceeds confidently (especially in *Medicin. Psychol.* and *Mikrokosm.*) to a closer investigation. The significance which the material processes have for mental life cannot — according to the conception Lotze has urged of the nature of interaction — consist in finished

sensations or representations being conveyed from them to the soul; they can only give signals, which the soul must translate into its own language, just as, conversely, the soul's inner states are not as such transferred to the material organs, but only become the occasion of these entering into function. The material organs work in the service of the higher spiritual activities by delivering and preparing the material upon which the soul is to exercise its powers, and in that their whole significance consists. The higher spiritual activities themselves — memory, thinking, aesthetic and moral feeling — take place, once the material is given, in the soul's interior, and it is wholly unnecessary to assume peculiar material processes corresponding to them. Lotze would in any case sooner interpret such material processes as effects of the mental states — brought about by these vibrations propagating themselves from soul to brain — than as their causes. So especially when we see a state of feeling persist long after the original occasion has fallen away.

Upon Lotze's special psychological theories we cannot here enter. Particular mention must be made of his brilliant doctrine of local signs, by which the development of the apprehension of space becomes more intelligible,[14] and of his assertion of feeling's importance as a fundamental element in mental life, over against Hegelian and Herbartian psychology. On a mass of particular points Lotze throws new light by his fine, spirited conception and his often happy manner of expression. It conflicts with good psychological method, on the other hand, when Lotze, instead of seeking the laws that connect the mental phenomena with one another, conceives the relation in the arising of the different mental phenomena in such a way that the representations react upon "the soul itself" and release the feelings out of it, which must in turn act upon "the soul itself" in order to release from it the expressions of will. Likewise when he lets a peculiar faculty of *[524]* thinking appear in opposition to the association of representations. Here his spiritualistic metaphysics has forced upon psychology an all too artificial and scholastic mode of explanation. It takes its revenge here that Lotze does not acknowledge psychology as an independent science of experience, but regards it as applied metaphysics. (*Logik und Encyclopädie* § 93.)

When Lotze's psychology recalls Descartes's on essential points, it must nevertheless be held fast that this likeness is present only so long as Lotze holds to the popular usage which he employs, in his psychology no less than in his philosophy of religion, to a great extent. The soul a thing for itself and the body a thing (or group of things)

[14] See on this *Reinhold Geijer*: *Hermann Lotzes lära om rummet* (Nyt svensk tidskrift 1880) and *Darstellung und Kritik der Lotzeschen Lehre von den Localzeichen* (Philosophische Monatshefte 1885). Cf. my paper *Lotze og den svenske Filosofi* (Letterstedtske Tidsskrift 1890) and my *Psykologi* V C, 7 c.

for itself — from many of Lotze's utterances that might seem to be his doctrine, as it was Descartes's and still is that of popular metaphysics. But it is only a provisional conception for Lotze. From Descartes he differs first and foremost by his doctrine of the subjectivity of extension and by his analysis of the concept of matter, which leads him to the assumption that matter is only a phenomenal form of the interaction of psychical elements. Thereby the dualistic conception of the relation between soul and body is abolished. The qualitative difference between the interacting elements is thereby considerably reduced: instead of an interaction of psychical and physical elements we get a mutual interaction of psychical elements. And this reduction Lotze makes — as must be well noted — not because he finds any difficulty in popular or in Cartesian metaphysics. The problem of the relation between soul and body really does not exist for him at all; the interaction between them presents to him no greater riddle than the relation of interaction that takes place between two bodies striking one another.[15] It is general philosophical considerations that guide him here — considerations that would retain their validity even if he had, instead of laying Cartesian dualism at the ground of his psychology, built upon Spinoza's identity hypothesis. Metaphysical idealism is independent of every more special psychological hypothesis. Since Lotze began as a
[525] Cartesian, those considerations lead him to a conception that may be described as a *monistic spiritualism*, in that it regards soul and body as two *different substances* which are nevertheless of *homogeneous* constitution. Herbart's theory, and a theory that appeared within the Wolffian school (see Note 1 to this volume), form predecessors of this peculiar theory.

On another point, and one more essential for his whole philosophy, Lotze also separates himself from Descartes — and here from Leibniz and Herbart as well. As we have seen, he regards the single elements in existence as moments or expressions (actions) of the primal substance, which is the one thing that is in truth real. The soul can then, no more than the physical atom, be substance in the stricter sense of the word. The independence of finite things is only apparent. "Our monistic conception" — says Lotze — "has *wholly and without reserve* laid the ordering of the world, and the existence and efficacy of every single thing, in the hands of the one infinite being upon which alone the possibility of all interactions rested." (*Drei Bücher d. M.* § 245, cf. §§ 72–73.) The words substance and being he therefore takes, when he uses them of the soul,

[15] *Mikrokosmus* Buch III, Kap. 1: "*Warum sollte nicht ein Atom des Nervensystems ebenso auf die Seele oder sie auf jenes stossen und drücken können, da doch jeder gemeine Stoss und Druck sich für die nähere Betrachtung nicht als ein Mittel zur Wirkung, sondern nur als die anschauliche Form eines viel zarteren Ereignissen zwischen den Elementen ausweist?*"

only in the sense of that which can act and suffer (ib. § 243) — and he admits at last (ib. § 307) that it might perhaps be best to refrain from using these words, which can easily mislead. Already in his paper *Seele und Seelenleben* (in Wagner's "Physiologisches Handwörterbuch", 1846; "Kleine Schriften" II, p. 198) Lotze said that every soul possesses only the reality that belongs to it on account of its significance in the whole of the world, and that its immortality depends not upon its nature but upon its place within the ethical world-order. In "Mikrokosmus" and in his principal work (*Drei Bücher der Metaphysik*) Lotze returns to the same thought. Just as it was not an interest in the soul's immortality that led him to his spiritualistic psychology, so he sees clearly that this question cannot be answered by scientific means at all. "No other fundamental proposition stands here at our disposal than the general conviction: all that has come to be will subsist, if its subsistence belongs to the world-plan, and so long as it belongs to it; all will perish *[526]* whose reality had its warranted place only in a passing phase of the world's course. That this fundamental proposition cannot be applied in detail by human hands scarcely needs saying; we certainly do not know the merits that could give one being a right to eternal subsistence, any more than the defects that could deny another that right." (*Drei Bücher d. M.* § 245.)

When one notices the agreement between Lotze's earlier (1846 and 1864) and later (1879) utterances upon this point, one will scarcely be led to assume, with some recent critics of Lotze,[16] that Lotze made essential alterations in his doctrine of the soul's substantiality and thereby approached Spinozism more nearly than before. But it would surely have been to the advantage of clarity if the excellent thinker — whose disposition assuredly permitted no unworthy accommodation — had already in his earlier presentations brought out his divergence from the current assumptions as definitely as in the latest utterances.

[16] *Kresto Krestoff*: *Lotzes metaphysischer Seelenbegriff.* Halle 1890, p. 46 f.; 73. — *Max Wentscher*: *Lotzes Gottesbegriff und dessen metaphysische Begründung.* Halle 1893, p. 11 ff. — The opposition in which Lotze's conception stands from the very first to Herbart's, despite all the points of contact it has with it, also testifies that no change of principle has taken place in Lotze's thinking in the course of the years. Compare on this *Max Nath*: *Die Psychologie Hermann Lotzes in ihrem Verhältniss zu Herbart.* Halle 1892. — With Lotze's ethical monism his optimism hangs together. Although he himself repeatedly emphasizes the reality of physical and of moral evil as an obstacle to the carrying through of his world-view, an otherwise well-disposed critic has nevertheless complained that he lacks interest in "the fundamental dogma of all religions, the atonement", because he does not take sufficient account of evil. G. *Vorbrodt*: *Principen der Ethik und Religionsphilosophie Lotzes.* Dessau und Leipzig 1891, p. 97.

b. Gustav Theodor Fechner

Fechner and Lotze stand as two Dioscuri in German philosophy in the last part of the century. They are kindred spirits in their ideal disposition, their great natural-scientific culture, their poetic sense and their need for unity in the conception of the world. They pursue kindred ends, but in part by different roads. A comparison between them is therefore interesting, especially as on decisive points one must choose between them. The choice will — since the fundamental proposition, an idealist world-view upon a realist foundation, is the same in both — depend upon which of them lets the realist foundation come most fully into its own. That is a measure Lotze has himself acknowledged, in the remark that idealism can be maintained only by investigations conducted in realism's spirit. Even if Lotze takes the first place as regards the working-through of the general philosophical principles, Fechner may well have the precedence as regards the
[527] consistent and resolute holding fast to the natural-scientific foundation. This is all the more remarkable in that Fechner's imagination has from the first a far greater influence upon his thinking than is possible in his critical, deliberate companion. Fechner is — like Kepler, of whom he strongly reminds one — an interesting example of how one may be led through bold and fantastic speculations to positive and exact results, provided only one holds fast to a definite fundamental thought and is able to extract it from its mystical wrapping. Just as Kepler was led successively from mystical speculations to the discovery of his famous laws, which satisfied his need to find determinate mathematical relations expressed in the universe, so Fechner was led through bold analogies to the conviction of a determinate quantitative relation between the spiritual and the material, and, in working that thought out more closely, came to found psychophysics, or experimental psychology.

Fechner was born in Lausitz in 1801 and in his youth pursued medical and physical studies. In 1835 he became professor of physics in Leipzig and made himself known by able work in his subject. The philosopher of religion C. H. Weisse belonged to his closest friends and had great influence upon his religious conception — as upon Lotze's. It is nevertheless doubtful whether Fechner would have come to play so great a part in the history of philosophy as he has, had he not, by studying subjective phenomena of light and color (in the winter of 1839–40), ruined his eyes, so that after long suffering he had to give up his professorship of physics. In philosophical thinking, in free fancy, in absorption in the inner world of the mind, he sought compensation for the bright and colourful world in which his weakened sight no longer allowed him to dwell as persistently as before, but which he did not forget. On the contrary, it became his life's task to work the two

worlds together and to find the law of their connection. In opposition to Romanticism's philosophy he would go from below upwards, not from above downwards. And where empirical inquiry gives out, he would rather let imagination range in the boldest analogies than let abstract thinking spin its slender threads. His bent for paradox he gave vent to in *[528]* a series of humorous writings, under the pseudonym Dr. Mises. His remaining writings divide naturally into two groups. The one — within which *Zendavesta* (1851), *Ueber die Seelenfrage* (1861) and *Die drei Motive des Glaubens* (1863) are especially to be named — develops his poetic-speculative world-view; to the other belong his most epoch-making works in the fields of natural philosophy and psychology: *Ueber die physikalische und die philosophische Atomenlehre* (1855), *Elemente der Psychophysik* (1860) and *Vorschule der Aesthetik* (1876).

Right into his advanced old age Fechner was intellectually active and productive. Even in his eighty-sixth year he made an interesting contribution to the discussion of his psychophysical theory (*Ueber die psychischen Massprincipien und das Weber'sche Gesetz*, Wundt's Philosophische Studien IV). Shortly afterwards he died, in 1887.[17]

α. Poetic-Speculative World-View

In the field of the world-view Fechner combats every conception which takes the rich and bright world of mental life to be borne by, or to be the product of, dark, intuitable things or beings. Materialism's "matter", spiritualism's "soul-substance" and Kant's "Ding an sich" are again and again objects of his polemic. No less does he combat those who separate God and world, spirit and nature. He rejects in equal measure belief in a natureless God and belief in a spiritless matter — orthodoxy as much as materialism. He reproaches the usual conception of the world with dualistically separating the infinite and the finite: "One sets the infinite over against, above, beyond, outside, behind the finite, perhaps even places an immense partition between them, as though they had nothing whatever to do with one another. And yet they by no means lie outside one another. On the contrary, the finite is the content of the infinite .... The infinite is therefore not incomprehensible either; it can be grasped by countless handles within the finite; only it cannot be encompassed." (Ueber die Seelenfrage, p. 111.)

[17] Interesting biographical notes in A. *Elsas*: *Zum Andenken Gustav Theodor Fechners* (Grensboten 1888). A fine characterization of his great predecessor was given by *Wundt* in his memorial words at the graveside (printed in "Philosophische Studien" IV) and later, more fully, in his address on the centenary of Fechner's birth (*Gustav Theodor Fechner.* Rede. Leipzig 1901). A rich biography has been given by J. E. *Kuntze* (*G. Th. Fechner.* Ein deutsches Gelehrtenleben. Leipzig 1892), and *Kurt Lasswitz* has given a critically searching presentation of Fechner's scientific standpoint (*G. Th. Fechner.* Stuttgart 1896, the first volume of "Frommanns Klassiker der Philosophie").

[529] Closely connected with the prejudices mentioned is, according to Fechner, the opinion that only human beings and animals are ensouled, and nothing else — at least upon our earth. It is thought that experience forbids extending ensoulment further. But only of my own soul have I direct experience; to the existence of other souls I must infer by way of analogy. And what then can prevent me from extending the analogy further, from human beings and animals to plants and to the heavenly bodies, when good grounds show themselves for it? That plants have no nerves is no counter-proof; for the lowest animal species have no nerves either. If it is said that the plant is no closed individual, the answer is that individuality is always relative, since no living being stands absolutely separated from its surroundings. There is so continuous a transition between animal and plant life that it is unwarranted to lay down an opposition between them as between the ensouled and the soulless. The plant's mental life might stand as far below the animal's as the animal's below man's. — And why should the heavenly bodies not be ensouled? Human beings and animals belong to the earth, and the earth-soul might stand to the individual human and animal souls as the earth's body does to the individual human and animal bodies. It is only an artificial abstraction when one sets human and animal life as ensouled in opposition to the life of the whole globe. The living does not proceed from the lifeless; rather the lifeless is a product of organic metabolism and of the perishing of organisms — a conception which according to Fechner is easier to defend than the opposite. In his work *Ideen zur Schöpfungs- und Entwickelungsgeschichte der Organismen* (1873) he develops especially the thought that variation and differentiation proceeded with greater rapidity in earlier periods of the earth than in the present one. — The lower souls must be supposed to stand to the higher as representations and motives stand to the individual soul. In the end all souls are parts of the highest, all-embracing soul, whose life and whose reality make themselves known in the law of causality — the principle of all particular laws of nature, and thereby of all connection and order in existence.

[530] As for Lotze, then, the law-governed connection in the world is the foundation upon which the religious-philosophical view is built. What for many makes the idea of God impossible or superfluous makes it, for Fechner and Lotze, precisely necessary. In that universal law they find the highest unity expressed — the eternal and unchangeable which embraces all. The concept of law is the last thing in all our knowledge; upon it, therefore, according to Fechner, our highest idea must be built. (See especially his paper *Ueber das Causalgesetz* in "Berichte der sächs. Societät der Wissenschaften" 1849. Cf. Seelenfrage p. 205 f.; Atomenlehre, 2. Aufl., p. 125.)

The concept of the world goes back for Fechner, as for Lotze, into the concept of

God. The widest conception of the concept of God Fechner regards as the best grounded; the narrower is in his eyes an abstraction. And since the world's life is God's life, this cannot be complete in itself, but develops and unfolds with and through the world's development. God's perfection consists not in finished completeness but in unlimited advance. The divine world-plan is not finished once for all, but is shaped in accordance with the world's events, which are at the same time divine experiences. — In this doctrine — in which Fechner follows his friend Weisse, while Lotze did not think he could ascribe validity to time for the Godhead — Fechner sees a carrying through of Christianity's declaration that God is spirit and shall be worshipped in spirit and in truth, and that in him we are, live and move: declarations which for the most part stand as mere words! He admits that his doctrine is certainly not Christian, if one reckons as an essential constituent of Christianity "belief in the bite of the apple in paradise and its mystical consequences, in the eternal damnation of the non-elect, in miracles against the laws of nature, in God's remoteness from the world, and in all the unedifying stuff out of which the theologians build up Christianity" (Seelenfrage p. 194). Fechner's bold theology makes it easier for him to understand how man can be God's fellow-worker, and how evil can be foreign to God's essence and yet a necessary member in the world's development.

β. Psychophysics

Early on, in accordance with his whole fundamental view, Fechner had arrived at the *[531]* conviction that the difference between the spiritual and the material could not be a difference between two beings, of which the one was immaterial and the other spiritless. The material world is the Godhead's outer, the spiritual world the Godhead's inner side. The difference is phenomenal, resting upon a difference in the spectator's standpoint, not upon a difference in substance. (Zendavesta II, p. 341.) Or, as he later expressed it (Elemente der Psychophysik I, p. 2), the difference between the spiritual and the material is to be likened to the difference between the concave and the convex side of one and the same circle: he who stands within the circle sees only the concave side, he who stands outside it only the convex — and he who can change his standpoint may perhaps believe that he has two different things before him. It is Spinoza's identity hypothesis that Fechner here arrives at, led by the physicist's interest in maintaining the continuity of the physical processes, and by the philosopher of religion's interest in thinking the Godhead, the all-embracing spiritual principle, in such a way that it expresses itself at every point in material phenomena, so that these are not loosened from it and do not fall outside it. Fechner arrives, so to speak, at a theophysics before he develops a

psychophysics.

A confirmation of this fundamental conception he found in the law of the conservation of energy, and he is the first to use the newly discovered law in this way. (Elemente der Psychophysik I, p. 21–45.) He admits that no proof has been given that this law holds for the material processes bound up with spiritual activities; but he holds that all experience points in this direction, and that we must therefore, until a counter-proof is produced, assume the law to be valid. Experience establishes that the material processes to which conscious life is bound stand in interaction with the other material processes within and outside the body, so that the total physical energy at our disposal may now be consumed predominantly by the former, now by the latter. Physical energy is drawn upon both
[532] when we think and when we chop wood — and therefore we cannot do both at once as well as we can do each of them separately.

From Spinoza he differs in this doctrine only by laying strong weight upon finding out a mathematical functional relation[18] between the two sides of existence. He first proceeded from the view that there must be a directly proportional relation between them. Later — he has noted the date and the situation: 22 October 1850, in the morning, in bed — he hit upon the thought that the spiritual did not rise and fall directly with the material, but that the changes of the former corresponded to the *proportional* changes of the latter, so that the change ($d\gamma$) in the intensity of a spiritual state is to be set equal to the ratio between the change ($d\beta$) in the energy of the corresponding material state and the energy (β) already present — that is, $d\gamma = d\beta/\beta$. With this he had come down from the region of indeterminate speculations to an assumption that could be checked experimentally. He found it at once confirmed by the fact that the strength of the sensation of light does not grow as fast as the physical intensity of the light. His own experiments and his researches in the literature gave him a material on the basis of which he set up the general law that a sensation's increase does not correspond directly to the increase of the physical impression, but to the ratio between that increase and the whole impression present — so that with one and the same increase of impression the sensation

[18] In my earlier presentation (*Filosofien i Tyskland.* 1872, p. 264 f.) I allowed myself to be misled by some utterances in *Zendavesta* (II, p. 352 ff.) into saying that Fechner assumes an interaction between spirit and
[600] matter. What Fechner there develops is, however, only the importance of changing standpoint and of finding out the formula for the mutual relation of the two sides. As little there as later does he assume a passage from the one side to the other. In *Elemente der Psychophysik* I, p. 8 it runs: "*Wir nennen das Psychische Function des Physischen, davon abhängig und umgekehrt, insofern eine derartige constante oder gesetzliche Beziehung zwischen beiden besteht, dass von dem Dasein und den Veränderungen des Einen auf die des Anderen geschlossen werden kann.*" For this kind of dependence Fechner often uses (for example Elemente I, p. 18; II, p. 393) the expressions *Simultanbedingungen, simultane oder Wechselabhängigkeit.* On his conception the brain processes corresponding to the activities of consciousness cannot — as Lotze and the spiritualists hold — be regarded as releasing impressions (*Reize*) upon the soul (Elemente I, p. 18).

grows the less, the stronger the impression already is. This law Fechner calls the Weber law, in memory of his teacher, the physiologist E. H. Weber, from whom some of the most significant experiments upon which it was built derived. — Of the law's validity, its limitation and its interpretation nothing further shall be said here; for that, reference must be made to more recent psychological works. Here it need only be remarked that by the step Fechner thus took, exact experiment became possible in the psychological field: he made — to use an expression of Galilei's — something measurable that had not hitherto been so. Through its relation to the outer impression the sensation, which in and for itself, as a purely subjective element, cannot be measured, can be determined quantitatively. Therewith the road was pointed out which *experimental psychology* had to *[533]* take: it had to seek points of connection between psychical processes and such processes as can be directly counted, weighed or measured.

What Fechner had first been seeking was certainly not what the new science came to occupy itself with. He had wanted to find the relation between the mental activities and the corresponding brain processes. But that relation is inaccessible to us. What we immediately apprehend is, from the subjective side, the mental state, and from the objective side, the physical impression. For the relation between these the law Fechner found holds, within certain limits; and it is most probable that brain process and mental process are directly proportional — which also agrees best with the identity hypothesis Fechner espoused. Psychophysics has so far become something other than Fechner himself had conceived. But this does not lessen the importance of his brilliant work *Elemente der Psychophysik*: between the human sciences and natural science there has been inserted — in consequence of the great law of the division of labor — a new discipline. And it was in Fechner's free spirit, rich in possibilities and in ideas, that this budding was accomplished. The investigator who after Fechner has done most in the new science's field said at his graveside: "Psychophysics, which he founded, was only the first conquest in a domain whose further appropriation could no longer present significant difficulties, once this beginning had been made" (Wilhelm Wundt: *Zur Erinnerung an Fechner.* Philos. Studien IV, p. 477).

In defense of the psychophysical law he had set up, Fechner composed a series of writings which are models of truth-loving and careful discussion and of chivalrous polemic; they are marked too by the author's freshness and humor. —

Akin to Fechner's endeavors toward the founding of an experimental psychology is his endeavor toward an aesthetics grounded upon experience. His *Vorschule der Aesthetik* (1876) is an epoch-making work, into whose special content we cannot here enter more

closely.

γ. Philosophy of Nature

[534] His opposition to Romanticism's philosophy Fechner has expressed above all in his "Atomenlehre". Speculation, he says, descends into nature as the bear into a beehive. It enjoys the coherent honey, but does not notice that it has been brought together by the activity of a multitude of small individual beings. Natural science shows — in physics and in chemistry alike — that the material processes can be understood only by being regarded as processes of interaction of small parts which for us are indivisible. Every part of the material world, however small, natural science conceives as a little world with globes and systems of globes. What seems to us a continuous mass is, like so many nebulous masses in the sky, a group of different parts. Natural science does not assume absolute atoms, but only that divisibility goes further than the eye and the microscope discover. It assumes only relative atoms — that is, atoms indissoluble for us, or by the forces of nature known to us — led by the need to refer every effect to a determinate place in space as its point of departure. Philosophical atomism goes to the end of the road along which physical atomism had reason to travel only part of the way. We obtain a philosophical conclusion to the conception of nature when we assume that the smaller one thinks the atoms, the more exact will be the results one can derive by thinking of matter as consisting of atoms. It was in order to have points of departure for the effects that natural science conceived atoms. The share with which every such point of departure contributes to the fulfilment of a law of nature we call the force proceeding from that point. What we know about the atoms is now precisely only that these contributions proceed from them: therefore philosophical atomism designates them as centers of force. There is thus no reason to ascribe extension to the atoms — the absolute, philosophical atoms — and the concept of matter is no longer materialistic. Philosophy stands here for Fechner as a hypothesis in which we think as absolute what in reality is always given only relatively.

[535] The foundation of our whole knowledge of existence is the law of the connection of phenomena; from the law we come to determine the forces in the co-operating elements. Fechner will not, with Leibniz, Herbart and Lotze, see psychical elements in the atoms. These are for him merely the last points at which spiritual beings arrive when they analyze the content of their consciousness.

A closer explanation Fechner regards as superfluous. The atom is the lower limit of our knowledge, just as the universal world-law, which testifies to the reality of an

all-embracing being, is the higher limit. Between these two limiting concepts lies all our knowledge. (Seelenfrage, p. 215–216.)

c. Eduard von Hartmann

In the year 1869 there appeared a book that caused a great stir and enjoyed a circulation rare for voluminous philosophical works. It was Eduard von Hartmann's *Philosophie des Unbewussten*. Its motto — "Speculative Resultate nach inductiv-wissenschaftlicher Methode" — shows that the author aimed at a grounding and development of philosophical ideas similar to Lotze's and Fechner's. By his clear and broad presentation, by his detailed use of natural-scientific examples, by meeting the reader at once with a realist foreground and a romantic-mystical background, and finally by taking up pessimism — which, under the influence partly of Schopenhauer's now much-read writings, partly of prevailing currents of the time, was a widespread train of thought — as a member of his system, Hartmann could find readers in wide circles. The attention he aroused is seen from the fact that in the course of twenty years ten editions of his principal work appeared, and that in the years 1870–1875 alone there appeared fifty-eight writings dealing with him.

Eduard von Hartmann was born in Berlin in 1842, was the son of a general, and himself entered upon a military career. In his free time he cultivated music, painting and philosophy. A knee injury compelled him in 1865 to resign his commission. And when he became clear that his real talent did not lie in the field of art, he gave himself — "bankrupt *[536]* in everything but thought" — to philosophy. He had philosophized and worked out philosophical essays earlier. He now proceeded to the working out of "Philosophie des Unbewussten", which must still be regarded as his principal work, notwithstanding that it has been followed by some thirty larger and smaller works. The most considerable of these is certainly *Phänomenologie des sittlichen Bewusstseins* (1879), which is on the whole Hartmann's best work. For the characterization of his philosophical standpoint I shall here keep to two principal points, in full consciousness that Hartmann's untiring activity in developing his ideas in all directions, and illuminating them from all points of view, does not thereby quite come into its own. Hartmann has complained that I have "halved" him by mentioning only the part of his activity that lies before 1880. That, however, lay in this book's plan; and I have besides, it must be admitted, the view that the two works named are and remain Hartmann's most considerable contributions to philosophy. Those who wish to supplement the short presentation of Hartmann's philosophy that I can give here I must refer to the detailed work, sprung from excessive admiration, of

Arthur Drews: *Hartmanns philosophisches System im Grundriss* (Heidelberg 1902).

α. Philosophy of Nature and Psychology

While Lotze and Fechner in principle laid the natural-scientific mode of explanation at the ground, and would ground their philosophy by drawing the consequences of its presuppositions and results, Hartmann's philosophy bears far more the stamp of a neo-Romantic reaction against natural science's realism. He sets out to show that the natural-scientific mode of explanation is not sufficient, but that together with the causes the mechanical view of nature assumes there must operate a spiritual principle which he — in order not to make it too anthropomorphic — calls *the Unconscious*. He appeals to
[537] this principle wherever he thinks the causes empirical science can exhibit do not suffice. It is thus that he thinks to arrive by "inductive" means at "speculative" results.

Hartmann agrees with Fechner that matter must be conceived as a system of atomic forces. No single atom can have mass, any more than one can speak of the magnitude of a unit. If, then, we keep to atomic force as the last element to which the analysis of material reality leads, there can — Hartmann holds — be determinate connection and consistency in the atomic force's striving only when that striving is conceived as a willing, with an unconscious representation of its goal. We understand force only by thinking of it as a willing. Matter itself is therefore representation and will — and the difference between matter and spirit falls away. In organic growth there likewise expresses itself an unconscious willing and representing: in the organism's coming to be a purpose is realized, a purpose in relation to which the single substances and processes are to be regarded as means, whose provision and combination become intelligible only through the purpose, notwithstanding that this is not an object of consciousness. There is only a difference of degree between organic formation and instinctive action. Instinct cannot, Hartmann claims, be explained by the material organization or rest upon a nerve-mechanism implanted once for all; for then it could not vary in its expressions when the organization does not change. As little does it rest upon conscious deliberation. There remains, then, only to conceive it as unconscious willing and representing. In the human mind the Unconscious expresses itself in sense-perception, in that the intuition of the outer object comes to be by an unconscious working-together of the sensations. The association of representations proceeds only through representations being brought forward, without conscious search, which stand in connection with the present ones. Feelings and motives arise on account of unconscious processes and are therefore often unintelligible to ourselves. Even where the will is bound up with consciousness, an

unconscious will is required if it is to lead to a willed movement: consciousness does not know which nerve-center in the brain must be affected for the movement to come about. In history the Unconscious is able to get individuals to work in the service of great world-purposes while they believe they are working only for their own limited ends. *[538]* What is fate or providence other than the Unconscious's dominion in men's actions, so long as their conscious understanding is not mature enough to make history's purposes its own? — The Unconscious is everywhere that which makes co-operation between individual beings possible. There are individuals of every possible degree — from the atom to universal nature — and it is one and the same Unconscious that stirs in them all. Plurality has only phenomenal, not metaphysical significance. Hartmann declares himself in principle in agreement with speculative theism (as it appears in Schelling, Weisse, Lotze and Fechner), his "Unconscious" being rather to be called "superconscious". But he prefers the negative expression in order to keep anthropomorphic representations at a distance. He thereby overlooks that even if the name he has chosen removes such representations, they are in high degree favored by the manner in which he lets "the Unconscious" operate.

It is an unscientific procedure that Hartmann employs when, wherever he thinks he finds a hole in the scientific explanation, he puts his "Unconscious" in as a magical means that works equally well everywhere. Of it holds what Galilei said of the appeal to an almighty will: it explains nothing, because it explains everything equally well. The real interest in the concept of the Unconscious has in Hartmann scarcely arisen from a striving after understanding of the phenomena either. It is due rather to a need for rest from reflection, analysis and criticism, and to a faith in the value of what emerges involuntarily and unprepared. Thus it runs in one place in Hartmann (Philosophie des Unbewussten, 3. Aufl., p. 369): "Conscious reason is only negating, criticizing, controlling, correcting, measuring, comparing, combining, subordinating, inducing and generalizing. Never is it creatively productive, never inventive; in this respect man is entirely dependent upon the Unconscious, and when he loses the capacity to receive the Unconscious's promptings he loses his source of life, without which he will drag out his life in monotonous fashion *[539]* under dry schematizing. Therefore the Unconscious is indispensable, and woe to the age that violently suppresses it and, in one-sided overvaluation of the consciously rational, will let that alone count!" Consciousness cannot be the purpose of development, but can only be a means, necessary at a certain stage of development. (Phil. d. Unbew., 3. Aufl., p. 739.) — It is this psychological-historical experience of the significance of unconscious life that is the warranted foundation of Hartmann's thinking. But he then

makes the Unconscious into a mythological being, a demon that everywhere intervenes in the natural connection, directs the atoms, displaces the brain molecules, adjusts the circumstances. Hartmann's philosophy takes on a mythological and dualistic character, which is certainly most prominent in the first editions of "Philosophie des Unbewussten", but is too closely bound up with his principle for it to be able, despite all correctives, to disappear entirely. Characteristic above all is his position on Darwinism, in that he seeks to show that the selection of qualities does not explain the arising of new forms, but is only a mechanical means which "the Unconscious" makes use of when it wants to prepare the way for such forms. Darwin keeps — Hartmann holds — to the conditions, but overlooks the properly producing force.

It was then no wonder that Hartmann was often attacked from the natural-scientific side. Now, in order to show that he himself had a very good command of the natural-scientific points of view, he wrote an anonymous critique of himself: *Das Unbewusste vom Standpunkt der Physiologie und der Descendenzlehre* (1872). The critique was so good and so telling that his opponents praised the anonymous work and used it against him. In the second edition Hartmann named himself and added notes in which he sought to refute the objections he had himself raised. It is doubtful, however, how far he has succeeded; he has conjured up spirits that will not vanish again. He admits that he had undervalued the mechanical causes and overlooked many intermediate links. *But* his procedure continues to be that of seeking out the holes which scientific inquiry always leaves, and filling them by the help of his mystical *deus ex machina*. Since there is
[540] always the possibility that the holes may be filled scientifically, it is a dubious method to employ. Lotze employed this method in his spiritualistic psychology; Hartmann extends it to his whole philosophy, without attending to Lotze's clear and energetic emphasis upon the significance of mechanism. His "speculative results" are not won by "inductive method". He accordingly admits (Das Unbewusste etc., 2. Aufl., p. 359 f.) the insufficiency of induction, and ascribes to it only the significance of serving to confirm what has been thought out by another way.

β. Pessimism and Ethics

Hartmann declares it a great misunderstanding that he should himself, in his personal temper, be a pessimist; he teaches pessimism as a *theoretical view*, after disappointments and sufferings in his youth opened his eyes to it. "With the chapter on pessimism [in "Philosophie des Unbewussten"] I have", he says, "written myself free once for all from world-sorrow as such, and purified it into an objective, calm *knowledge* of existence's

wretchedness — but thereby also won back for myself that undisturbed cheerfulness which may be found in the philosopher who floats in the ether of pure thought and from there regards the world, and his own pain, as an alien object of investigation." (*Mein Entwickelungsgang.* Gesammelte Studien und Aufsätze, p. 38.) He emphasizes as well that pessimism does not even theoretically constitute his whole philosophy, and does not give it its principal stamp. (*Die Stellung des Pessimismus in meinem philosophischen System.* Zur Geschichte und Begründung des Pessimismus, 2. Aufl., p. 18–28.) He describes his task as that of uniting pessimism — as the result of valuing the world by the standard of happiness — with the philosophy of development. He would thus reconcile Schopenhauer's philosophy with Hegel's, just as in his very concept of the Unconscious he has sought to combine Schopenhauer's absolute will with Hegel's absolute Idea. In a peculiar way Hartmann here finds a point of attachment in Kant. When Kant declared every attempt to solve the problem of evil hopeless, that was consistent from his standpoint, since he proceeded from the Christian concept of God, with which the *[541]* reality of evil stands in insuperable contradiction. But the problem must be posed thus: how must the world's absolute ground be thought, if it is, without contradicting itself, to have come to posit such a world? And when the question is so posed, Kant has prepared the way for its treatment, since he has given indications in a pessimistic and in an evolutionist direction alike. Schopenhauer pursued the one indication, Hegel the other: now the time has come to gather up what lies in both. (*Kant als Vater des modernen Pessimismus.* Zur Gesch. u. Begr. des Pess., 2. Aufl., p. 136 f.) (Compare Note 17 in the present volume.)

Since now — so Hartmann reasons — an unprejudiced consideration shows more unhappiness than happiness in the world, the world's coming to be cannot be derived from reason. It cannot be due to a rational but must be due to an irrational principle. The explanation must then be sought in the will-element within the Unconscious having separated itself from the representation-element. Hartmann takes up Böhme's and Schelling's mystical doctrine of a dark element in the Godhead whose breaking loose leads to the world's development. In a groundless manner the in itself blind willing has torn itself out of connection with representation, or the Idea. But this seeing element in the Unconscious continues to work in the world's development and seeks to lead that blind and refractory element to harmony and reconciliation. Hence the double-sided impression we get of the world. Both pessimism and evolutionism are right. There are two principles at work in the world. Hartmann's philosophy of religion comes here to recall Voltaire's, Rousseau's and Stuart Mill's — save that development is, on his

belief, a progressive redemption of the suffering God, since what matters is, through the abolition of the will, to undo what was done when the world became actual. Of great importance in this respect is that the knowledge of life's wretchedness should awaken, and Hartmann therefore holds that with Schopenhauer a new world-period has begun. (Phänomenologie des sittlichen Bewusstseins, p. 782.) Only slowly does this
[542] knowledge dawn in mankind. It happens through a successive abolition of illusions. First one expects happiness in the present life. When one discovers that the goods it can bring are only apparent, one sets one's hope upon another life. That is the illusion's second stage. Belief in the beyond cannot maintain itself; one gains fresh confidence in earthly life, but, taught by experience, one does not expect an immediate enjoyment of happiness, and instead hopes for a state of bliss for mankind at some time in the future, straining every nerve to prepare the way for it. That is the illusion's third and last stage, at which mankind will probably remain for long ages. But in the very fact that a small minority has seen through the illusion in all three of its forms, Hartmann sees sufficient warrant for the pessimistic theory; its correctness cannot be settled by a vote.

Hartmann's ethics hangs closely together with his pessimistic theory. There is for him an irreconcilable conflict between culture and happiness. Culture's advances are bound up with a decline in happiness. The more complicated life's apparatus becomes, the more occasion for displeasure is given. Susceptibility to pain grows greater, and the increasing reflection sees through the illusions more easily. Culture lets needs grow faster than the means of satisfying them. One must then choose between culture and happiness, between the doctrine of development and the doctrine of bliss. Happiness presupposes peace and rest — and will therefore bring stagnation and dissolution with it. Development leads ever further, until all possibilities are exhausted.[19] Through development dissatisfaction is increased and the illusions are abolished, until life at last comes to a stop. Then redemption is reached, and the blind will's breaking loose is annulled. The highest form of ethics is according to Hartmann that which builds not merely upon love of mankind and the urge to development, but makes compassion with God — the need to release the bound and suffering world-principle — the foundation of morality. One then makes the Unconscious's purposes one's own.

Hartmann's philosophy is theology inverted. He operates confidently with the thought of an absolute world-purpose, without seriously engaging the difficulties that
[543] thought contains. In his ethical work (Phän. d. sittl. Bew.) there are, however, excellent particulars, especially in the characterization and criticism of the different standpoints.

[19] On the relation between culture and happiness compare my *Etik*, Kap. 7, and my *Etiske Undersøgelser*, Kap. 2.

And in detail his train of thought is independent of pessimism's metaphysics. He has overcome that metaphysics to a still higher degree than he says. In practice he is an evolutionist and an optimist. For provisionally the affirmation of the will to life is the one right thing, since something can be accomplished for the world-process only by full devotion to life and its pains, not by cowardly renunciation and self-surrender. Pessimism's time really comes only after long ages — ages so long that one wonders doubly at Hartmann's confident knowledge of what will happen when they are past. By this long postponement pessimism loses the personal, practical significance it had in Schopenhauer. Hartmann stands to Schopenhauer as the modern Christians, for whom the day of judgment has been pushed out into the indefinite, stand to the first Christians, whose whole life was determined by the expectation of its speedy coming. Distant possibilities cannot be determining for our conduct of life, and cannot shake life's value. Only the life we know and live can we value, and within that life itself we must find what we are to be able to acknowledge as valuable.

4. Criticism and Positivism

a. Friedrich Albert Lange

It was natural that the dissolution of Romanticism's philosophy, the rise of natural science, and materialism's claim to be science's last word, should bring to greater influence the tendency which in the century's first half had been only an undercurrent. Already the greater attention and adherence which Schopenhauer's and Herbart's ideas now became the object of are signs in this direction. There is a need for a critical revision of the very foundation of our knowledge, in order to be able to take up the right attitude to the new material of knowledge. In Lotze and Fechner, and in part in Hartmann too, investigations *[544]* in this direction are indeed to be found, but the problem of knowledge proper is not taken up. They take realism as given, and then seek, more or less consistently, to find a foundation for their idealist constructions either in realism's presuppositions or in its holes. Great as the interest of Lotze's attempt in this direction is, it was nevertheless the problem of existence rather than the problem of knowledge that he treated. One of the first to draw attention to the necessity of taking up Kant's work upon the problem of knowledge again was the famous historian of philosophy Eduard Zeller, in a small work *Ueber die Bedeutung und Aufgabe der Erkenntnisstheorie* (1862), which caused all the more stir in that its author had come from Hegel's school. A few years later Friedrich Albert Lange, in his brilliant historical-philosophical work *Geschichte des Materialismus*

(1866), pointed to epistemology as a court of appeal overlooked not only by Romantic but also by materialist philosophy, and sought upon its basis to carry on the struggle for an idealist world-view.

Lange is one of the most attractive figures in the history of recent German philosophy. Ideal enthusiasm, sharp criticism, great fullness of knowledge and a vigorous realist eye have entered in him upon a rare union. He commands besides a great talent as a writer. Unfortunately his career was short. He was born in 1828 near Solingen in Rhenish Prussia, but lived his early youth in Zürich, whither his father, the well-known theologian J. P. Lange, was called as professor of theology in 1841. He thus received the foundation of his intellectual development in the free air of Switzerland, and it was to be traced through his whole life and striving. His university studies he pursued in Bonn. Educational work stood before him as his principal aim, and he studied philology as a means to it, without however neglecting natural-scientific and philosophical studies. His philosophical direction was early laid down. In a letter of 1851 (in *Ellissen*: *Friedrich Albert Lange.* Eine Lebensbeschreibung. Leipzig 1891, p. 69–73) he develops the view
[545] that all representations man can form of existence — of God and the world, of good and evil and so on — are developed according to the laws of the human mind: those laws are then the last foundation of our knowledge, beyond which we cannot get. He was here led to the psychological-epistemological standpoint which Ludwig Feuerbach had reached through criticism of the religious and speculative ideas, though Feuerbach does not seem to have influenced him. What influenced him philosophically in his youth was above all Hegel's "Phenomenology", Herbart's psychology and Schleiermacher's writings — of the last he speaks with enthusiasm in a letter of 1849.

After some time as lecturer and schoolmaster, Lange worked as editor and as a political and social agitator. He took part in the opposition to Bismarck's domestic policy and in the German workers' associations' efforts to win a firm organization. The work *Die Arbeiterfrage* (1865) derives from this period. It is one of the first writings — next after Mill's political economy — in which the social question is conceived from a more general point of view and in an impartial manner. It has interest besides in bringing the Darwinian ideas into connection with the workers' question, in that this is seen to spring from the struggle for existence in the social field. Lange conceives the matter thus, that what is required is to wage a struggle against the struggle for existence — a struggle which precisely constitutes man's higher spiritual vocation. (Since at this time only "The Origin of Species", and not "The Descent of Man", had appeared, Lange could not make use of the points of view Darwin has given for conceiving the ethical endeavors too as

a struggle for existence.) He saw the political significance of the workers' question in its exerting a constant pressure upon the conservative institutions, whose dykes will be broken through if canals are not dug in time. The liberal-minded men in the higher classes now have the choice between strengthening those dykes and helping to dig the canals. Lange's own choice was easily made. But in Germany men were busiest with the dykes, and Lange therefore emigrated to Switzerland. In the midst of his journalistic and agitational activity he had nevertheless found time to put the last hand to his principal work *Geschichte des Materialismus* (which according to Ellissen, p. 145, appeared in the *[546]* autumn of 1865, though the title page bears 1866). It is not a purely historical work. Through a spirited depiction of materialism's principal forms and of the development of the natural sciences, Lange seeks to illuminate the problem of the relation of the spiritual and the ideal to the material, and he sets the theoretical question at the same time in connection with the practical and social one. Although many parts of the work are now of course out of date, no one will read it without enrichment and stimulus. — Shortly afterwards there appeared a new social-philosophical work of Lange's: *Mill's Ansichten über die soziale Frage* (1866).

In Switzerland Lange worked as bookseller, teacher and editor in Winterthur, and unfolded an astonishingly many-sided activity. He took eager part in the agitation for a revision of the constitution of the canton of Zürich. Later he became professor in Zürich and could now live for his studies again. He was soon noticed, however, in his old fatherland, back to which he also longed, and in 1872 he became professor in Marburg. But by then he had already suffered for several years from cancer of the abdomen, and his last years were painful. His ideal disposition and his great enthusiasm for research and for freedom sustained him; his spiritual nourishment he found above all in Schiller's philosophical poems. He died in 1875. — An unfinished but interesting work on logic (*Logische Studien.* 1877) was published shortly after his death. Later (1897) Ellissen published *Einleitung und Kommentar zu Schillers Philosophischen Gedichten*, a work Lange had prepared during his stay in Zürich but laid aside for other labors. Schiller was, as Lange said in the preface to his principal work, the Kantian with whom he felt himself most akin. Schiller's poems contained Lange's religion.

As the title of Lange's book indicates, his problem is how one is to take up a position with respect to materialism. His standpoint here is the same as Fechner's. He would overcome materialism precisely by carrying it through. What he praises in materialism, and what for him constitutes its great historical significance, is its consistent demand for the exhibition of material causes for all natural phenomena. In idealism, by contrast, *[547]*

Lange sees a great danger in the theoretical field, since it easily leads to relaxing the strict mechanical explanation and to inserting subjectively constructed members into the series of objective causes. The functions of the nervous system and of the brain too must be explained according to the general physical and chemical laws. One must — relying upon the law of the conservation of matter and force — follow the influences from without through sense-organs, afferent nerves, brain, motor nerves and muscles, and seek a complete and unbroken connection in this whole process. It is unwarranted to assume the connection broken at any point. Consciousness itself is not a member of this connection. It is the subjective state of the individual in whom the process takes place — another side of the process, but not itself a part of it. Here materialism has its limit: precisely because the states of consciousness are not members or parts of the material processes, and cannot be explained by the law of the conservation of matter and force, precisely therefore do they fall outside materialism, and materialism comes into contradiction with itself when it thinks it has explained everything. It can at most explain the objective, material process, which is the way in which the conscious individual's subjective state appears to outward consideration. Lange regards it as conceivable that one will some day be able to determine that part of the material processes which coincides in time with a certain state of consciousness in the individual; but he does not think one will ever be able to give a closer determination of the relation between the subjective state of consciousness and the objective nerve process. It is a warranted conjecture that behind the two corresponding worlds — the world of matter and the world of consciousness — there lies an unknown third as their common cause.

It is remarkable to see how the great hypotheses of the seventeenth century are renewed after the rise of natural science at the middle of the nineteenth. As Vogt, Moleschott and Büchner renew Hobbes's standpoint, so Lotze renews Descartes's, and
[548] Fechner and Lange Spinoza's. It is a testimony that these hypotheses indicate the possibilities between which our thinking must here make its choice. On closer investigation one will find that it is not a simple resumption but a renewal of the older hypotheses that has taken place. Not only the natural-scientific foundation but also the philosophical elaboration is other than it was in the great dogmatists of the seventeenth century. There is greater critical circumspection, a clearer consciousness that what is at issue is finding the best and most consistent points of view in a discussion that can scarcely be brought to a close — not the setting up of definitive systems. One notices that the critical philosophy, whose breakthrough lies between the two epochs, has not worked in vain.

Lange's standpoint, which from one side points back to Spinoza, now precisely from

another side points back to Kant. He does not stop at the double form of existence — as motion and as consciousness — but calls attention to the fact that the whole of matter, the world outside us as well as our own organism, brain and sense-organs included, is only there for our consciousness, is a product of consciousness. It is in virtue of our organization that we apprehend the world as we do apprehend it. That Kant showed, and the later physiology of the senses has confirmed it. Here again materialism is overcome by being consistently carried through. For even if it be conceded to materialism that our whole conception of the world is a product of our material organization, that material organization is itself only an object of consciousness, and exists immediately only in and for consciousness. Matter and organization are our representations — even if they are representations we necessarily form, and form according to determinate laws. The dispute between matter and spirit is thus ended in the latter's favor. The naive belief in matter must vanish upon closer investigation, and it can vanish without natural research's campaign of conquest being in the least degree hindered thereby. — In the way Lange here takes up his position against materialism he has a predecessor in Schopenhauer. At about the same time as Lange, the famous Viennese pathologist *Rokitansky* expressed a similar view.[20]

There is in Lange's presentation in "Geschichte des Materialismus" a certain wavering *[549]* as to how far he will extend subjectivism.[21] Occasionally he says that the very opposition between the thing in itself and the phenomenon is a product of our organization, and that we therefore do not know whether that opposition has any significance whatever outside our experience. In other places — especially in the second edition — he speaks of an order of things corresponding to our knowledge. In a letter to the natural scientist Dohrn (in Ellissen, p. 258–262) Lange develops the view that with respect to this whole question four different stages must be distinguished: 1) the naive belief in sensible appearance; 2) the recognition that the senses do not give us things as they are in themselves, but vibrations of air as sound, vibrations of ether as color and so on — that is natural science's standpoint; 3) the recognition that our understanding as well as our senses belongs to our organization, and that therefore even the scientifically corrected

[20] *Karl Rokitansky's* spirited paper *Der selbständige Werth des Wissens* (Sitzungsberichte der Wiener-akademie 1867) did indeed appear more than a year after Gesch. d. Mat., but as early as 1862 he had in an address pointed to Kant's idealism as the true consequence of natural-scientific thinking. Compare Th. *Meynert*: *Karl Rokitansky.* Ein Nachruf. (Sammlung von populär-wissenschaftlichen Vorträgen. Wien und Leipzig 1892, p. 76 f.) — *Helmholtz* too (in his *Physiologische Optik.* 1867, p. 443 ff.) maintains that it is in the last instance unwarranted to ascribe to our representations any validity other than a practical and symbolic one. The physiologist A. *Fick* arrives at a similar result in his lecture *Die Welt als Vorstellung.* Würzburg 1870.

[21] *Max Heinze* has shown this in his interesting paper *Der Idealismus Friedrich Albert Langes* (Vierteljahrschrift für wissensch. Philos. I, p. 183–185).

apprehension of sense does not give us the thing in itself — that is Kant's standpoint; 4) finally the recognition that the very opposition between thing in itself and phenomenon is a product of our organization, so that its validity may be doubted. This fundamental doubt cannot, according to Lange, be removed by way of knowledge. It marks science's limit, beyond which only poetry, not thinking, can lead. Lange reproaches Kant for stopping at the third stage; Kant's Romantic successors he reproaches with having "tumbled out beyond the fourth stage into poetry, and mixed criticism and poetry in their systems instead of bringing them into a clearly delimited relation to one another".

In this conception Lange has, however, overlooked that when our knowledge applies at its very limit the same principles as within the limit, it must necessarily hold fast to the opposition between subject and object, knowledge and thing: in the contrary case it would in one form or another come to assume a creation out of nothing, a causality proceeding from an absolute unity. It is a law for all knowledge that what we understand is understood only by the help of *several* premisses or conditions.[22] If
[550] one does not acknowledge that law, one becomes entangled in difficulties which from Nicolaus Cusanus onwards emerge again and again in the history of thought.

Lange, like Schopenhauer, uses idealism — in the sense of subjectivism — to unseat materialism. But he also espouses idealism in the word's practical sense. It is not enough for him to maintain reality's subjective character; reality would not thereby become other than it is, bad or good. But he maintains that just as deeply grounded in human nature as the need to apprehend and explain reality according to natural-scientific laws is the need to form ideal images and to regard them as the expression of the highest reality. The producing power which expresses itself in sensation and in thinking does not rest content with what is given, but strives beyond it in order to form an ideal world. Man cannot do without a supplementing of reality. All religions and all speculative systems are products of this need. They differ from the scientific picture of reality in that it is not the organization common to all, the race's nature, that operates in their arising, but individual and personal tendencies and requirements — and also in that a complete confirmation by experience is not possible. It is of great importance to hold fast to this difference, so that one does not lay down as literal truth what is only a symbol. Every significance for the knowledge of reality must be denied to the religious and speculative ideas. But on the other hand a distinction must also be drawn between such ideas and mere figments. Although Lange in single places lets poetry begin where knowledge ends, he maintains in other places (1st ed. p. 539, 541; 2nd ed. II, p. 540) that speculation stands

[22] Compare my *Psykologi* V D, and *Filosofiske Problemer* (Universitetsprogram 1902), p. 48–52.

between empirical science and poetry: in the one the material rules, in the other free form; speculation seeks to unite the two. Clearer is the difference Lange finds in this, that the idea is to be judged by its value for the development of our spiritual life, not by its grounding and origin (1st ed. p. 346 f.; 2nd ed. II, p. 545). We do after all regard Cologne cathedral as something other and more than a heap of stones. Indeed, Lange even speaks of practical idealism as possibly hanging together with the unknown truth, though in *[551]* another way than materialism does: the idea is image and symbol for an unknowable Absolute. (1st ed. p. 346, 539, 541, 545. — 2nd ed. II, p. 494 and elsewhere.)

Here again it shows itself how difficult it is to place an absolute limit. Lange becomes here involuntarily a Platonist. How does he know that the ideas which for us express the highest value are images or symbols? And on the other hand — if his sharp distinction between ideas and scientific knowledge is held fast — who will then work at the development of the ideas, if they have no real significance at all? The speculative ideas can be maintained in their significance only by being seen as hypotheses, in which human thought draws the last consequences of what stands fast for it at the given time. It must then be left to closer testing what theoretical and practical significance they may have; but practical significance they can scarcely have if they have no theoretical significance whatever. As regards the practical significance itself, finally, Lange gives no definite criterion, no measure. Where he indicates his ethical conception, he says that the natural sympathy which for Auguste Comte was the foundation of ethics is not enough for an idealist ethics, and he regards the idea of a totality to which we belong as the a priori foundation of ethics, by which alone the moral law's inexorable strictness becomes intelligible. But at once he becomes doubtful again: does the idea not often lead astray, and may it not be asked, especially with respect to the religious systems, whether it is not better simply to give oneself over to the ennobling effect of natural sympathy than to listen to prophetic voices which only too often have led to the most terrible fanaticism? — A closer investigation of the relation between formalism and realism in ethics Lange has not given. In his ethical ideas he was a disciple of Kant, or rather of Schiller, though he saw the limitation of their conception.

His doctrine of the symbolic character and practical value of the religious ideas Lange applies not only to philosophical speculation but also to the positive religions. When materialism attacks Christianity as charcoal-burner's faith, it overlooks — Lange *[552]* maintains — the possibility of a freer, ideal conception of the content of the ecclesiastical traditions. One should look to the ideas' value and not to their literal formulation. In a certain sense religion's ideas are imperishable. Who will refute a mass of Palestrina's, or

establish that Raphael's Madonna is an error? In the inwardly moved spiritual life of the genuinely devout something stirs which the ideally minded freethinker must understand and must sympathize with. Lange desires a reinterpretation of the ecclesiastical doctrines, in the spirit of Kant, Fichte and Schleiermacher, so that Christianity's ethical and ideal content may be drawn out of its literal husk; Hegel's philosophy of religion was for him too conservative. Direct polemic against positive religion he does not want, lest the people's unity and the strength of its spiritual life should suffer. Of the two, he would rather sacrifice enlightenment for a time than strength, provided only that freedom of teaching is secured and priestly rule kept away. He has an open eye for the great oppositions of our time, and expects from philosophy, as a contribution to softening the struggle, above all a more penetrating understanding of the nature and the laws of human development. He closes his brilliant work by expressing the hope that serious work in this direction will bear its fruit. —

The great interest in Kant's philosophy, to which Lange's work partly testified and which it partly awakened, led to an eager study of the great philosopher whom Romanticism's philosophy believed it had overcome. On several points disagreement showed itself as to how Kant was to be understood. A rich Kant literature blossomed, which in several respects led to a more exact conception of Kant, his task, his method and his position in the history of philosophy. Even if one often lost oneself in investigations of more philological than philosophical significance, it was nevertheless of great importance that Kant should be understood more correctly than before. This is due especially to works by Cohen, Fr. Paulsen, Laas and Vaihinger, to which were later added the contributions to the illumination of Kant's development and thought which B. Erdmann, R. Reicke and others brought forward from Kant's unpublished notes and letters. In
[553] the great edition of Kant's works begun by the Berlin Academy, the interest in critical philosophy will set itself a fine memorial.

In a spirited paper *Was uns Kant sein kann* (Vierteljahrsschr. für wissensch. Philos. 1881) Friedrich Paulsen has sought to determine the significance Kant can have for our time. That significance rests not upon a direct resumption of his doctrine, but upon a treatment of the problems in his spirit. So-called *neo-Kantianism* therefore designates no closed school, but an endeavor toward epistemological testing of the concepts we operate with. Wilhelm Wundt, in his paper *Was soll uns Kant nicht sein?* (Philosophische Studien VII), has spoken against some neo-Kantians' unduly dogmatic adherence to the old master, and has shown why Kant's doctrine cannot be set up as an epistemology valid for the science of our own time.

b. Eugen Dühring

The year before Lange's "Geschichte des Materialismus" there appeared one of the most excellent epistemological works of the century's last half, namely Eugen Dühring's *Natürliche Dialektik*. It takes its point of departure in critical philosophy in so far as it means, in a more radical way, to take up again Kant's problem, which Romanticism's "deluge" had washed away. Among later German works it mentions only Trendelenburg's "Logische Untersuchungen" with great recognition as a predecessor; among foreign thinkers Comte and Stuart Mill are named, who have not, however, specially treated the book's subject. The book is already now a literary rarity, since the author has not republished it — perhaps because his standpoint has since altered in a more positivist direction. And yet it is surely the best of all Dühring's books, in form and in content alike. It is an interesting contribution to the illumination of the relation between critical philosophy and positivism, two tendencies which contend not only in Dühring but in the whole of the most recent philosophy. By two fundamental thoughts Dühring here already approaches positivism. Still more sharply than Kant and Schopenhauer he maintains that *[554]* the law of grounding — the principle of sufficient reason — is only a law for our thinking, not for reality, which is more comprehensive than our thinking. And he sets up, on the other hand, the principle of the lacking ground, whereby the burden of proof is laid upon whoever puts forward a new claim, what has hitherto been given being retained so long as ground is lacking for abandoning it. By the first fundamental thought he rejects rationalist dogmatism, by the second idealism and dualism; through both he works his way forward to building upon the ground of the given reality, without the admixture of representations not drawn from it. He continues Ludwig Feuerbach's work and is positivism's most important representative in Germany. He himself calls his philosophy the philosophy of reality. The great fundamental features of reality, which experience teaches us to know, are to form the basis both of a view of life and of the conduct of life. Philosophy is for him not merely theory, but the expression of a personal disposition.

In Dühring's view the nineteenth century has a decidedly reactionary character — the technical field alone forms an exception. The seventeenth century stood for him as the classical age in scientific respects. Serious science begins only with it, and our century has brought forth nothing that can eclipse it. Dühring is thinking here above all of Galilei, Huyghens and Newton in natural science, and of Bruno, Hobbes and Spinoza in philosophy. It is the fundamental thoughts of these great investigators that we still live upon. It is characteristic of Dühring that the great, vigorous and fruitful initiating thoughts have for him far greater significance than the special working-out and the

confirmation by experience. This hangs together with his historical eye and with his sense for what is inspired; but it leads him to misjudge the importance of the great thoughts finding their confirmation in particular experiences — an importance which a positivist of all people ought least to misjudge. This tendency in Dühring, which was surely nourished by his blindness preventing him from seeing and orienting himself for
[555] himself, increased in the course of his development and became of fateful significance not only for his philosophy but also for his outward circumstances.

While he praises the seventeenth century as the age of great thoughts, he praises the eighteenth for the reforming application of those thoughts. When his own time seems to him poor, he turns with all the greater hope toward the future. He feels himself the herald of a tendency in thought and in life which will be able to win currency only in a distant future. His great energy and fullness of thought are united with the reformer's self-feeling — unfortunately also with a bitterness and a suspiciousness which his fate has nourished, but which make his later writings in many parts unpleasant through coarse attacks upon those he calls his enemies. His writings have a strongly personal stamp, in the word's good and in its less good sense alike. He has himself depicted his personality and his life in the remarkable book *Sache, Leben und Feinde. Als Hauptwerk und Schlüssel zu seinen sämmtlichen Schriften* (Karlsruhe und Leipzig 1882). He is right that his cause and his life hang closely together, and to his honor it must be said that with him the cause stands before the life; less right is he perhaps in supposing that his enemies hang together with his cause and his life as closely as he suggests in the title.

Eugen Dühring was born in Berlin in 1833. His family came — a thing he set great store by — from Sweden. By his father he was brought up in Rousseau's spirit, under the influence of a liberal religious way of thinking. The upbringing developed independence and firmness in his thinking. Mathematics and astronomy early became principal interests. After his father's death he entered a Gymnasium which was also a boarding house, but where he was as little satisfied with the spiritual as with the bodily fare. He then studied law and worked for a time as a practicing lawyer. An eye disease which brought blindness with it made this activity impossible, and gave occasion for his resolving to use his great store of knowledge and thought to enter upon a scientific writer's career. His wife, and later his son, was his secretary. A series of philosophical,
[556] economic and scientific-historical works testify that the energetic thinker did not put his trust in vain in his inner resources when the outer ones were closed to him. Of his writings there are to be named, besides the one already mentioned, *Der Werth des Lebens* (1865), *Cursus der Philosophie* (1875) — reworked as *Wirklichkeitsphilosophie* (1895)

— *Logik und Wissenschaftstheorie* (1878) and *Ersatz der Religion durch Vollkommeneres* (1883). He worked as a Privatdocent at the University of Berlin, where his lectures were among the best attended. His indignation at having been passed over in the filling of professorships, and his criticism of conditions at the university, brought him into sharp opposition to the leading circles there; and when he came to believe that Helmholtz had usurped the credit for the discovery of the law of the conservation of energy, which belonged to Robert Mayer — in whom Dühring saw the Galilei of the nineteenth century — his indignation found particular vent in repeated violent attacks upon the famous physicist. The ministry, on the recommendation of the philosophical faculty, punished him by depriving him of the right to lecture. That this was an unfair harshness toward the blind philosopher seems, from everything available, to be clear, even if his polemic was improper in content and in tone alike. The charge Dühring brought against Helmholtz becomes intelligible only through his not having followed the literature carefully, on account of his blindness; for Helmholtz had in fact, as soon as he discovered that he had a predecessor in Mayer, neglected no opportunity of giving him the honor that was his due. But there operated here in Dühring as well that overvaluation of first ideas as against their exact carrying through which we have mentioned above; what let him place Robert Mayer so high above all his competitors was the same fundamental consideration that let him set the seventeenth century so far above the nineteenth. And finally his suspiciousness contributed, probably nourished by the isolation blindness brought with it, and which let him find "enemies" on every side. Next after the professors, the Social Democrats and the Jews were his "enemies". "Opponents" concerned with truth and not with their own advantage he does not think he has met upon his way. (Leben, Sache und Feinde, Kap. 14.) He is so certain of having all good powers on his side that he says at *[557]* the close of his autobiography: "*Ueberhaupt sind meine Feinde wesentlich keine andern Elemente gewesen, als diejenigen, welche auch Feinde der besseren Menscheit sind!*"

After working for some years by giving free lectures, Dühring has now withdrawn to a small town near Berlin, occupied with natural-scientific and literary-historical labors. A great and noble force has here been pushed aside, though not broken, by misfortune and by his own fault and others'.

α. Epistemology

In his investigation of knowledge Dühring works (especially in "Natürliche Dialektik") entirely in the spirit of critical philosophy. He seeks to exhibit our knowledge's peculiarity in form and in procedure, in order to be able to decide how far we may regard its forms

and its results as expressions of existence. He thus attempts a critical testing of the relation between thinking and reality.

A principal point for him here is that thinking always seeks *continuous connection* and always seeks to *continue* its course. In mathematics the concept of infinity arises thereby, since no limits can be set to a magnitude's decrease or increase. In logic we enter, by virtue of the principle of sufficient reason, into an infinite series, one Why constantly succeeding another. Against this tendency to continual connection and continuation Dühring sets what he calls *the law of the determinate number*. Every *actual* given, every *factual* result is limited, and exhibits a determinate number of parts in time and in space. Infinity and unlimitedness mean only the possibility of continuation — a possibility not always really warranted. And where the continuation actually comes, it happens through successive additions of single, determinate elements. — There appears here, then, according to Dühring, a characteristic difference between thought and reality.

Thought must be supple and untiring in order to be able to follow reality in all its
[558] oscillations, in order to be able to rise and sink with the phenomena and to discover all relations of dependence between them. But the law it must follow in order to fulfill its task must not — Dühring insists — be transferred to nature, to actual existence itself. This is seen clearly by comparing the real science with the formal. In astronomy the determinate given magnitudes stand in contrast to pure mathematics' unlimited possibility of production. In chemistry the atomic numbers are examples of the law of the determinate number. Indeed, all really given numbers and magnitudes are finite: there is a determinate number of globes in the heavens, a determinate number of atoms in matter; at every moment a wholly determinate number of parts of time has elapsed; the earth has moved round the sun a certain determinate number of times; the causal series consists of a finite number of members.

With great resoluteness Dühring draws from the law of the determinate number the conclusion that the process of nature — the sum of all changes in nature — must have had a beginning. An infinite regress cannot be thought. Prior to the time filled with changes there lies, then, an eternal being, in which the differences, the inner distinctions that bring succession and change with them, had not yet appeared *as successive*. Simultaneous multiplicity may very well have been present, and must have been present, for it contains the germs — or, as Dühring says, the roots — of all that we call process and apprehend as determinate existence (Wirklichkeitsphilosophie, p. 15). Time and the causal series first came to be with the transition from that homogeneous and unchangeable condition to the unfolding of differences and changes. — This regressive conclusion of the causal series

in an unchangeable condition is for Dühring satisfying not only from the standpoint of thought but also from that of feeling.

But Dühring has emphasized the opposition between thought's ideal continuity and the reality of experience — always given in separation and parceling — so strongly that he ends in an insoluble riddle. For how is a transition from absolute equilibrium and homogeneity to succession and change possible? He has here, in his positivist thinking, come to face the same problem that Böhme and Schelling had put forward in mythologizing and speculative form. He would soften the paradox by pointing out *[559]* that every time a new phenomenon [a new quality] arises within our experience, we really have an absolute beginning. (Cursus p. 79. Logik p. 191.) But the difference is that what is new and beginning in our experience awakens wonder and gives thought a problem that can occupy it, whereas that sharp opposition between absolute identity and change makes thought's work hopeless once for all. Not merely a difference but a conflict between thinking and reality is what Dühring here comes to lay down. — When one has the choice between assuming such a conflict and assuming that existence is as incapable of being concluded as thinking is, this latter possibility seems preferable. There is a constant interplay between thought and reality: as inquiry advances there shows itself new continuity behind the apparent discontinuity, and so far reality confirms thought's ideal; but new discontinuity is also discovered where there seemed to be continuity in what is given, and the problem of how far reality can be understood is then posed anew. In Dühring the dogmatist has got the upper hand of the critical philosopher, on account of his great zeal to let reality come into its own.

On the other hand Dühring lays very great weight upon the inner connection between thinking and reality. Existence itself continues on into our thinking, where this unfolds according to its true essence. To the relation between premisses and conclusion (ground and consequent) there corresponds, in the real processes, the relation between cause and effect. To logic's principle of identity there corresponds the single thing's identity with itself, notwithstanding that it occurs in different connections. To our thinking's combinations and deductions there corresponds nature's unity and the interplay of forces in the arising of phenomena. We do not thereby conceive nature on the analogy of human consciousness, but as operating in such a way that human knowledge gets a material to investigate that corresponds to its own constitution. And even if our analysis will never be able to exhaust the great connection in nature, the necessity does not cease *[560]* of maintaining the kinship between what operates in things and what operates in the understanding.

What appears in our scientific principles is according to Dühring the fruit of forces that have long worked in a determinate direction before coming to clear consciousness. Consciousness and thought rest upon something that is as little consciousness and thought as the moving mechanical force is itself a phenomenon of movement. But it is a warranted hypothesis to represent to ourselves the ground of our thinking activity — a ground not appearing in consciousness — as agreeing with the ground of nature's connection showing itself to us to be ordered in accordance with the understanding. (Nat. Dial. p. 155, 225. Logik p. 173. Leben, Sache etc. p. 303.) What makes us think is thus the same as what makes nature operate.

This last train of thought now gradually gains such preponderance over Dühring that in his later writings he conducts a violent polemic against the critical philosophy, which maintains that we do not know things in themselves, because our knowledge is determined not merely by the nature of things but also by our own nature. He sees in that conception a bogging-down of science, a hypocritical accommodation of thought's results to what the need to assume a world beyond demands. Here again it is the law of the determinate number that underlies the matter: number is an example of our commanding concepts which express what is, in such a way that between reality and the concept no insertion conditioned by the constitution of the I takes place at all. (Logik p. 165.)

As sharply as Dühring elsewhere emphasizes the opposition between thinking and being, so definitely does he here emphasize the possibility of their completely covering one another. There he has formal knowledge chiefly in view, here real knowledge. But they cannot be kept wholly apart, as is shown by the fact that the law of the determinate numbers precisely sets inquiry new tasks — namely, to find out why changes occur just when those determinate numbers are reached.

It is Dühring's merit with respect to recent epistemology that he sets the problem
[561] of the relation between quality and continuity, between the determinate turning points and the connected processes, in so clear and distinct a light. His whole standpoint is determined by a contention between criticism and positivism, and contains an invitation to a closer investigation of the relation between those two tendencies.

β. Conception of the World

Dühring lays great weight upon the development of a *world-schematism*, which is to give a fully formed picture of actual existence's general features. In this the task of the philosophy of reality consists. It is to lead to a system of experience, whereas the earlier

metaphysics was a kind of alchemy which sought by mystical ways to fathom existence's essence. — At this point he recalls Comte and Spencer. He is in his "schematism" more abstract than Spencer and less original than Comte.

My system, he says, knows not two realities but only a single one — nature and its parts. He approaches materialism in emphasizing that everything in the world has a material side, and that by following the material phenomena in their intuitable connection one must be able to find everything that has reality. But he will not stop at the mere phenomena. Thereby he distances himself from "the usual, rather coarse materialism". He often uses the word *matter* in the sense of *being, world-medium, the bearer of all that is actual.* But he expressly emphasizes that the concept of matter must be widened and deepened so that it becomes clear that the dispositions to the phenomena of consciousness are found in nature just as much as the possibilities of all other phenomena. *The philosophical concept of matter* contains more than the physicists' and the chemists', since it is to be the concept of what underlies all bodily and spiritual states and comprehends the whole and full reality. (Cursus p. 62, 75. Leben, Sache etc. p. 302 f.) Materialism is for Dühring only a "pedestal", a real foundation, an outer, mechanical frame for the conception of the world and the view of life — not the exhaustive, ample explanation of existence itself.

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A doctrine one does not expect to find in Dühring is the assumption of *purposes* in nature. To understand him rightly on this point one must notice that he distinguishes between purpose (*Zweck*) and intention (*Absicht*). (Logik p. 203.) When he ascribes validity to the concept of purpose for our conception of nature, it is because the forces and dispositions operating in nature always exhibit certain tendencies, work in certain directions, and above all because they *operate together* and thereby make nature appear as a systematic unity. Those tendencies do not always lead — as little in outer nature as in conscious human life — to completed results. But whether the goal is reached or not, *the combined operation of manifold elements* appears as a characteristic principal feature of all processes. In apprehending nature one must attend not merely to the general laws but also to the directions and tendencies of the processes that go on, and to the types and forms in which nature's activity appears. Nature forms a scale of stages, the lower forms of existence being the foundation of the higher. The measure by which we distinguish between lower and higher forms can be sought only in the results of the different tendencies, in the degree to which they lead to their natural goal. And the last goal must be the arising of beings which not only exist and operate but also become conscious of their existence and their operating: only as the foundation of the history

of conscious beings has the history of the mechanical whole of nature a meaning; a thoroughly unconscious world would — says Dühring (Cursus p. 104) — be a foolish half-measure, a theater without actors and without spectators. And that the phenomena of consciousness are, on the other hand, essentially to be regarded as purposes is seen from the fact that unconscious and conscious operation can accomplish one and the same thing; it would be mere luxury for men and animals to become conscious of their impulses and motives, if it were only a question of attaining outer results, and if we were not here facing something that was in itself purpose. Our feelings in particular — the feeling of love, the social feeling — do not have their value merely by being means or
[563] motives leading to actions necessary for life's subsistence; they have immediate value as expressions of a higher stage of life than the purely elementary one. (Cursus p. 158 f. Der Werth des Lebens, 2. Aufl., p. 241.)

If one were to object to this last thought that conscious life is nevertheless not always bound up with pleasure, but that pain occupies so great a place that conscious life does not seem to be a real "purpose", Dühring would point to what he calls *the law of difference* — a law which under different forms and names plays a great part in recent psychology. (The most suitable name would be: the law of ratio.) For Dühring it is not merely a psychological law but a world-law. The antagonism of forces is a principal feature in the world-schematism. All expression of force, all movement and all development is conditioned by distance, difference, opposition. All consciousness too presupposes difference. Without difference, no sensation. Within consciousness the same holds. Life's play would lose its attraction if there were no resistance and hindrance to overcome, and if need and satisfaction did not succeed one another. It is existence's rhythms and differences that make it valuable for us: our feeling of life is set in motion by the transitions between mutually different states. Without what is harsh, bitter and painful, the deep satisfaction in life would not be felt. — The same emphasis upon feeling's being determined by oppositions which made Schopenhauer a pessimist makes Dühring an optimist. Schopenhauer is, as regards the "law of difference", more consistent than Dühring, in so far as he assumes that unrest and movement lie as an original tendency in the world-ground's nature; whereas for Dühring's philosophy it becomes a great riddle how successive differences and rhythmically changing states could have succeeded the resting being which that philosophy assumes in virtue of the law of the determinate number.

The *combination* of the different forces is what makes development possible. Whether this combination leads to a state of equilibrium or to a new movement, the tendencies of

the several forces are preserved in the development's result. A constant strife between the forces would be an absurdity — but it is not found in nature. The avoidance of the *[564]* absurd is a natural-logical principle. Only in exceptional cases does this principle lead to untenable forms being annihilated and dissolved. As a rule nature takes the positive way. Therefore, on Dühring's view, the idea of the struggle for existence is false: it one-sidedly emphasizes the negative point of view and lays the weight upon antagonism instead of upon combination. Dühring gives Lamarck's theory the preference over Darwin's.

γ. Ethics

The ethical has according to Dühring its foundation in the human impulses. Only its foundation — for nature can err, and therefore a closer development and correction is necessary. The higher a being stands, the greater the possibility that it goes wrong. Besides, what matters ethically is to combine elements which from nature's hand are not yet connected, but are perhaps even in mutual strife.

The germs of the good Dühring finds, like Comte, in the sympathetic instincts. Nature has itself seen to it that another's suffering determines our own feeling in a painful way, and there is perhaps no feeling that has undergone so conspicuous a development in the course of culture as this one. It does not lead to subduing and abolishing our own individuality. On the contrary, ethical development consists both in an individualization and in a socialization; the two necessarily belong together, since the full development of the single individuality first becomes possible in a more highly developed society. Under more imperfect social conditions the free, distinctive development of individuals is hindered. But since our present states are essentially built upon compulsion, they constantly hamper that development. A free society — the germs of which are given in the organizations formed by free association — will first let both individuality and community come into their own. In such a society all relations of production and of consumption would be brought under social direction, so that the individual's whole interest can be attached to the work and not to the work's yield. Only thereby does a thoroughgoing *[565]* ennoblement of life become possible. From socialism Dühring's conception differs partly in that he thinks of free labor communes and not of an absolute centralization of production,[23] partly in that he does not regard it as necessary that development should be interrupted by a historical miracle: the building of the future will on his view arise, not — as Karl Marx holds — through evil increasing, but through the good growing quietly. On this point too Dühring is decidedly an optimist, however much baseness he

[23] On Dühring's position on the social question, and on the different standpoints he has taken here, I refer to B. *Friedländer*: *Die vier Hauptrichtungen der modernen socialen Bewegung.* Berlin 1901.

may think he can establish in the world. He is a definite opponent of pessimism. Only the pessimism of indignation (*Entrüstungspessimismus*), as represented by Byron and called forth by determinate social conditions, has his approval; whereas he despises the Romantic "pessimism of the beyond" and the "pessimism of rottenness" frequent in the higher classes.

Beyond the feeling for the good forces that work their way forward within the human world, there arises a *universal affect* when the gaze is raised to the great unity of which the human world is only a single ramification — when the thinker feels himself as belonging to that unity and as moved by its forces — and when the thought of his own fate is taken up into the thought of the great order of things, within which so many good germs have been able to unfold. —

Not only by the problems it poses, but also by the noble, antique style in which it is carried through, and by the connection it establishes between thinking and personality, Dühring's philosophy stands as one of the most characteristic attempts of thought in the generation that followed Romanticism's fall and natural science's new rise at the middle of the nineteenth century. The year 1880 forms here a characteristic turning point, in that a circle of the more considerable investigators conclude their activity about that time — namely, besides Dühring, Lotze, Fechner, Darwin and Spencer; shortly before, Lange had succumbed. In a separate work some principal forms of the history of philosophy in the following quarter-century are to be presented.

Notes to the Second Volume (71–112)

[71]) P. 295. While de *Maistre* by his principle of authority and by his emphasis on war recalls Hobbes, by his doctrine of causation he recalls Malebranche, whom he also mentions with great admiration, just as he also follows him in his mystical-pantheistic intellectual tendency, as when he speaks of the divine ocean that shall one day take up all things and all men. He feels himself that he stands here at the border of heresy, for he adds at once: Je me garde cependant de vouloir toucher à la personnalité, sans laquelle l'immortalité n'est rien. (*Les* Soirées de St. Petersbourgh. 7. éd. II. p. 203).

[72]) P. 302. Cf. on this question my *Psykologiske Undersøgelser.* (Det kgl. danske Videnskabernes Selskabs Skrifter. 6. Række) p. 68—81. — When Biran speaks of sensations that fall outside the self, he means by the self more or less what in my *Psykologi* (V B, 5) is called the real self, the central and more active part of the content of consciousness. Outside the formal self, which expresses the connection and the unity in consciousness, no sensations can fall. — In his criticism of Condillac Biran had a predecessor in Charles *Bonnet (Essai analytique sur les facultés de l'âme.* Copenhague 1760).

[73]) P. 306. See, on the phenomena of concretion, *Philosophie des deux Ampère.* p. 316 f. *Essai sur la philosophie des sciences* (1834) p. LVIII ff. — A. *Bertrand (La psychol. de l'effort* p. 62) uses *[587]* the expression *concrétion* wrongly, in that he uses it of free association between two different representations; for that kind of combination Ampére uses the expression *commémoration.* When I recognize the tree, it is concrétion; when I see the tree again and then come to think of the dog that lay at the root of the tree last time, it is commémoration. — I regret that I did not know Ampère's account when I wrote the chapter on immediate recognition in my "Psykologiske Undersøgelser". But it is a great joy to me to see how close an agreement there is between his conception and the one I have developed.

[74]) P. 308. Ampère, moreover, makes the difference between his theory and Kant's greater than it is, by ascribing to Kant the conception that the relations, the forms of coordination, stand in no connection whatever with things in themselves.

[75]) P. 310. Of Cousin's first appearance as a philosophical teacher *Jouffroy* gives an interesting characterization in his *Nouveaux Mélanges philosophiques* (2. éd. p. 85—95). — *Renan (Feuilles detachées,* 4. éd. 1892. p. 298 f.) has expressed himself about Cousin, and about what he owes him, in the following way: A travers une foule de défauts, quel haut sentiment de l'infini! quelle vue

juste du spontané et de l'inconscient! quel accent religieux, inoui depuis Malebranche, quand il parle de la raison! Que l'on comprend bien les traces que gardèrent de ce premier enseignement des hommes tel que Jouffroy! Je connus le cours de 1818 sous les ombrages d'Issy vers 1842. L'impression fut sur moi on ne peut pas plus profonde. J'ai la conscience que plusieurs des cadres de mon esprit viennent de là . . . M. Cousin a été non un de mes pères, mais un des excitateurs de ma pensée.

[76]) P. 316. On the relation between Saint-Simon and his school see *Paul Janet*: *Saint-Simon et le Saint-Simonisme.* Paris 1878. — In the characterization of Saint-Simon I rely especially on *George Weill*: *Saint-Simon et son oeuvre.* Paris 1894 and on Chr. *Adam*: *La philosophie en France* (Première moitié du 19e siècle). Paris 1894. — On the relation of the Polytechnic School to Saint-Simonism cf. G. *Pinet*: *L'Ecole polytechnique et les Saint-Simonisme.* (La Revue de Paris. 15 May 1894). — While Saint-Simon's writings are incoherent and make only constant fresh starts, his school produced a coherent development of doctrine, namely the one edited by *Hippolyte Carnot*, the *Doctrine de Saint-Simon* (1829), which gives the content of *Bazard's* lectures, and which in time lies before the fantastically extravagant period that ended with the dissolution of the school. The further development that the doctrine of critical and organic periods received in "Doctrine de Saint-Simon" surely took place, however, under the influence of *Auguste Comte's* first important work (Plan des travaux scientifiques nécessaires pour réorganiser la société. 1822); although he still called himself Saint-Simon's disciple, he was nonetheless fully emancipated.
[588] — In his Autobiography (Danish trans. p. 164—169) *Stuart Mill* speaks in an interesting way of the significance that the Saint-Simonians' doctrine of critical and organic periods had for him. — The whole doctrine had, as was shown earlier, first been developed by Kant and Fichte, the first thinkers who saw both that criticism and the Revolution had triumphed and that they could nevertheless not be the last word of history. In Rousseau and Lessing the same thought is dawning. — Moreover, Lessing's "Erziehung des Menschengeschlechts" was one of the Saint-Simonians' favorite books.

[77]) P. 321. Comte's disciples go a step further than the master, in that they maintain that it is rather Saint-Simon who learned from Comte than the latter from the former. *Littré*: *Auguste Comte et la philosophie positive.* 2 éd. p. 90 f. *Robinet*: *Notice sur l'oeuvre et sur la vie d'Auguste Comte.* 2. éd. p. 139. Even if Saint-Simon — as is natural — did not remain uninfluenced by the gifted younger men who worked together with him, it nevertheless stands firm that he had developed his main thoughts before he came into contact with Comte. The priority with respect to the idea of a positive philosophy as the new spiritual authority that was necessary after the age of criticism and revolution cannot be denied to Saint-Simon. But — as *Georges Weill* (Saint-Simon et son oeuvre. p. 208—210), who has clearly demonstrated this, rightly remarks — Comte's great significance is not in reality diminished by it.

[78]) P. 322. Instead of perfecting, *Comte* later (Cours de philosophie positive. 2. éd. IV. p. 262. 264) prefers to use the expression development (developpement), because a moral valuation

might lie in the word perfecting.

[79]) P. 326. In a letter to Stuart Mill of 27 June 1845 he says: Le trouble a consisté en insomnies opiniâtres, avec mélancolie douce, mais intense, et oppression profonde, longtemps mêlée d'une extrême faiblesse. He connects the attack with the fact that a few days before he had begun to work out his new work, and adds: *L'ensemble de ma composition aura beaucoup gagné à cette période exceptionelle,* où ma méditation était loin d'éprouver l'atonie de ma motilité. — The year after, when Chlotilde de Vaux had died, he confessed that the feeling that had arisen in him toward her had contributed to that nervous crisis. (Lettres d'Auguste Comte à Stuart Mill, p. 413 f.). — Madame Chlotilde de Vaux's husband had been sentenced to penal servitude for life for a degrading crime, but she regarded her marriage as indissoluble, and the relation between her and Comte therefore did not go beyond an intimate friendship.

[80]) P. 328. Of the two most important writings on Comte, Littré's maintains that Comte's real significance ends with the publication of "Cours de philosophie positive", while *Robinet's* regards "Politique positive" as the real conclusion and polemicizes against Littré as an ungrateful and uncomprehending apostate. — It follows of itself that Stuart Mill's recognition of Comte *[589]* applies only to the first period. The correspondence between them ceased when Comte stood at the entrance to the second. In his work *Auguste Comte and Positivism* (1865) Mill has given an interesting valuation of Comte's doctrine in both of its stages. — The schism that Littré caused has, moreover, later repeated itself within the circle of the "orthodox" positivists. Some of these (with *Lafitte* at their head) hold that the positivist priesthood is to work upon the heart through the intelligence, while others (with the Englishman *Congreve* at their head) hold that positivism must triumph, as Christianity did in its time, through direct appeal to proletarians and women, so that the heart must be worked upon without its happening through the intelligence. See on this *Caird*: *The social philosophy and religion of Comte.* Glasgow 1885. p. 171 ff.

[81]) P. 334. *Cours de philosophie positive.* 2. éd. VI. p. 612: Kant a réellement mérité une éternelle admiration en tendant, le premier, d'échapper directement à l'absolu philosophique par sa célèbre conception de la double réalité, à la fois objective et subjective, qui indique un si juste sentiment de la saine philosophie. Cf. *Discours sur l'esprit positif.* p. 24. *Catéchisme positiviste.* 2. éd. p. 8 and 150. — According to what Comte himself states in one place, he did not read Kant's chief work. Yet he early came to know his ideas at second hand. In a letter to Valat of 3 Nov. 1824 he states that in Kantianism there are "very good things alongside a mass of extravagances", and he protests against Cousin's account of Kant being regarded as valid: Cousin est bien loin de comprendre la portée des Idées-mères du philosophe de Koenigsberg. Yet he here counts Kant as belonging to the metaphysical stage, and draws a parallel between his own relation to Kant and Galilei's relation to the Peripatetic physics. — The only writing of Kant's that he knew is the little treatise "Idee zu einer allgemeinen Weltgeschichte", which his friend Gustave d'Eichthal translated for him (1824), and which he admired greatly. Had he known it six or seven years earlier, he could, he says, have spared himself the trouble of working out his treatises [of 1820 and

22]. Letter to Eichthal of 10 Dec. 1824. — On Comte's relation to "metaphysics" cf. the interesting discussion in *Bulletin de la société française de philosophie.* January 1903. (Report of the meeting of 27 Nov. 1902).

[82]) P. 343. The first time Comte touches on the question of the possibility of self-observation is in *Lettres à Valat* p. 89 (letter of 24 Sept. 1819). In his treatise *Examen du traité de Broussais sur l'irritation* (1828) (reprinted in the interesting collection *Opuscules de philosophie sociale,* Paris 1883, which contains the most important of Comte's works before the publication of Cours de philosophie positive) he praises Broussais for his polemic against self-observation and remarks: One can observe one's feelings when they are not too strong, because they depend on an organ other than that of observation; but one's own intellectual activity one cannot observe, for then
[590] observer and object would be identical, and who would then make the observation? What has won subjective psychology a large following in the last ten years [before 1828] is, he thinks, the justified criticism of the ideology of Condillac and Helvetius, which regarded our knowledge as the mere product of external influences, without taking into account the necessity of inner dispositions; but this criticism had been better given by Bonnet and Cabanis, and above all by Gall and Spurzheim. (Opuscules p. 293—296). In *Cours de philosophie positive* (I p. 32 f.; III p. 564; VI p. 605. 2. éd.) he repeats his polemic against self-observation, remarking as well that an elementary analysis cannot be applied to the most complicated phenomena of all. — Since he assumed that self-observation was nevertheless possible as regards the feelings, he has given a practical proof of it in one of his letters to Valat, in which he analyzes the motives that underlie his activity as an author (28 Sept. 1819). — On the whole question I refer to my Psykologi I, 8.

[83]) P. 345. Cf., as regards political economy, the interesting treatise by *Ingram*: *Nationaløkonomiens nuværende Stilling og Udsigter.* (Translated from the English). p. 16 ff. — To Mill, who had told him of his plan to write a political economy, Comte stated that although political economy properly belongs to sociology as a whole, it can nevertheless do provisional service by being treated in isolation, especially when the working out is undertaken by so good a head as Mill. *Lettres à Stuart Mill.* p. 231 f. 254 f. Cf., on the other hand, *Cours de la philosophie positive.* (2. éd.) IV. p. 255: Toute étude isolée des divers éléments sociaux est donc, par la nature de la science, essentiellement stérile, à l'exemple de notre économie politique, fût-elle même mieux cultivée.

[84]) P. 346. Cf. *Cours de philosophie positive.* 2. éd. IV p. 392. Also *Lettres à Stuart Mill.* p. 121: — cette noble école, qui . . . fut certainement la plus avancée de toutes celles du dernier siècle. — p. 275: C'est à l'école écossaise et non, comme beaucoup d'autres, à l'école germanique, que j'ai dû la première rectification des graves aberrations, à la fois morales et intellectuelles, propres à ce qu'on appelle l'école française: je n'oublierai jamais combien ma propre évolution a été d'abord redevable surtout à quelques lumineuses inspirations de Hume et d'Adam Smith. — (By the Scottish school Comte evidently means the whole tendency that began with Shaftesbury and Hutscheson, not what is called the Scottish school in a narrower sense: Reid and his successors). — I may perhaps add the remark here that the fundamental conception on which I build in my

various ethical writings developed in me especially in the study of Comte, as my treatise *Om Grundlaget for den humane Etik* (Copenhagen 1876) will also bear witness.

[85]) P. 354. It is therefore not correct to say with *Fiorentino*: *Manuale di storia della filosofia,* 2. ed. Napoli 1887. p. 581, that Comte takes account only of the relations between objects among themselves, not of the relations between object and subject. To be sure, he has not drawn the *[591]* consequences of this latter relation.

[86]) P. 357. See on this *Politique positive.* IV p. 187; 233 f. *Catéchisme positiviste.* p. 165 f. 222. — In his later period Comte also makes another characteristic change in his classification. He sums up in a series (15) of propositions the most general principles of the positive philosophy, and gives this "systematization of the positive dogma" the name of *philosophie première.* The name he takes from Bacon. Then comes *philosophie seconde:* the series of the seven fundamental sciences, which are, however, again divided into two groups: *philosophie naturelle* (the first five sciences) and *philosophie morale* (the last two sciences). Politique positive IV p. 226. — In the Catéch. pos. (p. 167 f.) he speaks a little differently, mathematics, astronomy, physics and chemistry making up *cosmology,* and biology, sociology (in the narrower sense) and ethics *sociology.*

[87]) P. 368. In his essay on Bentham (Dissertations and Discussions. I) — where the utterance quoted is found — *Stuart Mill* says: "Bentham conceives the world as a collection of persons, each pursuing his separate interest or pleasure, and who can be kept from colliding more than is necessary by hope and fear under the influence of the law, of religion and of public opinion". (p. 362). — *Leslie Stephen* (The *Utilitarians.* I. London 1900. p. 314 and 325) remarks that the manuscript of the "Deontology", which is now at University College in London, seems to show that Bentham was essentially the author of this work, and that the editor, who was probably not very intelligent, is responsible only for some rhetorical exaggerations, especially in the second part.

[88]) P. 368. By my distinction between the motive of valuation and the motive of action I have sought to avoid the difficulty that arises for Bentham when the question of the justification of the principle of utility comes up. Cf. my *Etik* ch. III, 1—2; 18 and the first treatise in my *Etiske Undersøgelser.*

[89]) P. 377. Cf. J. H. *Newmann*: *Apologia pro vita sua.* London 1879. p. 10. 189 f. — See also L. E. *Fløystrup*: *Den anglokatholske Bevægelse i det nittende Aarhundrede.* Copenhagen 1891. p. 28. 30 and *George Worley*: *The Catholic Revival of the 19th Century.* London 1894. p. 45—51. — Coleridge's philosophical ideas are perhaps best got to know in brief from his work *Church and State,* where Appendix B sets out his doctrine of reason and understanding, and where Appendix E develops his relation to earlier philosophers. In *Biographia literaria* I chap. 12 he develops his ideas in Schelling's terminology and with a *very* extensive use of Schelling's own sentences. — Sterling's utterances about Coleridge are found in *Journals of Caroline Fox* (28. january 1842) and in a letter to Carlyle (in the latter's *Life of Sterling* p. 273).

[90]) P. 379. See, besides *Froude*: *Thomas Carlyle* (London 1882— 84), also *Dilthey's* charac- *[592]*

terization of Carlyle in Archiv für Geschichte der Philosophie IV, and *Leslie Stephen's* in *The Utilitarians* III. p. 462—477.

[91]) P. 390. In my *Psykologiske Undersøgelser* p. 58 I have touched on Hamilton's merit in regard to the doctrine of association in setting up the law of the whole (law of redintegration). Of older writers there should have been named here, besides Kant and Fries, also *Beneke*: *Psychol. Skizzen.* I. p. 382 ff., and already Chr. *Wolff*: *Vernünftige Gedanken von Gott, der Welt und der Seele des Menschen.* § 238. Hamilton may have known both these writings. — In my *Formel Logik* § 18 I have touched on Hamilton's standpoint in logic. — In my work *Den engelske Filosofi i vor Tid* (Copenhagen 1874) there is a fuller discussion of Hamilton. — Among Hamilton's disciples and critics it has been much discussed how he conceives the knowledge of the so-called primary qualities of things (extension and form): whether it too is determined by the subject, or whether we are here to know things as they are in themselves. *Stuart* Mill found (in his *Examination of Sir William Hamiltons* Philosophy*)* here a contradiction of Hamilton's doctrine that all knowledge is relative, conditioned by the relation between object and subject. *Mansel (Philosophy of the Conditioned.* London and New-York 1866. p. 84) and *Veitch* (*Hamilton.* Edinburgh and London 1882. p. 179) interpret their teacher as assuming a relative knowledge of the primary qualities (which are thus primary only in relation to the secondary ones, the qualities of sense usually so called). *George Grote* (*Minor Works.* p. 293) thinks that Hamilton changed his standpoint with respect to this question, so that in the dissertations found in his edition of Reid's works he assumes a knowledge of the primary qualities in themselves, whereas earlier he assumed only a relative knowledge of them. *Leslie Stephen* expresses himself in a similar direction on this point in Hamilton (*The Utilitarians.* III. p. 385—401).

[92]) P. 393. Cf. *Masson*: *Recent British Philosophy.* 3. ed. London 1877. p. 252 f. — *Edinburgh Review.* July 1884. p. 218.

[93]) P. 396. Yet there was one among these whose early development and early learning is quite as remarkable as Mill's, namely *William Rowan* Hamilton (b. 1805, d. 1865), famous for his optical and mathematical discoveries (not to be confused with the philosopher William Hamilton). At three years old he could read and do sums; at four he was proficient in geography; at five he could translate from Latin, Greek and Hebrew and was familiar with Homer, Milton and Dryden; at eight he read Italian and French and improvised Latin poems; before he was ten he studied Arabic and Sanskrit, and when he was twelve he wrote a Syriac grammar. And all this with full preservation of bodily and mental health. He appropriated everything without effort and with understanding. He had a strong feeling for poetry, was a friend of Wordsworth, and was himself the author of fine poems. From his ninth year on he was already a good swimmer, so that his
[593] bodily development was not neglected. *Nature.* 3. May 1883. — If one compares his development with Mill's, and both with that of other children, one gets an impression of how very different the original foundation is on which the work of education builds.

[94]) P. 402. Cf. *Stuart Mill*: *Levned.* Danish trans. p. 190 f. 243. 245. 256. — *Bain* (*John Stuart*

Mill. A Criticism. London 1882. p. 163—174) gives an interesting characterization, grounded on various testimonies and on psychological investigation, of the relation between Mill and his wife. — *Theodor Gomperz* (*John Stuart Mill.* Ein Nachruf. Wien 1889. p. 18) describes a visit to Stuart Mill's house. Mill's wife takes part in the conversation, especially by interjecting "brilliant and telling words", and even when the discussion passes over into the purely philosophical domain, "her husband, listening almost with devotion, asked her opinion, which she put forward in clear and well-ordered form".

[95]) P. 403. See *Memories of old Friends from the Journals of Caroline Fox.* Edited by Horace S. Pym. (Tauchniz Edition. Leipzig 1881). I. p. 150 ff. II. p. 218 f. (cf. 77. 143). — These diaries give a most interesting glimpse into the spiritual life of a whole series of distinguished English thinkers, poets and men of science. For a long series of years Caroline Fox wrote down every evening what she remembered of the utterances of the eminent guests in her parents' house.

[96]) P. 407. In my book *Den engelske Filosofi i vor Tid.* Copenhagen 1874. p. 47 I was led by my criticism of some of Mill's views, — a criticism I still maintain, — to use disparaging expressions about Mill which I would not use now, which I already had struck out in the German translation (Einleitung in die englische Philosophie unserer Zeit. Uebers. von Dr. H. Kurella. Leipzig 1889), and which Mill's posthumous treatises on the philosophy of religion, which were not available at the time, moreover refute. Repeated occupation with Mill and his writings has increased my admiration for him. — An excellent characterization of Stuart Mill's personality and significance has recently been given by *Leslie Stephen* in the third volume of his work *The Utilitarians.* (London 1900).

[97]) P. 413. This circular reasoning has been demonstrated with great force and sharpness by *Stanley Jevons* in the fragments of the "Examination of J. S. Mills Philosophy", which his early death prevented him from completing. See *Pure Logic and other Minor Works.* London 1890. p. 254 ff. From his correct criticism of Mill on this point, however, Jevons draws all too far-reaching consequences with regard to Mill's significance as a thinker as a whole. — See also *Arne Löchen*: *Om Stuart Mills Logik.* Kristiania and Copenhagen 1885. p. 165 ff. Jevons's criticism had already been published in "Contemporary Review" 1877—79, so that Löchen was able to profit by it. But Löchen's work is rich in interesting and instructive remarks and investigations, and gives a good characterization of Mill as logician and epistemologist.

[98]) P. 414. On this I already built my criticism of Mill's epistemology in my earlier account. *[594]*
See *Den engelske Filosofi i vor Tid* p. 37 f. (On p. 37, instead of: "comparison is the condition of synthesis" there should stand: "synthesis is the condition of comparison"). — In his essay *Criteria of truth and error* (Mind 1900. p. 15) *Henry Sidgwick* remarks: I think it impossible to establish the general truths of the accepted science by processes of cogent inference on the basis of merely particular premisses; and I think the chief service that J. S. Mill rendered to philosophy, by his elaborate attempt to perform this task, was to make this impossibility as clear as day.

[99]) P. 418. In my *Psykologi i Omrids paa Grundlag af Erfaring* (Copenhagen 1882. 4. ed.

1898) I have tried to make "the unifying principle" the guiding point of view and to show how both the laws of association and the laws of sensations lead back to this principle, which Kant (under the name of synthesis) had already made the basis of the conception of consciousness. It has therefore surprised me more than once to be counted without further ado among the adherents of the English school. *Paul Carus* (*Primer of Philosophy.* Chicago 1893. p. 175) thus names me among "the psychologists who hold that the doctrine of association gives a key to all the problems of the soul". *Wundt* (*Physiologische Psychologie.* 4. Aufl. Leipzig 1893. II. p. 482) has named me among "the representatives of association psychology", who hold that "all mental processes can be derived from the associations". Already in my book on English philosophy (1874) I criticized association psychology, the conception that makes the sensations and representations in their independence the real in consciousness and regards association as an external connection between them, inessential to themselves. And through the whole of my Psykologi there runs a criticism of this conception, combined, to be sure, with attempts to preserve the truths it has led to discovering. I also believe that association psychology can be corrected only by seeking out its own hidden presuppositions, not by assuming a special faculty of thought or "apperception" which should cancel or limit the associations. Cf. my *Psykologiske Undersøgelser* p. 5 — 6 and 59 — 67.

[100]) P. 436. A survey of the attempts in the philosophy of religion in England before and after Mill's time is given by O. *Pfleiderer*: *Die Entwickelung der protestantischen Theologie in Deutschland seit Kant und in Grossbritannien seit 1825.* Freiburg 1891. (The account of Mill's views in this work is, however, not accurate. Pfleiderer commits a bad historical error on p. 407 by having Hamilton, who had died 9 years before Mill's "Examination" appeared, reply to Mill's criticism!!). — Cf. also N. H. *Marshall*: *Die gegenwärtigen Richtungen der Religionsphilosophie in England und ihre erkenntnisstheoretische Grundlage.* Berlin 1902. — The somewhat naive and often not very profound character of Mill's treatises on the philosophy of religion speaks for
[595] the correctness of *Leslie Stephen's* conjecture (*The Utilitarians.* III. p. 86) that Mill did not know Hume's writings on the philosophy of religion, any more than he seems to have known the "Treatise".

[101]) P. 442. A somewhat fuller biography and characterization of Darwin I have given in my popular sketch: *Charles Darwin.* Copenhagen 1889. (Studentersamfundets Smaaskrifter Nr. 91—92). Some passages of this little book will be found again in the account. — The chief source is *The Life and Letters of Charles Darwin, including an autobiographical chapter.* Edited by Francis Darwin. London 1887. (Translated in "Bibliotek for de tusen hjem"). To this must be added the later published *More Letters of Charles Darwin.* A record of his work in a series of hitherto unpublished letters. Edited by Francis Darwin and A. C. Seward. London 1903.

[102]) P. 452. It is not impossible that this utterance is owed to the recollection of the well-known scene at the meeting of naturalists in Oxford in 1860 between Huxley and the Bishop of Oxford (see Life and Letters. II. p. 320 —22), — except that Huxley preferred the ape as ancestor not to a

savage but to — the Bishop of Oxford.

[103]) P. 452. Cf. my remarks on the relation between Darwinism and ethics in *Grundlaget for den humane Etik* (1876). p. 30—39, and in *Etiske Undersøgelser* (1891). p. 22 f.; 32—38.

[104]) P. 455. For Spencer's biography and course of development I use, besides utterances in his writings (especially the interesting remarks in the essay *The Classification of the Sciences, with Reasons for Dissenting from the Philosophy of M. Comte.* 3. ed. p. 31, 34, 46), above all a short biography by Prof. *Youmans*, based on information from Spencer, in the American periodical *The Popular Science Monthly* (March 1876), which I translated at the time in my translation of "Mindre Afhandlinger af Herbert Spencer" (Copenhagen 1877), and a draft of a speech by Youmans, found in the report of a celebration for Spencer in New York in the autumn of 1882. (*Herbert Spencer on the Americans, and the Americans on Herbert Spencer.* New York 1882. p. 68—76).

[105]) P. 461. This first part of "First Principles" has been translated into Danish under the title *Religion og Filosofi* by S. *Hansen* (Copenhagen 1887).

[106]) P. 463. Already in my book *Den engelske Filosofi i vor Tid* (1874) p. 123 f., 132, 145 f. I put forward this objection.

[107]) P. 470. In the year 1876 I drew Spencer's attention to these contradictions in a private letter, and in his answer he explained them in the way stated, remarking that he had not until then noticed them.

[108]) P. 470. Cf. *First Principles.* 3. ed. p. 318 f. "Although the evolution of the various products of human activity cannot be said directly to furnish examples of the accumulation of matter and the dissipation of motion [which Spencer assumes in every evolution, the parts of the whole that is forming losing their independent motion], they are nevertheless *indirect* examples of *[596]* it." p. 391: "The phenomena that are subjectively apprehended as changes in consciousness are objectively apprehended as nervous processes, which science now interprets as processes of motion." — *Principles of Psychology.* 2. ed. I. p. 508: "Although the evolution of mind itself cannot be explained by a series of deductions from the conservation of force, it is nevertheless possible that its correlate (its obverse). The evolution of physical changes in a physical organ, can be so explained."

[108 b]) P. 474. Strangely enough, even the most acute of Spencer's English critics, *James Ward*, has overlooked this in his work *Naturalism and Agnosticism* (London 1899), interesting in epistemological respects. He puts, e.g., "the universe" without further ado in place of "a universe". Cf. the work named, I. p. 192 with 187.

[109]) P. 481. Already in *Den engelske Filosofi i vor Tid* (1874). p. 171—173 I put forward this criticism of Spencer's theory of knowledge. In my *Psykologi* VII C, 4 I have discussed this subject again.

[110]) P. 486. An interesting criticism of Spencer's sociology is found in *Émile Durckheim*: *De la division du travail social.* Paris 1893. p. 218—247.

[111]) P. 488. Cf., concerning the difficulties for a scientific ethics, my work *Etiske Undersøgelser.*

Copenhagen 1891. — An instructive criticism of Spencer's attempt to ground ethics immediately on biology (on the organic force and breadth of life instead of on its value as experienced in pleasure and pain) — and thereby of Spencer's inclination to apply a deductive method in ethics, is given in *Henry Sidgwick's Lectures on the Ethics of Green, Spencer and Martineau.* (London 1902). (Thus Sidgwick states: So long as life is partially painful, it may still be true in any particular case that conduct conducive to maximum quantity of life is not conducive to maximum of pleasure. p. 148. — Sidgwick also emphasizes the possibility of a conflict between what the development of the individual organism and what the development of the social organism requires. p. 143 f).

[112]) P. 492. In the concluding section of the German translation of my book on modern English philosophy I have already named the works of Boole, Jevons and Sidgwick, besides several others that I must pass over here. (*Einleitung in die englische Philosophie unserer Zeit.* Leipzig 1889. p. 243—249).